



Service

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Service Operations and Processes

Service Operations and Processes includes service order management to plan and deliver services, service request management to process services requests originating from various sources, in-house service to manage services in-house at service centers, and case management to process issues that involve multiple processing steps.

[Service Order Management](#)

Service Order Management provides a variety of functions that allow you to set up end-to-end service processes. It is integrated with Service Contract Management as well as with logistics, inventory management, pricing, billing, and Finance.

[Service Request Management](#)

You use Service Request Management to monitor and fulfill requests for service by your customers. Service requests can be used internally by companies where a department delivers services as well as in external customer-facing scenarios. They enable you to measure and evaluate the quality of services provided.

[In-House Service](#)

In-House Service supports companies that offer maintenance and repair services for service objects (in-house service objects).

[Service with Advanced Execution](#)

Service with Advanced Execution is a process that is suitable for services which require extensive planning and execution. They generally have longer service times with a requirement to bill the time and service parts used for the service. The core characteristic of this process is the integration of the commercial aspects of Service with the planning and execution capabilities of Maintenance Management.

[Case Management](#)

Case Management enables you to consolidate, manage, and process information about a complex problem or issue in a central collection point, the case. Within a case, you can group diverse information, such as business partners, transactions, products, and documents. This information can reside in different physical systems.

Service Order Management

Service Order Management provides a variety of functions that allow you to set up end-to-end service processes. It is integrated with Service Contract Management as well as with logistics, inventory management, pricing, billing, and Finance.

i Note

The following images contain links to documentation you might be most interested in. Click on each of the areas to learn more.

Service Orders



Please note that image maps are not interactive in PDF outputs.

Use service orders to provide short-term agreements and add planning

Service Quotations



Please note that image maps are not interactive in PDF outputs.

Use service quotations to provide binding offers and cost estimates to

Service Order Templates



Please note that image maps are not interactive in PDF outputs.

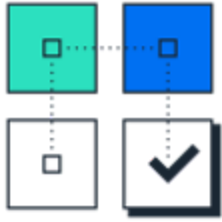
Use service order templates to specify reusable sets of service-related data and

data for the provision of one-off services to your customers.

your customers and quickly create follow-up service orders.

minimize the time to create service orders or service quotations.

Service Confirmations



Please note that image maps are not interactive in PDF outputs.

Use service confirmations to precisely record time and materials consumed to provide services.

Cross Functions



Please note that image maps are not interactive in PDF outputs.

Use functions that are available in more than one service transaction throughout Service Order Management.

Apps



Please note that image maps are not interactive in PDF outputs.

Use the apps in Service Order Management to support your service processes.

Service Orders

A service order is a short-term agreement between a service provider and a service recipient, in which the service recipient orders one-off services from the service provider.

Use

You can use a service order to plan and execute services for customers. A service order contains all the information needed to plan, execute, and bill for a service such as customer details, material or service to be delivered, dates and times for service activities, and any related documents or attachments. Service orders help in the planning and scheduling of resources, monitoring service progress, and generating invoices for completed services.

Use of Items in Service Orders

A service order consists of a header and one or more items. You use items in a service order to reflect the purposes or needs of a service and to create account assignment objects so that cost and revenue from the service order can be posted to Financial Accounting. You can add items to a service order and edit them on the [Items](#) tab. For more information on different types of items and how to use items in service orders, see [Items in Service Orders](#).

Basic Settings

The standard transaction type of a service order is predefined in the system as SRV0. Transaction types define the characteristics and attributes of a business transaction and determine how the transaction is processed, for example, the profiles and procedures used to determine texts, business partners, statuses, and organizational data. If the standard settings do not meet your business requirements, you can configure the transaction types using the Customizing activities under [Service > Transactions > Basic Settings](#). For more information, see [Business Transaction](#).

Life Cycle Statuses

A service order and each item in a service order can have the following life cycle statuses:

Life Cycle Status	Description
Open	This is the default status that all service orders or service order items have after they're created. It indicates that you're free to edit them.
In Process	You can manually set the status of a service order or a service order item to In Process to indicate that a service is taking place. For example, an executing service employee has been scheduled to visit a customer.
Released	This status indicates that you have finished planning a service and the service execution can be started. When this status is set, a service order or a service order item has been finalized and no further changes are required. It's possible to create a service confirmation against the service order that has the status Released .
Completed	This status indicates that a service has been executed. You can no longer create service confirmations based on a service order that has the status Completed .

i Note

- If a service order item is set to the status **Released** or **Completed**, it's not possible to perform the following actions:
 - Change the item category of the item.
 - Change the service contract that has been assigned to the item.

Key Features

Reference Objects

You can use reference objects in a service order at header or item level to simplify the process of entering certain data for a service order or an item. The details that has been predefined for the reference objects can be transferred from the master data to the service order or service order item. For more information, see [Reference Objects for Service Orders](#).

Credit Checks in Service Orders

You can perform credit checks in service orders. For more information, see [Credit Checks in Service Orders](#).

Multilevel Categorization

You can use the function **Multilevel Categorization** in service orders. For example, you can create a categorization for a reported defect. For more information, see [Multilevel Categorization](#).

You can also use **Auto Suggest**, which is part of the functionality of multilevel categorization, for service orders. You can create a category in a service order and assign a service order template to the category. Next time when you create a service order, the system can automatically fill the service order with the data from the service order template. For more information, see [Auto Suggest](#).

Knowledge Articles

You can obtain information from knowledge articles to solve issues related to service orders. For more information, see [Knowledge Articles](#).

Questionnaires

You can display a list of relevant questionnaires and assign questionnaires to the order depending on the transaction type of the service order.

Billing

You can release service orders for billing. After you've released the service order for billing, the system creates the required billing document requests (BDRs). For more information on billing, see [Billing](#).

You can manually schedule jobs to create billing document requests for billable service order items, billable service confirmation items, and items with ad hoc billing plan using the report [Generate BDRs for Service Orders](#) (CRMS4_SERVORD_BDR_GENERATE). You can find the report on the [SAP Easy Access](#) menu under [►Service ► Service Processes ►](#).

Ad Hoc Billing for Service Orders

You can use ad hoc billing to define billing values and dates freely for fixed price service order items. For more information, see [Ad Hoc Billing for Service Orders](#).

Pricing

The system determines price information when you create or edit a service order. For more information, see [Pricing](#).

Accounting Indicator

You can use an accounting indicator in service orders with the pricing condition types DAI1 and DAI2 to influence pricing procedure and differentiate items based on discount or surcharge rates.

To set the condition types for accounting indicators, you use the activity [Set Condition Types for Pricing in Sales](#) in Customizing for [Sales and Distribution](#) under [►Basic Functions ► Pricing ► Pricing Control ► Define Condition Types ►](#). To enable the transfer of the value of accounting indicators from billing to accounting, you also need to make sure that the accounting indicator is taken into account in account assignment by entering **B** in the [Rel.Acc.Assig](#) field in the [Control Data 2](#) section in the same Customizing activity [Define Condition Types](#).

Trade Compliance Check

The integration with SAP Global Trade Services (GTS) allows you to perform embargo checks, sanctioned party list screening, and legal controls for service orders and service quotations. For more information, see [Integration with SAP Global Trade Services](#).

Release of Service Orders

You can set a service order and each item in the service order to the status [Released](#) when you edit the service order. You can also use the [Release](#) button on service order header to release a service order along with all the items in the service order, provided that the service order and the items don't have any errors.

i Note

The [Release](#) button is only active when the service order is in display mode.

Rejection of Service Orders

You can reject a service order by choosing **Reject All Items**. You can also reject single items in a service order by assigning a rejection reason to each item. You can revert the rejection when you remove the rejection reason of single items or choose **Reset Item Rejection** to remove the rejection for all rejected items.

You can define your own rejection reasons, group them in a rejection profile, and then assign the rejection profile to a transaction type and a sales organization for service orders. With this, you can select from the predefined reasons in a profile when you reject a service order or a service order item. You can do so with the activities in Customizing under **Service > Transactions > Settings for Service Transactions > Settings for Rejection Reasons for Service Transactions**.

i Note

If a service quotation has been rejected, you cannot create a follow-up service order for the service quotation. If you create a follow-up service order for a service quotation, the rejected items in the service quotation are not copied to the service order. For more information, see [Rejecting Service Quotations](#).

Business Completion of Service Orders

You can set a service order to **Business Completed** to indicate that no cost and revenue can be posted from the service order to Financial Accounting anymore. The **Business Completed** status is a prerequisite for archiving service orders.

After you have set a service order to **Business Completed**, you can still manually set the service order to **Business Reopened** to revise the business completion of the service order. You might reopen the service order for business, for example, if further costs that are related to the service order have been occurred subsequently and have to be posted to Financial Accounting and Controlling. For more information, see [Finalizing the Service Order](#).

i Note

You can only set a service order to **Business Completed** or **Business Reopened** when item-based accounting is active.

Qualification Requirements

You can define qualification requirements, such as skills, knowledge, or abilities, necessary to perform services specified in a service transaction. For more information, see [Qualifications in Service Transactions](#).

Involved Apps

[Manage Service Orders](#)

Items in Service Orders

You can learn about items in service orders in this topic.

Use

A service order consists of a header and one or more items. You use items in a service order to reflect the purposes of a service and to create account assignment objects so that cost and revenue from the service order can be posted to Financial Accounting.

The following table provides you with an overview of what types of items you can create in a service order:

Item Type	Description
Service Item/Service Product Item	You can use a service item to document customer wishes about how the service is to be performed, or to enter the planning-relevant data such as the service teams, estimated duration of performance, and dates.
External Service Item	You can use an external service item to provide details of a procured service performed by an external workforce.
Service Part Item	A spare part that is required to perform a service. It can be a spare part that is available in stock and can be picked up by a technician, or one that is not in stock and needs to be procured.
Expense Item	You can use an expense item to specify expenses that may be incurred during a service process, for example, travel costs of an executing service employee.
Sales Item	You can use a sales item to trigger the creation of a follow-up sales order.
Intercompany Service Item	You can use an intercompany service item to trigger the creation of an intercompany service order, which is a business transaction used to distribute a service between two companies. For more information, see Intercompany Service Orders .
Execution Order Item	You can add an execution order item to a service order. An execution order item is associated with a corresponding billable maintenance order. It is used to define the scope and terms of a service requirement as part of the Service with Advanced Execution process. For more information, see Service with Advanced Execution .
Price Item	You can use a price item in a service order to reflect a surcharge for additional service provided. For more information, see Price Items in Service Orders and Service Quotations .
Tool Item	An item within a service order that represents a tool needed during the service execution to fulfill the service and that can be reused for the next service. For more information, see Tool Items in Service Transactions .

Item Categories

When you add an item to a service order, you assign an item category to the item. An item category specifies the properties and attributes of each item. You use item categories to determine how items are processed, for example, the manner in which the items are billed.

The following table gives you an overview of which item categories you can use in a service order:

Item Type	Item Category
Service Item/Service Product Item	• SRVP: Service Item • SF0S: Fixed Price Service Item • SB0S: Service Item (Ad Hoc Billing)

Item Type	Item Category
External Service Item	SRVW
Service Part Item	• SRVM: Service Part Item • SF0M: Fixed Price Service Part Item • SB0M: Service Part (Ad Hoc Billing) • SPRS: Stock Transfer Service Part • SRVG: Service Part Manual Goods Issue • SRVU: Advance Shipment of Procured Service Parts • SRVV: Advance Shipment of Stock Service Parts
Expense Item	• SRVE: Expense Item • SF0E: Fixed Price Expense Item • SB0E: Expense Item (Ad Hoc Billing)
Sales Item	SRVS
Intercompany Service Item	• SVAI: Intercompany Service Item (Ad Hoc Billing) • SRVI: Intercompany Service Item (Fixed Price)
Execution Order Item	MSI1
Price Item	SVPI
Tool Item	• SRVT • SB0T: Service Order Tool Item (Ad Hoc Billing) • SF0T: Fixed Price Tool Item (Order)

i Note

You can only change the item category of a service order item when the following conditions are met:

- The item has the status **Open** or **In Process**.
- You have not created any follow-up documents such as billing documents for the item.

Settings

The item categories described in the overview table are predefined in the system.

If the standard settings do not meet your business requirements, you can also configure your own item categories using the Customizing activities under **Service > Transactions > Basic Settings**.

Products and Item Category Groups

Products are used in service order items to represent which type of items you sell to customers. Item category groups are attributes in product master data that are used to control how item categories are determined in service orders. An item category group contains different item categories. You can assign an item category group to a product in the product master data. This way the item categories in the item category group can be determined for the product in a service order item.

You can use the following item category groups in products:

Item Category Group	Description
SRVP	Service Product
SRVE	Expense Item
SRVM	Service Material
NORM	Standard Item
DIEN	Execution Order Item (Service with Delivery)
LEIS	Execution Order Item (Service without Delivery)
SRVT	Service Tool

→ Remember

Make sure that the product that you enter for a service order item has the correct item category group assigned in the product master data. For example, if you want to use the product **Service Product** in an item, which represents the service item, you should assign the item category group SRVP (service product) to the product.

Key Features

You can use the following functions at service order item level:

Copy from Template

You can add predefined items to service orders by choosing **Copy from Template** when you edit items. You can create and display templates for service order items. Service order templates contain reusable sets of service-related data. They do not contain any data that describes the actual execution. When you copy the templates to service orders, only templates that have the status **Released** are displayed. For more information, see [Service Order Templates](#).

Product Proposals for Service Order Items

You can use the function Product Proposal in the items list of a service order to identify the service parts, expenses, and services relevant for a service order without entering the service part IDs. For more information, see [Product Proposals for Service Order Items](#).

Configurable Products

You can use configurable products or product variants to customize complex products with single-level variant configuration for your service processes. For more information, see [Configurable Products and Product Variants](#).

Related Information

[Service Orders](#)

Service Orders with Other Service Transactions

You can link service orders to other service transactions such as service order templates, service quotations, and service confirmations with the **Create Follow-Up** function. This allows for efficient creation of service transactions and provides a history of related transactions.

Use

Depending on your Customizing settings, you can create a service order as the follow-up transaction of another service transaction. For example, if you create a service order as a follow-up transaction of a service order template, you can select items from the template and copy them to the service order.

You can also create other service transactions as the follow-up transactions of a service order. When you choose **Create Follow-Up** and select a follow-up transaction that you want to create for a service order, the system proposes items for selection in a popup and the item you select is automatically copied to the service order.

A follow-up transaction is a business object that refers to an existing business object. When you create a new transaction that relates to an existing transaction, you can create the new transaction as a follow-up transaction. When you do this, data from the original transaction is copied to the new transaction and the transactions are therefore linked to each other. This linking provides you with information about the history of a particular transaction. You can view this history on the **Transaction History** tab in each service transaction. This is useful for managing and tracking service-related data and documents relevant for service transactions, for example, billing documents. For more information, see [Follow-Up Transactions](#).

Example

The following diagram illustrates an example of how you can use a service order with other service transactions.



Please note that image maps are not interactive in PDF outputs.

Reference Objects for Service Orders

Use

When you create or edit a service order, you can specify a product, a functional location, or a piece of equipment as the reference object in the service order header and items.

Features

- Search for equipment and functional locations

When specifying a product, functional location or a piece of equipment as the reference object, you can search for existing products, functional locations or equipment.

Reference Object	Search Criteria
Functional location	◦ Functional location ID ◦ Functional location description ◦ Functional location category ◦ Functional location structure indicator ◦ Object type ◦ Partner function ◦ Business partner ID ◦ Superior functional location
Equipment	◦ Equipment ID ◦ Equipment description ◦ Equipment category ◦ Functional location structure indicator ◦ Business partner ID ◦ Partner function ◦ Reference product ID ◦ Serial number

Reference Object	Search Criteria
Product	○ Product ID ○ Product description ○ Product type ○ Product hierarchy ○ Configurable product ○ Sales organization ID ○ Distribution channel ID ○ Product type group ○ GTIN

i Note

Functional location, equipment, and product are mutually exclusive when you specify them as a reference object.

When you specify a functional location or equipment as the reference object and save the service order:

- The service order can be saved with errors if the equipment or functional location has been deactivated or marked for deletion.
- The service order cannot be saved if the equipment or functional location has been archived (equipment and functional locations that have already been used in service transactions cannot be archived).

• Warranty determination

If a warranty is assigned to a functional location or equipment which has been specified as the reference object in a service order, the warranty can be automatically determined in the service order.

○ Main reference object

You can specify functional locations or equipment as reference objects in both the service order header and items, and these specified functional locations or equipment will be displayed in the **Reference Objects** assignment block. In this assignment block, you can add new reference objects and specify the main reference object. In this way, you can check and view the warranty assigned to your desired product, equipment, and functional location by specifying the product, equipment, or functional location as the main reference object.

i Note

- Only the warranty assigned to the main reference object will be automatically determined in the service order header.
- Warranty is determined on an item only when the item's service product is a part of the master warranty that is assigned to the main reference object.
- The determined warranties cannot be changed.

• Navigate to the Fiori app to view details on equipment and functional locations

After you have defined a functional location or a piece of equipment as the reference object, the description of the functional location or equipment is automatically displayed. You can choose the description to navigate to the Fiori app to

Intercompany Service Orders

An intercompany service order is a business transaction used to distribute a service between two companies with different company codes that belong to the same organization.

Use

A company that sells a service to a customer can use an intercompany service order to distribute the service to another company. The company that receives the intercompany service order executes the service and posts the revenues from the service as costs (accounts payable) to the company that sells the service.

i Note

You can only use fixed price service items for intercompany service orders. The items that are billed based on the consumption of time and materials cannot be used for intercompany service orders.

Intercompany Item Categories

You can use the following item categories for intercompany service orders:

- [Intercompany Service Item \(Fixed Price\) \(SRVI\)](#)
- [Intercompany Service Item \(Ad Hoc Billing\) \(SVAI\)](#)
- [Intercompany Service Item Collective Accounting \(SRHI\)](#)

Prerequisites

To use intercompany service orders, you need to fulfill the following prerequisites:

- Activate item-based accounting in the Customizing activity [Enable Item-Based Accounting for Service](#) in Customizing for [Service](#) under [►►Transactions►Settings for Service Transactions►Integration►](#). For more information, see [Setting Up Item-Based Accounting](#).
- Define the relevant data required for creating intercompany service orders using the Customizing activity under [►►Service►Transactions►Settings for Service Transactions►Intercompany Billing►Define Settings for Creation of Intercompany Service Orders►](#).
- Define the business partners for sales organizations in the Customizing activity [Define Internal Customer Number By Sales Organization](#) under [►►Sales and Distribution►Billing►Intercompany Billing►](#).
- Specify that the item category (BUS2000140) used for a fixed price service item is relevant for intercompany service orders. You can do so in the Customizing activity under [►►Service►Transactions►Basic Settings►Define Item Categories►](#).
- Specify the receiving sales organization, which is the sales organization that receives an intercompany service order and executes a service for the sales organization that sells the service to a customer.
- Set up automatic posting to supplier account for intercompany billing with SAP Electronic Data Interchange (EDI) and set up invoice receipt via EDI. For more information on how to make these settings, see the section [Electronic Data Exchange \(EDI\) for Intercompany Service Orders](#).

Electronic Data Exchange (EDI) for Intercompany Service Orders

To send billing documents from one company code to another company code and post them to financial accounting via EDI, you need to follow the steps as follows:

1. Generate IDocs.

- If you use the output management framework, do the following: In the [Output Parameter Determination](#) app, choose **IDoc** as the channel for the output type **Billing Document**.
- If you use the SAP S/4HANA output control (CA - GTF - OC) framework, do the following: Choose a billing type and assign the output determination procedure **V40000** to the billing type. You can then create an output condition record with transaction **VV31** for the output type **RD04** to trigger the use of IDocs.

2. Set up automatic posting to supplier account with EDI.

To do this, perform the Customizing activity in SAP Customizing Implementation Guide under [Sales and Distribution > Billing > Intercompany Billing > Automatic Posting to Supplier Account \(SAP-EDI\)](#).

i Note

Only choose intercompany service orders as the recipients.

3. Set up invoice receipt via EDI.

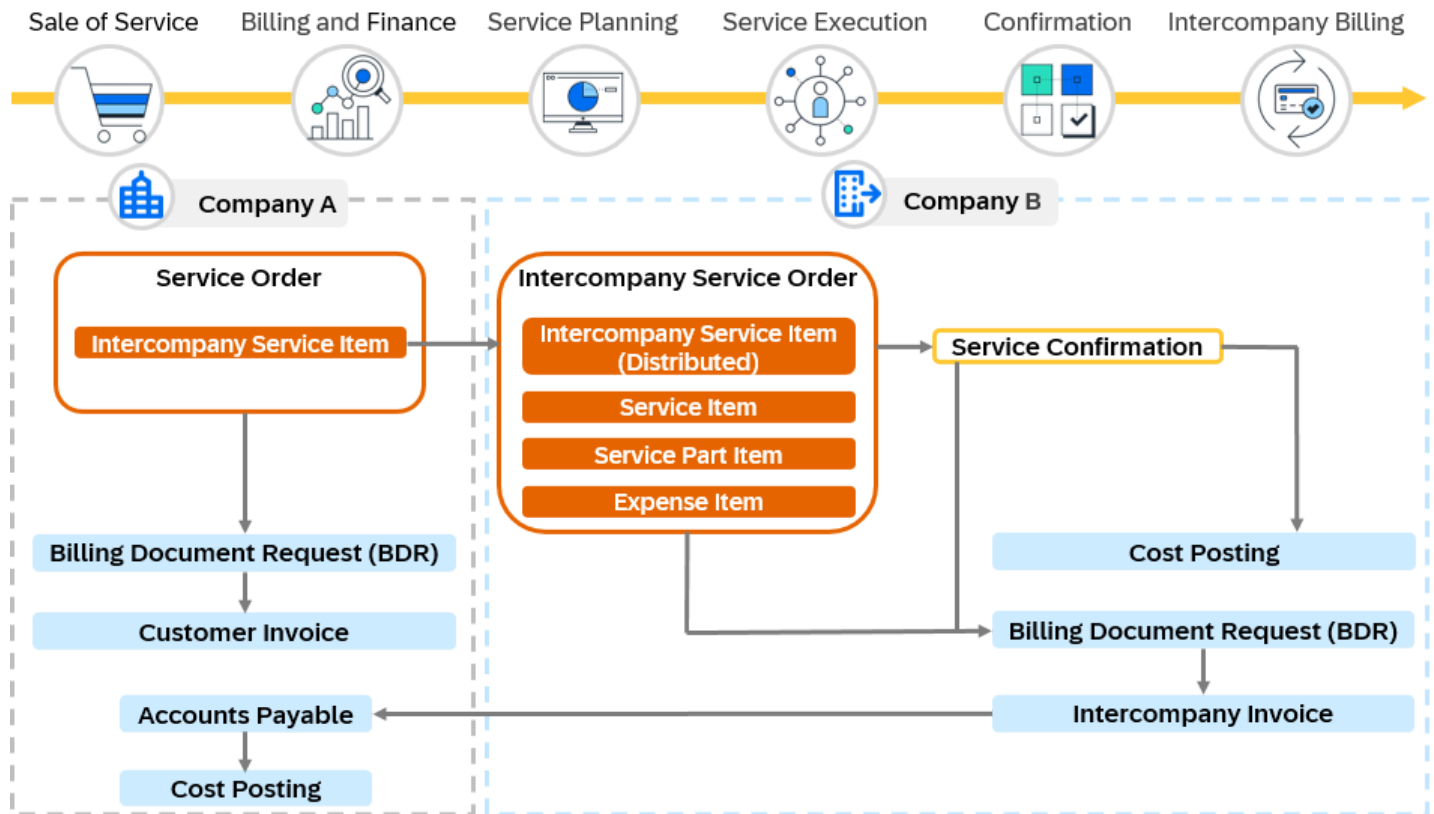
To do this, carry out the Customizing activities in the order specified under [Financial Accounting > Accounts Receivable and Accounts Payable > Business Transactions > Incoming Invoices/Credit Memos > EDI](#).

Process

The following diagram illustrates a sample process of how to use intercompany service orders.

i Note

- Company A: the company that sells and distributes a service to another company.
- Company B: the company that receives the intercompany service order and executes the service.



→ Tip

You can create more than one intercompany item in a service order. Items with the same receiving sales organizations will be created in the same intercompany service order.

Copying of Data During Intercompany Service Order Creation

In the intercompany scenarios, the transition from a service order of the company that sells and distributes the service to an intercompany service order is automated to ensure operational continuity. When a fixed price intercompany service item is released, the system creates a corresponding intercompany service order in the receiving company. During this process, all essential operational data is transferred, while sensitive or company-specific information is recalculated or redefined in the receiving organization.

The following data is **copied** from the original service order to the intercompany service order:

- Product ID and quantity including unit of measure
- Attachments (GOS) and notes
- Begin and end dates (both requested and planned)
- Reference objects (including equipment data, functional location, serial number, and the **In-House Service-Related** indicator)
- Contact person and ship-to party (including partner address data if it was changed in the transaction)
- Multi-level categorization (if profiles match)
- Configuration data for configurable service products
- Purchase order number, date, and reference field data (both for sold-to party and ship-to party)

The following data is **not copied** and is instead recalculated or newly defined in the receiving company:

- Service contract assignment

- Billing data (payment method, terms, tax type, etc.)
- Pricing data (pricing conditions are recalculated)
- Service level agreement, credit status, profit center
- Employee responsible, service employee group, executing employee
- Organizational data (redefined based on the sold-to party in the intercompany order)

Once the intercompany service order is created, key fields such as product ID and quantity become non-editable, preserving the integrity of the original request. Any changes made in the distributed item of the intercompany order do not retroactively update the source service order. The relationship between the two orders is maintained in the transaction history and process flow, providing full traceability and transparency throughout the lifecycle of the service.

This approach minimizes manual intervention and ensures that operational requirements are met during the handover of service execution.

Intercompany Reference Item Number

This is the number of the intercompany item that is used as a reference for manually entered items in an intercompany service order. You need to enter an intercompany reference item number for any items that you have entered manually in an intercompany service order.

When an intercompany service order is created, an intercompany item is added to it automatically. This intercompany item refers to the intercompany item created in the service order by the company that sells and distributes the service. When there are multiple intercompany items in the intercompany service order and you add items to the intercompany service order, you need to use **intercompany reference item number** to indicate to which intercompany item the added items belong. This ensures that the revenues from the items in an intercompany service order are posted as costs (accounts payable) to the correct account assignment objects of the intercompany items created by the company that sells and distributes the service.

i Note

If the intercompany service order contains only one intercompany item, the number of this intercompany item is entered as the intercompany reference item number automatically.

Cascade Intercompany Items

When you receive an intercompany item from a company that sells a service, you can forward this intercompany item to another company so that they can execute the service.

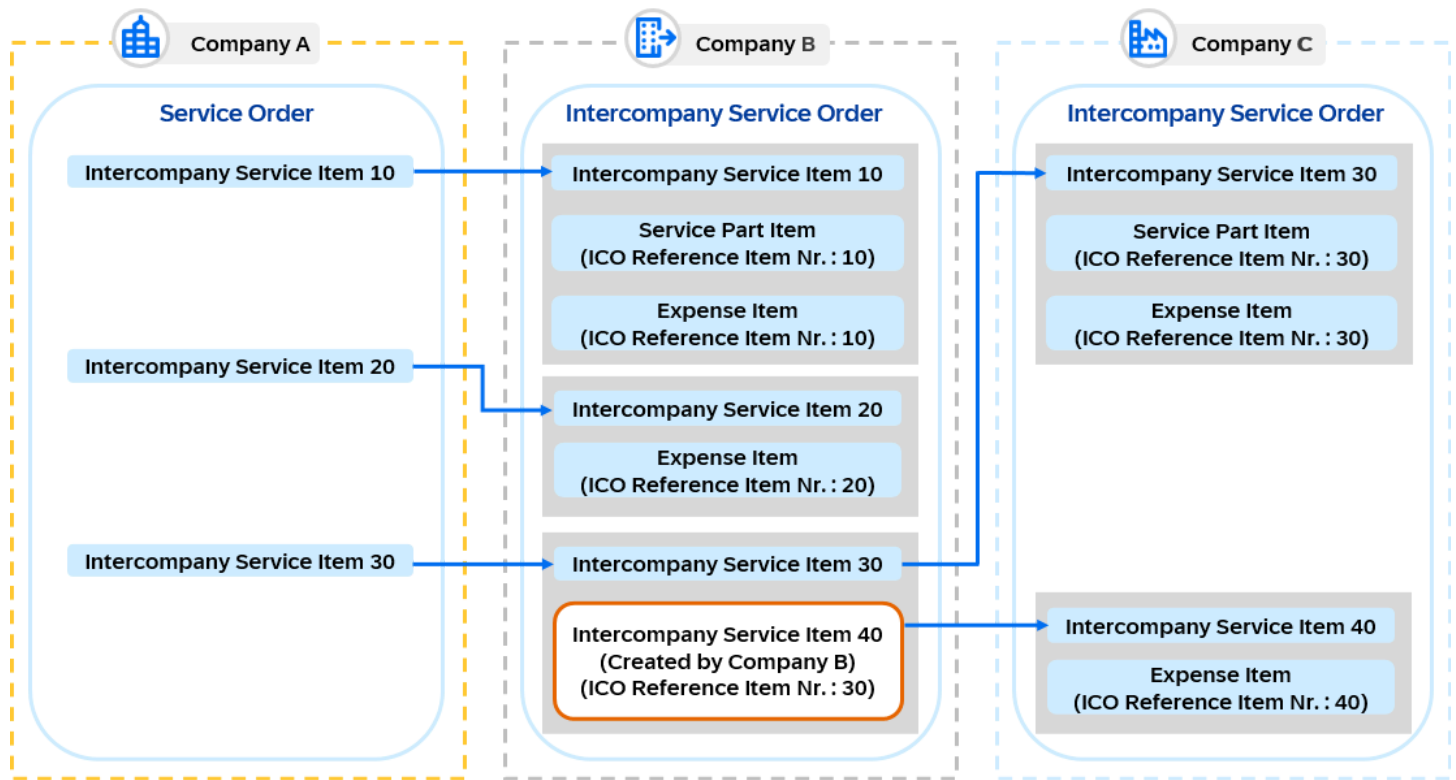
You can also add new intercompany items in an intercompany service order and distribute the new intercompany items to other companies for execution.

Example

The following diagram shows you an example of how you can cascade intercompany items and use intercompany reference item number.

i Note

- The arrows in the diagram represent the action of releasing an item.
- ICO Reference Item Nr.: Intercompany Reference Item Number



Service Hierarchy in Intercompany Service Process

You can use the concept of service hierarchy with collective accounting in intercompany service processes. The feature lets you use intercompany items as main and subitems in a service hierarchy. Costs incurred by the intercompany service order are collected under the account assignment object of the intercompany item specified as the main item of a service hierarchy. A service hierarchy can be used both in the service order of the company that sells the service and in the service order of the company that executes the service. That means that it's possible to have a service hierarchy only in the first service order, only in the second service order, or in both. Both service orders follow their respective accounting logic.

To specify an intercompany item as a main item in a service hierarchy, you use the item category **Intercompany Service Item Collective Accounting** (SRHI).

i Note

You use SRHI in the service order of the company that sells the service. In the intercompany service order of the company that executes the service, you use **Service Item with Collective Accounting** (SHSV) and **Service Item with Collective Accounting (Fixed Price)** (SHSF).

If you manually add items as subitems in a service hierarchy of an intercompany service order, these subitems automatically inherit the intercompany reference item number from the main item of a service hierarchy.

For more information about service hierarchy, see [Service Hierarchy with Collective Accounting](#).

Planned Costs and Revenues for Intercompany Service Order Item

Planned costs and planned revenues are now calculated for intercompany service order items in the service order. This generates more accurate financial forecasting and efficient resource allocation.

The expected cost is determined by adding together:

1. The net value of the item in the intercompany order.
2. The net value of any manually created items within the same order that reference it using the ICO reference number.

3. The net value of any sub-items within the service hierarchy of the aforementioned items in the intercompany order.

i Note

No planned costs and revenues are calculated when the item of the customer-facing service order is part of a service hierarchy (whether it is a main item or a sub-item).

Integration

Event-Based Revenue Recognition

The integration of Event-Based Revenue Recognition with intercompany service orders is supported. For more information, see [Event-Based Revenue Recognition for Service Orders](#).

Service with Advanced Execution

You can add an execution order item to an intercompany service order, release the execution order item, and trigger the creation of a maintenance order. In this scenario, the debit memos from the maintenance order are transferred from the intercompany service order to the customer-facing service order. You can view the debit memos in the transaction history of the service order and navigate to the debit memo details as well. For more information, see [Service with Advanced Execution](#).

Product Proposals for Service Order Items

You can use the function Product Proposal in the items list of a service order to identify the service parts, expenses, and services relevant for a service order without entering the service part IDs.

Use

When you choose this function, the system searches for the product or the equipment that was created as main reference object, and checks whether it has a BOM (Bill of Material) assigned or not. If an assignment exists, and if the BOM has been defined with usage **S** (Service Management), the **Product Proposal** displays the relevant service parts and the BOM items defined for the equipment. For more information, see [Maintain Bill of Material](#).

Master Data Maintenance

To use the function **Product Proposal** in a service order, you need to fulfill the following prerequisites in master data maintenance:

Create Material BOM

1. Log on to the SAP Fiori launchpad as BOM Engineer (Role: SAP_BR_BOM_ENGINEER).
2. In the app **Maintain Bill of Material**, choose **Create BOM**. Create a BOM with usage **S** and assign the relevant components (service parts and optionally service products) to it.
3. Enter BOM item category, product ID, and quantity, and then save your changes.

i Note

Use BOM item category **L** for service parts or pick-from-stock items and **N** for other items.

Create Equipment BOM

1. Log on to the SAP Fiori launchpad as Maintenance Planner (Role: SAP_BR_MAINTENANCE_PLANNER).

2. Create a technical object and choose **Equipment** as the technical object type. Assign a material ID as construction type for this equipment in the field **Structure**, or as reference product for this equipment in the field **SerData**.
3. Alternatively, you can create or change Material BOM as the construction type or reference product for this equipment in the App **Create Equipment BOM** or **Change Equipment BOM**.

Process

1. When you create a service order, enter the equipment ID or product ID that you've created as main reference object.
2. In the items list of the service order, choose **Product Proposal**. In the dialog box you can see a list where the equipment, the construction product, the reference products, and the BOM items are displayed in a hierarchy.
3. Select the products you want to use for the service order.

i Note

The quantities of the selected BOM items are copied to the service order items as well.

Execution Statuses of Service Orders

You can process service orders in systems beyond Service such as in Maintenance Management and in SAP Field Service Management. You can check the status of service execution in Service with Advanced Execution in Maintenance Management and in SAP Field Service Management.

Maintenance Management

Service is integrated with Maintenance Management in the process Service with Advanced Execution.

When you set the status of a service order to **Released**, the system creates a maintenance order as a follow-up to the execution order item. The service order is then processed in the maintenance order. The execution status shows the progress of the service order in the maintenance order.

The following execution statuses are available:

Execution Status in Service Order	What Happens in Service	System Status of the Maintenance Order
Not Available	All service order items have the status Not Available , or at least one item has the status Not Available and all other items have the status Not Relevant .	Not applicable
Execution Planning	At least one service order item has the status Execution Planning .	Created
Planning Completed	The planning process for a service has been completed.	Order Approved
In Execution	At least one service order item has the status In Execution , or at least one service order item has the status Execution Done .	Released
Execution Done	The service process has been finalized in the maintenance order.	Technically Completed

Execution Status in Service Order	What Happens in Service	System Status of the Maintenance Order
Canceled	All service order items have the execution status Not Available , or at least one item has the status Not Available , and all others have the status Not Relevant .	Do Not Execute
Not Relevant	No execution order items have been added to the service order. The service order is not processed in Maintenance Management.	Not applicable

Execution Status in Service Order Item	What Happens in Maintenance Management	System Status of the Maintenance Order
Not Available	No maintenance order has been created. The process in Maintenance Management has not yet started.	Not applicable
Execution Planning	The maintenance order has been created as a follow-up to the service order once the status of the execution order item has been set to Released in the service order.	Created
Planning Completed	The planning process for a service has been completed.	Order Approved
In Execution	In the maintenance order, the planning was completed and the service can now be performed in the maintenance order. Confirmations can also be created for the maintenance order.	Released
Execution Done	The service that has been performed in the maintenance order has been completed. The labor and material consumption have been estimated, planning has been completed, and the items have been approved in the maintenance order.	Technically Completed
Canceled	The execution order item has been rejected or the maintenance order has been manually set to the system status Do Not Execute .	Do Not Execute
Not Relevant	The execution order item has been rejected in the service order even before the maintenance order was created.	Not applicable

SAP Field Service Management

In SAP Field Service Management you execute the service in service calls.

When you set the status of the service order to **Released**, the system creates a service call and the corresponding activities. The service is then executed in the service call and in the activities. The execution status shows the progress of the service execution in the service call.

For more information, see [Integration Scenarios for Transactional Data](#).

The following execution statuses are available:

Execution Status in Service Order	What Happens in Service	Activity Status in SAP Field Service Management
Not Available	All service order items have the status Not Available , or at least one item has the status Not Available and all other items have the status Not Relevant .	Not applicable
Execution Planning	At least one service order item has the status In Planning .	Dispatching
In Execution	At least one service order item has the status In Execution , or at least one service order item has the status Execution Done .	Execution
Execution Done	The service process has been finalized in SAP Field Service Management.	Closed
Canceled	All service order items have the execution status Not Available , or at least one item has the status Not Available , and all others have the status Not Relevant .	Canceled
Not Relevant	No activity has been created. The service is not performed in SAP Field Service Management.	Not applicable

Execution Status in Service Order Item	What Happens in SAP Field Service Management	Activity Status in SAP Field Service Management
Not Available	No activity has been created. The process in SAP Field Service Management has not yet started.	Not applicable
Execution Planning	An activity has been created by the system when the status of the service order item has been set to Released . The activity is visible in the planning list or on the planning board. It is ready to be planned or is already assigned to a technician but is not yet released for the technician. The assignment of the service employee in the service order item is removed.	Dispatching
In Execution	The activity has been released to the technician and can now be viewed and completed.	Execution
Execution Done	The activity has been completed and closed by the assigned technician. All time efforts, expenses, materials, and mileage related to the activity are approved and transferred from SAP Field Service Management to Service.	Closed

Execution Status in Service Order Item	What Happens in SAP Field Service Management	Activity Status in SAP Field Service Management
Canceled	The activity has been canceled.	Canceled

i Note

Service items that have the life cycle statuses **Released** or **Completed** and do not have execution statuses are not considered for the calculation of the execution status on header level. In this case, it is considered that Service is not integrated with SAP Field Service Management.

Credit Checks in Service Orders

Perform credit checks in service orders.

Use

You can use Credit Management to minimize financial risks for your organization. To do so, you perform a credit check when you save service orders in status **Released**.

The credit check determines the creditworthiness of the payers (business partners) who are assigned to the service order items. You can check, for example, if the gross value of the items within a service order exceeds the available credit limit that has been defined in the credit segment data for the business partner (see also [Defining Credit Master Data in the Business Partner](#)).

If the payer passes the credit check, a commitment that is equal to the gross value that was checked is passed to the credit exposure in Credit Management.

If the credit check fails, the credit blocked service orders need to be approved in Credit Management. For more information, see [Perform Credit Checks in Service Orders](#).

i Note

Payers are business partners that are maintained in the partner function **Payer** under **Parties Involved**.

Prerequisites

i Note

The following settings also apply if you want to use the credit check in service quotations and to simulate the credit check for service order items and service quotation items.

Settings in Customizing

Perform the following Customizing activities in Customizing for **Financial Supply Chain Management** under **Credit Management** > **Integration with Accounts Receivable Accounting and Sales and Distribution** > **Integration mit Service** >

- **Deactivate SAP Credit Management**

i Note

If you want to use the credit check functionality, make sure that the **Deactivate Credit Check** checkbox is not selected.

- [Define Credit Control Area](#)
- [Define Credit Segment](#)
- [Assign Credit Control Area and Credit Segment](#)
- [Assign Sales Area to Credit Control Area](#)
- [Activate Credit Management in Service Transactions](#)

You can also activate the credit check for the transaction type in the [Define Transaction Types](#) Customizing activity in the [Customizing header](#) view for the business transaction category [Sales](#).

- [Activate Credit Check on Items](#)

You can also activate the credit check for the item category in the [Define Item Categories](#) Customizing activity in the [Customizing header](#) view for the business transaction category [Sales](#).

You can check whether the credit exposure categories that are provided as default values meet your business requirements. You can make your own settings in Customizing for [Financial Supply Chain Management](#) under [Credit Management](#) > [Credit Risk Monitoring](#) > [Credit Exposure Update](#) > [Define Liability Categories](#) >. For more information about credit exposure categories, see [Managing Credit Exposures](#).

Settings to Perform Credit Checks

If you want to perform a credit check for a payer, make sure that the following steps are performed for the business user in Credit Management:

- Assign the Credit Management role (UKM000) to the business partner.

You have the following options:

- Assign the role to the business partner in the global credit data for the business partner.

For more information, see [Manage Credit Accounts](#).

- Perform the job [Assign Credit Management Role to Business Partners](#) in the [Schedule Credit Management Jobs](#) app.

- In the [Manage Credit Accounts](#) app, create the credit segments and define the credit limit for the business partner.

i Note

If you change the settings for the credit segments, you need to rebuild your credit management data afterwards in the [Rebuild Credit Management Data](#) app.

Note the following when you set up credit checks for business partners in Credit Management:

- If no credit segment is assigned to the sales area in a service order, no credit check is performed. The status of the service order items doesn't change.
- If the Credit Management role isn't assigned to a payer in a service order item, no credit check is performed for the service order items of this payer.

If you initially don't assign the Credit Management role to a payer but assign it after the service order items have been posted to financial accounting and invoiced, you need to use the [Rebuild Credit Management Data](#) app to update the corresponding credit exposure.

For more information, see [Rebuild Credit Management Data](#) and [Rebuild Credit Exposure](#).

Credit Check in a Distributed Landscape

You can now perform credit checks in a distributed landscape, using one system for the service transaction and a separate system for credit management. You no longer need to maintain the credit management role for the business partner in the system that processes the service transaction. A separate system can be used for all credit check-related decisions.

The BAdI method [IF_EX_UKM_R3_ACTIVATE~GET RFCDEST_FSCM](#) needs to be implemented to specify the logical destination of Credit Management.

For Financial Supply Chain Management (FSCM) APIs to perform successful credit management operations, the following steps must be completed:

- Establish connections between S/4HANA Service and Credit Management system
- Create Service definition bindings
- Define a communication user

For more information, see [Connecting Sales and Distribution in SAP S/4HANA Systems or as of SAP ERP 6.0](#).

Further Information

For more information about Credit Management, see [SAP Credit Management \(FIN-FSCM-CR\)](#).

Perform Credit Checks in Service Orders

Perform credit checks in service orders.

Use

When you save a service order with the status **Released**, the system performs a credit check to check the creditworthiness of the payers (business partners) who are assigned to the service order items.

Performing Credit Checks in Service Orders

Credit checks are automatically triggered for service order items when you change the status of the service order to **Released** and then save the service order. The system then ensures that the credit exposure that derives from the service order doesn't exceed the available credit limit for the payers.

The credit status is displayed for each item and for the service order.

→ Recommendation

For more information about the credit checks that are performed by Credit Management for service order items, see [Credit Check](#).

For more information about the credit exposures that derive from service orders, see [Managing Credit Exposures](#).

Credit Status for Service Order Items

The following statuses apply to items:

Credit Status	What It Means
Not relevant for credit check	Credit Management hasn't been enabled for service orders.
Relevant for credit check, but credit check not executed	The credit check hasn't been executed for the service order item yet.
Credit check not OK	The item has failed the credit check. The status of the item remains In Process. If the status of the item was previously set to Released, the item status remains the same. i Note This status is also set if the item has not been released in the Manage Documented Credit Decisions app in Credit Management.
Credit check OK	The item passed the credit check in one of the following ways: - You changed the status of the item to Released and saved the service order. - The service order has been rechecked in the Manage Documented Credit Decisions app in Credit Management. The status of the service order item changes to Released.
Credit released	The item has been released in the Manage Documented Credit Decisions app. The status of the service order item changes to Released.

Credit Status for Service Orders

The following statuses apply to service orders:

Credit Status	What It Means
No item is relevant for credit check	None of the items added to the service order are relevant for the credit check.
Credit check pending for all relevant items	All items that are relevant for the credit check have the credit status Relevant for credit check, but credit check not executed. The service order has the status Open or In Process.
Credit check OK for all relevant items	All items that are relevant for the credit check have the credit status Credit check OK or Credit released. The service order has the status Released.
Credit check not OK for all relevant items	All items that are relevant for the credit check have the credit status Credit check not OK. The service order has the status In Process. You can process service orders with this credit status as described in the section <i>Approving Service Orders with a Credit Block</i>.
Credit check not OK for some items	Contains items that have the credit status Credit check not OK and items that aren't credit blocked (Credit check OK and Credit

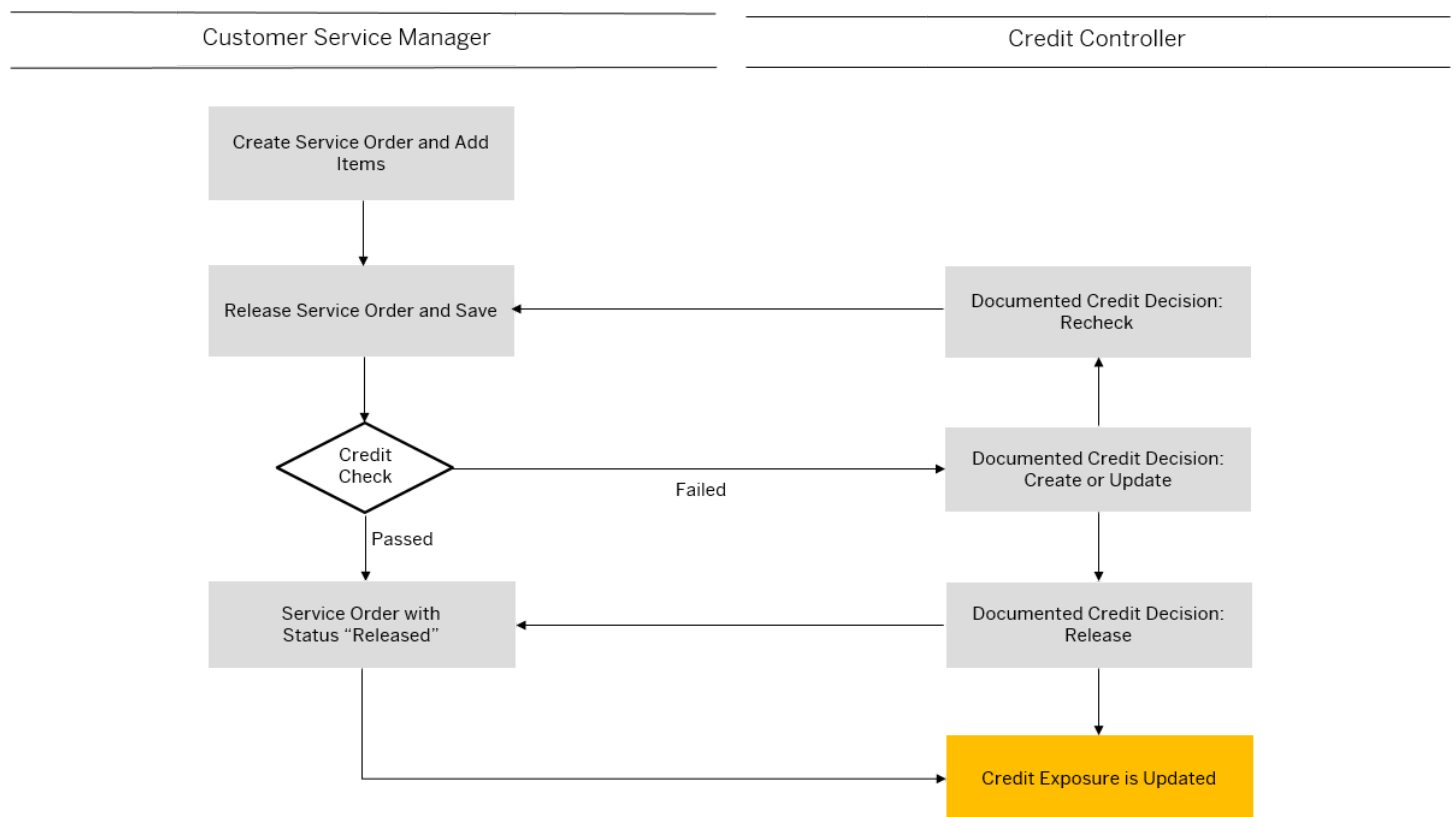
Credit Status	What It Means
	released). The service order has the status In Process. You can process service orders that are credit blocked as described in the section <i>Approving Service Orders with a Credit Block</i>.
Credit check OK for some items and pending for some items	Contains items that aren't credit blocked (Credit check OK and Credit released) and items that have the status Relevant for credit check, but credit check not executed. The service order has the status In Process.

i Note

If the service order is saved again with the status **Released**, the credit check is triggered for all items that are credit blocked from a credit check run that has previously failed. If the items pass the credit check, the credit block is removed.

Credit Check Process

The figure shows the credit check process for service orders.



Credit Checks for Sales Items

The credit check in service orders is also performed for sales items that have been added to the service order. If the sales item passes the credit check, the system creates a sales order and updates the credit exposure that is used for sales orders. If the credit check has failed for the sales item, the system does not create a sales order and you first need to approve the sales item in Credit Management (see the section [Approving Service Orders with a Credit Block](#)).

Credit Checks for Execution Order Items

Credit checks are performed for execution order items that have the billing relevance **Billing on Completion** (fixed price billing) and **Billing via Billing Plan After Release** assigned.

The system performs the credit check once the execution order item is released. If the execution order item passes the credit check, the system creates the follow-up maintenance order.

If the credit check has failed for the execution order item, the system does not create a maintenance order and you first need to approve the execution order item in Credit Management (see the section [Approving Service Orders with a Credit Block](#)).

i Note

For more information, see [Billing](#).

Simulating Credit Checks

You can use the credit check simulation to see the possible credit check result before saving the service order with the status **Released**.

The result of the simulated credit check does not change the credit status that is displayed for the service order, items and the credit exposure, or prevent further processing of the service order.

The following applies to the credit check simulation:

- The result of the simulation is displayed for the service order but not for the related items.
- The simulation allows you to check the possible credit result when the simulation is being performed. If additional changes are made before the service order is saved, the actual result of the credit check might differ from the simulation result.
- Unlike the credit check that is performed when you save the service order with the status **Released**, you can simulate the credit check for items with the status **Open** and **In Process**. If you perform the simulation for items with the status **Released**, only items with a change in the gross value are considered.
- You cannot perform the simulation for items with the status **Completed**.

Approving Service Orders with a Credit Block

If the credit check has failed for at least one service order item and the status of the service order remains **In Process**, you can approve the items in Credit Management as follows:

- Release or recheck the service order items in the [Manage Documented Credit Decisions](#) app.

The system creates cases (documented credit decisions) for service orders that contain credit blocked items. The items to be checked are grouped by the assigned payers.

- Perform the job **Service Transaction Credit Recheck** (RMS4_SRV_CREDIT_RECHECK) to repeat the credit check.

For more information, see [Check Credit for Service Transactions](#).

Once the credit recheck is passed, the status of the service order changes to **Released** and the relevant credit exposures are updated accordingly for the business partner.

Performing Credit Checks for Released Service Orders

The system automatically triggers a credit check for service orders with the status **Released** if the following applies:

- You've added service order items to the service order and then changed the status of the service order items to **Released**.

- You've changed the status of service order items with the status **Open** or **In Process** to **Released**.
- You've increased or decreased the gross value of the service order items with the status **Released**.

If the credit recheck fails for the service order items, their status remains **Released**, but their credit status is set to **Credit check not OK**.

The system doesn't perform a credit recheck for service orders with the status **Released** if the following applies:

- You've changed the status of some of the service order items with the status **Released** to **In Process**.
- You've rejected all service order items.
- You've changed the status of a service order item to **Completed**.

If the credit check fails for the service order items, you can approve the service order as described in the section [Approving Service Orders with a Credit Block](#).

Managing Credit Exposures

Use credit exposure categories to report and analyze a payer's credit situation.

Use

Credit exposure is the combined value of all items (and the corresponding credit exposure categories) that are related to a payer (business partner). You can use the [Display Credit Exposure](#) app to see an overview of the total credit exposure for a business partner according to credit segment.

The credit check only observes the total credit exposure for a business partner.

Credit Exposures for Service Orders

You can use the following credit exposure categories that are provided as default values in the [Define Liability Categories](#) Customizing activity:

Credit Exposure Category (Default)	Name of the Credit Exposure Category	Used for
100	Open Orders	Sales orders that are created as a follow-up to sales order items
110	Service Order & Service Contract	• Service order items with the status Released for which service confirmations haven't been created or • This also applies to Service with Advanced Execution • Service confirmations have been created that have a total gross value that is less than the gross value of the related service order item
200	Open Invoices	Invoices that have not yet been paid

Credit Exposure Category (Default)	Name of the Credit Exposure Category	Used for
410	Service Confirmations	Service confirmation items with the status Completed that are ready for billing
500	Billing Document Value	Invoices, credit memos, debit memos, pro forma invoices, or cancellation documents

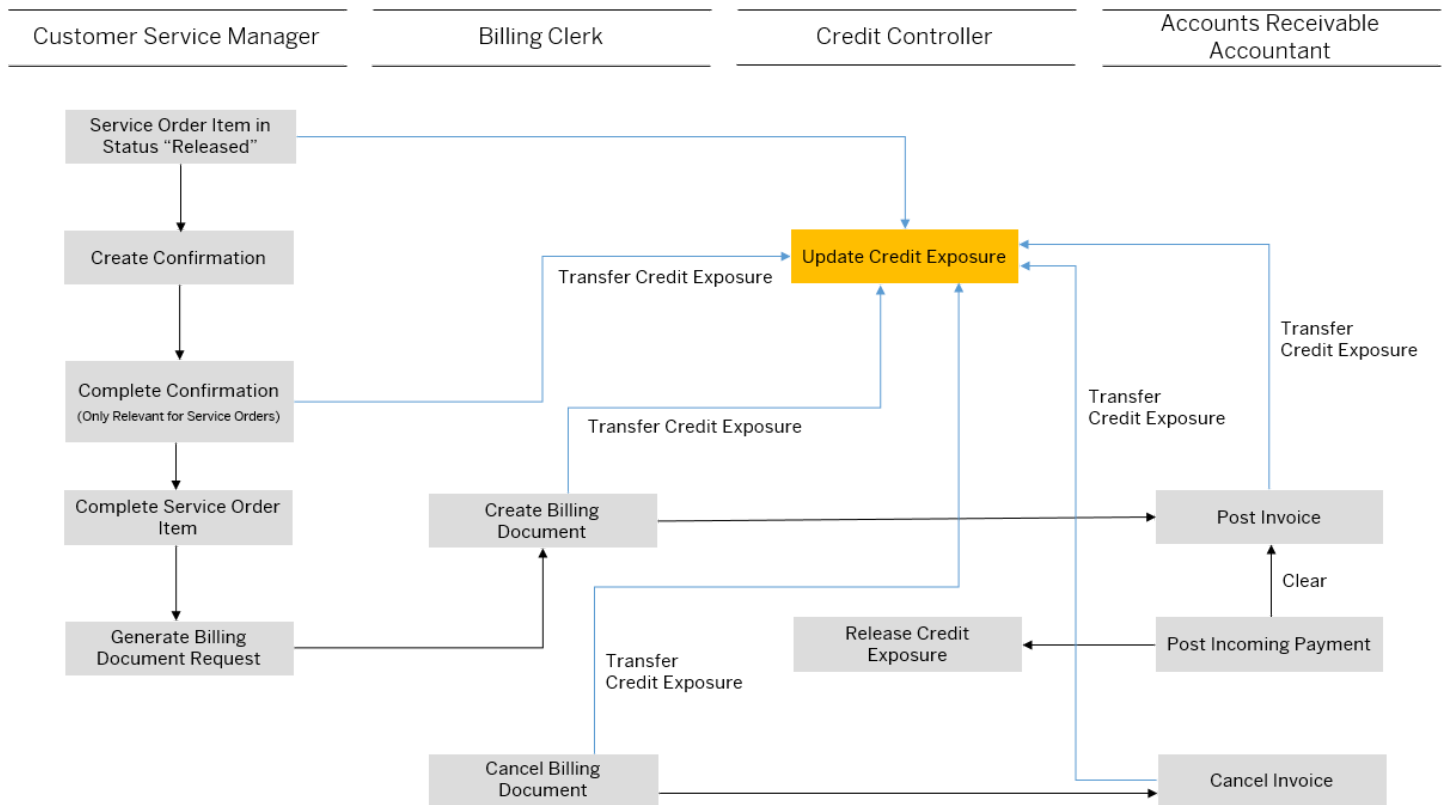
Posting Credit Exposures

If the credit check is successful, the gross value of all items is transferred to the credit exposure category **110**, which is used for service orders. If a sales order is created as a follow-up to a sales order item, the gross value is transferred to the credit exposure category **100**.

The gross value of the items is posted to the relevant credit exposure categories as follows:

Process Step	Credit Exposure Category Used (Default)
The service order items pass the credit check when being saved with the status Released .	110
The status of the service confirmation items that have been created as follow-ups for service orders changes to Completed .	Transfer from 110 or to 410
The service confirmations are cancelled.	Transfer from 410 to 110 i Note If the gross value of the service confirmation is higher than the original gross value that has been posted for the related service order item, the system uses the gross value that was originally posted for the service order item.
A billing document is created for the service confirmation items.	If the billing document is created and not posted automatically: transfer from 410 to 500 If the billing document is created and posted automatically: transfer from 410 to 200
The billing document is cancelled.	If the billing document has not been posted: transfer from 500 to 410 or 110 If the billing document has been posted: transfer from 200 to 410 or 110

Credit Exposure Processing



Managing Inconsistencies in Credit Exposure

If there are inconsistencies in the credit exposure, you can use the job [Rebuild Credit Exposure](#) in the [Rebuild Credit Management Data](#) app to rebuild the credit management data.

If Credit Management is activated in the system for the first time, but service orders exist that were created before the activation, this can lead to inconsistencies in the credit exposure: If a credit check is performed for new service orders created and no changes have been made to the existing service orders, you need to rebuild your credit management data. This ensures that the credit exposure is updated with the correct value for both existing service orders and any new service orders created.

Once you've performed the job to rebuild the credit management data, the credit exposure is cleared to zero if the following applies:

- Credit Management has not been activated (globally or for service orders).
- Credit Management has been activated, but the credit segment has not been defined for the business partner.
- Credit Management has been activated, but the Credit Management role (UKM000) has not been assigned to the relevant business partner (payer).

If credit management data is being rebuilt and a service order or a service confirmation is locked by another user, the job clears the credit exposure for that service order to zero. As soon as the service order or service confirmation is unlocked, the correct value of the credit exposure is updated.

Service Quotations

A service quotation is a legally binding offer with fixed conditions, made to a customer for the performance of services or for the delivery of service parts.

Use

Quotations provide customers with information about prices and delivery conditions before placing an order. They also enable providers to plan early for expected services orders and to get an overview of expected costs and revenues. Within the validity period of a quotation, customers have the option of agreeing to the quotation.

You can use the [Manage Service Quotations](#) app to create and process service quotations.

Depending on your Customizing settings, you can do the following:

- Create a service order from a service quotation as a follow-up transaction
- Create a service quotation as a follow-up transaction of a service order template (see [Service Order Templates](#))

Solution quotations are variants of service quotations, and enable you to create a quotation for combinations of tangible products and services. For more information, see [Solution Quotation Management](#).

i Note

The user interface for service quotations is a configuration of the component for service orders with the object type BT116_SRVQ.

Prerequisites

You can find the settings listed below in Customizing for [Service](#) under [► Transactions ► Basic Settings ►](#).

Business Transaction Customizing

The business transaction for service quotations is predefined as follows in the standard system:

- Transaction type SRVQ
- Item categories:
 - SRQP for service items
 - SRQM for service part items
 - SRQS for sales items
 - SRQE for expense items
- Fixed price item categories:
 - SFQS for fixed price service items
 - SFQM for fixed price service part items
 - SFQE for fixed price expense items
 - SBQT for tool items with ad hoc billing
 - SFQT for fixed price tool items

If the settings for these Customizing objects don't reflect your business requirements, you can copy them and modify the copies.

You also need to set up copy control to determine the follow-up transactions that are created from the service quotation. The copy control customizing is checked for every item to determine the transaction type of the follow-up transaction. If multiple transaction types are selected, the first entry applies.

Quotation Items

You define whether an item is a service quotation in the Customizing activity [Define Item Categories](#) by choosing the required item categories and the business transaction category [Sales](#). In the [Customizing Item](#) view, specify in the [Relevance for Quotation](#) field whether the item is an initial quotation.

In the above Customizing activity, ensure that the value in the [Subsequ. Processing](#) field is [Quotation is copied to subsequent document](#).

Features

Service quotations contain the same type of data and have the same features as service orders (see [Service Orders](#)).

Trade Compliance Check

The integration with SAP Global Trade Services (GTS) allows you to perform embargo checks, sanctioned party list screening, and legal controls for service quotations. For more information, see [Integration with SAP Global Trade Services](#).

Specify Ad Hoc Billing in Service Quotations

In service quotations, you can assign billing plan templates to predetermine the billing conditions for individual items. This enables you, for example, to include the billing conditions defined in the billing plan template in the output document for the service quotation. When you create a follow-up service order to a service quotation to which items with billing plan templates are assigned, the system automatically generates the billing request lines in the related service order items.

For more information, such as the settings that enable billing plan templates to be assigned to service quotations, and creating billing request lines in the follow-up service orders, see [Billing Plan Templates in Service Quotations](#).

Qualification Requirements

You can define qualification requirements, such as skills, knowledge, or abilities, necessary to perform services specified in a service transaction. For more information, see [Qualifications in Service Transactions](#).

Using Service Hierarchies with Collective Accounting

i Note

You can only use service hierarchies with collective accounting if item-based accounting is active.

You can add service hierarchies with collective accounting to service quotations as main items. Sub-items that you add to a service hierarchy inherit the service contract assigned to the main item. If the service hierarchy is copied to a follow-up service order, you can use the features of the service hierarchy in the service order as described under [Service Hierarchy with Collective Accounting](#).

Using Service Quotations in the In-House Service Process

Service quotations allow you to provide a cost estimate for services that are related to the in-house service process for an in-house service object (service object). The [In-House Service Objects](#) assignment block displays the service objects and the related in-house services to which the service quotation refers. You can also assign additional service objects or remove service objects.

For more information about processing service objects in service quotations, see [In-House Service Objects in Service Quotations and Service Orders](#).

Performing and Simulating Credit Checks

In service quotations, you can perform credit checks and simulate credit checks to determine the creditworthiness of the payers (business partners) who are assigned to the related items.

The credit check is automatically performed once the service quotation is sent to the customer and the status changes to **Released**. The result of the credit check is displayed in the credit status.

i Note

The actual credit check is performed in the service order that is created once the service quotation is accepted.

If you want to know the credit check result while the service quotation is in process, you can simulate a credit check for the related items.

i Note

Unlike the credit check that is performed in service orders, the result of the checks and the simulations in service quotations doesn't change the credit exposure and doesn't block further processing of the service quotation.

i Note

Prerequisites

You can perform credit checks and credit check simulations for quotation items if the payer who is assigned to the quotation items has the credit management role assigned and credit checks have been activated for service quotations.

For more information about setting up credit checks in Customizing, see [Credit Checks in Service Orders](#).

Service Orders as Follow-Up Transactions

You can create service orders as follow-up transactions from service quotations with the following options:

- **Create Follow-Up**

When a customer accepts quotation items, you can create a follow-up service order by choosing this button and copy the accepted items into the follow-up service order. When you copy items to a service order, they're automatically marked as accepted in the quotation.

- **Accept**

You can use this button to accept a service quotation. The system then automatically creates a service order as a follow-up transaction that contains all the items from the service quotation. This function provides you with the option to accept all items in a service quotation and create a follow-up service order in one go.

i Note

- You can only choose **Accept** when the service quotation and the items in the quotation are error-free and set to the status **Released**.
- If there's more than one follow-up transaction available, you can choose the service transaction types from the popup. For example, a service order or a fixed price service order.
- After you accept the service quotation, the status of the service quotation is set to **Completed** and the status of the items in the quotation are set to **Quotation Accepted**.

- **Requote**

You can create a requote to a service quotation if you require additional service items, expense items, or service parts to perform a service, which haven't already been specified in the accepted service quotation. For more information, see [Creating Requotes](#).

Rejecting Service Quotations

You can reject service quotations or single items of service quotations. For more information, see [Rejecting Service Quotations](#).

Completion of a Quotation

Quotations are automatically set to **Completed** when:

- The validity of the quotation has expired
- All quotation items have been copied to follow-up service orders

You can also manually set the status **Completed** if the quotation is rejected by the customer.

Creating Requotes

Create requotes for service quotations that have the status **Completed** but need to be adapted subsequently.

Use

If you require additional service items, service parts, or expense items to perform a service, which haven't been specified in the accepted service quotation (status **Completed**), you can resume the quotation process by creating a requote. In the requote, you add the additional items. Having done this, you send this updated service quotation, or **requote**, to the customer.

The following terms are used when processing requotes:

- **Initial quotations** are service quotations that aren't related to a predecessor quotation or a predecessor order. When you accept an initial quotation, the system creates a service order as a follow-up.

i Note

You can't create requotes for initial quotations that have been rejected. Instead, create a new service quotation.

- **Requotes** are created as a follow-up to the initial quotation. Consequently, the service order that has been created as a follow-up to the initial quotation, is related to this requote. When you accept the requote, the system copies the items that have been added to the requote to the related service order.

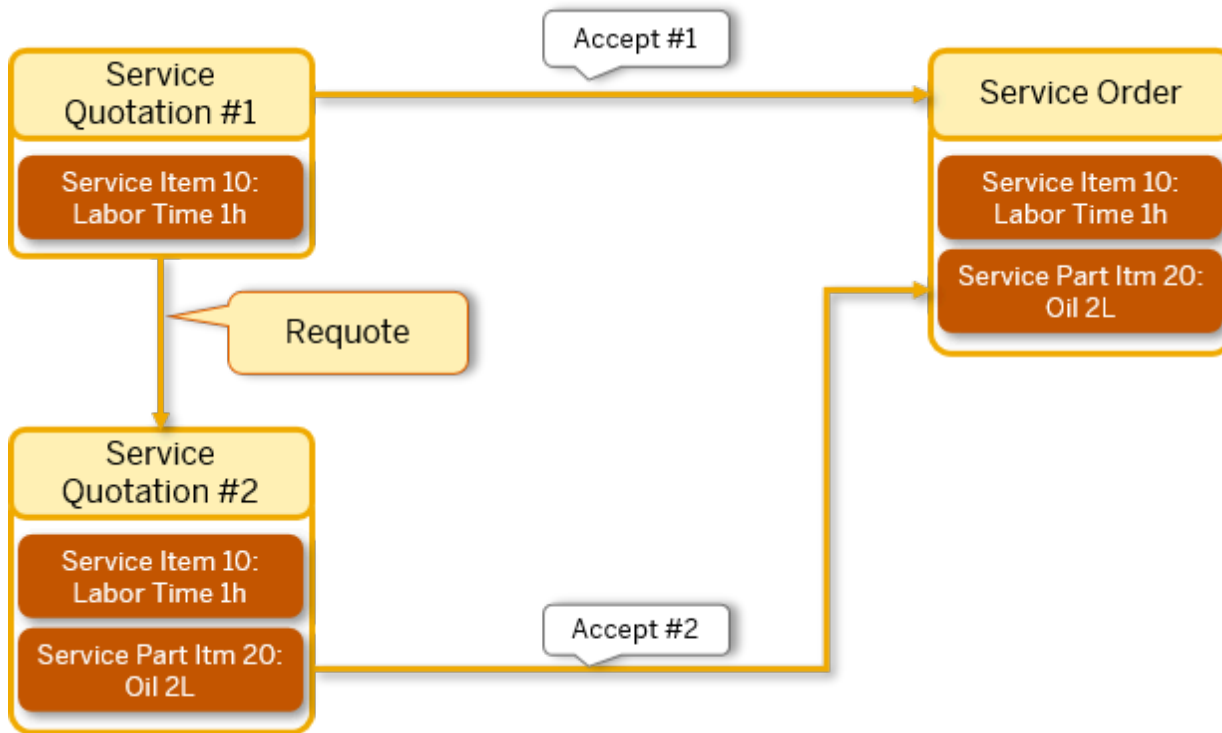
Prerequisites

You can create a requote if the following applies:

- The initial quotation has been accepted and has no predecessor service quotation or service order assigned.
- The latest quotation that can result from an initial quotation or from a requote has been accepted or rejected.

In addition, it's related to only one service order that hasn't yet been completed.

Process Overview



Example of Using Requotes

You can use requotes if you first want to plan and perform a diagnosis to figure out the basic needs of the service and then want to provide a labor and material consumption estimate to the customer:

1. Create a service quotation, that is the initial quotation, and that includes an item to perform the diagnosis.

This item can be a fixed price item.

2. Accept the service quotation.

The system creates a follow-up service order that contains the accepted item from the service quotation.

3. Plan and perform the diagnosis as defined in the items. Create service confirmations to record the actual service effort.

Do not complete the service order and do not define a service confirmation as the **Final Confirmation**.

4. Open the service quotation (initial quotation) that is displayed in the transaction history for the service order.

5. Create a requote from the initial quotation.

The requote has its own service quotation ID assigned. The accepted items of the initial quotation are copied to the requote.

The initial quotation and the related service order are displayed in the transaction history for the requote.

6. In the requote, enter the services and service parts that are required to plan and perform the service.

You can add a price item to a higher-level item that has already been completed to reflect a price surcharge for additional services to be provided.

For more information about price items, see [Price Items in Service Orders and Service Quotations](#).

i Note

Do not make changes in the service quotation items that have been copied from the predecessor quotation. Changes such as changed quantities, aren't copied to the related service order when you accept the service quotation and the status of the service quotation is set to **Completed**.

Having done this, process the service quotation as usual. When you accept the requote, the new items added are copied to the related service order. You can now plan and perform the service in the service order.

i Note

You can create additional requotes at any time until the service is completed, provided that the prerequisites for creating requotes are met.

Processing Related Service Orders

When you process the service order that is related to the initial quotation and the requote, the following applies:

- You can complete the service order if all related service quotations have been accepted or rejected, and consequently have the status **Completed**.
- If you complete the service order, you can no longer create a requote to the related service quotation . Therefore, do not define a service confirmation as the **Final Confirmation** if you're planning to create requotes.

Rejecting Service Quotations

Reject service quotations or single items of service quotations.

Use

If service quotations or single items of a service quotation have become obsolete during the quotation process, you can reject them.

You can reject service quotations and items of any status and if no follow-up service order has been created for the service quotation. If a follow-up service order already exists, you must first reject it and the existing service order items.

Rejecting Service Quotations

If you want to reject a service quotation, you have the following options:

- You can choose the button **Reject All Items**.

All items within the service quotation are rejected. The status of the service quotation and the related items changes to **Completed**.

If required, you can assign a rejection reason to the items of the service quotation.

- You can reject single items in the service quotation by assigning a rejection reason to each item to be rejected. The status of the rejected items then changes to **Completed**.

Once you've rejected all items of the service quotation, the status of the service quotation changes to **Completed** as well.

i Note

You can't create a follow-up service order for service quotations that have been rejected.

Rejecting Items in Service Quotations

If you want to reject single items in the service quotation, assign a rejection reason to the items to be rejected. The status of the rejected items then changes to **Completed**.

If you create a follow-up service order for the related service quotation, the rejected items aren't copied to the service order.

Reverting Your Decision

You can revert the rejection when you remove the rejection reason of single items or choose the button **Reset Item Rejection** to remove the rejection for all rejected items.

Once you've reverted the rejection, the status of the service quotations and/or the items changes to **In Process**.

Service Order Templates

A service order template is a service transaction used to define reusable sets of service-related data that minimize the amount of time required to create a service transaction. A service order template describes only the scope of planned services and not the actual execution.

Integration

Business Transactions

Depending on your Customizing settings, you can do the following:

- Create a service order on the basis of the service order template, as a follow-up transaction
- Create a service order template on the basis of a service order, as a follow-up transaction
- Copy items from service order templates when creating the following:
 - Service order
 - Service quotation

Prerequisites

You have configured your service order template transaction(s) in Customizing for **Service** under **Transactions** **Basic Settings**.

i Note

The user interface for service order templates is a configuration of the component for service orders with the object type BT116_SRVT.

The service order template is predefined as follows in the standard system:

- Transaction type SRVT
- Item categories:
 - SRTP for service items
 - SRTM for service part items
 - SRTS for sales items

If the settings for these Customizing objects do not reflect your business requirements, you can copy them and modify the copies.

Features

You can create hierarchies of templates by referencing a template from within another template. For more information, see [Template References](#).

Qualification Requirements

You can define qualification requirements, such as skills, knowledge, or abilities, necessary to perform services specified in a service transaction. For more information, see [Qualifications in Service Transactions](#).

Service Order Templates for Planned Recurring Service

Use

You use service order templates for planned recurring service to create service orders.

Integration

Business Transactions

Depending on your Customizing settings, you can do the following:

- Create a service order on the basis of the service order template, as a follow-up transaction
- Copy items from service order templates when creating a service order

Prerequisites

You have configured your service order template transactions in Customizing for **Service** under **Transactions** > **Basic Settings**.

The service order template for planned recurring service is predefined as follows in the standard system:

- Transaction type MPTL
- Item categories:
 - MPTP for service items (maintenance plan)
 - MPTM for service part items (maintenance plan)

If the settings for these Customizing objects do not reflect your business requirements, you can copy them and modify the copies.

Template References

Use

You can build hierarchies of templates by assigning templates to items in other templates. Since the templates in a hierarchy are only linked by means of reference, you only need to make changes to templates once and they are automatically available wherever referenced in the hierarchy.

❖ Example

You define a service order template for an oil change and assign it to the item “Oil Change” in the templates for a standard inspection and complete inspection. The inspection templates, therefore, implicitly contain all the template items for the oil change.

You can also reference templates in service order and service quotation items, if you have customized the item category correspondingly (see “Prerequisites” below).

Prerequisites

You have assigned the object reference profile **TEMPLATE** to those item categories from which you want to allow template references. You do this in Customizing for **Service**, by choosing ► **Transactions** ► **Basic Settings** ► **Define Item Categories** ►.

Features

If you have assigned a template to a (template) item, when you create a follow-up transaction one of the following occurs:

- The template reference is copied to the new item.
- Items from the template are automatically copied as subitems of the follow-up item.

If you want new items to be created from the referenced template, you have specified this in Customizing for **Service**, by choosing ► **Transactions** ► **Basic Settings** ► **Copying Control for Business Transactions** ► **Define Copying Control for Transaction Types** ►.

i Note

The items from all templates referenced in the hierarchy are copied as subitems on an equal level. The hierarchy of items in the template hierarchy is not reproduced.

Service Confirmations

A service confirmation is a service transaction used to report working time, expenses incurred, and service parts used while performing a service.

Use

You can use service confirmations to confirm service processes for cost postings. The actual service duration, service parts used, and expenses incurred during service execution can be used as a basis for keeping track of working time, updating stock levels for service parts, processing invoices, and capturing actual costs. For example, after a service part item in a service order has been released, a purchase requisition is triggered against this item and the relevant procurement costs are updated. You use service confirmations to confirm items from the service order and to record the actual consumption of time and materials during service execution. This way, the actual costs of a service can be posted.

Use of Items in Service Confirmations

A service confirmation consists of a header and one or more items. You use items in a service confirmation to confirm items from a service order. This in turn creates account assignment objects so that cost and revenue can be posted to Financial Accounting. You can edit items in a service confirmation on the **Items** tab. For more information on different types of items and how to use items in service confirmations, see [Items in Service Confirmations](#).

Basic Settings

The standard transaction type of a service confirmation is predefined in the system as SRVC. Transaction types define the characteristics and attributes of a business transaction and determine how the transaction is processed, for example, the profiles and procedures used to determine texts, business partners, statuses, and organizational data. If the standard settings do not meet your business requirements, you can configure the transaction types using the Customizing activities under [Service Transactions](#) [Basic Settings](#). For more information, see [Business Transaction](#).

Life Cycle Statuses

A service confirmation and a service confirmation item can have the following life cycle statuses:

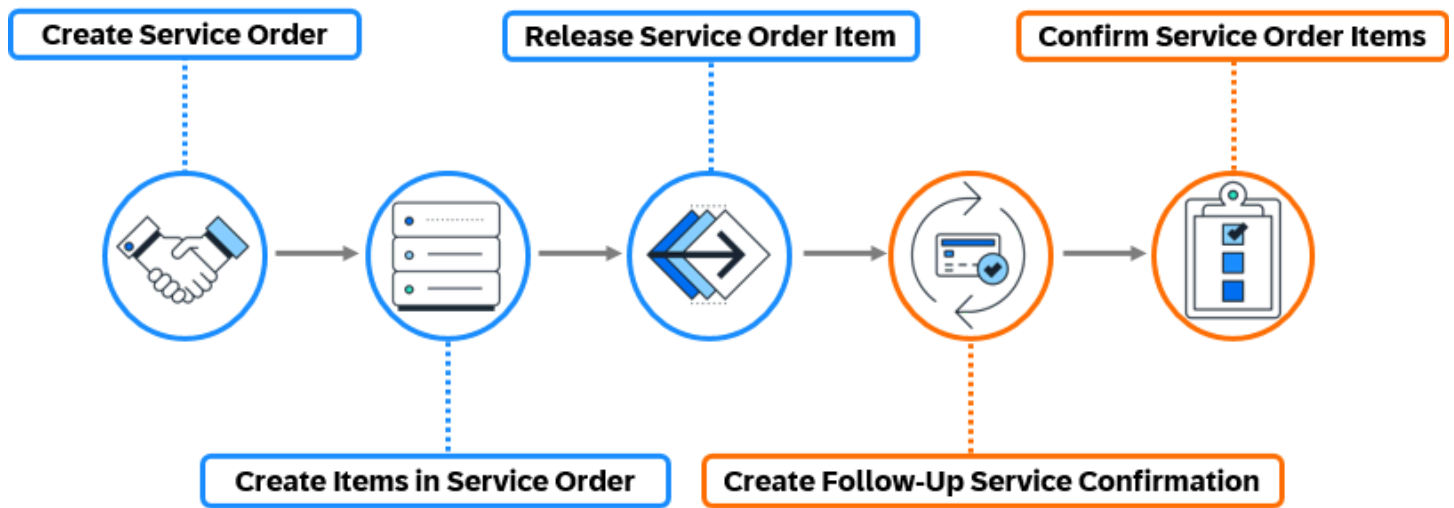
Status	Description
Open	The default status that all service confirmations have after they're created. It indicates that you can edit or cancel the confirmation.
Completed	This status indicates that no further actions are required. You can release a service confirmation for billing if the following conditions are met: - The header status of the service confirmation is Completed. - The items from the predecessor service order also have the status Completed. i Note You cannot edit a service confirmation that has this status anymore, but you can still cancel a service confirmation that has this status.

Activities

Create Service Confirmations as Follow-Up Transactions

You can create service confirmations as follow-up transactions of service orders to confirm service order items.

- Prerequisites:** To create service confirmations as follow-up transactions, you must configure the respective transaction flow in Customizing. For more information, see [Follow-Up Transactions](#).
- When you create a follow-up service confirmation for a service order, the system automatically proposes the items planned in the service order for selection. The items that you select are copied to the service confirmation. The relevant data from the item, such as prices, reference objects, subjects, conditions, and working times for assignments, is also copied to the service confirmation.
- Sample process:** The following diagram illustrates a sample process of creating a service confirmation from a service order.



i Note

- You can only create service confirmations for service order items that are error-free and have the status **Released**. The **Create Follow-Up** function is only active when a service order item has been set to the status **Released**.
- You need to use a separate service confirmation to confirm each service order.
- If you create a service confirmation as a follow-up, the system always suggests to take over quantity of the service order item. Any quantity that has already been confirmed will not be considered.

Create Standalone Service Confirmations

You can also create service confirmations without reference to service orders with the **Create Service Confirmation** app. For instance, you can create standalone service confirmations for unplanned services that don't relate to an existing transaction, such as additional services completed at the customer site.

i Note

You cannot create standalone service confirmations if you have activated item-based accounting.

Create Partial and Final Service Confirmations

You can create multiple service confirmations for a service order, for example, if you're required to make confirmations on a daily basis but require more than one day to complete an assignment.

- If you're working with such partial service confirmations or if multiple service confirmations are assigned to a service order, the system sets the service order to **Completed** automatically when the following conditions are met:
 - All service confirmations based on this service order have been set to **Completed**.
 - One of the completed service confirmations has been defined as the **Final Confirmation**.
 - There are no execution order items in the service order.

i Note

If the service order contains an execution order item, neither the execution order item, nor the service order, nor any other items in the service order can be set to **Completed** automatically.

- If all service confirmations have been set to **Completed** and no confirmation is defined as the **Final Confirmation**, you have to manually set the service order to **Completed**.

Cancel Service Confirmations

You can cancel a service confirmation if you have entered an incorrect service duration or a wrong service part. After you cancel a service confirmation, all the CATS postings or postings of goods issue are canceled as well. There are two ways to cancel a service confirmation:

- You can navigate to a service confirmation from the service order UI. On the service confirmation UI, choose **Cancel Confirmation** to cancel the service confirmation.
- You can also search for service confirmations. Then select the service confirmation you want to cancel and choose **Cancel Confirmation**.

i Note

If a service confirmation has the status **Completed** and has already been released for billing, you can only cancel the service confirmation after you delete or reject the billing document request (BDR) that has been created as the follow-up document of this service confirmation.

Key Features

Pricing

You can use an accounting indicator in a service confirmation item to determine different prices for products, based on absolute or percentage discounts and surcharges for services performed under warranty. You can define discounts and surcharges in pricing conditions. For more information, see [Pricing](#).

Cost Allocation

You can use an accounting indicator in a service confirmation item to determine the settlement receiver for costs. For example, if a service is covered by a warranty, you bill the quality department, but for a free-of-charge repair, you want to bill the sales department instead of the executing cost center.

Billing

You can initiate billing based on the service order items confirmed in service confirmations. For more information, see [Billing](#).

You can specify a service type and valuation type in a service product item confirmed in a service confirmation. This helps determine different rates for billing such as weekend work, overtime, travel time, senior or junior service representative.

Attachments

You can attach documents to service confirmations at header and item level. In Service, a full range of functions are available for you to manage document attachments. For more information, see [CRM Content Management \(Legacy\)](#).

Printing Service Reports

You can print reports that provide a summary of confirmed service activities using **Output** on the overview page. In Service, you can use output determination to set up the relevant output methods such as e-mail, fax, or letter. For more information, see [Output Determination](#).

Categorization

You can use categorization to define predefined content, which makes confirmation entry faster and allows automatic analysis. For more information, see [Multilevel Categorization](#).

Related Information

[Service Orders](#)

Items in Service Confirmations

You can learn about items in service confirmations in this topic.

Use

When you create a service confirmation as a follow-up transaction of a service order, you can select which item from the service order you want to confirm. Each item in a service confirmation corresponds to an item from the preceding service order. You use service confirmations to confirm items from the service order and to record the actual consumption of time and materials during service execution. This way, the actual costs of a service can be posted. For example, when you confirm a service item, time sheet entry is posted and the actual labor cost is updated.

Item Categories

The following table provides you with an overview of which item categories you can use in a service confirmation:

Item Type	Item Category
Service Item/Service Product Item	• SVCP: Service Item • SFCS: Fixed Price Service Item • SPCS: Service Item (Partial Billing) i Note The date profile SRV_RM_ITEM1 (SAP Confirmation Service Item) is assigned to the item category SVCP. This date profile contains date types such as SRVC_ACTUAL (Start of Work) and SRVC_ACT_T0 (Date Work Ends) that are used to record working time.
External Service Item	SVCW
Service Part Item	• SVCM: Service Part Item • SFCM: Fixed Price Service Part Item • SVCS: Stock Transfer Service Part • SVCG: Service Part Manual Goods Issue • SVCU: Advance Shipment of Procured Service Parts • SVCV: Advance Shipment of Stock Service Parts
Expense Item	• SVCE: Expense Item • SFCE: Fixed Price Expense Item • SPCS: Expense Item (Partial Billing)

Item Type	Item Category
Tool Item	• SVCT: Service Confirmation Tool Item • SFCT: Fixed Price Tool Item (Confirmation)

Key Features

Unplanned Items

You can add additional items that have not been created in a service order to a service confirmation using the **Unplanned Items** function. For more information, see [Unplanned Items](#).

Predecessor Items

You can add items to a service confirmation from the respective preceding service order using the **Predecessor Items** function. For example, if you have planned and created multiple items in a service order, but you have only created a service confirmation for one item and you still want to confirm other items from the service order, you can still take over other items from the service order into the same existing service confirmation. You can choose **Predecessor Items** when you edit the item list on the **Items** tab on the UI of **Service Confirmations**. When you choose **Predecessor Items**, the system automatically proposes the items that have been planned and released in the service order for selection and the item that you select is copied to the service confirmation.

Configurable Products

You can use configurable products or product variants to customize complex products with single-level variant configuration for your service processes. For more information, see [Configurable Products and Product Variants](#).

Service Rendered Date

You can use **Service Rendered Date** to record the date on which the expenses were incurred or a goods issue was posted for a stock service part. You can enter the date of the cost postings for the relevant service confirmation item types such as expense items or stock service part items. For service product items, the start of work date is used as the service rendered date. On the following UIs the service rendered date is displayed as the **Posting Date**: Material Document, Display Line Items, and Display Line Items in General Ledger.

If you enter a service rendered date, this date is used to determine how taxes are calculated for pricing-relevant items in service confirmations. When a service confirmation is released for billing, the service rendered date is used for billing document request items.

→ Remember

This date must be after the requested start date specified in the respective service order.

i Note

The service rendered date is usually a date in the past, but it can also be a date in the future. For example, as a technician, you can create a service confirmation before you execute the service. You can enter the date in the future when you'll execute the service and then record the actual time and materials consumed after you complete the service. In this case, you can ignore the warning message that asks you to check the service rendered date.

Batch Numbers for Service Parts

If you use batch-managed service parts, you can enter the batch number on the [Service Part Item Details](#) tab. Batches can only be entered at item level. Only one batch number can be entered for each item. The batch numbers entered are transferred to the goods movement posting.

Material Valuation for Service Parts

If you use materials with split valuation, you must select the valuation type on the [Service Part Item Details](#) tab in the [Material Valuation](#) field. Only one valuation type can be entered for each item. The valuation type is transferred to the goods movement posting.

i Note

The [Material Valuation](#) function is not the same as the [Valuation Type](#) that is available for the service product item and used to determine different rates for billing (such as senior or junior service representative). [Material Valuation](#) is used for costing purposes, for example, to differentiate costs of a material.

For more information about split valuation, see [Split Valuation](#).

Serial Numbers for Service Parts

You can enter the serial numbers for service parts on the [Serial Numbers](#) tab if the service parts have a serial number profile assigned with the serialization procedure MMSL in the product master data.

When you create service confirmations for service parts, you enter the serial number for each service part that is consumed. Once you set the status of the service confirmation to [Completed](#), the serial numbers are posted to the goods movement that has been created for the service part.

You control the usage of serial numbers for service parts in the [Define Serial Number Profiles](#) Customizing activity in Customizing for [Logistics - General](#):

- Assign the serialization procedure MMSL to the serial number profile. This ensures that the serial numbers are posted to the goods movement once the service confirmation is completed. For more information, see [Serial Numbers in Physical Inventory](#).
- Activate the usage of serial numbers by assigning the serial number usage 02 or 03 to the serialization procedure.

For more information about managing serial numbers, see [Serial Number Management \(LO-MD-SN\)](#).

Related Information

[Service Confirmations](#)

Unplanned Items

The [Unplanned Items](#) function enables an executing service employee to add additional items, that have not been created in a service order, to a service confirmation.

Use

Items that are created in a service order are planned items. While performing a service, an executing service employee might need to provide extra services, incur additional expenses, or use additional service parts that were not planned in the service order. In this case, you can use the [Unplanned Items](#) function to record these additional services in the service confirmation directly. When

you add an unplanned item in a service confirmation, the same unplanned item is created automatically in the preceding service order.

Unplanned items can be service items, expense items, stock service parts, or tool items.

i Note

Sales items and non-stock service parts cannot be used as unplanned items.

Unplanned items are pricing-relevant. That means that planned cost and revenue – both baseline and ongoing – can be calculated for unplanned items in a service order. This is especially useful if you use the event-based revenue recognition method **Recognize Revenue on Cost-Based Percentage of Completion (POC)**, and the unplanned items are created as subitems of the main item in a service hierarchy.

Prerequisites

To use the **Unplanned Items** function, the following prerequisites must be fulfilled:

- Activate item-based accounting in Customizing for **Service** under ► **Transactions** ► **Settings for Service Transactions** ► **Integration** ► **Enable Item-Based Accounting for Service** ►.
- Assign the item category usage UNPL (Unplanned Item) when you determine item categories in Customizing for **Service** under ► **Transactions** ► **Basic Settings** ► **Define Item Category Determination** ►.
- The service confirmation to which you want to add an unplanned item must be a follow-up transaction of a service order.
- Specify a **reference service order item** for an unplanned item in a service confirmation. The **reference service order item** must be a planned service item in the preceding service order of the service confirmation.

Reference Service Order Item

When creating an unplanned item in a service confirmation, a reference item is needed in the service order for controlling purposes. This reference item is called **reference service order item**. Only a **service product item** can be a reference service order item. When an unplanned item is created in a service confirmation and this same unplanned item is automatically created in the preceding service order, an account assignment object for the newly created unplanned service item is created.

i Note

You cannot use a price item as the **reference service order item** for an unplanned item.

Features

Service Contract Determination

You can use contract determination for unplanned items for the transaction type **Service Confirmation**. For more information on how to customize contract determination for service transactions, see [Service Contract Determination](#).

Executing Service Employee

The executing service employee cannot be determined automatically. You need to enter this manually.

Integration

The **Unplanned Items** function is integrated with SAP Field Service Management, logistics integration, and billing. However, the integration with procurement, material reservation, or credit checks is not supported.

Profit Center

The profit center of an unplanned item in the service confirmation is copied from the **reference service order item**.

Deletion of Unplanned Items

- You can delete an unplanned item in a service confirmation if the unplanned item hasn't been set to **Completed**.
- When you delete an unplanned item in a service confirmation, the corresponding unplanned item in the preceding service order is deleted automatically as well.

Cross Functions in Service Order Management

Cross functions include features that are available in more than one service transaction throughout service order management.

Configurable Products and Product Variants

You can use configurable products or product variants to customize complex products with single-level variant configuration for your service processes.

Terminology

- **Configurable Product:** A product that can be configured whose characteristics can be changed.
- **Product Variant:** A configurable product that has been configured whose characteristics have been predefined and cannot be changed.

❖ Example

The following is an example of how to use a configurable product:

Your company offers car wash services and you can use a configurable product to customize a car wash service for your customers.

1. Create a service order and a service item, and choose **Car Wash Configurable** (CAR-WASH_CONF) as the service product for the service item.
2. Choose the icon  *Configure Product* in the **Actions** column and navigate to the integrated advanced variant configurator that supports single-level variant configuration.
3. You can then configure the service product and customize the car wash service, such as by choosing the car size for the car wash, the car wash type, or whether it needs extra polish or not.
4. Complete the configuration and navigate back to the service order. The **Net Value** column displays a specific price for the service item based on your configuration.

Prerequisites

A service product or service part must fulfill the following requirements in the product master data to be configurable:

- The item category of the product must be relevant for Service Product Item or Service Part Item.
- The product must be marked as **Configurable**.
- The classification must be **Variants** (Class type: 300) and a configuration class must be associated with it.
- Only one configuration profile can exist. When you configure the profile, you must select the process **Plnd/Prod. Order** and choose **None** for BOM explosion. The processing mode must be **A for Advanced Variant Configuration**.
- New prices must be defined in pricing condition records for configurable products. This way, all configurable products can be assigned with certain prices, which are then used to determine the prices for items.

Use

You can use configurable products or product variants as service products in the following service transactions:

- Service Order
- Service Confirmation
- Service Quotation
- Service Order Template

i Note

The configuration of a product in a service order template is optional.

You can use configurable products or product variants with the following item categories:

Item Category	Description
SRVP	Service Item
SF0S	Fixed Price Service Item
SRVW	External Service Item
SVCP	Service Item (Service Confirmation)
SFCS	Fixed Price Service Item (Service Confirmation)
SVCW	External Service Item (Service Confirmation)
SRQW	External Service Item (Service Quotation)
SRTW	External Service Item (Service Order Template)
SRVM	Stock Service Part Item
SRPM	Service Part Item (Purchase Requisition)
TAC	Sales Item
MSI1	Execution Order Item
SLAC	Configurable Service Part for Advance Shipment

i Note

If you add a configurable product as the product to a **stock service part item** (SRVM), the configurable product you added must be replaced by another product that is a product variant through variant matching. This product variant must be based on the characteristics that you have selected for the configurable product you added.

Configurable Product

When you create a service item or a service part in a service transaction, you can use a configurable product as a product and then configure the product by choosing the icon  *Configure Product* on the **Items** tab. You can select the characteristics and the values that are defined in the product master data. After you finish configuring the product, the configuration status is changed to **Released**.

i Note

Consider the following when using configurable products:

- In a service order, service confirmation, or a service quotation, if you add a configurable product as a product to a service item or service part, you must configure the product before you can set the item to **Released** or **Completed**.
- You cannot change the configuration of a configurable product in a released or completed service item.
- You can only display the configuration of a service part item in a service confirmation that is taken over from the service order and you cannot configure the service part item in the service confirmation.
- If you configure a service part item in a service order, the configuration of the service part item is forwarded to the follow-up service confirmation. You can display the configuration in the service confirmation but you cannot configure the service part item in the service confirmation.

Product Variant

You can add a product variant as product to a service item or service part and display the details of the product variant by choosing the icon  *Configure Product* on the **Items** tab.

i Note

You can use a product variant as an unplanned item if you have activated item-based accounting. For more information, see [Unplanned Items](#).

Price Details

You can see the impact on the price at header or item level of a service transaction after you configure a configurable product based on the selected characteristics and characteristic values for the defined variant conditions.

Related Information

[Advanced Variant Configuration](#)

Dates in Service Orders and Service Quotations

Use

Date settings that you make in Customizing enable the relevant dates for business transactions to be calculated automatically. You use a date rule to define date types and assign these date types to a date profile.

Configuration

You set up date management for service orders and service quotations in Customizing for **Service** under ► **Basic Functions** ► **Date Management** ►.

You can modify the standard definitions for dates as needed in the **Define Date Profile for Service Transactions** Customizing activity. In the **Assign Date Profile to Transaction Type** and **Assign Date Profile to Item Category** Customizing activities, you assign these date profiles to the transaction types and item categories. Follow the instructions in the documentation of the Customizing activities.

The following table shows the most important predefined date types for the standard service transaction types:

Transaction Type	Date Profile	Date Type
Service order template (SRVT)	SRV_TEMPLATE	Valid From (VALIDFROM) Valid To (VALIDTO)
Service quotation (SRVQ)	SRV_QUOT	Valid From (QUOTSTART) Valid To (QUOTEND)
Service order (SRVO)	SRV_HEADER01	Requested Start (SRV_CUST_BEG) Requested End (SRV_CUST_END)

Processing Dates

For more information about processing dates in service orders and service quotations, see [Dates in Business Transactions](#).

Related Information

[Dates in Business Transactions](#)

[Date Management](#)

Prediction of Service Parts and Service Duration

Predict service parts and time required to perform a service.

Use

You can use values from past service confirmations to predict the service parts and time required to perform a service. In the standard system, predicted values are provided based on service confirmation data stored in **in-memory database**. You can only predict service parts and service duration for service items in service orders and service quotations.

Prerequisites

You need to perform the following settings in Customizing for Service in order to predict service parts and service duration for service items:

- Activate prediction for the relevant item categories.

You can activate prediction for item categories with the item object type BUS2000140 (Service Product Item) under ► **Transactions** ► **Basic Settings** ► **Define Item Categories** ►.

- Define the minimum number of data records that must exist in the database for prediction to take place for an item.

You define this threshold value under ► [Transactions](#) ► [Settings for Service Transactions](#) ► [Prediction of Service Parts and Service Duration](#) ► [Specify Minimum Number of Data Records Required for Prediction](#) ►.

- To set **in-memory database** as the data source, you need to activate the enhancement implementation `SERVICE_PREDICTION_DIRECT` in Customizing for Service. You can do this under ► [Transactions](#) ► [Settings for Service Transactions](#) ► [Prediction of Service Parts and Service Duration](#) ► [BAdI: Prediction of Service Parts and Service Duration](#) ►.

To use the prediction function, all the following data must be available in a service transaction:

- Service product
- Reference object for which product master data can be determined
- First level of categorization

Features

You can predict the relevant service parts and the service duration for service items.

You can overwrite the predicted values after the values have been retrieved from the system in a service transaction.

Service Part Prediction

In the **All Items** view of the **Items** tab, you can select a service item and display the predicted service parts with their quantities and relevance in percent. By default, the 5 most relevant service parts are shown in the prediction based on previously defined data. You can select the relevant service parts for the service item. The selected service parts are then added to the service item as subitems.

Service Duration Prediction

The predicted duration is automatically retrieved when you create a new service item in the **Items** tab. If data cannot be retrieved from the data source that you have defined, the duration is retrieved from the master data of the object or product in the item.

Related Information

[Service Quotations](#)

[Service Orders](#)

Price Items in Service Orders and Service Quotations

You can use a price item in a service order and a service quotation to reflect the surcharge for additional service provided, for example, if you discover that the fixed price agreed upon cannot cover all service costs and you need to charge your customers more.

Use

Price items (item category `SVPI`) can only be used as the subitems of service items or execution order items. When you add a price item to a service order or a service quotation, the price item inherits the account assignment object and service contract from its higher-level item. The price item also inherits the external reference from its higher-level item. You can still change the external reference of the price item manually when you edit the price item, unlike the account assignment object and the service contract, which cannot be changed.

Prerequisites

The following prerequisites must be fulfilled when using price items:

- You have activated item-based accounting. For more information, see [Item-Based Accounting](#).
- You must specify that the price item is relevant for billing if it is used in a service order.
- You must specify that the price item is relevant for pricing.
- You have selected the checkbox **Inherit from Higher-Level Item** in the Customizing activity under **► Service ► Transactions ► Basic Settings ► Define Item Categories ►**.

i Note

- You cannot change the item category of items that have the status **Released** or **Completed** to **Price Item**.
- You cannot use a price item as the **reference service order item** for an unplanned item. For more information, see [Unplanned Items](#).

Features

Event-Based Revenue Recognition

You can conduct Event-Based Revenue Recognition (EBRR) for service items in a service order that have price items as subitems. The following conditions must be met:

- The service items must have the status **Completed**.
- You must have billed all the price items used as subitems.
- You must have posted all the invoices created from the service order to financial accounting.

For more information, see [Event-Based Revenue Recognition for Service Orders](#).

Planned Cost and Revenue

Planned cost and revenue are not calculated for a price item since it inherits the account assignment object of its higher-level item. However, the net value of a price item is considered and added to the planned revenue of the higher-level item of the price item.

Intercompany Service Orders

If you have selected **Price Item** as the item category for an item, you cannot specify that the item is relevant for intercompany service orders. Similarly, if you have specified that an item is relevant for intercompany service orders, you cannot use the item as a price item.

Service Contract Assignment

A price item inherits the service contract from its higher-level item for pricing purpose. You cannot manually assign a service contract to a price item. If the price item has been billed, you cannot change the service contract of its higher-level item anymore.

In-House Service Objects in Service Quotations and Service Orders

You can use service quotations to provide a cost estimate, and service orders to plan and perform the service for in-house service objects (service objects).

The **In-House Service Objects** assignment block in service quotations and service orders (service transactions) displays the service objects for which the service is planned and performed. You can also assign additional service objects or remove service objects.

Assigning Service Objects

In service quotations and service orders (service transactions), you can assign service objects that are in process at either the header or item level as follows:

Add Existing Service Objects

The system proposes those service objects that are already related to an in-house service and that are in process. If an in-house service is already related to the service transaction, only service objects from this in-house service are proposed.

When you save your entries, the relationship between the added service objects and the service transaction is established.

If the service object to be added is a higher-level service object, the related subordinate service objects are also selected (see also [Possible Actions and Logistics-Related Process Steps for Service Objects](#)).

Insert

You can insert equipment, and products with a serial number or without a serial number (non-serialized products) assigned. The system then proposes entries to be inserted regardless of whether they're related to an existing in-house service. When you save your entries, the relationship between the inserted service objects and the service transaction is established.

You can insert a service object in more than one service transaction. If you insert a service object in a service transaction that is already related to an in-house service, the service object is added to that in-house service.

Create from Document

You can create service objects by extracting the required data from documents without typing them in manually. This data can be products (ID and description), serial numbers, or equipment IDs.

To do so, upload the file that contains the data for the service objects in the **Create Service Objects from Document** dialog box. The data is then extracted from the document and enriched by product master data. The result of this data extraction and data enrichment is then displayed in the corresponding fields and can be reviewed.

If the data extraction and data enrichment fails or no unique values can be determined, messages are displayed for the relevant field in the message log. Edit each entry as described under [Messages Displayed During Data Extraction and After Data Enrichment](#). You can overwrite the results displayed in the fields and delete service objects. If all messages have been solved, you can create the service objects by confirming the entries.

For more information about creating service object from documents, see [AI-Assisted In-House Service Initiation](#).

Specifics when Assigning Service Objects

The following information applies to service objects assigned to service transactions:

- You can assign only service objects that have the same sold-to party as the related service transaction.

- If you add an existing service object, the related in-house service appears as a reference. The system does not create a new in-house service.
- Service objects assigned at the header level are not copied to the related service transaction items.

However, you can later assign these service objects to specific service transaction items by selecting the [Assign to Item](#) button at the header level.

❖ Example

In service transactions to which multiple service objects are assigned, you can only assign service objects to specific service transaction items if a subset of the assigned service objects should be considered for individual service transaction items.

- Service objects assigned to specific service transaction items also appear at the header level. The numbers of the related items are shown in the [In-House Service Objects](#) assignment block.
- If you copy a service transaction, the service objects and related reference objects are not copied to the new service transaction.
- You cannot assign service objects to service transaction items of item category [Sales Item](#) (BUS2000131). This is because this item category does not support reference objects, including reference objects that are related to the in-house service.
- For service objects created from a non-serialized product, you can adjust the quantity in the returns order created for the service object. For more information, see [Processing Split Quantities in Customer Returns](#).

Service Objects as Reference Objects

Service objects also appear in the [Reference Objects](#) assignment block and are marked as **in-house service related**. The relationship between reference objects and service objects allows you to use reference object-specific functions, such as warranty and service contract determination for service objects. For more information about managing reference objects in service transactions, see [Reference Objects for Service Orders](#).

When you assign a service object that is already listed in the [Reference Objects](#) assignment block, the entry is marked as **in-house service related** retroactively.

For service objects with have multiple serial numbers assigned, the serial numbers are shown in the [Reference Objects](#) assignment block and are marked with **In-House Service Related**.

Restructured Service Objects (Split and Merge)

In the [Manage In-House Service](#) and [Manage In-House Service Objects](#) apps, you can restructure service objects by splitting and merging their quantities and assigned serial numbers to newly created service objects. The preceding service objects are then set to **Obsolete** and have the status **Completed**. That means that they cannot be processed further. For example, they can no longer be added to service transactions.

If a service transaction was created for a service object **before** it was split or merged, the succeeding service objects do not inherit these service transactions. However, you can add them to the service transactions afterward. However, in the [In-House Service Objects](#) assignment block, service objects set to **Obsolete** are still shown.

For more information about splitting and merging service objects, see [Restructuring Service Objects: Split and Merge](#)

Processing Service Objects in Intercompany Service Orders

You can launch the intercompany process for service objects in service orders to delegate the service execution to a service center in another company code. To do, proceed as follows:

1. Assign the service object to an intercompany service item in a service order that has not been released yet.
2. In the intercompany service item, enter the receiving sales organization that specifies the executing service center.
3. Release the service order.

The system creates the following:

- An in-house service in the current company code if no in-house service process has yet been launched for the service object
- An intercompany service order and an intercompany in-house service that both have the service object assigned

If an intercompany in-house service exists for the target company code, the system adds the service object to it.

The intercompany service order is displayed in the process flow for the related service object in the respective intercompany in-house service.

You process the intercompany service order as described under [Intercompany Service Orders](#).

The following applies when assigning service objects to intercompany service items:

- If the service object has not yet been received at the service center and its availability is **Not Tracked**, you can create a returns order or an inbound delivery for the service object related to the original in-house service or the intercompany in-house service.
- You can add existing service objects. But you cannot add service objects of an intercompany in-house service.

This is because in the In-House Service apps, you can create service transactions only for service objects related to the original in-house service but not to an intercompany in-house service.

For more information about intercompany in-house services, see [Processing Service Objects in Intercompany In-House Services](#).

Deleting In-House Service Objects

If you delete a service object at header level, the entry at item level is also deleted. However, if you delete a service object at item level, the entry at header level remains. The reference objects that are marked as **in-house service related** are also deleted.

If you've created an in-house service from a service transaction and then delete the related service objects, the relationship to the service transaction is deleted. However, the in-house service and the related service objects remain.

Setting Up Your System

If you want to create in-house services from service transactions, you have to set up the [Specify In-House Service as Follow-Up for Service Quotations and Orders](#) Customizing activity in Customizing for [Service](#) under [Transactions](#) > [Settings for In-House Service](#).

Where to Find More

Related Apps

- [Manage Service Quotations](#)
- [Manage Service Orders](#)

Related Information

[In-House Service Object \(Service Object\)](#)

[In-House Service](#)

[AI-Assisted In-House Service Initiation](#)

[Intercompany Service Orders](#)

Qualifications in Service Transactions

In Service, you can define qualification requirements, such as skills, knowledge, or abilities, necessary to perform services specified in service transactions.

To provide top-quality service, you must ensure that the right service personnel are matched to the appropriate tasks. By clearly outlining qualifications necessary for each job, you can better meet the needs of your customers and ensure that tasks are completed efficiently and to the customers' satisfaction.

You can define qualification requirements in the following service transactions, both on header and item level:

- Service orders
- Service quotations
- Service order templates

On item level, you can define qualification requirements only for service items.

It's possible to enter qualifications manually or have the system automatically identify them based on the master data, such as ship-to party or equipment, specified in a service transaction. You can set up the system accordingly using the settings described in [Settings for Qualification Requirements](#).

i Note

If the system identifies the same qualification requirement as you added manually, but the one it finds is mandatory or has a higher proficiency, it override the existing qualification.

If the system identifies new qualification requirements when you update a service transaction, it won't remove qualification requirements identified in the service transaction earlier.

The system identifies qualifications only for service transactions in status **Open** or **In Process**. Once identified, they are displayed in the **Qualification Requirements** assignment block of the corresponding service transaction.

Copying Qualifications to Follow-Up Transactions

If a service transaction is created as a follow-up of another service transaction, the qualifications from the header and/or item of the preceding service transaction are copied to the follow-up transaction. For example, the qualifications are copied from a service quotation to a service order, from a service quotation to another service quotation, or from a service order template to a service order.

Handling of Duplicate Qualifications

If the system identifies duplicate qualifications, it assigns the instance with the highest proficiency level to the service transaction. For example, if servicing the equipment specified in a service order requires basic English proficiency, but the sold-to party in the service order demands an advanced level of English, the system will prioritize the advanced qualification and assign it to the service transaction.

Also, if one instance has the mandatory flag activated while the other does not, the qualification with the activated flag is assigned to the service transaction or item.

Dates for Qualification Validity Check

During the qualification identification process, the system checks the validity of qualifications against a reference date, for example, a requested start date in service orders and service quotations or the valid from date in service order templates. If these dates have not been specified, the system uses the current date as a reference.

As an alternative to the standard reference dates, you can also define your own date types to be used as a reference date to check the validity of a qualification. The custom date types you define override the standard date types. If, however, no value is defined for the custom date type, the system uses a standard date type as a reference date. You define your custom date types using the **Date Type for Qualification Requirements** parameter in the **Define Parameters for Qualification Requirements in Transaction** or **Define Parameters for Qualification Requirements for Item Category** Customizing activities.

Related Information

[Settings for Qualification Requirements](#)

Settings for Qualification Requirements

Customizing settings for qualification requirements allow you to control display and inheritance of qualifications in service transactions. You can activate parameters for identification of qualification requirements, edit qualifications, as well as implement custom logic using BAdI.

To configure how qualification requirements are identified, displayed, and inherited, you can use the following Customizing activities for **Service** under **Basic Functions > Qualifications**:

- **Activate Qualification Requirement Parameters:** In this activity, you activate the qualification requirement parameters to be displayed in the service transactions. The parameter *proficiency* is active by default.
- **Edit Qualifications Catalog:** Using this activity, you can create new or edit existing qualifications and qualification groups of your interest.
- **Define Parameters for Qualification Requirements in Transaction** and **Define Parameters for Qualification Requirements for Item Category:** These activities define the parameters, based on which the system identifies qualifications required in a service transaction. For example, you can define that the system should consider parameters such as the sold-to party or the functional location to automatically identify qualifications needed to perform the service requested. To assign qualification requirements to master data such as sold-to party, you can use the ppm transaction or go to **SAP Menu** and change the profile under **Human Resources > Personnel Management > Personnel Development > Profile > Change**. For all master data objects that can be used for the identification of qualification requirements, see the documentation of the activities in Customizing.

In these activities, you can also define your own date type for checking qualification validity.

Furthermore, you can define whether qualification requirements should be inherited, for example, from header to item or within an item hierarchy of a service transaction. If qualification requirements are to be inherited within an item hierarchy, you can also define for which items the inheritance of qualifications should work. A prerequisite for qualification inheritance is item status **Open** or **In Process**. For more information, see the documentation of the activities in Customizing.

To identify qualification requirements in service transactions, you can also implement your own custom logic using **BAdI: Qualification Determination in Transactions**. For more information, see the BAdI documentation in Customizing for **Service** under **Basic Functions > Qualifications > BAdI: Qualification Determination in Transactions**.

Related Information

[Qualifications in Service Transactions](#)

[Create Profile](#)

Service Hierarchy with Collective Accounting

You can use the function Service Hierarchy with Collective Accounting to customize the inheritance of the account assignment object of a main item in an item hierarchy.

Use

In a service hierarchy with collective accounting, you specify an item as the main item in the hierarchy where only this main item, but not its subitems, creates an account assignment object. The subitems inherit the account assignment object and the relevant attributes such as the profit center and the service contract assignment from the main item. With this, all the costs and revenue triggered by the subitems are only posted to the account assignment object of the main item. A service hierarchy with collective accounting makes it possible to collect costs and revenue under one account assignment object. For example, if one item only incurs costs and another item only brings in revenue, the hierarchy ensures that event-based revenue recognition conducts accurately.

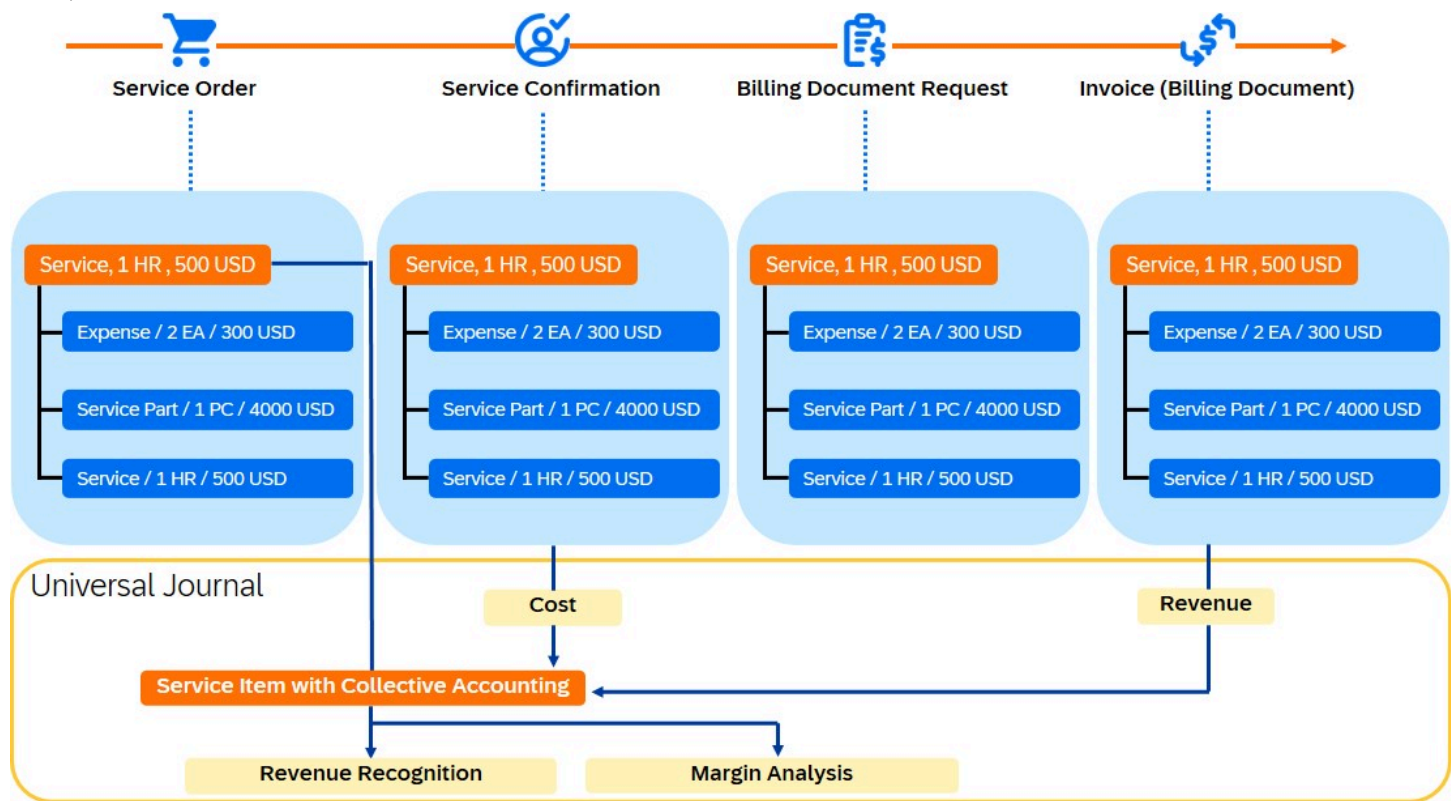
You can use the following item categories for service hierarchy with collective accounting:

- [Service Item with Collective Accounting](#) (SHSV)
- [Service Item with Collective Accounting \(Fixed Price\)](#) (SHSF)
- [Intercompany Service Item Collective Accounting](#) (SRHI)

i Note

You use SRHI in the service order of the company that sells the service. In the intercompany service order of the company that executes the service, you use [Service Item with Collective Accounting](#) (SHSV) and [Service Item with Collective Accounting \(Fixed Price\)](#) (SHSF).

The following diagram illustrates a sample process of how to use **Service Hierarchy with Collective Accounting**:



Prerequisites

To use **Service Hierarchy with Collective Accounting**, you need to fulfill the following prerequisites:

- Activate item-based accounting in the Customizing activity **Enable Item-Based Accounting for Service** in Customizing for **Service** under **Transactions > Settings for Service Transactions > Integration**. For more information, see [Item-Based Accounting](#).
- Select the value **Inherit to Subitems** for the item category for service hierarchy with collective accounting (SHSV, SHSF, or SRHI) in Customizing under **Service > Transactions > Basic Settings > Define Item Categories**.

i Note

You can select this value from the drop-down list in the field **Inheritance of Account Assignment Object** under **Item-Based Accounting** in the **Define Item Categories** Customizing activity.

- Assign the item category for service hierarchy with collective accounting (SHSV, SHSF, or SRHI) to an item to designate the item as the main item in the hierarchy.

Service Order and Service Confirmation

The following features are available when you use a service hierarchy with collective accounting in a service order and a service confirmation:

Service Contract Assignment

Service contract assignment is an attribute of an account assignment object. You can only assign a service contract item to the main item in a service hierarchy with collective accounting. However, the price agreements from a service contract item are directly copied to the subitems in the hierarchy. In the popup **Contract Assignment**, the service contracts that you can select are filtered according to the equipment specified for the main item.

Unplanned Items

You can add an unplanned item to a service confirmation in a service hierarchy with collective accounting. An unplanned item can only be created with a **reference service order item** in the preceding service order of the service confirmation. This reference service order item can be the main item or any subitem whose item type is **Service Item** in the hierarchy. With this, the unplanned item in the service confirmation inherits the account assignment object from the main item of the service hierarchy with collective accounting. The costs from this unplanned item in the service confirmation is then posted to the inherited account assignment object.

The corresponding item in the preceding service order has the status **Released** and the added unplanned item has the status **Open** in the follow-up service confirmation. You can still delete the unplanned item as long as it has not been set to **Completed**.

For more information, see [Unplanned Items](#).

Price Items

You can use a price item as a subitem in a service hierarchy with collective accounting. For more information, see [Price Items in Service Orders and Service Quotations](#).

Sales Items

You can use a sales item as a subitem in a service hierarchy with collective accounting. For this, you need to choose the Controlling scenario **Service** in Customizing for the item category of the sales item. This way, the sales item inherits the account assignment object of the main item in the hierarchy. The account assignment object is then transferred to the follow-up sales order as well. To do so, you can use the Customizing activity **Define Item Categories** under **Service > Transactions > Settings for Service Transactions > Basic Settings**.

Calculation of planned cost and revenue and event-based revenue recognition can be conducted for a sales item in a service hierarchy with collective accounting.

Planned Cost and Revenue

When you use a service hierarchy with collective accounting, the planned cost and revenue are calculated on the level of the main item. This means that the planned cost and revenue of all the subitems and the main item itself in the hierarchy are accumulated and displayed on main item level. You can view the sum of planned cost and revenue in the item details of the main item under the **Planned Cost and Revenue** tab.

i Note

- Planned costs and revenue are not calculated for unplanned items in a service hierarchy with collective accounting.
- If an execution order item is the subitem in a service hierarchy with collective accounting, the planned cost can be different after you choose **Update Baseline**. The reason for this is that the maintenance order related to the execution order item triggers its own planned cost separately from the service order.

For more information, see [Service Planning: Managing Planned Cost and Revenue](#).

Event-Based Revenue Recognition (EBRR)

Event-based revenue recognition (EBRR) can be conducted for a service hierarchy with collective accounting. To enable this, a revenue recognition key must be assigned to the item category of the main item in the hierarchy. To conduct revenue recognition on the main item level, the following revenue recognition methods are supported:

- Completed Contract (Method 15)
- Cost-Based Percentage of Completion (POC) (Method 3)

If the status **TECO** is set for the main item in the service hierarchy with collective accounting, all the costs and revenue are recognized and the EBRR accounts are cleared regardless of which method is chosen. To set the TECO status, the main item and all of its subitems must fulfill the prerequisites, for example, completed follow-on processes, billing processes, or cost postings.

i Note

Since only the main item in a service hierarchy with collective accounting creates an account assignment object, only the account assignment object of the main item can be set to the status **TECO**. For more information, see [TECO Status of Account Assignment Objects in Service](#).

For more information, see [Event-Based Revenue Recognition for Service Orders](#).

Intercompany Service Orders

You can use a service hierarchy with collective accounting in intercompany service orders. For more information, see [Intercompany Service Orders](#).

Service Quotation

You can use a service hierarchy with collective accounting in a service quotation in the following ways depending on your Customizing:

- Add **Service Item with Collective Accounting** as the main item of the service hierarchy with collective accounting to a service quotation.
- Add subitems to the main item in the hierarchy. With this, the subitems inherit the service contracts from the main item.
- Create a follow-up service order and copy the service hierarchy with collective accounting to the service order.
- Create a requote and if the requote is accepted, the accepted items from the hierarchy are copied to the existing service order.

i Note

The relations of the items to the higher-level items are also copied to the service order.

For more information, see [Service Quotations](#).

Service with Advanced Execution

Execution order item is an item type used in the process of service with advanced execution. You can add an execution order item as a subitem to a service hierarchy with collective accounting in a service quotation or a service order.

Note the following when using execution order items in a service hierarchy with collective accounting:

- No account assignment object is created for the execution order item.
- An execution order item cannot be used as the main item of a service hierarchy with collective accounting.
- All the planned and actual costs from the corresponding billable maintenance order are collected to the account assignment object of the main item of the hierarchy.
- Itemized billing and summarized billing for execution order items in service with advanced execution are **not** supported in this context. Depending on the business requirement, the execution order item can have the following billing relevance: **Not Relevant for Billing**, **Billing on Completion** or **Billing via Billing Plan After Release**.

- Features of service with advanced execution such as resource-related quotation and distributed execution are supported.

For more information, see [Service with Advanced Execution](#).

Warranty and Maintenance Plans

If you use warranty management and maintenance plans with a service hierarchy with collective accounting, note the following:

Warranty

In a service hierarchy with collective accounting, warranty is determined for each item respectively in the hierarchy. The main item and each subitem can have a different main reference object, so warranty can be different among the main item and subitems.

Maintenance Plans

i Note

When a maintenance plan is scheduled, the service order and the corresponding billable maintenance order are generated as call objects.

- Scenario A:
 - A service order template is assigned to a maintenance plan item and this service order template contains a service hierarchy with collective accounting.
 - In this scenario, if you create a follow-up service quotation or a follow-up service order from the service order template, the service hierarchy with collective accounting from the template is copied to the call objects (the service order or the service quotation) in the maintenance plan as well.
- Scenario B:
 - A service contract is assigned to a maintenance plan item and this service contract contains a service hierarchy with collective accounting. Only the main item in the hierarchy is referred in the maintenance plan item.
 - In this scenario, if you create a follow-up service order from the service contract, only the main item, but not the subitems, in the service hierarchy with collective accounting is copied to the service order.

Execution Statuses

You can process service orders in systems beyond Service such as in Maintenance Management and in SAP Field Service Management and you can check the status of service execution in Service with Advanced Execution in Maintenance Management and in SAP Field Service Management.

When you use a service hierarchy with collective accounting, the execution status is determined for the main item and each subitem respectively in the hierarchy.

For more information, see [Execution Statuses of Service Orders](#).

Service Level Agreements (SLA)

You and the customer can use the SLA to make an agreement regarding the methods used to render a guaranteed service.

Use

The SLAs define the attributes (such as content, scope, and means) for the service products (such as maintenance or hotline) that have been agreed on with the customer. The SLAs that are available in the standard system are the service profile and the response profile.

❖ Example

Examples of parameters within an SLA include the following:

- Response profile (the technician's response time, in other words, the time by which the technician must start the service laid down in the service transaction item, as well as the time period in which the service must be completed)
- Service profile (time period in which the services that were defined can be provided)

Integration

Default Values in Master Data

The **Service Levels** assignment block is available in the following master data and can be used in SLA determination to assign the correct service profile and response profile to service request transactions such as service requests.

- Service contracts
- Service product items and products
- Business partners (sold-to party)
- Service organizational units

In the **Service Levels** assignment block, you can specify values for the above-mentioned SLA parameters, service profile, and response profile.

Pricing

An SLA influences pricing and the business transactions in Service. Therefore, for example, the monthly price of a service can change depending on the response time defined. SLAs also influence date calculation in the service process.

Escalation Management

If the services agreed on in the SLA cannot be performed within the time specified, a predefined escalation process is started.

SAP delivers a standard procedure for escalation management that you can use and enhance if required. You can make settings for escalation management in Customizing for **Service**, under ► **Transactions** ► **Settings for Service Transactions** ► **SLA Escalation Management** ►.

❖ Example

An escalation for a missed deadline could proceed as follows:

- The responsible processor is informed per e-mail when the **first response time** is exceeded and the transaction is **still not** in process.
- The responsible department manager is informed per e-mail when the first response time is exceeded by two hours and the transaction is still not in process.
- The responsible service manager is informed per e-mail when the first response time is exceeded by eight hours and the transaction is still not in process.

A similar escalation could occur if the **completion date** is exceeded and the transaction is **still** in process.

Features

By default, you can use the following SLA parameters:

- Service profile
- Response profile

Activities

You maintain SLA parameters on the [SAP Easy Access](#) menu, under [Service](#) > [Service Processes](#) > [Maintain Availability and Response Times](#) .

In the service order items, you can select values for the SLA parameters.

Response Profile

In the [Maintain Availability and Response Times](#) transaction, you can use any combination of the following indicator keys to create as many time frames with duration and time unit as you want:

- Category
- Priority
- Catalog
- Code Group
- Code

In this way, you can define, for example, a duration to first response, a duration to completion, and time frames that can be established in between, such as:

- Time until the arrival of a technician at the customer site
- Duration of repair
- Downtime of machines at the customer site

To maintain the response times, choose a key combination in the transaction by selecting the required line and switching to the view for response times. There you can enter any number of time frames for the selected key combination.

Choose a duration of time for each period that was created in Customizing for date management under [Service](#) > [Basic Functions](#) > [Date Management](#) > [Define Date Types, Duration Types and Date Rules](#) . Specify the duration and time unit for these.

When you maintain service products in the product master, you can enter default values for the response profile in the assignment block [Service Levels](#).

❖ Example

You receive a malfunction report from a customer on Monday at 10:00 a.m. and a service order with priority 1 is created.

In the response profile, a duration to first response of two hours and a duration to completion of eight hours is defined for service processes with priority 1. The standard service profile is defined in the service profile, which contains the following

availability times: Monday to Friday from 8:00 a.m. to 5:00 p.m.

The following times/dates result from this: First response time 12:00 noon, completion time 9:00 a.m. on the following day.

This means you must have responded to the malfunction report by 12:00 noon on Monday, for example, by calling the customer to discuss or clarify the problem. The transaction must be completed by Tuesday at 9:00 a.m., for example by a technician on-site.

Service Profile

You can stipulate a service profile for each service order item during the maintenance of service orders.

When you maintain service products in the product master, you can enter default values for the response profile in the **Service Levels** assignment block on the **Service** page. You can overwrite the default values in the service order item in which the service product was stipulated.

❖ Example

You define a standard service profile with the following availability times: Monday to Friday from 8:00 a.m. until 5:00 p.m. Central European Time (CET).

You assign the service profile **Weekend** to the service product **Weekend Hotline**, in which you define Saturday 8:00 a.m. CET to Sunday 7:00 p.m. CET as availability time.

Related Information

[Service Level Agreement \(SLA\) Determination](#)

Service Level Agreement (SLA) Determination

You can activate the automatic determination of service level agreements in a service order. For example, a service level agreement can be determined by attributes such as product, product category, or customer group.

Use

You can use service level agreement (SLA) determination to assign the correct service profile and response profile to a service transaction. This ensures that the service team processes, for example, the incident according to the agreed SLA.

Integration

You can assign SLA information to the following master data in the **Service Levels** assignment block, unless otherwise indicated.

- Products (defined as service products and identified as being part of a service order)
- Accounts with role **Sold-To Party**
- Service organizational units (**Service Profile** and **Response Profile** are added as attributes)

Prerequisites

- You have defined SLA profiles in the **SAP Easy Access** menu under **►Service ►Service Processes ►Maintain Availability and Response Times ►** (transaction CRMD_SERV_SLA).

- For the master data to which you want to assign SLAs, you have assigned a service profile and a response profile in the [Service Levels](#) assignment block in the WebClient UI in Service .
- You have defined an SLA determination procedure with an access sequence in Customizing for [Service](#) under [Transactions](#) [Settings for Service Requests](#) [Define SLA Determination Procedures](#) . The access sequence tells the system where and in what order to find the appropriate SLA information.
- You have assigned the SLA determination procedure to a transaction type in Customizing for [Service](#) under [Transactions](#) [Basic Settings](#) [Define Transaction Types](#) in the Customizing header for the transaction category [BUS2000116 Service Process](#).
- You can also use the [Business Add-In for SLA Determination](#) to define your own logic for determining SLAs. You can find this BAdI in Customizing for [Service](#) under [Transactions](#) [Settings for Service Requests](#) [Business Add-Ins](#) .

Features

SLA determination uses an access sequence to tell the system where and in what order to find the appropriate SLA information. The system finds the first source identified in the access sequence that has both a service profile and a response profile and assigns the profiles to the service transaction item. Examples of SLA sources include service contracts, service product items, and service organizational units.

More Information

[Service Level Agreements \(SLA\)](#)

Related Information

[Service Level Agreements \(SLA\)](#)

Tool Items in Service Transactions

Definition

An item within a service order that represents a tool needed during the service execution to fulfill the service and that can be reused for the next service.

Use

You can enter tools for the following service transactions in order to bill them to your customers and determine and allocate internal costs: Service orders, service order templates, service order quotations, and service confirmations.

Serialized Tools

You can enter the tool's equipment and serial number for serialized tool materials in tool items. This is available for service orders and service confirmations under [Equipment](#) on the [Details](#) page of the respective service transaction. The tool's serial number field will be filled automatically. Additional checks for quantity and equipment fields ensure accurate data entry.

Prerequisites

The following configurations need to be done in order to use tools in service transactions:

- Under ►► [Integration with Other SAP Components](#) ► [Customer Relationship Management](#) ► [Settings for Service Processing](#) ► [Controlling Integration](#) ► [Determine Cost Center and Activity Type for Tool](#) ► you need to maintain the [ActTyp](#) for tool items.
- In transaction KP26, you need to maintain the cost rates for the tool item for the respective cost center and the activity type.
- Under ►► [Logistics - General](#) ► [Material Master](#) ► [Basic Settings](#) ► [Material Types](#) ► [Define Attributes of Material Types](#) ►, you need to enable the user department [Sales](#) for the material type FHMI.
- Under ►► [Logistics - General](#) ► [Material Master](#) ► [Configuring the Material Master](#) ► [Define Structure or Data Screen for Each Screen Sequence](#) ► you need to maintain the program SAPLCRMS4_PRUDUCT_SUBSCREEN for the [Screen sequence](#) 21 ([Std ind.\(short\) tab pages](#)) and the [Data screen](#) 17 ([Production Resources/Tools](#)).
- Under ►► [Logistics - General](#) ► [Material Master](#) ► [Field Selection](#) ► [Assign Fields to Field Selection Groups](#) ► you need to maintain the [Maint. Status](#) VF for the field MARA - SRV_DURA ([Duration of Work](#)).
- Under ►► [Logistics - General](#) ► [Material Master](#) ► [Field Selection](#) ► [Assign Fields to Field Selection Groups](#) ► you need to maintain the [Maint. Status](#) VF for the field MARA - SRV_DURA_UOM ([Unit for Duration of Work](#)).
- Under ►► [Logistics - General](#) ► [Material Master](#) ► [Field Selection](#) ► [Assign Fields to Field Selection Groups](#) ► choose a new field selection group for the two fields MARA - SRV_ESCAL ([Response Profile](#)) and MARA - SRV_SERWI ([Service Profile](#)). Since these two fields MARA - SRV_ESCAL and MARA - SRV_SERWI are not relevant for tool items, you can hide them for the field reference that is assigned to the material type FHMI.

Service Process Flow

A process flow of service order processing is available in the [Manage Service Quotations](#) and the [Manage Service Orders](#) app. This process flow provides you with a graphical overview of the sequential steps related to service transactions such as service quotations, service orders, service confirmations, or maintenance orders and to documents such as billing documents. You can choose each transaction in the process flow to navigate to all related detail pages or apps. The process flow also provides status information for each shown transaction at a glance.

This is an example of what a process flow can look like:



For each service transaction, you can check the process flow in the system and learn more about its preceding transactions (the transactions that come before a specific service transaction in the process chain) and its subsequent transactions (the transactions that come after this service transaction in the process chain).

Prerequisites

In order to use the process flow functionality, you have to active the related service. For details on how to do this, refer to SAP note [3705125](#) .

The process flow is enabled as a new assignment block for the overview pages of service quotation and service order. This new assignment block can be added to the overview page via personalization.

Related Information

[Manage Service Orders](#)

[Manage Service Quotations](#)

Apps Used in Service Order Management

You can use the apps in service order management to support your service processes.

Manage Service Orders

App ID: TBT116MSR

With this app, you can search for service orders and display the search results. You can navigate to service orders and display and edit service order details. You can also use this app to directly create service orders and process them. You can add items to service orders and edit them to detail the information needed to plan, execute, and bill for a service.

Key Features

You can use this app to do the following:

- Search for service orders using different filter criteria such as service order description, life cycle status, sold-to party, or posting date.
- Display your search results in the result list and search service orders within the result list.
- Customize the view of the result list. For example, you can personalize how many columns of data relevant for service orders are displayed in the list. You can add columns to display relevant data of service orders such as transaction type, employee responsible, or sales organization.
- Save or archive your searches.
- Navigate to service order details.
- Create service orders and add items to service orders.
- Edit all the data relevant for service order details at header level and item level.
- Trigger follow-up actions, for example, create service confirmations to confirm service execution.
- Copy service orders or reject service order items.

Supported Device Types

- Desktop
- Tablet

Situation Handling

This app uses Situation Handling to inform specific users about the following situation:

No executing service employee has been assigned to the service order item.

For more information, see [Situation Template: Missing Executing Service Employee](#).

Using automatic notifications, you can navigate to the critical service orders directly and take the appropriate action. You can also use the **My Situations** app to display your open situations in an overview table and close situation instances by setting the appropriate status manually.

For more information, see [Situation Handling](#).

Related Information

[Service Orders](#)

Create Service Order

App ID: TBT116MCR

With this app, you can create and process service orders. You can add items to service orders and edit them to detail the information needed to plan, execute, and bill for a service.

Key Features

You can use this app to do the following:

- Create service orders and add items to service orders.
- Edit all the data relevant for service order details at header level and item level.
- Display service order details and navigate to every items added to a service order and display the item details.
- Trigger follow-up actions, for example, create service confirmations to confirm service execution.
- Copy service orders or reject service order items.

i Note

You can also use the **Manage Service Orders** app to create and process service quotations. For more information, see [Manage Service Orders](#).

Supported Device Types

- Desktop

- Tablet

Situation Handling

This app uses Situation Handling to inform specific users about the following situation:

No executing service employee has been assigned to the service order item.

For more information, see [Situation Template: Missing Executing Service Employee](#).

Using automatic notifications, you can navigate to the critical service orders directly and take the appropriate action. You can also use the **My Situations** app to display your open situations in an overview table and close situation instances by setting the appropriate status manually.

For more information, see [Situation Handling](#).

Related Information

[Service Orders](#)

[Manage Service Orders](#)

Manage Service Quotations

App ID: TBT116QMSR

You can use this app to create, display, and edit service quotations.

Key Features

You can use this app to do the following:

- Search for service quotations using different filter criteria, such as the description of the service quotation, life cycle status, sold-to party, or posting date.
- Display your search results in the result list and search for service quotations within the result list.
- Customize the view of the result list. For example, you can personalize how many columns containing service quotation-relevant data are displayed in the list. You can add columns to display service quotation-relevant data, such as the transaction type, employee responsible, or sales organization.
- Save or archive your searches.
- Create service quotations and add the required items.
- Navigate to the details of the service quotation and the related items.
- Add execution order items to service quotations to trigger the planning of labor (operations) and material (components) consumption that is performed in the follow-up maintenance orders.
- Add price items to service items and execution order items as sub-items, which are used to reflect the surcharge for any additional services provided.
- Perform credit checks and simulate credit checks to determine whether the gross value of the items might exceed the available credit limit.

- Trigger follow-up actions, such as creating service orders.
- Send the service quotation to the customer.
- Modify the service quotation and send it to the customer again.
- Create a requote if additional service items or service parts are required to perform a service, which have not been specified in the accepted service quotation.
- Record whether the customer accepts or rejects the service quotation.

Supported Device Types

- Desktop
- Tablet

Create Service Quotation

App ID: TBT116QMCR

With this app, you can create and process service quotations.

Key Features

You can use this app to do the following:

- Create service quotations and add items to the service quotations.
- Edit the service quotation and the related items.
- Display the details of the service quotations and the related items.
- Add execution order items to service quotations to trigger the planning of labor (operations) and material (components) consumption that is performed in the follow-up maintenance orders.
- Add price items as sub-items to service items and execution order items that are used to reflect the surcharge for additional service provided.
- Perform credit checks and simulate credit checks to determine whether the gross value of the items might exceed the available credit limit.
- Trigger follow-up actions, for example, create service orders.
- Send the service quotation to the customer.
- Modify the service quotation and send it to the customer again.
- Create a requote if additional service items or service parts are required to perform a service and which have not been specified in the accepted service quotation.
- Record if the customer accepts or rejects the service quotation.

i Note

You can also use the [Manage Service Quotations](#) app to create and process service quotations. For more information, see [Manage Service Quotations](#).

Supported Device Types

- Desktop
- Tablet

Situation Template: Missing Executing Service Employee

Technical Name: SRVC_ORDER_MISSING_SERV_EMP

Consuming App: [Service Orders](#)

Template used for situation handling informs the service requester that no executing service employee has been assigned to the service order item.

i Note

This situation doesn't support service confirmation items.

For background information about the Situation Handling framework, see the Product Assistance for [Situation Handling](#).

Business Value

As a configuration expert for business processes (SAP business role SAP_BR_BPC_EXPERT), you can use this template for the following situation:

Customer service managers are informed automatically if the executing service employee is missing even though the requested start date is approaching for the service order item.

The manager can then instruct the Service Planner in SAP Field Service Management to assign the executing service employee, ensuring that the service is completed on time.

This standard template is useful when SAP Field Service Management integration is used, because in the non-SAP Field Service Management integration scenario, the executing service employee is assigned as a mandatory field in the service confirmation and is not updated in the service order.

Default Settings

The template comes with predefined settings. For the settings that aren't visible or self-explanatory, you can find information in the following sections. For generic information about how to configure situations based on this template, refer to the documentation of the [Manage Situation Types](#) app with which you can display and use the template.

For more information, see [Manage Situation Types](#).

Situation Trigger Object and Anchor Object

These technical settings define for which object a situation is displayed (anchor object) and the object based on which a situation is triggered (trigger object):

Trigger Type: Batch

Trigger Object: Service Order

Trigger Object View: I_SRVCORDITMMISGEXECTGEMPLSITN

Trigger Object Unique ID: ServiceDocumentItemUUID

Anchor Object: Service Order

Anchor Object View: I_SRVCORDITMEJECTGEMPLSITN

Anchor Object Unique ID: ServiceDocumentItemUUID

Description: The system checks for service orders and informs the manager that no executing service employee has been assigned to the service order item.

A batch job removes all situations not determined by the latest run every 24 hours, considering them resolved.

Conditions:

The condition in this standard template defines that a situation instance with the status **Open** is created and a notification is sent to the defined recipients if the system detects a service order that meets the criteria specified above.

This template provides the following filters that you can use to define your own conditions when you create situation types:

Filter	Description
Created By	The user that created the transaction.
Distribution Channel	The distribution channel defined for the service order.
Division	The division defined for the service order.
Exec. Service Emp.	The ID of the executing service employee.
Item Category	The category of the service order item or the service confirmation item for which the billing document was created.
Item Number in Doc.	The service order item number.
Sales Group	The sales group defined for the service order.
Sales Office	The sales office defined for the service order.
Sales Org. ID	The ID of the sales organization.
Sales Organization	The sales organization defined for the service order.
Service Organization	The service organization defined for the service order.
Transaction ID	The transaction ID of the service order.
Transaction Type	The transaction type of the service order. Possible values are: SV01 SV02
Responsible User ID	The ID of the service responsible employee.
Days until Requested Start Date	The difference between the requested start date and the current system date. Valid values are from 0 to 14.

For more information, see [Conditions](#).

Situation Display

When situations occur, users are informed about them by various texts, such as in-app situation messages or notifications on SAP Fiori launchpad. In this section, you can see all preconfigured texts.

For more information, see [Situation Display](#).

i Note

When you copy the template and adapt it to your needs, you can define the notification settings and change all text elements in this section. After making the changes in the original language, please remember to translate these changes into the other languages that you want to use. The template comes with translations that can be changed directly when you copy the template.

For more information, see [Translate Situation Types](#).

Recipients

In this section, you see the predefined settings for determining who is responsible for a situation instance. Based on these, you can define who receives a notification when a situation instance is triggered (if enabled) or who sees the instance in the [My Situations](#) app.

In this template, the recipient of the notification is determined using the rule **Notify responsible employee for the service order**. This means that if you have procured service parts or third-party service performers to provide a service and these part or services have not been reported as completely consumed, the employee responsible assigned to the service order is notified.

For more information, see [Recipients](#).

Situation Monitoring

Monitor Instances: Monitoring (the tracking of changes to situation instances) isn't enabled by default for situation templates. When you copy the template to create a situation type, you need to enable this option if you want to capture data for this situation type and enable the creation of [business situation events](#). You can view the data in the [Monitor Situations](#) app.

CORS

In the hub scenario, where the frontend and the backend systems are separate, the call to the reuse library from the backend to the frontend server may be blocked due to **Cross-Origin Resource Sharing (CORS)** settings. If this is the case, you need to maintain the **CORS** setting in the frontend server to allow request from the backend server in the transaction UCONCOCKPIT.

CORS should be enabled in UCONCOCKPIT transaction as per below settings:

- **Scenario:** HTTP Whitelist Scenario
 - **Context Type :** 4 - Cross-origin Resource Sharing
 - **Mode:** Active Check
1. Login to [Fiori Launchpad](#).
 2. Navigate to [Libraries](#).
 3. Open transaction UNCOCKPIT.

4. Enable the following settings for context type Cross Origin Resource Sharing(CORS) whitelist rule:

- HTTP Service Path: /sap/
- Origin Host Rule: It must contain a matching of the back end server origin. (e.g. *.wdf.sap.corp)
- Get: X
- Head: X
- Options: X
- Connect: X
- HTTP CORS Credentials: X
- Allowed Methods: GET,HEAD,OPTIONS,CONNECT
- Allowed Headers: X-XHR-LOGON,ACCESS-CONTROL-REQUEST-METHOD,ACCESS-CONTROL-REQUEST-HEADERS,ORIGIN,X-REQUESTED-WITH
- Exposed Headers: ACCESS-CONTROL-ALLOW-HEADERS,ACCESS-CONTROL-ALLOW-ORIGIN,ACCESS-CONTROL-ALLOW-METHODS,ACCESS-CONTROL-EXPOSE-HEADERS,X-XHR-LOGON
- Max Age: 60

i Note

Instructions may be valid for SAP or at the customer site.

For more information , please refer to [UI Integration With Situation Handling](#)

[Using the UCON Cockpit](#)

Related Information

[My Situations](#)

Service Request Management

You use Service Request Management to monitor and fulfill requests for service by your customers. Service requests can be used internally by companies where a department delivers services as well as in external customer-facing scenarios. They enable you to measure and evaluate the quality of services provided.

Service personnel or users can log service requests in the system through a service desk and then, if necessary, send them to the appropriate personnel for service delivery.

Integration

- You can use the integration with E-Mail Request Management System (ERMS). E-mails sent to the service desk through ERMS automatically generate a service request, and subsequent replies are also automatically linked to that service request. For more information, see [E-Mail Response Management System](#)
- You can connect **SAP Service Ticket Intelligence** to use machine learning to automatically categorize service requests.

Prerequisites

You have defined the following in Customizing for **Service**:

- Service request-related transactions under **► Transactions ► Basic Settings ► Define Transaction Types ►**. There is a predefined transaction type named **Service Request** (SRVR) provided in the standard system that you can use as a base.
- Settings for service requests under **► Transactions ► Settings for Service Requests ►**

Features

The following features are available for Service Request Management:

i Note

These features are available for the technical object type BUS2000223 — **Customer Management Service Request**

- You can set up your business transactions to meet the specific requirements of Service Request Management.

❖ Example

- You can use expanded capabilities of multilevel categorization.
- You can define templates to facilitate the entry of commonly reported requests.
- You can optimize the information flow to keep your service personnel informed about the status of service requests, the customers and contacts and the reference objects.
- You can use helpful tools that enable you to monitor the service requests that are assigned to your organizations and service teams.
- You can use several functions that support you when processing service requests.

❖ Example

- Checklists for step-by-step task processing by one or more users
- Text management capabilities to provide text information for the internal and external documentation of the service request processing
- Forwarding functions that you can adapt to your requirements
- Processing log to record all processing steps
- You can monitor the processing of service requests.

❖ Example

- You can control the compliance with the agreements you have made with the partners involved in the performance of services.
- You can escalate service requests if they are not processed within a specific period of time.

Service Request

Definition

A user-related request for a defined service or for a new service.

Use

You can use this object to monitor and fulfill internal and customer service requests.

Service requests can be logged directly by users themselves or by service personnel and can, when necessary, be sent to the appropriate personnel for service delivery. They can also be monitored and analyzed to evaluate the performance of your service organization.

Integration

Service requests can be integrated with knowledge articles. Once a solution to a problem has been found, you can create and release a knowledge article to make that solution (or workaround) available for future incidents and service requests.

More Information

[Service Request Management](#)

[Incident](#)

[Knowledge Article](#)

Incident

Definition

An event that is not part of the standard operation of a service and that interrupts or reduces the quality of that service.

The business transaction used to log such an event.

Use

You use this object to report an incident to the service desk when something in the infrastructure is not working as expected. Either the user or the service desk agent creates the incident transaction.

A service desk agent then prioritizes the incident and searches for a related knowledge article that might provide a solution or workaround. If the agent cannot find an appropriate solution or workaround, they dispatch and track the incident until a solution is found.

Within service management, incident transactions are based on the service request business object. For more information, see [Service Request](#).

Integration

Incidents can be integrated with the same objects as service requests.

More Information

[Knowledge Article](#)

Service Request Search

You can use the Enterprise Search to search for service requests using various search criteria, such as the life cycle status, sold-to party, or service team.

Use

To find a specific service request, you use the Enterprise Search in the header bar of the SAP Fiori launchpad. Before you enter any attributes in the search input field, you select the category [Service Requests](#).

Search Criteria

To search for service requests, you can use the following attributes:

- Service request ID
- Service request description
- Life cycle status
- Sold-to party description
- Employee responsible (first name and/or last name)
- Service team

You can also combine several attributes (you need to separate them by a space in the input field): For example, you can search for all service requests with the life cycle status [Open](#) that have been created by a specific employee.

Search Result

A list of service requests is displayed as a result. For the first service request, the display area includes the header and the details. To display the details of other service requests, you can expand the display area. You can also sort the search results according to your needs, export them to Excel, or share them by email.

To narrow down your search results, you can apply one or more filters, such as a life cycle status, employee responsible, or transaction type. The filters can be visualized as a list or a chart.

From the search results, you can choose [Display Service Request](#) in the display area of a service request to navigate to the selected service request. You can also use the [Search Service Requests](#) app to search for a specific service request.

Information for Key Users

Relevant Business Catalog

Your user must be assigned the business catalog `SAP_SERV_BC_EMP_ORDERS_T`.

Related Information

[Search](#)

[Searching Apps or Business Objects](#)

Item Determination

Use

The system creates an item for the service request and assigns a service product to the item based on the selected categories. The service product can be used to determine, for example, service level profiles and dates.

Prerequisites

You have made the following settings in Customizing for **Service**:

- Mapped transaction types to catalog categories in [Cross-Application Components > Multilevel Categorization > Assign Transaction Types to Catalog Categories](#)
- Configured item determination in [Transactions > Settings for Service Requests > Define Item Determination](#)

You have defined service products for categories on the UI in Service under [Service Operations > Categorization Schemas](#). At the schema level, you can assign schemas to application areas. At the category level, you can assign service products to categories.

Features

Depending on the Customizing settings, the system determines the service product using multilevel categorization or by using the Business Add-In (BAI) [Product Assignment for Creation of Service Items](#). You can find this BAI in Customizing for **Service** under [Transactions > Settings for Service Requests > Business Add-Ins](#).

- If item determination has not been set up in Customizing and in the UI category modeler in Service, the BAI is used.
- If item determination has been set up for multilevel categorization but the service product cannot be determined based on the selected categories, the BAI is used.
- If item determination is done using multilevel categorization, the service product is assigned to the item when you save the service request.

If multiple service products are determined based on multilevel categorization, the system chooses the service product assigned to the lowest level category. If there is no service product specified for the lowest level category, the system searches for a service product at the next highest category level.

More Information

[Multilevel Categorization](#)

Business Context

Use

You can use this function to link documents, such as customer invoices, to a service request.

Integration

The business context function is integrated with the service request.

Prerequisites

You have made the settings in Customizing for **Service** under **Transactions** > **Settings for Service Requests** > **Define Settings for Business Context**.

Features

You can use the business context to do the following:

- Display and maintain a list of links in service processes in Service
- Create and view links to objects in Service such as e-mail and knowledge articles
- Unlink processes and objects, depending on whether you have made the necessary settings in Customizing
- Navigate directly to linked back-end objects and processes

Objects that you want to link to the business context can be found or created in the following ways:

- Enter a known object ID to link an object in the list of **Business Context** objects.
- You can create a new object with or without dialog by selecting an object type and choosing **Create**.

Whether or not a dialog is used depends on the settings you make in Customizing or in the implementation of the object link class.

- You can search for objects in the agent inbox.
- Objects from the global data context of the IC can be automatically linked to transactions created during the current session.

If the user role does not have the object type configured for the business context, the object appears in the business context but you cannot view the object details.

Example

You are working on a service request and want to link it to a customer invoice. In the **Business Context** assignment block, you search for the invoice and confirm it. The system links the customer invoice to your service request and displays it under **Business Context**. From here, you can navigate to the customer invoice and review its details.

Setting up Service Request Transactions

Use

There are several features available that make it easier for your customers or your service teams to complete a service request.

Features

Multilevel categorization

You can use the expanded capabilities of multilevel categorization to categorize your service requests quickly and accurately and, for example, to determine service products automatically based on the selected categorization.

To use multilevel categorization, you must have made settings related to multilevel categorization under ► **Service** ► **Cross-Application Components** ► **Multilevel Categorization** ►

- **Define Application Areas for Categorization**
- **Assign Transaction Types to Catalog Categories**
- **Assign Item Categories to Catalog Categories**

For more information, see [Multilevel Categorization](#)

Templates

You can define templates to use for frequently reported requests.

Service Teams

You can automatically assign a service team to a service request based on a ship-to party specified in this service request. To be able to do this, you first define a service team profile using the Customizing activity **Define Service Team Profile and Assign Service Team Types** under ► **Service** ► **Master Data** ► **Service Teams** ►. You then assign this service team profile to the transaction type SRVR (service request) using the **Define Transaction Types** activity under ► **Service** ► **Transactions** ► **Basic Settings** ►.

In addition, if you use the enterprise organizational model, you have to assign partner determination procedure IT000002 to service requests using the **Define Transaction Types** activity. Compared to the previously used partner determination procedure IT000001, the new partner determination procedure includes the partner function **Ship-To Party** and excludes **Service Employee Group**.

Processing Service Requests

Use

For the processing of service requests, several functions are available to help you to establish and optimize your internal processes and the communication with your customers.

Features

- You can customize the processing log that presents changes made to a transaction in a structured manner. For more information, see [Processing Log](#).
- You can use checklist functions, which you can use to define step-by-step processing of a task by one or more users. For more information, see [Checklist](#).

Checklist

Use

You can use checklists for step-by-step task processing by one or more users of transactions such as service requests. A checklist is a list of steps to be performed by assigned step partners. The checklist profile consists of a number of checklist-related settings, including the checklists themselves.

There are two types of checklists: simple and decision-based. Simple checklists are static and allow only a linear processing of their steps, while the flow of decision-based checklists is influenced by user input.

Integration

Checklists can be integrated with the following business transactions:

- Service request
- Service order

Prerequisites

You have completed Customizing in [Service](#), under [Transactions](#) [Basic Settings](#) [Checklist Settings](#).

Features

For processing the checklist in the appropriate assignment block, you have the following options (for both simple and decision-based checklists):

- Select the appropriate checklist from a list of all available checklists for the transaction type that you are creating or editing.
- When using simple checklists, you can insert steps manually into the selected checklist: both sequential and parallel processing of checklist steps is possible. To define a sequential processing manually, enter the previous step number.

When using decision-based checklists, adding manual steps is not possible. Also, parallel processing of checklist steps is not possible.

- Define checklist step details. For example, you can add the following information to the single checklist steps:
 - Enter work instructions and internal notes
 - Assign the parties involved
 - Schedule single steps
 - Add attachments with their properties and reference to URLs

Changes to a checklist step are logged in a change history.

- When using decision-based checklists you can trigger actions, for example, launch a new transaction, from individual checklist steps.
- Changing of the service request's attributes can result in the determination of another checklist. However, in this case, the system removes the information contained in the previous checklist from the transaction. That applies only until the checklist selection has been locked. For more information, see [Automatic Checklist Determination](#).

Processing Log

The most important actions that users perform on a service request are logged in the [Processing Log](#) assignment block. For example, the system logs changes to the step status or the partner. Actions that the users perform on a checklist step are logged in the change history of the checklist step.

Example

You have an employee who is moving from one office to another. To facilitate this move, several parties, for example, IT management and facilities management, must carry out specific tasks. You can create a checklist that describes each of these tasks in separate steps, and assign each step to the appropriate step partner. Those responsible can then see what tasks they need to carry out, and can indicate when each step is completed.

Customizing of Checklists

Use

In Customizing, you can define different checklists for all transactions types for which a step-by-step processing is reasonable.

You make the necessary settings in Customizing for **Service** under **Transactions** > **Basic Settings** > **Checklist Settings**.

Features

To configure the checklists, you have the following options:

- You can define checklist steps, simple and decision-based checklists, and checklist profiles independently. This enables you to reuse these objects for different purposes.
- You can create checklist profiles that contain a set of all checklists that you need for a special transaction type.
- You can create checklists that contain a list of steps that the responsible employee, the contact person or other predefined role shall process.
- You can create checklist steps that you can assign to different checklists. Additionally, when simple checklists are used, the steps are available in the input help of checklist steps in the checklist assignment block.

When defining checklist steps, you have the following options:

- You can define the sequence, in which the steps shall be carried out.
- You can define the objects that the users can maintain for every step. For example, you can define the following objects:

- Different text types for internal and external documentation and information.

If required, you can add fields to the checklist assignment block that contains these text types. The texts are displayed as a tooltip, so you can read the texts without opening the checklist step.

By mapping the checklist step texts with the text types of the business transaction header, you can integrate the texts in the notes assignment block. This allows you to monitor the message flow of checklist steps.

- Options to categorize the step progress, for example for reporting purposes.
- You can define rules to determine checklists automatically. For more information, see [Automatic Checklist Determination](#).
- You can define rules to determine the start and end date of the checklist steps.

For more information, see [Calculation of Checklist Step Duration](#).

- You can integrate checklists with SAP Business Workflow.

Additional Customizing of Simple Checklists

In addition to general customizing, you can use certain features and functions for simple checklists.

For editing the checklist in the appropriate assignment block, you have the following options:

This is custom documentation. For more information, please visit [SAP Help Portal](#).

- Define the sequence of the steps by changing the step number.
- Cancel steps that are not in status **Completed** and which are not set as **Mandatory** in Customizing.

When defining checklist steps, you can select whether a step should be mandatory or optional.

Additional Customizing of Decision-Based Checklists

In addition to the general customizing, you can use the following features and functions for decision-based checklists:

- Define so-called triggers and assign them to a checklist step. A trigger appears as a button on the UI and can change the status of the checklist step to **Completed**. It can also start actions, for example, sending a status report via e-mail. Actions are defined in the Post Processing Framework.
- Add logical links to checklist steps. Logical links are either web links or links to other internal systems. You can use the Transaction Launcher to enable navigation to other systems.

Calculation of Checklist Step Duration

Use

For the steps of the checklist, you can create rules that determine the planned start and planned end dates automatically.

You can define rules to fill these dates with default values. The start date and the planned end are then determined automatically.

Features

- The system calculates the duration of the checklist steps according to the response times that you have defined in the **Response Profile** field.
- The determined dates are assigned to every step of the checklist. If required, you can overwrite these dates to schedule single steps in the time frame of the given duration. Especially for follow-on steps, manual scheduling can be helpful. The end date must not be higher than the automatically calculated value.
- The system uses the creation date of the checklist to calculate the dates.

i Note

The start date of the transaction itself does not influence the calculation of the dates in the Checklist steps.

- By editing the transaction, the system automatically recalculates the dates in the following situations:
 - When you change one of the attributes in the transaction that are used as key fields in the **Response Profile** field, for example **priority**.
 - When you change the response profile.

Automatic Checklist Determination

Use

You can use the automatic checklist determination to provide predefined checklists for transaction types such as service requests.

This feature is intended to support you during the initial phase of processing a service request.

Prerequisites

You have defined the fields for the automatic checklist determination in Customizing of [Service](#) under [Transactions](#) [Basic Settings](#) [Checklist Settings](#) [Define Fields for Triggering Checklist Determination](#).

Features

- You can define rules for the automatic checklist determination by using attributes for the transaction type, for example for the following:
 - Priority
 - Category
 - SLA response profile

Depending on the transaction details the users have entered, the system inserts the appropriate checklist or provides the users with a list from which they can select one.

- You define the rules for the checklist determination in the rule policy in the context [Checklist Determination](#) (CHECKLIST_ID).

If your rules determine more than one checklist, the checklists are listed in an input help.

- For technical reasons, the checklist selection must be locked at an early stage of transaction processing. This means that after this moment the system no longer calls the automatic checklist determination. If the rule has returned more than one checklist, the users are no longer able to change the selected checklist.

In Customizing, you can choose between the following options for determining this point in time:

- Locking the checklist after the transaction is saved for the first time.
- Locking the checklist after the first step is completed.

For more information, see Customizing activity [Transactions](#) [Basic Settings](#) [Checklist Settings](#) [Define Checklist Profiles](#).

Text Management in Service Requests

Use

When processing service request transactions, you can use specific functions to add and provide text information. These functions support you in managing and monitoring the communication between the several partners involved in the process.

Reporters or specialized partners can add and change notes to describe problems or add additional information. For example, service personnel can use the notes for the following:

- Checking back on a problem
- Informing the reporters about the progress of a service request
- Providing a solution to a problem.

Prerequisites

- You have made the settings for text management in Customizing for **Service** under **Basic Functions > Text Management**. In the standard system, the text object CRM_ORDERH and the text determination procedure **Service Request** (IT000001) are used for Service Request Management.
- The notes assignment block with ID GSTEXT is visible in the following overview pages:
 - Service Request/Incident (UI component SRQM_INCIDENT_H, View ID IncidentOV)
- If required, you have removed the other text and text log assignment blocks (UI Component ID GSTEXTEDIT).

Features

To map the particular requirements of service request management, the following functions for the **Notes** assignment block (UI Component GSTEXT) are available. For more information, see

- [Editing Notes](#)
- [Handling of Notes](#)

More Information

For more information about the standard text management, see [Text Management](#)

Editing Notes

Use

When editing notes, there are some functions available to meet the special requirements of text management in the context of service request management.

Features

Classifying texts

For editing and administration of notes, you classify texts in two ways:

- You create different text types. For example, you create text types that describe the following usages of a text:
 - Problem description
 - Solution description
 - Reply to requestor
 - Memo
 - Internal note
- You create keywords that the users can use to classify and to filter texts.

❖ Example

A user enters a deep technical analysis of a problem in a text with the type **Internal note**. Several internal notes are assigned to the problem. For this reason, the user adds the keyword **technical analysis** to that note. Now the processors of the service request are able to easily filter the list of texts to find the text that contains the technical analysis.

For technical reasons, the number of keywords that the users can assign to a note is restricted to 8. Therefore, you should consider how you want to design the usage of your text types and keywords.

❖ Example

You create different text types for the different usages and target groups of texts:

- Text types **Problem Description** and **Solution Description** to classify the communication at a special phase of the service request processing
- Text type **Internal note** for the communication between the different processors of a service request.
- Keywords **Technical Analysis** and **Test Case** to classify the content of a text type.

In Customizing, you can define the keywords for every combination of text object and text determination procedure. For more information, see Customizing for **Service** under **Basic Functions > Text Management > Define Keywords**.

Logging texts

For every text type, you can define in the Customizing for the text determination procedure whether it should be logged. For more information, see Customizing for **Service** under **Basic Functions > Text Management > Define Text Determination Procedure**. Select map **Definition of Procedure** and set the value of field **Changes (Control for Text Changes)** to P (= Log).

Text types that are not logged have the following features:

- You can only create them once for each language.
- You can change and delete them.

Text Templates

By editing a text type, you can include text templates. You can use this function to provide standardized texts for situations that occur frequently. After including a text template, you can continue editing the text and add further information.

You can create a text template by editing a text and saving it as a template. A text template requires the following attributes:

- ID and description
- Language
- Visibility level. You have to classify every text template whether it is to be used only by you or by all users (system).

Adding URLs

You can add URLs to the text body. The following URL schemas are supported and rendered as hyperlinks in the notes:

- http: and https: for adding web-sites or call of a Web application. Additional information in web-sites or by calling a Web application. The URL schemas http and https are supported for this function.
- mailto: for sending e-mails
- ftp: using the file transfer protocol

The system can automatically identify strings that you use to create a reference. To customize the URL recognition pattern, you need to develop your own class derived from class CL_CRM_TEXT_FORMAT_CONVERSION with its method SET_REGEX_PATTERNS and assign it in Customizing. For more information, see Customizing for **Service** under **Basic Functions > Text Management > Define Text Format**.

Handling Notes

Use

For the handling of notes the following features are available.

Features

Handling of note items

- The notes assignment contains the following basic functions:
 - Editing of texts
 - Displaying the notes history (log)

For this reason, you do not need to assign an additional text log assignment block to your service request overview pages.

- For tracking the whole notes history (log), you can sort and filter the texts that are assigned to a request by several categories such as the following:
 - Type
 - Creation Date
 - Created by
 - Keyword
 - Language

If for some notes multiple keywords are assigned, the users can enter wildcards in the custom filter to filter the notes.

- You can personalize the size of the notes assignment block quickly with the **Rows** pushbutton. This simplifies editing and displaying of long texts.
- You can simultaneously add notes even if the transaction is edited by another user (parallel text editing mode).

Hyperlinks

Even if a user is only allowed to display a note, he or she can use assigned hyperlinks to open the reference. For example, they can open a URL in the browser or in an application window.

Authorizations

With the authorization object CRM_TXT_ID, you can control which users are allowed to display or edit a text type.

Authorization object CRM_TXT_ID

Field	Description	Value
ACTVT	Activity	02Change, 03 Display
TEXTID	Text type from Customizing of Text Management	<Text type>
TEXTOBJECT	Text object from Customizing of Text Management	CRM_ORDERH or CRM_ORDERI

Visibility of Attachments

Use

You can use this function to control which user roles are allowed to display and edit documents that are attached to a business transaction.

❖ Example

For Service Request Management, you want to use this function to control the access to documents that are attached in the **Attachment** assignment block. The service requester should be able to do the following:

- Attach documents
- Open attached documents that are classified as **Information for customer**

He or she should not be able to open documents that are allowed for internal use only.

A button with a paperclip icon in the work area toolbar indicates that attachments are assigned to a service request. Choose the icon to navigate directly to the attachments assignment block.

Prerequisites

- For the authorization domain ATTACHMENT - BTX, you have defined different authorization scopes in Customizing of **Service** under ► **Transactions** ► **Basic Settings** ► **Define Authorization Scopes for attachments.** ►

In the standard, no authorization scopes are assigned to the authorization domain ATTACHMENT - BTX. If you do not define the authorizations scopes, the system does not check the authorizations. This also applies if you delete the authorization domain.

i Note

This function does not call the Business Add-In CRM_DOC_AUTHORITY that is described in Customizing of **Service** under ► **Basic Functions** ► **Content Management** ► **Business Add-Ins (BAIs)** ► **BAI: Authorization Check on Document Level.** ►

- You have added the authorization object CRM_AUTHSC to the required user roles (transaction PFCG).

Features

To control the access to attachments, you define in Customizing different authorization scopes for the authorization domain ATTACHMENT - BTX. By editing an attachment in the **Attachments** assignment block, the users assign the appropriate authorization scope.

❖ Example

You create the following authorizations scopes related to the different needs of the target groups:

- Information for agent
- Information for customers
- Internal message

- Confidential

In the authorization object CRM_AUTHSC, you can define for each authorization scope, which activities the users are allowed to carry out.

❖ Example

For Service Management purposes, you create the authorization scope **Information for agent**. The service requester can use attachments with this authorization scope to add additional information for the service personnel. The service requester should be able to create, change, edit, and delete this attachment.

The following authorization levels are available with authorization object CRM_AUTHSC:

- Display: The user can open attached documents.
- Create or generate: The user can attach a document and define its authorization scope.
- Change: The user is able to change the properties of an attachment, including the authorization scope.
- Delete: The user can remove the reference to the attached document.

Authorization object CRM_AUTHSC

The following table describes the authorization object **Authorization Scope** (CRM_AUTHSC).

Field	Description
Activity	Describes, what the users are allowed to do.
Authorization domain	ATTACHMENT-BTX
Authorization Scope	Use the value help to get the authorization scopes that you have defined in Customizing.

If a user does not have the required authorization, the system works as follows:

- Display: The reference to the document is not displayed in the attachment's list.
- Create or generate, change, and delete: The authorization scope is not available in the input help of the field.

Processing Log

Use

This function provides you with an overview of all updates that have been made to a service transaction, such as changes to field values, links to other transactions, and executed actions. You can access all of this information in one area, making it easier to monitor the processing of a transaction.

Prerequisites

You have made the settings in Customizing for **Service** under **► Transactions ► Settings for Service Requests ► Settings for Processing Log ►**.

Features

Processing Log

- Customizable content

You can customize the processing log to display different types of information, for example, old and new values for changed fields, who performed the update, and when the update occurred.

You can assign the required log types on the level of the transaction type and on the level of the category type.

- Default log type

You can set a default log type to be displayed for each transaction type. For example, when you open an incident, the processing log displays the change history. When you open a problem, the processing log displays executed actions.

- Navigational links

When there are changes in the form of linked transactions, attachments, or executed actions, the system provides a hyperlink to the corresponding objects.

- Assigned objects

You can log changes to the assigned service profiles or response profile.

- Checklists

You can log actions performed on a checklist, for example adding of a checklist, or changing the status of a checklist step, or automatic checklist determination.

- Authorizations

The users can display the log only for these notes or attachments that they are authorized for.

Data Determination Log

While processing a service request, you can use logs to analyze or trace the automatic determination of organizational data and partners.

- While determining organizational data, the system logs the organizational management objects by the sequence it has run through the organizational hierarchy. If you assign the user parameter CRM_ORGDETER_LOG to your user, the button **Determination Log** is available in the organizational data assignment block. By pressing the button, you can display the log.
- If changing a business transaction results in a new partner determination, the system logs the partner determination. This applies to the current session only.

If you select field **Log** in the partner determination procedure, the button **Determination Log** is available in the parties involved assignment block. In edit mode, press this button to display the log.

For more information, see Customizing of **Service** under **Basic Functions > Partner Processing > Define Partner Determination Procedure.**

Management of Objectives

Features

- Ability to classify transactions based on their urgency and impact to determine their recommended priority, which allows for accurate processing in accordance with agreed delivery times

- Escalation of service requests that sets escalation levels and dates to identify when the processing time of a service request has been exceeded. For more information, see [Escalation of Service Requests](#).
- Automatic calculation of the time spent on each stage of service request processing. For more information, see [Duration and Date Recording for Service Requests](#).
- Automatic determination of service level agreements (SLAs) to transactions based on determination procedures. For more information, see [Service Level Agreements \(SLA\)](#).

Duration and Date Recording for Service Requests

Use

This function allows you to define how to calculate how you record time spent on processing service requests, for example, an incident. You can record date information to evaluate if you are meeting the service profiles defined in the service level agreement (SLA).

Prerequisites

You have made the following settings in Customizing for **Service**:

- Defined date types and durations under ► **Basic Functions** ► **Date Management** ► **Define Date Types, Duration Types and Date Rules** ►.
- Defined status profiles under ► **Transactions** ► **Basic Settings** ► **Status Management** ► **Define Status Profile for User Status** ►.
- Assigned date types and durations to a date profile under ► **Basic Functions** ► **Date Management** ► **Define Date Profile** ►.
- Assigned status profiles and date profiles to transactions under ► **Transactions** ► **Basic Settings** ► **Define Transaction Types** ►.
- Defined date profiles for item categories under ► **Transactions** ► **Basic Settings** ► **Define Item Categories** ►.
- Assigned item categories to transaction types under ► **Transactions** ► **Basic Settings** ► **Define Item Category Determination** ►.
- Mapped the relationship between the user status, duration type, and date type under ► **Transactions** ► **Settings for Service Requests** ► **Define Settings for Durations** ►.

Features

- You can view the duration and date information for a service request in the **Service Level Agreements** and the **Dates** assignment blocks.
- You can define the durations that you want to record. There can be a one-to-one relationship between a user status and a duration or multiple user statuses can be combined together to calculate one single duration.
- The total duration is recorded. This means that if a transaction is in the same user status more than once, the duration is the sum of all these durations.
- You can set dates for date types when you change the user status of a service request. For example, the system sets the **Closing Date** of the incident when the user status is set to **Completed**.

Example

You want to record the total duration that a service transaction is in status **New**, **In Process**, and **Customer Action**. The time (duration) spent in status **New** and **In Process** represents the duration **Processing Duration**. The time (duration) spent in status **Customer Action** represents the duration **Customer Duration**. You combine the times spent in status **New**, **In Process**, and **Customer Action** to calculate the **Total Duration**.

Time Recording

Use

You can use this function to record your working time in business transactions such as service requests.

i Note

This function is not supported if you have activated item-based accounting.

Integration

The time recording function is integrated with the service request.

Prerequisites

You have assigned a service confirmation transaction type to the service request. The service confirmation transaction type will be used to automatically create a service confirmation during time recording. You can make the settings in Customizing for **Service** under ► **Transactions** ► **Basic Settings** ► **Define Transaction Types** ► by doing the following:

- Select the service request transaction type and choose the business transaction category **116 - Service Process**.
- Enter the service confirmation transaction type used for time recording in the **Trans. Type Confirm.** (Transaction Type Confirmation) field.

Features

- When you enter your time spent working on a business transaction, the service product available in the transaction's item is automatically copied to the time record as read-only information. You can enter start date and time, work duration, employee responsible, as well as additional data such as service and valuation type.
- Once the first time record is saved, the system automatically creates a service confirmation as a follow-up transaction of the service request. The confirmation is created according to the copy-control settings defined in Customizing. Additional time records in the same service request will be captured as items within the existing service confirmation.
- You can enter time records at any time as long as the transaction does not have the status **Completed**.
- You can edit a time record as long as it has not been saved. After saving the time record, you must apply any necessary changes to the service confirmation to which the relevant service confirmation item is linked.

Escalation of Service Requests

Use

You can escalate service requests if they are not processed within a specific period of time. For example, incidents that are not resolved within 48 hours after the initial response can be escalated to the first escalation level.

Two different escalation levels can be used to notify the designated responsible parties that the response deadline has passed. The first escalation step can be triggered when the deadline is exceeded the first time, and the second escalation step can be triggered when the first escalation has exceeded its predefined deadline.

You use actions to control escalation and de-escalation.

Prerequisites

- You have defined an action profile in Customizing for **Service** under [Basic Functions](#) > [Actions](#) > [Actions in Transaction](#) > [Change Actions and Conditions](#) > [Define Action Profiles and Actions](#).
- You have ensured that the following actions are available in each action profile:
 - IT_INCIDENT_DE_ESCALATION
 - IT_INCIDENT_ESCALATION
- You have assigned the action profile to a transaction type in Customizing for **Service** under [Basic Functions](#) > [Actions](#) > [Actions in Transaction](#) > [Assign Action Profile to the Business Transaction Type](#).

Features

- Service request response times are controlled by the service level agreement (SLA) profiles assigned to the service request. For more information, see [Service Level Agreements \(SLA\)](#).
- You can schedule a batch job to escalate service requests automatically, for example, to set the escalation level and escalation dates. At the defined time, the batch job is automatically started and the action identified in the variant of RSPPFPROCESS is executed for all of the identified service requests.
- You can execute the de-escalation of a service request directly from the **Scheduled Actions** assignment block. This lowers the **Escalation Level** and removes the corresponding escalation date.

Activities

To set up the batch job to automatically escalate service requests that are exceeding their due dates, do the following:

1. Run transaction SE38.

For each action, create a variant for the program RSPPFPROCESS to identify the actions and service requests you want to include in the batch job.

- In **Application Key**, identify the range of objects that you want to include in the variant.
- Choose the relevant action profile, for example, IT_INCIDENT_MANAGMENT.
- Choose the relevant action definition, for example, IT_INCIDENT_ESCALATION
- Select the **Processing Without Dialog** checkbox.

2. For each variant (for example, escalation and de-escalation), define a batch job using transaction SM36 and specify when to run the job.

More Information

[Service Level Agreements \(SLA\)](#)

In-House Service

In-House Service supports companies that offer maintenance and repair services for service objects (in-house service objects).

In-House Service helps you to plan and perform service activities for service objects by using the planning and execution capabilities provided by the service quotation and the service order. This allows you to handle the in-house service and the field service process for a service object in the same service transactions. In addition to the in-house service process, you can launch logistics-related process steps for the service objects that are sent in for service.

Alternatively, you can use Service with Advanced Execution, which combines the execution capabilities of maintenance management with the commercial capabilities of the service quotation and the service order.

i Note

The **In-House Repair** solution capability has been renamed **In-House Service**. This change affects the names of all objects containing the terms **in-house repair** and **repair**.

Features

In-House Service supports the following main features:

- Enter service objects: Start the in-house service process for the service objects in In-House Service, Service Order Management, or customer returns.
- Perform precheck: Decide on the follow-up actions to be performed within the in-house service process for each service object.
- Process service quotations: Edit and send out service quotations for the service object, and record whether the customer has accepted or rejected the service quotation.

If you use Service with Advanced Execution, you plan the service for the service object in maintenance management. The service quotation that is sent to the customer contains the prices determined in the related maintenance order. You then record the customer's decision in the service quotation.

- Plan and perform the service: Plan and perform the service of the service object using service orders. Record the actual consumption of services, service parts, and expenses in service confirmations.

If you use Service with Advanced Execution, plan and execute the service for the service object in maintenance management and use the commercial capabilities that are provided by the service order.

- Manage diagnosis: Plan and perform the diagnosis of the service object in the service order and record the actual consumption of services, service parts, and expenses in service confirmations. Update the service order with a requote and add items that were identified while the service is performed and that require acceptance by the customer.
- Release for billing: Trigger the billing process for the service performed.
- Launch logistics-related process steps that are performed in customer returns, Delivery Management (LE-SHP), and Inventory Management and Inventory:
 - Create returns orders or inbound deliveries for service objects that you receive from customers and for which a service is intended.
 - Create outbound deliveries to transfer the service objects to another service center where the service is performed.
 - Create outbound deliveries to send the service objects back to the customer.

Setting Up Your System

Follow the instructions outlined in Customizing for [Service](#) under **In-House Service** to perform the specific settings for In-House Service in addition to the generic settings for Service.

Basic Concepts for In-House Service

This section explains the basic concepts used in In-House Service.

In-House Service

In-house services are records that contain one or more in-house service objects (service objects), which are to be maintained or repaired at service centers. You use them to monitor the service and logistics activities for the related service objects.

Managing In-House Services

You create in-house services for single or for multiple service objects (see [Launch the In-House Service Process](#)). Having done this, you can process the in-house services in the **Manage In-House Services** app.

In-House Service Statuses

In-house services have statuses assigned that show their progress. The following statuses are provided for in-house services:

Status	Description
Open	This is the initial status that all in-house services have after they've been created in the Manage In-House Services app. It indicates that you can add and delete service objects. Once you've added the service objects to the in-house service, you need to confirm the in-house service. The status of the in-house service then changes to In Process.
In Process	Indicates that the service objects in the in-house service are in process. This status is set when you confirm the in-house service in the Manage In-House Services app or when you've created the in-house service from a service quotation, a service order, or a returns order.
Completed	Once all service objects in the in-house service have been completed (see Set Service Objects to Completed), the status of the in-house service automatically changes to Completed.

In-House Service Details

Under **In-House Service Details**, you enter the transaction type and the business partners who are involved in the in-house service process.

Shipping

Under **Shipping**, the ship-to party and the delivery terms, for example, the shipping conditions and the incoterms, are displayed. This data is first determined from the sold-to party that has been entered for the in-house service or the service transaction if the

in-house service process is started in a service quotation or service order. However, you can subsequently change the ship-to party and the delivery terms for the in-house service.

You can make the following changes with regard to the ship-to party:

- Change the business partner under **Ship-To Party**.

You can only use business partners to which the business partner role **Ship-to Party** is assigned in the business partner master data.

- Select a different **ship-to party address** for the specified ship-to party that is maintained for the ship-to party in the business partner data.
- Change the address under **Alternative Shipping Address**.

This address is then displayed under **Ship-To Party Address** and is used exclusively for this in-house service. It is inherited by the service objects and follow-up transactions, but not stored in the business partner data for the ship-to party.

i Note

You can change the shipping address for in-house services if the **Changeable Address** checkbox is selected in the **Define Partner Determination Procedure in Service** Customizing activity.

When adding service objects to the in-house service in the **Manage In-House Services** app, the shipping information is inherited by the service objects.

The following applies to the shipping information (ship-to party and delivery terms) that is managed in in-house services:

- If the shipping information for the in-house service and the related service objects contains the same values and the values in the in-house service are changed subsequently, these changes are inherited by the service object.
- If the shipping information for the service object is changed and therefore then differs from the values maintained for the in-house service, subsequent changes to the in-house service are no longer inherited by the service objects.
- Service quotations and service orders (service transactions) that are created for service objects take the shipping information that has been specified in the in-house service.

Changes that are subsequently made to the in-house service are **not** inherited by the service transaction. As a result, a service object can have different shipping information assigned than the related service transactions.

- Outbound deliveries that are created to send the service objects back to the customer, take the shipping information that has been specified in the service objects.

Service Contract Determination

In the **Manage In-House Services** app, the system determines the number of applicable service contracts when the in-house service is created or opened in the app.

For more information about the criteria that need to be fulfilled for service contract determination and the required settings in Customizing, see [Service Contract Determination](#).

Attachments

You can upload attachments, for example text documents or images, to your in-house service. If you create an in-house service from a document (see [AI-Assisted In-House Service Initiation](#)), this document is automatically added to the in-house service as an attachment.

For more information about setting up attachments, see [Harmonized Document Management](#).

Notes in In-House Services

You can enter notes, such as an instruction, that are provided by the customer and that support the in-house service process for the service objects.

The following applies to notes:

- When you enter notes for an in-house service and then add service objects, the notes are copied to the service objects.

You can then edit the notes for the service objects in the [Manage In-House Service Objects](#) app.

- When you enter or change notes for an in-house service to which service objects have already been added, the changes **aren't** transferred to the service object.

Instead, you can enter or edit the notes for each service object in the [Manage In-House Service Objects](#) app.

- You can't enter notes for in-house services that have the status **Completed**.

You can set up notes for in-house services in Customizing for [Service](#) under [► In-House Service ► Define Text Management ►](#).

Service Objects Overview

Within an in-house service, you can track the service objects' processing stage under [Service Objects](#). Depending on the related service transactions (service quotations, service orders, and service confirmations) and their status, a service object can be managed in more than one process step at the same time. For example, you can process a requote (see [Creating Requotes](#)) for a service object whilst you're already performing the service using a service order and service confirmations ([Plan and Perform the Service](#)).

The following process steps exist in the service objects overview:

Process Step	What it Means
Precheck	No service quotations or service orders have been created for the service object. In this process step, you decide on the follow-up transactions to be performed for the service object.
Quotation	At least one service quotation exists for the service object that hasn't been completed. In this process step, you edit and send out service quotations for the service object, and record whether the customer has accepted or rejected the service quotations.
Service	At least one service order or service confirmation exists for the service object that hasn't been completed. In this process step, you plan and perform the service of the service object in the service orders. You use the service confirmations to record the actual consumption of services, service parts, and expenses.
Completed	This is the final process step for the service object. The service object has been completed.

Processing Service Objects

From an in-house service, you can navigate to the service objects in the [Manage In-House Service Objects](#) app to process the service objects individually and to decide, for example, on the follow-up actions (see [Perform Precheck](#)) or which logistics-related process steps you want to launch for the service object (see [Logistics Process for In-House Service Objects](#)).

Alternatively, you can do this for multiple service objects simultaneously within an in-house service (*mass decision*). To do so, open the in-house service and select the relevant service objects within the in-house service. You then have the following options:

- Perform one of the actions that is provided when you click the **Decide** button.

i Note

An action is only displayed if it applies to all selected service objects. If no action is displayed, change the service objects selection.

The service transactions that are related to a service object are displayed in the [Manage In-House Service Objects](#) app under [Process Flow \(In-House Service\)](#).

- Use one of the buttons to launch a logistics related process step.

The logistics-relevant documents that are related to a service object are displayed in the [Manage In-House Service Objects](#) app under [Logistics](#) > [Process Flow \(Logistics\)](#) >

For more information about the possible actions and the logistics-related process steps that can be launched, see [Possible Actions and Logistics-Related Process Steps for Service Objects](#).

Processing Service Objects in Intercompany In-House Services

An **intercompany in-house service** is an in-house service with service objects that are processed in a service center in a different company code.

Creating Intercompany In-House Services

You create intercompany in-house services as follows:

- Launch an intercompany service for a service object by assigning it to an intercompany service item in a service order. When you release the service order, the system creates an intercompany service order and an intercompany in-house service.

For more information, see [Processing Service Objects in Intercompany Service Orders](#) and [Plan and Perform the Service](#).

- Launch an intercompany transfer of a service object to a service center (specified by a plant and a storage location) that belongs to another company code than the sending service center. The system then creates an intercompany in-house service.

For more information, see [Intercompany Transfer for Service Objects](#).

Managing Service Objects in the Intercompany Process

When creating an intercompany in-house service, the system copies the following data from the preceding in-house service:

- Business partner entered for the ship-to party
- Business partner displayed as the end customer

The end customer is the business partner that originally sent the object to the service and typically owns it.

- Serialized product, equipment, or non-serialized product related to the service object, and the entered quantity

- Serial numbers assigned to the service object
- Delivery terms in the service object and the in-house service
- Notes in the service object and the in-house service
- Attachments in the in-house service

Intercompany in-house services are displayed for the service object under [Process Flow \(In-House Service\)](#).

The following is displayed for the sold-to party:

- In the **original in-house service**, the sold-to party is the business partner that originally sent the object to the service and typically owns it.

The sold-to party and the end customer are the same.

- In the **intercompany in-house service**, the sold-to party is the business partner of the sales organization that launched the intercompany process.

In-House Service Object (Service Object)

You use in-house service objects (service objects) to collect and manage all service transactions and documents, and to keep the overview of the in-house service and the logistics process.

Service objects are the objects that are marked for service, such as maintenance or repair, and that can be derived from equipment, serialized products, or non-serialized products. Each service object is related to an in-house service (see [In-House Service](#)).

You can access service objects as follows:

- In the [Manage In-House Service Objects](#) app: Open the service objects individually.
- In the [Manage In-House Services](#) app: Get an overview of all service objects that are related to an in-house service. You can then navigate to the individual service objects in the [Manage In-House Service Objects](#) app.
- In service quotations and service orders in the [In-House Service Objects](#) assignment block: Get an overview of the service objects that are related to the service transaction. You can then navigate to the individual service objects in the [Manage In-House Service Objects](#) app or assign further service objects to the service transaction.

Creating Service Objects

Service objects are created when you launch the in-house service process (see [Launch the In-House Service Process](#)) for a piece of equipment, a serialized product, or a non-serialized product:

- **Equipment:** An individual, physical object that is processed as an autonomous unit and that has a serial number assigned.

For more information about setting up equipment, see [Equipment](#) and [Serialization Data in the Piece of Equipment](#).

- **Serialized product:** A product that has a serial number assigned and that isn't related to a piece of equipment.

You define a product as a serialized product by adding a serial number profile to the product. Having done this, you can then create serial numbers for the product (see [Serial Numbers in Physical Inventory](#)).

- **Non-serialized product:** A product that isn't distinguishable. Non-serialized products don't have a serial number profile assigned in the product master data.

❖ Example

A customer sends 250 identical valves that are marked for service. These valves aren't serialized. The customer expects to get the same number of valves once the service has been performed.

Service objects can have a quantity greater than one. If the service object is related to a serialized product or equipment, you have to add further serial numbers to it. For more information, see [Processing Serial Numbers](#).

Processing Service Objects

For each service object, you need to decide which follow-up transactions and documents are required to plan and perform the service. Furthermore, you can launch the logistics-related process steps. To do this, you can process the service objects individually or multiple service objects simultaneously:

- If you want to process the service objects individually, navigate to the object page for the service object in the [Manage In-House Service Objects](#) app. There, you can perform the required action that is provided on the [Decide](#) tab, or use the buttons to launch the logistics-related process steps.
- If you want to do this for multiple service objects simultaneously, perform a *mass decision* in the list report in one of the following apps:
 - In the [Manage In-House Service Objects](#) app
 - In the [Manage In-House Services](#) app, within an in-house service in the service objects overview

First, select the relevant service objects and then perform one of the actions that is provided when you click the [Decide](#) button or launch one of the logistics-related process steps using the corresponding buttons.

i Note

An action is only displayed if it applies to all selected service objects. If no action is displayed or the buttons for the logistics-related process steps are grayed out, change the service objects selection.

Process Flow (In-House Service)

All service transactions and documents such as service orders or billing document requests that are used to perform the service process and that have been created for the service object in the respective in-house service are displayed under [Process Flow \(In-House Service\)](#). From here, you can navigate to them to display their details.

Status of the Service Object

System statuses apply to service objects. However, you can define your own user statuses in Customizing for [Service](#) under ► [Transactions](#) ► [Basic Settings](#) ► [Status Management](#) ►.

Logistics

If the logistics process is performed for the service object in parallel to the in-house service process, the following logistics relevant information are displayed under [Logistics](#):

- The availability of the service object at the service center (see also [Tracking the Availability at the Service Center](#))
- The customer RMA number if the logistics process is managed in customer returns
- Information related to the inventory, such as the special stock indicator, the stock type, the plant, and the storage location

- **Shipping:** Contains the ship-to party and the delivery terms, such as the shipping conditions and the incoterms. This information is initially taken from the in-house service and can then be changed individually (see also [Managing In-House Services](#) under *Shipping*).

You can change the shipping address in the service object as follows:

i Note

You can change the shipping address in in-house service objects if the **Changeable Address** checkbox is selected in the **Define Partner Determination Procedure in Service** Customizing activity.

- Change the business partner under **Ship-To Party**.

You can only use business partners to which the business partner role **Ship-to Party** is assigned in the business partner master data.

- Select a different **ship-to party address** for the specified ship-to party that is maintained for the ship-to party in the business partner data.
- Enter a new address under **Alternative Shipping Address**.

This address is then displayed under **Ship-To Party Address** and is used exclusively for this service object. It is not stored in the business partner data for the ship-to party.

Outbound deliveries that are created to send the service objects back to the customer take over the shipping information from the service object. If you subsequently change the shipping information for the service object, it is not updated in the outbound delivery.

- **Process Flow (Logistics):** Contains the documents that are relevant for the logistics process of the service object

For more information, see [Logistics Process for In-House Service Objects](#)

Notes in Service Objects

You can enter further information that supports the in-house service process for the service object. To do this, open the service object in the edit mode.

When you enter notes in the related in-house service, the following applies to the service object:

- When you first enter notes for an in-house service in the **Manage In-House Services** app and then add service objects, the service objects use the customer instructions for the in-house service.
- When you enter notes for an in-house service to which service objects have already been added, the changes **aren't** transferred to the notes for the service object.

You set up notes in Customizing for **Service** under **► In-House Service ► Define Text Management ►**.

Service Contract Determination

In the **Manage In-House Service Objects** app, the system determines the applicable service contract items for the service object when the service object is opened in the app.

For more information about the criteria that need to be fulfilled for service contract determination, see [Service Contract Determination](#).

Processing Serial Numbers

Under **Serial Numbers**, you can view all serial numbers and pieces of equipment assigned to a service object. You can add more serial numbers or pieces of equipment to service objects with a quantity greater than one if the service object and the related in-house service are in **Open** status.

To add more serial numbers and pieces of equipment to a service object, follow these steps:

1. In the **Manage In-House Services** app, enter the quantity for the service objects where you want to add serial numbers and pieces of equipment.

→ Tip

You can change the quantity only if the in-house service is in edit mode.

2. Do not confirm the in-house service.

You can change the quantity of the service objects and add serial numbers and equipment only if both the in-house service and the service objects are in **Open** status.

3. Navigate to the **In-House Service Object Details** screen for the service object in the **Manage In-House Service Objects** app.

4. Enter the serial numbers or pieces of equipment under **Serial Numbers**.

The subordinate serialized product or equipment is also displayed under **Serial Numbers** and is added to the total quantity entered for the service object.

5. Confirm the in-house service in the **Manage In-House Services** app to continue the in-house service process.

Restructuring Service Objects: Split and Merge

You can restructure service objects by splitting and merging their quantities and assigned serial numbers to newly created service objects. This helps you to optimize the in-house service process for service objects.

Split and Merge in In-House Service

You can restructure service objects in the **Manage In-House Services** app within an in-house service in the service objects overview and in the list report in the **Manage In-House Service Objects** app in the following ways:

- **Split:** Distribute the quantity entered for the service object to several newly created service objects.
 - If the service object is a non-serialized product, distribute the total quantity specified for the service object to the newly created service objects.
 - If the service object is a serialized product or equipment, distribute all related serial numbers to the newly created or succeeding service objects.
- **Merge:** Combine multiple service objects into one new service object. As a result, the quantities and/or serial numbers of all service objects are consolidated.

If the service objects are serialized products or equipment, all serial numbers are combined in this service object as well.

If the quantity for non-serialized products can be specified in **decimal values**, you can use decimal values during splitting and merging. This means that the quantities to be split and merged can be less than or equal to one.

Splitting and merging service objects only changes the structure of the service objects within the in-house service process. It has no influence on the physical inventory and the stock postings for the service objects.

Prerequisites for Splitting and Merging

The service objects to be split or merged must fulfill the following criteria:

- They must have the status **In Process** assigned.
- The special stock indicator must be **Customer Stock (B)**.
- You can split or merge service objects only if they have one of the following availabilities at the service center:
 - **Not Tracked**
 - **Received by Service Center**
 - **Delivery to Service Center Canceled**

The following prerequisites must be met by service objects that you want to split:

- They must not contain subordinate service objects.
- They must not be related to intercompany in-house services.

The following prerequisites must be met by service objects that you want to merge:

- The service objects must belong to the same higher-level service object.
However, you cannot merge higher-level service objects.
- The service objects must contain a reference to the same product.
- You cannot merge serialized and non-serialized objects, and service objects related to intercompany in-house services.
- The following must be identical for the service objects to be merged:
 - Availability of the service objects at the service center
 - Logistics-related information: Plant and storage location, special stock indicator, stock type, customer RMA

Processing Split and Merged Service Objects

After you have split or merged service objects, the newly created or succeeding service objects are added to the in-house service of the preceding service objects. The preceding service objects are set to **Obsolete** and have the status **Completed**. This means that you can no longer perform the service or launch logistics-related process steps for them. However, this is to be performed for the succeeding service objects that have the status **In Process**.

In the service object's details under **Related Service Objects**, all service objects related to this one through splitting or merging are displayed:

- **Preceding service objects** are the ones from which the current service objects were created.
If a service object was split multiple times, only the most recent preceding service objects are displayed.
- **Succeeding service objects** are the ones created from the current or preceding service object.
Only the direct succeeding service objects are displayed.

i Note

Service objects that are set to **Obsolete** are not displayed in the service objects overview for an in-house service in the **Manage In-House Services** app.

In the [Manage In-House Service Objects](#) app, you can filter for service objects that are set to **Obsolete**. If you group by in-house services, you can see all service objects related to specific in-house services.

The following applies to the newly created service objects:

- The system copies the data from the preceding service objects to the newly created ones.
- Service orders and service quotations that have been created for the preceding service objects are not inherited by the new service objects, but still can be processed.

However, you can add the new service objects to the service quotations and service orders created for the preceding service object.

- The availability displayed for the preceding service objects is taken over by succeeding service objects.

The documents created for the logistics-related process step are displayed in the preceding and the succeeding service objects. You launch further logistics-related process steps in the succeeding service object.

Split Quantities of Service Objects in Customer Returns

For service objects for which a quantity that is greater than 1 has been entered, you can split the total quantity in customer returns (see [Manage Customer Returns for Service Objects](#)) as follows:

- **Non-serialized products only:** In the outbound delivery, you can record the quantity actually returned that can differ from the quantity that has been entered in the returns order. That means that you can split the quantity into different outbound deliveries.
- **Non-serialized products or serialized products:** In the material inspection, you can split the total quantity of the inspected service objects into items. Having done this, the system creates **subordinate service objects** for each item that can have its own logistical follow-up activity.

The initial service object, which is the **higher-level service object**, summarizes the total number of subordinate service objects.

In the list report in the [Manage In-House Service Objects](#) app and in the service objects overview in the [Manage In-House Services](#) app, the subordinate service objects are grouped by their higher-level service objects. However, you can adjust the grouping and sorting in the column settings for the [Higher-Level Service Object](#) and the [Service Object](#) columns in the apps.

i Note

If you are grouping for higher-level service objects, make sure that the display of the [Service Object](#) column also displays the related subordinate service objects.

Where to Find More

Related Information

[Equipment \(CS-BD/PM-EQM-FL\)](#)

[Serial Number Management \(LO-MD-SN\)](#)

Possible Actions and Logistics-Related Process Steps for Service Objects

For each service object, you need to decide which follow-up transactions and documents are required to plan and perform the in-house service, and which logistics-related process steps are to be launched.

You perform the actions and launch the logistics-related process steps for the service objects in the following apps:

- [Manage In-House Services](#)
- [Manage In-House Service Objects](#)

In-House Service Process: Possible Actions

You can perform the following actions for single service objects or multiple service objects simultaneously:

Possible Action	Prerequisites	Follow-Up Step
Create service quotation	None	Process Service Quotations
Add to service quotation	At least one service quotation exists that has already been created for a service object in the same in-house service.	
Create service order	None	Plan and Perform the Service
Add to service order	At least one service order exists that has already been created for a service object in the same in-house service.	
Set service object to Completed	• All related service quotations have been completed. • All related service orders and the follow-up service confirmations have been completed. • The service objects (including higher-level service objects and subordinate service objects) have been sent back to the customer, or the delivery of the service objects to the service center has been canceled. The material inspection has been reversed for subordinate service objects. • The logistics process has not been launched and the availability is set to Not Tracked. i Note You can set subordinate service objects to Completed automatically by completing the related higher-level service object. The prerequisite for this is that the higher-level service object and the related subordinate service	Set Service Objects to Completed

Possible Action	Prerequisites	Follow-Up Step
	objects fulfill the prerequisites mentioned above.	

Logistics-Related Process Steps: Relevant Buttons

Possible Logistics-Related Process Steps	Prerequisites	Follow-Up Step
Create returns order (under Receive from Customer)	The logistics process has not yet been launched for the service objects: No returns orders or inbound deliveries have been created for the service objects.	Manage Customer Returns for Service Objects
Create inbound delivery (under Receive from Customer)	The logistics process has not yet been launched for the service objects: No inbound deliveries or returns orders have been created for the service objects.	Manage Inbound Deliveries for Service Objects
Transfer service object to another service center	The logistics process has been launched for the service objects, which have been received by the service center.	Transfer Service Objects to Another Service Center
Send back to customer	The logistics process has been launched for the service objects, which have been received by the service center.	Send Service Objects Back to the Customer

Processing Higher-Level Service Objects and Subordinate Service Objects

If the total quantity for a service object has been split in customer returns (see [Processing Split Quantities in Customer Returns](#)), the following applies to the higher-level service object and the related subordinate service objects when performing an action or logistics-related step on the basis of the mass decision:

- If you select a higher-level service object, you can't select the related subordinate service objects at the same time.
When you create a service transaction or a document, it's only displayed in the process flow for the higher-level service object. However, in the **In-House Service Objects** assignment block in the service transaction, the subordinate service objects are displayed in addition to the higher-level service object.
- You can select individual subordinate service objects for the same higher-level service object. However, you can't select the higher-level service object.
- You can select higher-level service objects and subordinate service objects that aren't related to each other.

Related Information

[Tracking the Availability at the Service Center](#)

AI-Assisted In-House Service Initiation

AI-Assisted In-House Service Initiation enables you to create in-house services and in-house service objects (service objects) from documents using AI technology. This makes it easier to launch the in-house service process for service objects from service quotations or service orders or in the **Manage In-House Services** app without typing data manually in your system.

i Note

AI-Assisted In-House Service Initiation is only available in SAP S/4HANA Cloud Private Edition.

Use of this AI-assisted feature may require additional entitlement and authorization. Please consult your SAP account executive for more information.

AI-Assisted In-House Service Initiation is available in the following apps:

- **Manage In-House Services:** Create an in-house service using the header data (sold-to party and, if applicable, ship-to party) from a document and add the service objects listed in the document.

For more information, see [Creating In-House Services from Documents \(AI-Assisted In-House Service Initiation\)](#).

- **Manage Service Orders** and **Manage Service Quotations** in the **In-House Service Objects** assignment block: Add the service objects listed in the document to the service transaction. In this case, the header data for the document is not considered because the sold-to party and, if applicable, the ship-to party have already been entered for the service transaction.

For more information, see [Assigning Service Objects](#).

When you click the **Create from Document** button, a dialog box to upload the documents and to review the extracted and enriched data appears.

i Note

A prerequisite for using AI-Assisted In-House Service Initiation is that the document exists as an electronic document. These documents can be purchase orders or packing lists that contain the submitted service objects and information about the sold-to party and/or ship-to party of the in-house service. Also, make sure that the file meets the required criteria, for example, the maximum file size of 50 MB and that it has the supported document file format (see [Supported File Types \(Document Information Extraction\)](#)).

After uploading the document into your system, the content of the document is imported into the in-house service process using the BTP service **Document Information Extraction** (see [What Is Document Information Extraction?](#)). Generative AI technology supports the extraction of the data required for the creation of the in-house service and the service objects.

Based on the extracted data, the system then searches for and identifies the relevant business partner or product master data in In-House Service. The result of this data extraction and data enrichment is then displayed in the corresponding fields in the apps and can be reviewed there. Having reviewed the result, you can create the in-house service and/or the service objects.

i Note

The service objects to be added can result from equipment, serialized products, and non-serialized products. For more information about the specifics that apply to service objects, see [In-House Service Object \(Service Object\)](#).

Ensuring the Quality of the Extraction Result

The extraction of the data required for the creation of the in-house service and the service objects is supported by generative AI technology. When uploading documents into your system, make sure that you always validate the quality of the extraction result as described under [What Is Document Information Extraction?](#).

For more information about improving the quality of the extraction and enrichment results, see also [Optical Character Recognition \(OCR\): Best Practices \(Document Information Extraction\)](#).

Data Extraction Process

Once you have uploaded the document in the app, the following steps are performed by Document Information Extraction and In-House Service:

1. The document is submitted to Document Information Extraction.
2. Document Information Extraction uses generative AI to extract the header and item data from the document and to transfer the extraction results to the corresponding fields in the app:
 - The following header data can be used to create an in-house service in the [Manage In-House Services](#) app:
 - **Sold-to party:** It is mandatory to specify the sold-to party and the related sales organization for an in-house service.

The following entries can be used to identify the sold-to party: ID, name, address data, bank account number, tax ID
 - **Ship-to party:** It is optional to specify the ship-to party for an in-house service.

The following entries can be used to identify the ship-to party: name, address data
 - The following entries in the item data can be used to identify a service object:
 - Product (ID and description)
 - Serial number
 - Equipment ID
 - The following entries in the item data are only considered if the identified service object is a non-serialized product:
 - Quantity of the service object

The quantity can only be greater than 1 for non-serialized products.
 - Unit of measure related to the quantity

Only considered if the unit of measure cannot be derived from the product master data.

For more information about the specifics for service objects, see [Creating Service Objects](#).
3. The system performs data enrichment.

i Note

Data enrichment means that business partner and product master data are added to the data extracted from the document by Document Information Extraction.

In-House Service compares the header and item data extracted from the document for the in-house service and the service objects with the corresponding product and business partner master data in In-House Service. The master data is then added to the extracted data.

4. Once the data extraction and data enrichment has been finished, the system provides the status for each field related to the in-house service or service objects. Possible statuses of the data extraction are:
 - **Extracted value:** The value displayed corresponds to the value extracted from the document.
 - **Derived value:** The value extracted from the document has been replaced by the value specified for the related master data.
 - **Extracted value; verified:** The value extracted from the document matches master data in In-House Service.
 - **Multiple values found:** Multiple entries were found in the master data for the value extracted from the document. The most suitable entry is displayed.

- **Extracted value; not found:** No master data was found that matches the value extracted from the document
- **Manually entered value:** The value extracted from the document has been overwritten manually.
- **Value could not be converted:** The quantity extracted from the document could not be converted into the unit of measure that has been specified in the product master data.

5. Review the results of the data extraction and data enrichment

If the data enrichment fails or no unique values can be determined, additional messages are displayed for the relevant field in the message log. Edit each entry for the header data and item data as described under [Messages Displayed During Data Extraction and After Data Enrichment](#). You can overwrite the results displayed in the fields and delete service objects.

Once all messages that block the creation of the in-house service or the service objects have been resolved, you can start the in-house service process.

Prerequisites

You can use AI-Assisted In-House Service Initiation if you have set up Intelligent Scenario Lifecycle Management Framework (ISLM) described under [Setting Up the System](#). Otherwise, the feature is not activated in your system.

Related Information

[Create In-House Services and Add In-House Service Objects](#)

[In-House Service Objects in Service Quotations and Service Orders](#)

[What Is Document Information Extraction?](#)

Setting Up the System

As a system administrator, you need to perform some preparatory tasks to enable the system to create in-house services and in-house service objects (service object) out of documents in the [Manage In-House Services](#), [Manage Service Quotations](#), and [Manage Service Orders](#) apps. To do so, you create a connection between the intelligent scenario for AI-Assisted In-House Service Initiation and Document Information Extraction.

i Note

Use of this AI-assisted feature may require additional entitlement and authorization. Please consult your SAP account executive for more information.

Required Users and Authorizations

For the steps to be performed in the apps used to process the intelligent scenario, make sure that your user has the authorizations assigned to the SAP business role SAP_BR_ANALYTICS_SPECIALIST.

Setting Up Intelligent Scenario Lifecycle Management Framework (ISLM)

The integration with Document Information Extraction is handled by means of the Intelligent Scenario Lifecycle Management Framework (ISLM). This framework enables you to set up and manage connections to AI services in your system. For AI-Assisted In-House Service Initiation, you have to configure the intelligent scenario CRMS4_DOX_IHR provided by SAP with specific connection and usage information.

i Note

You can use AI-Assisted In-House Service Initiation only if Intelligent Scenario Lifecycle Management Framework has been set up as described in the following.

Prerequisites

You have performed the following steps to set up Intelligent Scenario Lifecycle Management:

- [Setting up the OData Services](#)
- [Setting up the bgRFC Queue](#)
- [Configuring Business Roles for the Backend](#)
- [Configuring Business Roles for the Frontend](#)

Request Access to SAP Business Technology Platform (SAP BTP)

Document Information Extraction is a service provided on the SAP Business Technology Platform (SAP BTP). To use this service, contact SAP to request a service instance for Document Information Extraction and to be onboarded to the intelligent scenario CRMS4_DOX_IHR.

For more information about the onboarding process, see [Requesting Access to SAP AI Services on SAP Business Technology Platform \(SAP BTP\)](#).

Setting Up the Connection to Intelligent Scenario Lifecycle Management

For more information, see also [Maintain Connection for an Intelligent Scenario](#).

1. Make the following settings in the [Maintain Connection for an Intelligent Scenario](#) Customizing activity (in Customizing for [ABAP Platform](#) under [Application Server](#) [Basis Services](#) [Intelligent Scenario Lifecycle Management](#) [Service Connections for Machine Learning Infrastructure](#)):
2. Create a connection for the intelligent scenario CRMS4_DOX_IHR. To do so, enter the following:
 - **OAuth 2.0 Client Profile:** CRMS4_DOX_IHR
 - **Service Key (JSON format):** Enter the service key (as a whole) of the instance that has been provided in the service request on the ECS Customer Service Request Portal.

i Note

If you are using Apple MacOS, you may encounter parsing or pasting issues in SAP GUI. As a result, the JSON file may become corrupted. In this case, perform this step on remote desktop or using the Microsoft Windows operating system.

3. Press [Next](#).

ISLM generates the connection artifacts.

4. Test the connection.

If the test is successful, the connection for the intelligent scenario has been established. You can then save your entries.

If the connection test fails due to an HTTP error, save your entries. Then, perform the error correction as described in the section below, *Testing the Connection to Intelligent Scenario Lifecycle Management*.

Testing the Connection to Intelligent Scenario Lifecycle Management

i Note

If the system is hosted in an Hyperscaler Cloud Infrastructure, you must update the proxy host and port in RFC destination and the configuration for the OAuth 2.0 client.

- **Connection Name:** Check if the name of the RFC connection is correct

You can check the settings, for example, the proxy settings for the RFC connection under [RFC Destinations \(Display/Edit\)](#) (transaction code SM59).

- **OAuth Profile:** CRMS4_D0X_IHR

Check the OAuth 2.0 client configuration in the [Maintain Connection for an Intelligent Scenario](#) Customizing activity (transaction code ISLM_CONN_MAP) and under [OAuth 2.0 Clients](#) (transaction code OA2C_CONFIG). Ensure that the same proxy settings have been maintained for the OAuth 2.0 Client and for the RFC connection.

Once you have corrected the errors, check the connection to the ISLM scenario in the [Maintain Connection for an Intelligent Scenario](#) Customizing activity again.

Activating the Intelligent Scenario

Once you have set up the connection to ISLM, activate the intelligent scenario CRMS4_D0X_IHR in your frontend system. To do so, proceed as follows:

1. In the [Intelligent Scenario Management](#) app, filter for the intelligent scenario CRMS4_D0X_IHR.
2. Check if the changes that have been performed in the [Maintain Connection for an Intelligent Scenario](#) Customizing activity have been transferred to the [Intelligent Scenario Management](#) app. If so, the deactivated version of the intelligent scenario is displayed.

i Note

The setup of the connection is validated and populated using a background job which runs regularly. However, it can take some time until the changes made in Customizing can be viewed in the app. Wait until the synchronization is complete before continuing.

3. Open the intelligent scenario CRMS4_D0X_IHR.

You can ignore the information about training the intelligent scenario because no training is required for Document Information Extraction.

4. On the [Deployments](#) tab, activate the deployment for all users ([For All](#)).

i Note

Make sure that your user has the authorization object S_ISM_DEPL with the 02 ([Change](#)) value assigned. Otherwise, you are not able to change the deployment.

Once you have activated the intelligent scenario CRMS4_D0X_IHR and setup the connection to Document Information Extraction, you can use the AI-assisted in-house service initiation. That means that the button [Create from Document](#) is displayed in the [Manage In-House Services](#), [Manage Service Quotations](#), and [Manage Service Orders](#) apps is displayed.

For more information about activating intelligent scenarios, see [Activate an Intelligent Scenario](#).

Attachments

Documents that are used to create in-house services in the [Manage In-House Services](#) app, are added to the in-house service as an attachment. To do so, you have to implement attachments to be processed in the [Manage In-House Services](#) app. For more

Related Information

[Intelligent Scenario Lifecycle Management](#)

[Maintain Connection for an Intelligent Scenario](#)

[SAP Delivered Document Information Extraction Scenario](#)

Messages Displayed During Data Extraction and After Data Enrichment

During the data extraction and the data enrichment, errors and warnings might occur. The related messages are displayed in the message log for the app. You can only create the in-house service and the in-house service objects (service objects) if all error messages have been resolved. This topic provides examples of errors and warnings and how to handle them.

The Data Extraction Has Failed

When you submit a document for extraction or retrieve the extraction results, errors can occur. Try the data extraction again later. If the problem persists, contact SAP Support.

The file upload or data extraction might fail due to the following reasons:

- An error occurred while sending the document to document extraction.

Check if the uploaded file meets the required criteria, for example, the maximum file size of 50 MB and if it has the supported document file format (see also [Supported File Types \(Document Information Extraction\)](#)).

- You can only upload files with a maximum size of 50 MB.
- An error occurred while extracting data from the document.

This message is displayed, for example, if the data extraction takes longer than usual. In this case, upload the file again.

- Can't upload document due to missing information.

Check if the file has a file name, or if the file contains information and is not empty.

- A technical error occurred during document extraction request.

Upload the file again.

- The system couldn't retrieve the parameters for the metering call.

Your system is not authorized to use AI-Assisted In-House Service Initiation. Consult your account executive for more information.

- Couldn't obtain http/rest client to connect for document extraction.

Intelligent Scenario Lifecycle Management Framework (ISLM) is not activated in your system. For more information about setting up ISLM for AI-Assisted In-House Service Initiation, see [Setting Up the System](#).

- A timeout occurred during the document information extraction process.

Messages Requiring Further Actions

Even if the data extraction is completed successfully from a technical perspective, some of the data might not be extracted correctly by Document Information Extraction and therefore cannot be processed in subsequent steps, such as identifying the master data in In-House Service.

The following messages can occur for header and item data extracted from the document. Once you have checked and corrected all data and no error messages are displayed any more, you can create the in-house service and the service objects.

The extracted number is assigned to multiple products as a serial number/manufacturer serial number.

Use the value help (F4 help) to select the appropriate combination of serial number and product.

The extracted product was found using the fuzzy search.

The fuzzy search checks the entries in the product descriptions. Use the value help (F4 help) to select the required product.

Multiple products were found using the fuzzy search.

The fuzzy search checks the entries in the product descriptions. Use the value help (F4 help) to select the required product.

Multiple sold-to parties/ship-to parties were found.

Multiple sold-to parties/ship-to parties were found during data determination. The most suitable entry has been selected.

Check if the proposed sold-to party/ship-to party is correct. You can do the following:

- Evaluate corresponding messages in the message log.
- Check the information in the uploaded document.
- Use the value help (F4 help) to select the appropriate entry.

Multiple products were found.

The extracted product is associated with multiple sales organizations and distribution channels. Use the value help (F4 help) to select the appropriate product.

Cannot convert the extracted quantity.

This message is displayed for non-serialized products only.

If the extracted unit of measure and the base unit of measure of the product specified in the product master data are different, the system cannot convert the extracted quantity to the base unit of measure of the product specified in the product master data.

The system can only convert the unit of measure to units of measure that have been specified as alternative units of measure in the product master data.

The organizational data for the extracted equipment/serial number and the in-house service differ.

The organizational data for the extracted equipment/serial number does not match the organizational data specified for the in-house service.

Make sure that the following organizational data entered in the in-house service matches the corresponding data for the extracted equipment/serial number:

- Sales organization

- Distribution channel

Organizational data for the extracted sold-to party/ship-to party and the in-house service differ.

The organizational data for the extracted sold-to party/ship-to party does not match the organizational data entered in the in-house service. Make sure that the following organizational data for the in-house service matches the corresponding data for the extracted sold-to party/ship-to party:

- Sales organization
- Division
- Distribution channel

Mismatch for the organizational data of the extracted sales organization unit

The following data associated with the extracted sales organization unit does not match the organizational data for the in-house service:

- Sales organization
- Distribution channel
- Division

Use the value help (F4 help) to select the entry for the sales organization unit that matches the organizational data for the in-house service.

The extracted product was not found with the specified organizational data.

The product which was extracted from the document could not be found. You have the following options to identify the appropriate product:

- Ensure that the information about the product is clearly readable in the uploaded document.

For more information, see [Optical Character Recognition \(OCR\): Best Practices](#).

- Ensure that the product is available in the system.

If this is the case, check the settings for the product in the product master data. You might need to adapt the assigned sales organizations.

- Use the value help (F4 help) to select the product.

The equipment does not match the extracted serial number.

The entry for the equipment has been deleted because the value specified for the equipment does not match the extracted serial number.

The extracted equipment was not found.

The equipment extracted from the document could not be found in your system.

You have the following options to identify the appropriate piece of equipment:

- Ensure that the information is clearly readable in the uploaded document.

For more information, see [Optical Character Recognition \(OCR\): Best Practices](#).

- Ensure that the equipment is available in your system.

If this is the case, check the settings for the equipment in the product master data. You might need to adapt the assigned sales organizations.

- Use the value help (F4 help) to select the appropriate piece of equipment.

The extracted serial number was not found.

The serial number extracted from the document could not be found in your system.

You have the following options to identify the appropriate serial number:

- Ensure that the serial number is clearly readable in the uploaded document.

For more information, see [Optical Character Recognition \(OCR\): Best Practices](#).

- Ensure that the serial number is available in your system.
- Use the value help (F4 help) to select the appropriate serial number.

The extracted sold-to party/ship-to party was not found.

The sold-to party/ship-to party extracted from the document could not be found in your system.

You have the following options to identify the appropriate sold-to party/ship-to party:

- Ensure that the information about the sold-to party/ship-to party is clearly readable in the uploaded document.

For more information, see [Optical Character Recognition \(OCR\): Best Practices](#).

- Ensure that the sold-to party/ship-to party is available in your system.
- Use the value help (F4 help) to select the appropriate sold-to party/ship-to party.

The extracted unit of measure was not found.

The unit of measure extracted from the document could not be found. You have the following options to identify the appropriate unit of measure:

- Ensure that the unit of measure is clearly readable in the uploaded document and is specified in English.

For more information, see [Optical Character Recognition \(OCR\): Best Practices](#).

- Ensure that the unit of measure is available in your system.
- Use the value help (F4 help) to select the appropriate unit of measure.

Informative Messages

The following messages indicate that the extracted data no longer needs to be post processed after data enrichment:

- The extracted product/equipment/serial number was found.
- The extracted product, serial number, unit of measure are derived from the equipment.
- The extracted product and unit of measure are derived from the serial number.
- The extracted unit of measure is derived from the product.

- The extracted equipment is derived from the serial number.
- The extracted sold-to party is derived from the extracted business partner ID.
- The extracted equipment/serial number is derived from the extracted manufacturer's serial number.
- The extracted sold-to party is derived from the extracted customer bank account/IBAN/tax registration ID/address/name.
- The extracted ship-to party is derived from the extracted address/name.

The following messages indicate that it is not possible to process the extracted data:

- Cannot consider, for example, the equipment or the sold-to party due to an invalid data format.

The data is not used to create the in-house service or a service object.

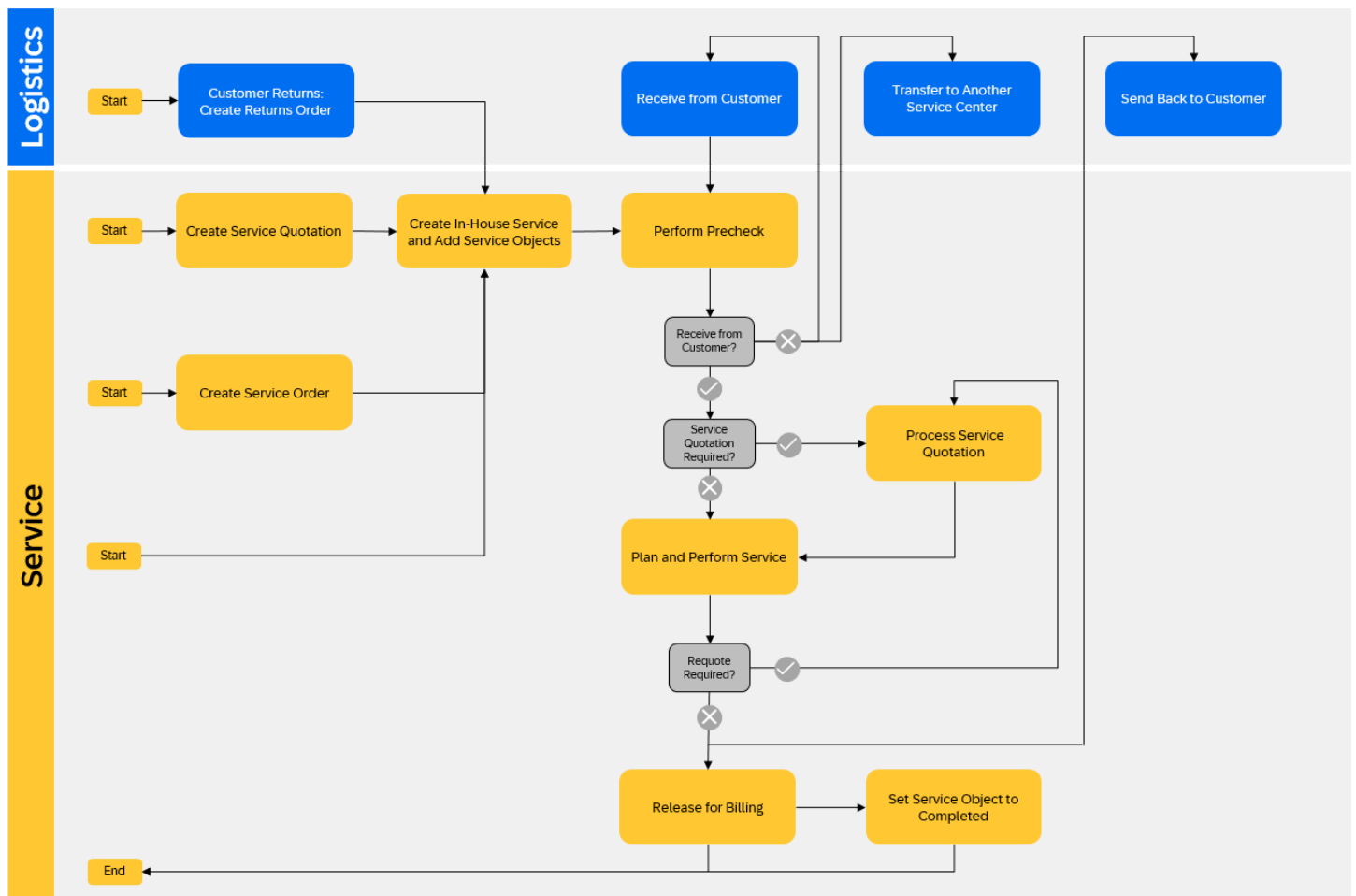
- Quantity is set to 1 for the displayed equipment/serial number as a default.

You can only enter the quantity 1 for equipment/serial numbers. For more information, see [Creating Service Objects](#).

Processes in In-House Service

Introduction to the process steps that are performed in In-House Service

In-House Service consists of several process steps that are described in the following.



Launch the In-House Service Process

When customers send in-house service objects (service object) that are marked for service, such as a maintenance or repair, you create an in-house service and add the service object to it.

You have the following options to start the in-house service process for the service objects:

- [Create In-House Services from Service Quotations and Service Orders](#): In service quotations and service orders (service transactions), you can assign the service objects in the **In-House Service Objects** assignment block. When you save your entries, the system establishes a link between the in-house service you created and the service transaction.
- [Create In-House Services and Add In-House Service Objects](#): Use the **Manage In-House Services** app to create in-house services and to add the service objects to them. AI-Assisted In-House Service Initiation supports you in creating in-house services and service objects from documents.
- [Create In-House Services from Returns Orders](#): When you select the follow-up activity **In-House Service** for returns order items that are related to equipment, serialized products, or non-serialized products, the system creates an in-house service and assigns the items to the in-house service as service objects.

Related Information

[In-House Service](#)

[In-House Service Object \(Service Object\)](#)

Create In-House Services from Service Quotations and Service Orders

You can launch the in-house service process for in-house service objects (service object) from service quotations and service orders by inserting the service objects as reference objects.

Creating In-House Services

Proceed as follows to create in-house services in service quotations and service orders (service transactions):

1. In the service transaction, insert equipment, serialized products, or non-serialized products as service objects in the **In-House Service Objects** assignment block at header or item level.

For more information about the assignment options, see [In-House Service Objects in Service Quotations and Service Orders](#).

i Note

Instead of entering the data for the service objects manually in the service quotations and service orders, you can record the information for the service objects from a document that you upload into your system.

2. Save your entries.

The system establishes a link between the in-house service that you created and the service transaction. The in-house service has the status **In Process** and contains the assigned service objects. The organizational data and the parties involved that have been entered in the service transaction are copied to the in-house service and are displayed under **In-House Service Details**.

3. Within the in-house service process, continue the service for the service object in the following steps:
 - If you've created the in-house service from a service quotation: [Process Service Quotations](#)
 - If you've created the in-house service from a service order: [Plan and Perform the Service](#)

Involved Apps

- [Manage Service Quotations](#)
- [Manage Service Orders](#)

Related Information

[Service Quotations](#)

[In-House Service Objects in Service Quotations and Service Orders](#)

[Service Orders](#)

[In-House Service Object \(Service Object\)](#)

[In-House Service](#)

Create In-House Services and Add In-House Service Objects

You can launch the in-house service process for in-house service objects (service objects) in the **Manage In-House Services** app by creating in-house services and adding the service objects to them. If you have implemented AI-Assisted In-House Service Initiation, you can speed up this step by creating in-house services and service objects from data that has been transferred from a document into the app.

Creating In-House Services Manually

Proceed as follows to create an in-house service in the **Manage In-House Services** app:

1. In the **Manage In-House Services** app, choose **Create**.

The **New In-House Service** screen appears.

2. Under **In-House Service Details**, enter the transaction data, the sold-to party (under **Account**), the organizational data, and if necessary the employee responsible (under **Responsibility**).

i Note

Make sure that the sold-to party and the sales organization already exist in the system, because this information is mandatory.

3. Under **Shipping**, you can enter the ship-to party to which the service objects are sent (see [Send Service Objects Back to the Customer](#)), and the delivery terms.

4. Then choose **Create**.

The **Manage In-House Service Details** screen for the in-house service appears.

5. Add the service objects to the in-house service.

You can add equipment, serialized products, and non-serialized products to the in-house service (see also [Creating Service Objects](#)).

When you add equipment and serialized products, you can use the filter **Current Customer** to restrict the selection to entries that have been assigned to a specific current customer. As a default, the system filters by the current customer that has been assigned to the sold-to party for the in-house service.

i Note

Service objects that have been added to the in-house service, are initially assigned the status **Open**.

Creating In-House Services from Documents (AI-Assisted In-House Service Initiation)

Instead of entering the data for the in-house service and the service objects manually in the [Manage In-House Services](#) app, you can extract the data for the in-house service and the service objects from a document that you upload into your system.

For detailed information about this feature, see [AI-Assisted In-House Service Initiation](#).

i Note

AI-Assisted In-House Service Initiation is only available in SAP S/4HANA Cloud Private Edition.

Use of this AI-assisted feature may require additional entitlement and authorization. Please consult your SAP account executive for more information.

For more information about the settings specific to AI-Assisted In-House Service Initiation, see [Setting Up the System](#).

Proceed as follows to create an in-house service and the related service objects from a document:

1. In the [Manage In-House Services](#) app, choose **Create from Document**.

The [Create In-House Service from Document](#) dialog box appears.

2. Upload the file that contains the data for the in-house service and the service objects to be added to the in-house service.

Make sure that the file meets the required criteria, for example, the maximum file size of 50 MB and that it has the supported document file format (see also [Supported File Types \(Document Information Extraction\)](#)).

- o The following header data can be used to create the in-house service:

- Sold-to party: It is mandatory to specify the sold-to party and the related sales organization for an in-house service.

The following entries can be used to identify the sold-to party: ID, name, address data, bank account number, tax ID

- Ship-to party: It is optional to specify the ship-to party for an in-house service.

The following entries can be used to identify the ship-to party: name, address data

Header data that you have entered manually before launching data extraction is **not** overwritten by the extracted header data.

- o The following entries in the item data can be used to identify a service object:

- Product (ID and description)
- Serial number
- Equipment ID

- o The following entries in the item data are only considered if the identified service object is a non-serialized product:

- Quantity of the service object

The quantity can only be greater than 1 for non-serialized products.

- Unit of measure related to the quantity

Only considered if the unit of measure cannot be derived from the product master data.

3. Once the data has been extracted and enriched by the product and business partner master data, check the extraction result under **New In-House Service** and **Service Objects**:

- You can manually change the entries or use the value help (F4 help) if multiple entries are proposed.
- Check the messages for the extracted data in the message log and make any corrections required, as described under [Messages Displayed During Data Extraction and After Data Enrichment](#).
- You can overwrite the results displayed in the fields and delete service objects.

4. If all errors stated in messages have been solved and the messages have disappeared from the message log, create the in-house service.

The **Manage In-House Service Details** screen for the in-house service appears.

Confirming In-House Services

As soon as you've added all of the service objects, you need to confirm the in-house service. The status of the in-house service then changes to **In Process**.

If an in-house service has the status **In Process**, you can't add any more service objects to it in the **Manage In-House Services** app, or delete service objects. However, you can still edit the in-house service details that aren't mandatory.

You can now perform a precheck for the service objects in the **Manage In-House Service Objects** app. In this step, you decide on the follow-up steps that are to be performed for the service objects (see [Perform Precheck](#)).

Involved Apps

- [Manage In-House Services](#)
- [Manage In-House Service Objects](#)

Related Information

[In-House Service](#)

[In-House Service Object \(Service Object\)](#)

Create In-House Services from Returns Orders

You can launch the in-house service process for service objects in customer returns by creating in-house services as a follow-up activity to returns orders. To do so, you enter equipment, serialized products, or non-serialized products in the returns order items that are then processed as service objects in In-House Service.

In customer returns, make sure that the following applies to the returns order if you want to launch the in-house service process for returns order items:

- In the returns order, you can enter equipment, serialized products, and non-serialized products as service objects (see [In-House Service Object \(Service Object\)](#)).

If you enter an equipment or a serialized product, you must specify the serial numbers for the specified quantity.

- Under **Sales**, select the order reason **Standard Return**.
- Under **Returns**, define the follow-up activity (see [Logistical Follow-Up Activities](#)) and the refund control that are necessary to create in-house services:

You can create an in-house service from the returns order as follows:

- If you've selected the follow-up activity **Receive into Plant** and the refund control **Decide later**, perform a material inspection in the warehouse for the items (service objects). In the material inspection, select the follow-up activity **In-House Service**.
- If you've selected the follow-up activity **In-House Service** and the refund control **No Refund**, the system creates the documents that are required to perform the logistics process in customer returns and also generates an in-house service as a follow-up.

The returns order items are assigned to the in-house service as service objects. The serial numbers entered for the returns order item are displayed for the service object under **Serial Numbers**. If configured, the system posts a goods movement for the items (service objects). The service objects are then transferred from the returns storage location to a storage location that has been defined for in-house services.

Once the in-house service with the service objects has been created, you can perform a precheck for the service objects in the **Manage In-House Service Objects** app and decide on the follow-up steps to be performed for the service object (see [Perform Precheck](#)). Furthermore, you can perform further logistics-related process steps (see [Logistics Process for In-House Service Objects](#)).

i Note

The following information related to the customer returns process is displayed for the service object:

- The customer RMA (Return Material Authorization) that has been created with the returns order.
- The availability (see [Tracking the Availability at the Service Center](#)) that displays the status of the service object within the logistics process.
- The returns order that appears under **Logistics > Process Flow (Logistics)**.

Setting Up Your System

Follow the instructions outlined in Customizing for **Service** under **In-House Service > Integrate In-House Service with Customer Returns** to perform the specific settings for running the logistics process for service objects.

Involved Apps

- [Manage Customer Returns](#)
- [Manage In-House Service Objects](#)
- [Manage In-House Services](#)

Related Information

[Customer Returns Processing](#)

[Create Customer Return Without Reference - VA01](#)

[Logistical Follow-Up Activities](#)

[Customer Compensation](#)

[Enter Inspection Results - From Warehouse](#)

[Cancellation of Returns Steps](#)

Perform Precheck

Perform a preliminary check (precheck) for the service objects and decide on the follow-up steps to be performed. You can, for example, create a service order that is used to plan and perform the service or provide a service quotation because the customer must first accept the estimated service costs for the service object.

Deciding On Follow-Up Steps

In the [Manage In-House Service Objects](#) app, you can decide which follow-up steps are to be performed for each service object:

- [Process Service Quotations](#)

Create a service quotation or add the service object to an existing service quotation, if you want to inform the customer about the estimated costs that are required to perform the service.

- [Plan and Perform the Service](#)

Create a service order or add the service object to an existing service order, if you want to plan and perform the service.

- [Set Service Objects to Completed](#)

Set the service object to **Completed**, if you want to finalize or cancel the service for the service object.

The service transactions that are created as follow-ups are displayed under [Process Flow \(In-House Service\)](#).

If an **intercompany process** has been launched for the service object, you can create service quotations and service orders only in the related original in-house service.

i Note

In the worklist that's available in the [Manage In-House Service Objects](#) app, or in the service objects overview for an in-house service in the [Manage In-House Services](#) app, you can decide which follow-up steps are to be performed for multiple service objects simultaneously if they're related to the same in-house service. For more information about the possible actions and the prerequisites to be met, see [Possible Actions and Logistics-Related Process Steps for Service Objects](#).

Filtering for Service Objects to Be Prechecked

In the [Manage In-House Service Objects](#) app, you can filter for the service objects for which service quotations or service orders haven't been created. To do so, select **No** in the filters [Order Exists](#) and [Quotation Exists](#).

i Note

You can save your filter settings in the [Manage In-House Service Objects](#) app that you use to display the service objects to be prechecked. For more information about personalizing your own views in SAP Fiori apps, see [Save Your Personal Settings as a View](#).

Involved Apps

- [Manage In-House Service Objects](#)
- [Manage In-House Services](#)

Related Information

[Possible Actions and Logistics-Related Process Steps for Service Objects](#)

Service with Advanced Execution for Service Objects

You can use Service with Advanced Execution to plan and perform the service for service object in Maintenance Management.

Within the in-house service process, Service with Advanced Execution enables you to do the following:

- [Process Service Quotations](#): In the service quotation, trigger the cost estimates for the service of the service objects, and more specifically, the planning of the service. Use the follow-up maintenance order to plan the labor (operations) and material (components) consumption.
- [Plan and Perform the Service](#): In the service order, trigger the planning and execution of the service. Use the follow-up maintenance order to execute the actual service by using the functionalities that are provided by Maintenance Management.
- [Release for Billing](#): Trigger the billing of the consumed and confirmed operations and components by creating billing document requests or debit memo requests.

The service quotations and service orders that are created during Service with Advanced Execution are displayed in the process flow for the service object in the [Manage In-House Service Objects](#) app. The maintenance orders that are created during this process are displayed in the transaction history for the service quotations and the service orders.

The execution status that is displayed in the service quotations and service orders informs you about the current status of the service that is performed in the maintenance order. For more information, see [Execution Statuses of Service Orders](#).

Related Information

[Service with Advanced Execution](#)

Process Service Quotations

You can use service quotations to provide a cost estimate to the customer for the requested service.

Processing the Service Quotation

1. Open a service quotation to be processed in one of the following apps:

- [Manage Service Quotations](#)
- [Manage In-House Service Objects](#): The service quotations are displayed in the process flow for the in-house service object (service object). From here, you can navigate to the service quotations.

You can filter for all service objects for which at least one service quotation is currently in process. To do so, select the relevant statuses in the filter [Quotation Status](#). Alternatively, you can filter for service objects for which a service quotation exists, independently of the service quotation status.

i Note

You can save your filter settings in the [Manage In-House Service Objects](#) app. For more information about personalizing your own views in SAP Fiori apps, see [Save Your Personal Settings as a View](#).

2. Make the required entries in the service quotation.

- Enter the estimated labor and material consumption (for example, service items, service parts, and expense items) involved in performing the service for the referenced service objects. Process the service quotation as described under [Service Quotations](#).

- If you use **Service with Advanced Execution** to perform the quotation process for the service object, proceed as described under [Service with Advanced Execution in Service Quotation](#).
- The service objects to which the service quotation is related to, are listed in the **In-House Service Objects** assignment block. For more information about managing service objects in service quotations, see [In-House Service Objects in Service Quotations and Service Orders](#).

3. Finalize the quotation process.

Once the customer agrees to the service quotation and you've accepted the customer's decision, the service quotation is completed and the system creates a follow-up service order that contains the accepted service quotation items and the related service objects.

You can now use this service order to plan and perform the service for the related service objects (see [Plan and Perform the Service](#)).

Involved Apps

- [Manage In-House Service Objects](#)
- [Manage Service Quotations](#)

Related Information

[Service with Advanced Execution for Service Objects](#)

[Possible Actions and Logistics-Related Process Steps for Service Objects](#)

Plan and Perform the Service

You can plan and perform the service for the service object in service orders. Having done this, you confirm the actual consumed resources in the service confirmations.

Planning the Service

1. Open a service order that you use to plan the service for the service object in one of the following apps:

- **Manage Service Orders**
- **Manage In-House Service Objects:** The service orders are displayed in the process flow for the service object. From here, you can navigate to the service orders.

You can use the filter **Order Status** to display all service objects that are related to at least one service order that has a specific status. Alternatively, you can filter for service objects for which a service order exists, independently of the service order status.

i Note

You can save your filter settings in the **Manage In-House Service Objects** app. For more information about personalizing your own views in SAP Fiori apps, see [Save Your Personal Settings as a View](#).

2. Make the required entries in the service order.

Define the details that are relevant for planning the service, for example, the requested start date and end date, and the service employee (executing service employee). Edit the existing service order items or add further service order items.

Process the service order as described under [Service Orders](#).

The service objects to which the service order is related to, are listed in the **In-House Service Objects** assignment block. For more information about managing service objects in service orders, see [In-House Service Objects in Service Quotations and Service Orders](#).

3. Finalize the service planning.

Once you've defined the details in the service order items that are relevant to perform the service, release the service order.

Performing the Service

Once you've planned the service and released the service order, you can perform the actual service. To do so, create service confirmations as follow-ups to the service order that you use to record the actual consumption of services (such as the working hours), service parts used, and the expenses that have occurred.

You process and complete the service confirmations as described under [Service Confirmations](#).

In the **Manage In-House Service Objects** app, you can filter for all service objects for which at least one service order has been released. To do so, select the relevant status in the filter **Order Status**.

i Note

You can save your filter settings in the **Manage In-House Service Objects** app. For more information about personalizing your own views in SAP Fiori apps, see [Save Your Personal Settings as a View](#).

The service confirmations are displayed in the process flow for the service object. You can navigate to the service confirmations from here.

Once you've completed all service confirmations and the related service orders, trigger the billing process for the billing-relevant items in the **Release for Billing** app (see [Release for Billing](#)).

i Note

In the **Manage Service Orders** and **Manage Service Confirmations** apps, you can filter for service orders and service confirmations that haven't been completed yet. This helps you to get an overview of all items that might be ready to be completed.

Creating and Processing Intercompany Service Orders

If you want to perform an **intercompany process** for a service object in a service order, add an intercompany service item to the service order and assign the service object to the item. Then, proceed as described under [Intercompany Service Orders](#).

For more information about the specifics when processing service objects in intercompany service orders, see [Processing Service Objects in Intercompany Service Orders](#).

Creating and Processing Requotes

i Note

You can only create requotes if specific requirements have been met. For more information, see the [Prerequisites](#) section under [Creating Requotes](#).

If you require additional service items, service parts, or expense items to perform a service and which haven't been specified in the accepted service quotation, you can create a requote. In the requote, you add the additional items that have been identified whilst the service was being performed, and that have to be accepted by the customer. Having done this, the accepted items are copied to the service order that has already been created for the initial quotation.

Creating and Processing Billable Maintenance Orders

If you use **Service with Advanced Execution** to plan the service for the service object, proceed as described under [Service with Advanced Execution in Service Order](#).

Where to Find More

Involved Apps

- [Manage In-House Service Objects](#)
- [Manage Service Orders](#)
- [Manage Service Confirmations](#)

Related Information

[Service Orders](#)

[Service Confirmations](#)

[Service with Advanced Execution for Service Objects](#)

[Possible Actions and Logistics-Related Process Steps for Service Objects](#)

Release for Billing

Once the billing-relevant items that are related to the service object have been completed, you release them for billing and trigger the billing process for the service performed.

Releasing the Items for Billing

1. Open the [Release for Billing](#) app that you use to create billing document requests for the billing-relevant items (related to service orders or service confirmations).

You can filter for the service orders or service confirmations for which you want to trigger the billing process.

2. In the result list of the app, select the relevant items and release them for billing.

The billing document requests are then created for the items.

In the [Manage In-House Service Objects](#) app, the billing document requests and the billing documents that have been created during the billing process are displayed in the process flow for the service objects.

For more information about the billing process for service transactions, see [Billing of Service Transactions](#).

You can now finalize the service for the service object in the [Manage In-House Service Objects](#) app. For more information and about the prerequisites that are to be met, see [Set Service Objects to Completed](#).

If you've planned and performed the service for the service object in **Service with Advanced Execution**, you can trigger the billing process once the service has been completed in the maintenance order and according to the billing relevance that has been specified in the service quotation and/or in the service order. For more information, see [Billing](#).

i Note

In the [Manage Service Orders](#) and [Manage Service Confirmation](#) apps, you can filter for service orders and service confirmations that haven't been completed yet. This gives you an overview of all items that can't be released for billing yet.

Involved Apps

- [Manage In-House Service Objects](#)
- [Manage Service Orders](#)
- [Manage Service Confirmations](#)
- [Release for Billing](#)
- [Create Resource-Related Billing Request](#)

Related Information

[Billing of Service Transactions](#)

[Manage Customer Returns for Service Objects](#)

[Possible Actions and Logistics-Related Process Steps for Service Objects](#)

Set Service Objects to Completed

Once all service activities have been finalized for the service object, you can complete the service object and the related in-house service.

Finalizing Service Objects

You can complete **service objects** (also higher-level service objects and subordinate service objects) if the following applies:

- All related service quotations have been completed.
- All related service orders and the follow-up service confirmations have been completed.
- If the logistics process has not been launched for the service object.
- If the service objects have been sent back to the customers and the outbound delivery has been created.
- If the delivery of the service objects to the service center has been canceled.
- If the material inspection for a subordinate service object has been reversed.

To do so, open the service object in the [Manage In-House Service Objects](#) app, and choose **Set Service Object to Completed**. You can also perform this step in the [Manage In-House Services](#) app for multiple service objects simultaneously within an in-house service (see [Processing Service Objects](#)).

You can set subordinate service objects that have been created as a result of an inspection split (see [Processing Split Quantities in Customer Returns](#)) to **Completed** automatically by completing the related higher-level service object. In this case, the higher-level service object and all related subordinate service objects must fulfill the prerequisites mentioned.

Finalizing In-House Services

As soon as all service objects for the in-house service have been completed, the status of the in-house service changes to **Completed** automatically.

Any in-house services with this status can no longer be edited.

Accessing the In-House Service Objects to Be Completed

In the [Manage In-House Service Objects](#) app, you can search for all service objects that can be completed. To do so, use the following filters in the app:

- Set the filter [Availability](#) to [Not Tracked](#), [Delivery to Service Center Canceled](#), and [Sent Back by Service Center](#).
- Set the filter [Quotation and Order Completed](#) to [Yes](#).

i Note

You can save your filter settings in the [Manage In-House Service Objects](#) app that you use to display the service objects that can be completed. For more information about personalizing your own views in SAP Fiori apps, see [Save Your Personal Settings as a View](#).

Involved Apps

- [Manage In-House Service Objects](#)
- [Manage In-House Services](#)

Related Information

[Tracking the Availability at the Service Center](#)

[Send Service Objects Back to the Customer](#)

Logistics Process for In-House Service Objects

In addition to the in-house service process, you can launch logistics-related process steps for the service objects in In-House Service.

The logistics process for service objects runs in addition to the in-house service process. In In-House Service, you can launch logistics-related process steps for service objects that are in process. The follow-up steps are then performed in the respective solutions.

You can launch the following logistics-related process steps for service objects:

- If the return of the service object is to be managed in customer returns, create a returns order. The follow-up steps are performed in customer returns and in Delivery Management (LE-SHP).

For more information, see [Manage Customer Returns for Service Objects](#).

- If you want to start the logistics process directly for the service object, create an inbound delivery. The follow-up steps are performed in Delivery Management (LE-SHP).

For more information, see [Manage Inbound Deliveries for Service Objects](#).

- If you want to transfer the service object to another service center, an outbound delivery is created. The follow-up steps are performed in Delivery Management (LE-SHP).

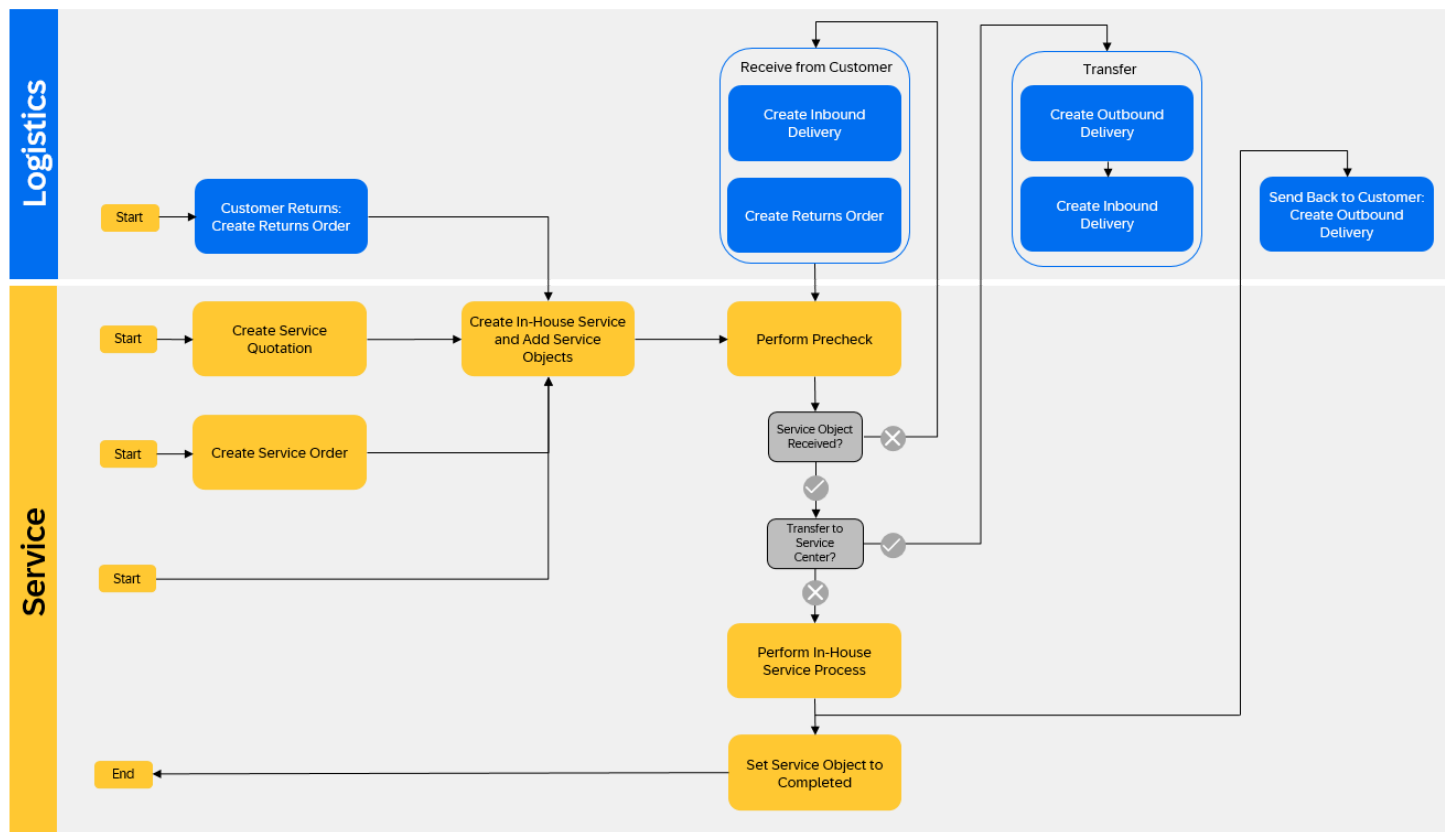
For more information, see [Transfer Service Objects to Another Service Center](#).

- If you want to send the service object back to the customer, an outbound delivery is created. The follow-up steps are performed in Delivery Management (LE-SHP).

For more information, see [Send Service Objects Back to the Customer](#).

In Inventory Management and Inventory, the stock information for the product, serialized product, or equipment related to the service object is managed.

Process Overview



Using Special Stock Indicators for Service Objects

Service objects are considered as special stock because they still belong to the customer, but their stock is managed by the service provider at storage location level. Furthermore, service objects are considered as non-valuated stock. That means that goods receipt postings to non-valuated stock do not lead to a posting in financial accounting.

The following special stock types or indicators apply to service objects:

- **Customer Stock (B):** The service objects' stock is directly allocated to a customer.

Service objects for which the logistics process is launched in an inbound delivery, have this special stock indicator assigned by default.

- **Orders on Hand (E) or sales order stock:** The service objects' stock is directly allocated to a returns order (see also [Working With Sales Order Stock](#)).

Service objects for which the logistics process is launched in customer returns have this special stock indicator assigned by default.

In the [Specify Settings for Returns Orders as Follow-Up](#) Customizing activity, you can specify the special stock indicator **Customer Stock (B)** to be used if the logistics process is launched in customer returns. Once the quality inspection has been performed successfully for the service object and its availability is set to **Received by Service Center**, the system transfers the stock for the service object from **sales order stock** to **customer stock**.

The special stock indicator for a service object is displayed in the [Manage In-House Service Objects](#) app under [Logistics > General Data](#).

Setting Up the System

Integration with Customer Returns

Follow the instructions outlined in Customizing for [Service](#) under [In-House Service > Integrate In-House Service with Customer Returns](#) to perform the specific settings for integrating In-House Service with customer returns in addition to the generic settings for Service.

Integration with Delivery Management (LE-SHP), and Inventory Management and Inventory

Follow the instructions outlined in Customizing for [Service](#) under [In-House Service > Integrate In-House Service with Inventory and Delivery Management](#) to perform the specific settings for integrating In-House Service with Delivery Management (LE-SHP), and Inventory Management and Inventory in addition to the generic settings for Service.

Filtering For In-House Service Objects Without Logistics Process

In the [Manage In-House Service Objects](#) app, you can filter for service objects for which no logistics process exists yet. To do so, filter for the availability **Not Tracked**.

i Note

You can save your filter settings in the [Manage In-House Service Objects](#) app that you use to display the service objects for which no returns order has been created yet. For more information about personalizing your own views in SAP Fiori apps, see [Save Your Personal Settings as a View](#).

Involved Apps

- [Manage In-House Service Objects](#)
- [Manage In-House Services](#)

Related Information

[In-House Service Object \(Service Object\)](#)

[Customer Returns Processing](#)

[Logistical Follow-Up Activities](#)

[Delivery Management \(LE-SHP\)](#)

[Inventory Management and Physical Inventory \(MM-IM\)](#)

Tracking the Availability at the Service Center

In In-House Service, you can track the availability of the service objects at the service center to see their status within the logistics process.

For **higher-level service objects** that have been created in the customer returns process, the availability displayed derives from the availability of all related subordinate service objects. The availability of the **subordinate service objects** derives from the logistical follow-up activity that has been defined for the related inspection item in the material inspection.

Service objects can have the following availability:

Availability at Service Center	What It Means
Not Tracked (a system status isn't applicable)	No logistics process has been launched for the service object. This means that no inbound delivery or returns order has been created for the service object.
Delivery to Service Center Pending (system status: I1212)	The logistics process has been launched for the service object: An inbound delivery or a returns order has been created for the service object, but a goods receipt hasn't been posted. If the logistics process has been launched in customer returns, no material inspection has been performed yet. Item split in the material inspection: Displayed for the higher-level service object if only a partial quantity of the planned delivery has received at the service center. However, more deliveries of subordinate service objects are expected.
Partial Delivery to Service Center (system status: I1395)	Only displayed for higher-level service objects if the service object quantity has been split in the material inspection - All subordinate service objects have the availability Received by Service Center and the total number of subordinate service objects doesn't match the total number in the higher-level service object. However, no more deliveries of subordinate service objects are expected. - At least one subordinate service object has the availability Received by Service Center and at least one subordinate service object has the availability Delivery to Service Center Canceled. No more deliveries of subordinate service objects are expected.
Material Inspection Reversed (system status: I1396)	Only displayed for subordinate service objects if the service object quantity has been split in the material inspection The result that has been stated in the material inspection has been reverted (see Cancellation of Returns Steps) and consequently, the inspection items have been deleted. The subordinate service object can no longer be processed in In-House Service and the quantity entered isn't observed when calculating the quantity for the higher-level service object. You can't create service quotations and service orders for the subordinate service object here.
Received by Service Center (system status: I1213)	The service object has arrived at the service center. A goods receipt has been posted. If the logistics process has been launched in customer returns: - The material inspection has been performed successfully. - Item split in the material inspection: Subordinate service object: The material inspection has been performed successfully.

Availability at Service Center	What It Means
	o Higher-level service object: The material inspection has been performed for all subordinate service objects successfully.
Transfer to Another Service Center Pending (system status: I1397)	The transfer of the service object to the plant or storage location at which the service is to be performed has been launched. An outbound delivery has been created for the service object, but a goods issue hasn't been posted yet.
Transfer to Another Service Center (system status: I1398)	The service object is currently being transferred to the plant or storage location at which the service is to be performed. A goods issue has now been posted and an inbound delivery has been created for the service object.
Processed by Another Service Center (system status I1430)	Only displayed for service objects that are involved in an intercompany transfer • If the intercompany transfer has been launched, this availability is displayed for the service object of the intercompany in-house service until the goods issue has been performed. • If the service object related to an intercompany in-house service has been received by the service center and the transfer is completed, this availability is displayed for the service object of the preceding in-house service.
Delivery to Service Center Canceled (system status: I1214)	The logistics process has been canceled for the service object in the returns order or in the material inspection. Item split in the material inspection: • Subordinate service object: The in-house service process has been canceled in the material inspection. • Higher-level service object: No more deliveries of subordinate service objects are expected. At least one subordinate service object has the availability Delivery to Service Center Canceled and others can have the availability Material Inspection Reversed.
Sent Back by Service Center (system status: I1215)	An outbound delivery has been created to send the service object back to the end customer who owns the service object. Item split in the material inspection: • Subordinate service object: The outbound delivery has been created. • Higher-level service object: Outbound deliveries have been created for all related subordinate service objects.

Receive Service Objects from Customers

You can launch the logistics process for service objects from customer returns or from an inbound delivery.

You have the following options to launch the logistics process for service objects in the In-House Service apps:

- Receive the service object using customer returns. In this case, you first create a returns order: [Manage Customer Returns for Service Objects](#)
- Receive the service object directly from the customer. In this case, you first create an inbound delivery: [Manage Inbound Deliveries for Service Objects](#)

You can launch the logistics process for service objects that are in process and that have the availability **Not Tracked**.

The availability at the service center that is displayed for each service object shows the current status of the service object within the logistics process (see [Tracking the Availability at the Service Center](#)).

Manage Customer Returns for Service Objects

When launching the logistics process for the service objects by creating a returns order, the follow-up steps that are relevant for the returns process are managed in customer returns.

i Note

You can also launch the logistics process in returns orders and create in-house services as a follow-up activity to returns order items. The items are then assigned to the in-house service and processed as services in In-House Service. For more information, see [Create In-House Services from Returns Orders](#).

Creating Returns Orders

You can launch the customer returns process in the **Manage In-House Service Objects** app by creating a returns order for individual service objects, or for multiple service objects by means of the mass decision in the **Manage In-House Service Objects** or the **Manage In-House Services** app.

To do so, choose **Receive from Customer** > **Create Returns Order** and select a plant and a storage location that specifies the service center to which the service object is posted to. The system proposes only those plants that are assigned to the sales organization of the in-house service. If you do not enter a plant and storage location, the system uses the plant and the storage location that have been configured for customer returns.

The service objects are then added to the returns order as items that have the next follow-up activity **Continue In-House Service** assigned (see [Logistics-Related Process Steps: Relevant Buttons](#)). In addition, the system creates a customer RMA (Return Material Authorization) number for the service object that is used to track the in-house service process.

Once the returns order has been created, it is displayed under **Logistics** > **Process Flow (Logistics)**. The availability of the service object at the service center is set to **Delivery to Service Center Pending** (see also [Tracking the Availability at the Service Center](#)).

i Note

The in-house service that is related to the service objects is referenced in the returns order.

Processing Service Objects in Customer Returns

Once you've created the returns order, the following steps are performed for each service object in customer returns to complete the returns process (see also [Customer Returns Processing](#)).

Goods Receipt Posting

In the outbound delivery that is created as a follow-up to the returns order, perform the picking for the returns delivery and post the goods receipt for the returned service objects to add them to the stock (see [Outbound Delivery Process](#)).

i Note

Although, the customer returns process models receiving goods, it technically uses the apps and processes for the goods issue process. Therefore, some of the terminology used applies to the outbound process, even though this is an inbound process. For example, the term **picking** is used where **put away** is meant.

Material Inspection

Perform the quality inspection for the returned service object and record the inspection result (see [Enter Inspection Results - From Warehouse](#)):

- If you want to continue with the in-house service process for the service object, select the logistical follow-up activity **Continue In-House Service**.

The availability at the service center that is displayed for the service object is set to **Received by Service Center**.

- If you select a follow-up activity that is **not** used to continue with the in-house service process, the delivery to the service center is canceled and the availability at the service center that is displayed for the service object is set to **Delivery to Service Center Canceled**.

In this case, you can still process the service object within the in-house service process, and create service quotations and service orders, or finalize the in-house service process for the service object (see [Set Service Objects to Completed](#)).

If configured, the system posts a goods movement for the service objects. The service objects are then transferred from the returns storage location to a storage location that has been defined for in-house services.

Processing Split Quantities in Customer Returns

If the service object has a delivery quantity greater than 1, you can split the quantities in the outbound delivery (**partial delivery**) and/or in the material inspection (**item split**). You can do this if the service object hasn't been completed (see [Set Service Objects to Completed](#)).

Partial Delivery for Service Objects Created from Non-Serialized Products

When you split the delivery of the service object into several partial deliveries in the outbound delivery, you enter the actual quantity delivered for each partial delivery and perform the picking for this quantity (see also [Subsequent Outbound Delivery Split](#)).

Make sure that the following applies in the case of multiple partial deliveries:

- The service object has been created from a non-serialized product (see [Creating Service Objects](#)).
- The total number of service objects entered in the partial deliveries doesn't exceed the total number that has been entered for the service object in the returns order.
- If the number of service objects that have been stated in the outbound deliveries differs from the number of service objects that have been stated in the related returns order, the returns order remains as not completed.

Inspection Split for Service Objects

During the material inspection, you can perform an **item split** (see [Creating Inspection Splits](#) for the details). For each item, the system creates a **subordinate service object** for the initial service object, which is consequently defined as the **higher-level service object**. If the service objects have multiple serial numbers assigned, you must assign them to the corresponding items.

You can revert the result of the material inspection by changing the logistical follow-up activity, for example, to **Transfer to Scrap** or deleting the item in the material inspection. Having done this, the availability that is displayed for the subordinate service object changes to **Delivery to Service Center Canceled**. You can still process the subordinate service object within the in-house service process, and create service quotations and service orders, or finalize the in-house service process for the service object.

❖ Example

You can perform an **item split** if the entire quantity of the physically returned service objects doesn't have the same quality status and only a subquantity of the delivered service objects is handed over to the in-house service process. In this case, the material inspection consists of inspection items with different logistical follow-up activities, **Transfer to Scrap** and **Continue In-House Service**. For each inspection item, the system creates a subordinate service object that has its own availability assigned.

If you've created a service transaction (service order and/or a service quotation) for the higher-level service object or added it to a service transaction item, the subordinate service objects are displayed in the **In-House Service Objects** assignment block. When you create a service transaction for one or multiple subordinate service objects, only these particular subordinate service objects are displayed in the assignment block.

Adjust the Actual Delivered Quantity for Service Objects Created from Non-Serialized Products

In the returns order, you can retroactively change the quantity of the service objects to be returned if it turns out that a different number of service objects than originally announced is returned by the customer.

You can change the quantity in a returns order item if the following applies:

- The service object has been created from a non-serialized product (see [Creating Service Objects](#)).
- If you want to change the quantity but have already posted a good receipt with a lower quantity than originally stated in the returns order, make sure that partial deliveries are allowed for the returns order item.

This setting is made on the **Shipping** tab in the returns order item.

- If a goods receipt has not been yet posted for the actual delivered quantity: Delete the returns delivery.
- If you have already posted goods receipts for a lower quantity of service objects than originally stated in the returns order.

Having done this, you can change the quantity in the returns order. The system then creates a returns delivery for the quantity for which a goods receipt has not been posted yet.

Note the following when adjusting the quantity in returns orders:

- The total quantity of all subordinate service objects can be less than or equal to the quantity in the higher-level service object. However, the total quantity that has been stated in the partial deliveries must not exceed the quantity that has been stated in the related returns order item.
- Subordinate service objects that have the availability **Material Inspection Reversed** assigned, aren't included when calculating the quantity in the higher-level service object.
- You can't change the item quantity in the returns order if the related higher-level service object has been set to **Completed** (see [Set Service Objects to Completed](#)).
- The availability of the higher-level service object at the service center is adjusted accordingly.
- If you change the quantity in the returns order after outbound deliveries (see [Send Service Objects Back to the Customer](#)) have been created for all subordinate service objects and no more deliveries are expected: As soon as the adjusted total

quantity of the higher-level service object matches the total quantity of all subordinate service objects that have the availability **Sent Back by Service Center** assigned, the availability of the higher-level service object is set to **Sent Back by Service Center**.

- If service transactions (service quotations and service orders) have already been created for the service objects, the quantity also changes in the service transactions.

Where to Find More

Related Information

[Subsequent Outbound Delivery Split](#)

Manage Inbound Deliveries for Service Objects

When launching the logistics process for the service objects in In-House Service by creating an inbound delivery, the follow-up steps that are relevant for the inbound process are managed in Delivery Management (LE-SHP).

Creating Inbound Deliveries

You can launch the logistics process in the **Manage In-House Service Objects** app by creating an inbound delivery for individual service objects, or for multiple service objects using the mass decision in the **Manage In-House Service Objects** or the **Manage In-House Services** app (see [Logistics-Related Process Steps: Relevant Buttons](#)).

To do so, choose ► **Receive from Customer** ► **Create Inbound Delivery** ►. You must then select a plant and a storage location that indicate the service center at which the in-house service is to be performed.

i Note

The system proposes only those plants that are assigned to the sales organization for the in-house service.

Once the inbound delivery has been created, it is displayed under ► **Logistics** ► **Process Flow (Logistics)** ►. The availability of the service object at the service center is set to **Delivery to Service Center Pending**.

Processing Service Objects in the Inbound Delivery Process

The follow-up process steps for the inbound delivery are performed in Delivery Management (LE-SHP) (see [Delivery Management \(LE-SHP\)](#)).

In the inbound delivery, perform the putaway for the service objects that are displayed as items by entering the actual delivered quantity of the service objects. Having done this, post the goods receipt to add the service objects to the storage location that you have specified as the service center. The availability of the service object at the service center is then set to **Received by Service Center**.

Related Information

[Inbound Delivery Process](#)

[Tracking the Availability at the Service Center](#)

[Possible Actions and Logistics-Related Process Steps for Service Objects](#)

Transfer Service Objects to Another Service Center

If the in-house service for the service object is to be performed at another service center, it must be physically transferred to this service center. The follow-up process steps that are required for the transfer are performed in Delivery Management (LE-SHP).

You can transfer service objects that are in process to other service centers and that have the availability **Received by Service Center**.

The receiving service center can have the same company code as the sending service center or a different one. The following types of transfers exist:

- **Intracompany transfer:** The plant assigned to the receiving service center belongs to the *same* company code as the sales organization specified for the preceding in-house service.
- **Intercompany transfer:** The plant assigned to the receiving service center belongs to a *different* company code than the sales organization specified for the preceding in-house service.

→ Remember

You can only perform an intercompany transfer for service objects that have the special stock indicator **Customer Stock (B)** assigned. During the transfer, the ownership of the service object is assigned to the customer stock and must remain consistent throughout the transfer.

Launching the Transfer Process

You can launch the transfer of a service object to another service center in the **Manage In-House Service Objects** app for individual service objects or for multiple service objects of the same in-house service using the mass decision in the **Manage In-House Service Objects** or the **Manage In-House Services** app (see [Logistics-Related Process Steps: Relevant Buttons](#)).

To do so, choose **Transfer**. You then have to enter a plant and a storage location that specify the service center to which the service is to be transferred to. Having done this, the system creates an outbound delivery which is displayed under **Logistics > Process Flow (Logistics)**. The availability of the service object at the service center is set to **Transfer to Another Service Center Pending**.

Transfer Process in Delivery Management

The follow-up process steps that are required to transfer service objects to another service center are performed in Delivery Management (LE-SHP). For more information, see [Delivery Management \(LE-SHP\)](#).

The steps described in the following apply to both, intracompany and intercompany processing. For more information about the specifics of intercompany processing, see [Intercompany Transfer for Service Objects](#).

Outbound Delivery Processing

In the outbound delivery, perform the picking for the service objects that are displayed as items by entering the quantity of the service objects to be transferred. Having done this, post the goods issue to remove the service object from the current service center. The system then creates an inbound delivery based on the target plant and storage location and that is used to post the service object to the new service center.

Once all steps in the outbound delivery process have been performed, the availability of the service object at the service center is set to **Transfer to Another Service Center**.

Inbound Delivery Processing

As soon as the service object has arrived at the service center at which the service is to be performed, perform the putaway for the service objects in the inbound delivery. To do this, enter the delivered quantity of the service objects as the putaway quantity. Then, post the goods receipt to move the service objects to the new service center.

Once all steps in the inbound delivery process have been performed, the availability of the service object at the service center is set to **Received by Service Center**.

Intercompany Transfer for Service Objects

When you launch the transfer of service objects to a service center within the same organization but that belongs to a different company code, the system creates an *intercompany in-house service* in addition to the already existing in-house service.

The following information is displayed for the service objects in the preceding in-house service in which the transfer is launched and in the intercompany in-house service:

- After the system creates the outbound delivery, it sets the availability of the service object in the preceding in-house service to **Transfer to Another Service Center Pending**.

In the intercompany in-house service that is created in parallel, the service object has the availability **Processed by Another Service Center**.

- When the service object arrives at the service center with a different company code, the system updates the availability in the preceding in-house service to **Processed by Another Service Center**.

In the intercompany in-house service, the service object now has the availability **Received by Service Center**.

- You can view the preceding in-house service for which the transfer was launched and the in-house service to which the service object has been transferred under **Process Flow (In-House Service)**.

You can process the service objects in both, the original and intercompany in-house services as follows:

- You can launch further logistics-related process steps, such as a transfer, only for service objects with the availability **Received by Service Center**. Logistics information are displayed only for these service objects.
- If an intercompany in-house service already exists for a specific company code because other service objects of the original in-house service were transferred to it, the system adds the service object to the existing intercompany in-house service.

If this in-house service has the status **Completed**, the system sets it to **In Process** again.

- If you transfer a service object to a service center where it was previously processed, the system updates only the logistics data for the service object in the relevant in-house service.
- You can create service quotations and service orders for the service object in the original in-house service. However, you cannot link intercompany in-house services to service orders. To do so, use an intercompany service order.

For more information about intercompany processing in service orders, see [Intercompany Service Orders](#).

- Changes you make in the intercompany in-house service do not update the preceding in-house service retroactively.

Intercompany in-house services can be created from existing intercompany in-house services via cascading. In this case, the preceding in-house service is the in-house service where the intercompany process is launched. This can be one of the following:

- The **original in-house service** in which the sold-to party is also the owner of the service displayed under **End Customer**.

This is the in-house service in which the intercompany service was originally launched.

- Another **intercompany in-house service** to which the service object has been transferred. The sold-to party is derived from the sales organization from which the intercompany process was launched.

Where to Find More

Related Information

[Outbound Delivery Process](#)

[Inbound Delivery Process](#)

[Tracking the Availability at the Service Center](#)

Send Service Objects Back to the Customer

When you send the service object back to the customer, an outbound delivery is created for the service object that is then processed in Delivery Management (LE-SHP).

Outbound deliveries can only be created for service objects that are in process and to which the availability **Received by Service Center** is assigned.

Creating Outbound Deliveries

You can create outbound deliveries in the **Manage In-House Service Objects** app for individual service objects, or for multiple service objects related to the same in-house service using the mass decision in the **Manage In-House Service Objects** or the **Manage In-House Services** app (see [Possible Actions and Logistics-Related Process Steps for Service Objects](#)).

To do so, choose **Send Back to Customer**. The system then creates the outbound delivery, which is displayed under **Logistics > Process Flow (Logistics)**. The availability of the service object is set to **Sent Back by Service Center**. The ship-to party and the delivery terms that are used for the outbound delivery are copied from the service object, but can be changed individually (see also [Processing Service Objects](#) under *Shipping*).

i Note

If you use the mass decision to send multiple service objects back to the customer, the system creates an outbound delivery for the service objects to which the same ship-to party is assigned.

If the service object has been created by splitting or merging, you can only create an outbound delivery if the service object is **not** set to **Obsolete**.

The stock data that contains the plant, storage location, stock type, and special stock indicator and that is displayed under **Logistics > General Data** is removed.

Creating Outbound Deliveries for Split Quantities

If the quantity of the service objects has been split in customer returns (see [Processing Split Quantities in Customer Returns](#)), you can create outbound deliveries for the higher-level service object or for individual subordinate service objects if the following applies:

- The subordinate service object has the availability **Received by Service Center**.
- The higher-level service object has the availability **Received by Service Center** or **Partial Delivery to Service Center**.

When you create an outbound delivery for a higher-level service object, the system creates one outbound delivery that contains all subordinate service objects that have the availability **Received by Service Center**. Subordinate service objects with the availability **Material Inspection Reversed** and **Delivery to Service Center Canceled** aren't observed here.

Once you've created outbound deliveries and goods issues for all subordinate service objects, the availability of the related higher-level service object automatically changes to [Sent Back by Service Center](#).

Selecting the Service Objects to Be Sent Back

In the [Manage In-House Service Objects](#) app, you can search for all service objects that can be sent back to the customer and for which you can create an outbound delivery. To do so, use the following filters in the app:

- Set the filter [Availability](#) to [Received by Service Center](#).
- Set the filter [Quotations and Orders Completed](#) to [Yes](#).

i Note

You can save your filter settings in the [Manage In-House Service Objects](#) app that you use to display the service objects that can be sent back. For more information about personalizing your own views in SAP Fiori apps, see [Save Your Personal Settings as a View](#).

Involved Apps

[Manage In-House Service Objects](#)

[Manage In-House Services](#)

Related Information

[Possible Actions and Logistics-Related Process Steps for Service Objects](#)

[Outbound Delivery Process](#)

[Delivery Management \(LE-SHP\)](#)

Apps Used in In-House Service

In In-House Service, you use apps to support your in-house service process.

Manage In-House Services

App ID: F4025

You can use this app to create and process in-house services, and add in-house objects (service object) for which you want to perform maintenance or repair. You can filter, sort, and group a list of all in-house services in the system. You can also call up in-house services, and change their data.

Key Features

i Note

An in-house service is a record that contains one or more in-house service objects that have been sent in by a customer for service.

You can use this app to do the following:

- Create in-house services and add the equipment, serialized products, and non-serialized products to be processed as in-house service objects.
 - Use AI-Assisted In-House Service Initiation to create in-house services and service objects from documents.
 - Display and edit the details of the in-house service.
 - Enter and edit notes for the in-house service.
 - Check the number of applicable service contracts that have been determined for the in-house service.
 - Confirm the added service objects to trigger the service.
 - Search for in-house services using filter criteria, for example, the employee responsible or the in-house service status.
 - Within an in-house service filter for service objects that are currently managed in a particular process step within the in-house service process.
 - Trigger actions (for example, create service order) for multiple service objects in the same in-house service simultaneously.
 - Launch logistics-related process steps for multiple service objects in the same in-house service simultaneously.
- You can, for example, create returns orders or transfer the service objects to other service centers at which the service is to be performed.
- Navigate to the service objects and display their details.

Situation Handling

This app uses Situation Handling to inform specific users about the following situations:

- An in-house service hasn't been confirmed within a defined number of days after its creation.
- Processing of an in-house service took longer than expected.

Using automatic notifications, you can navigate to the critical in-house services directly and take the appropriate action. You can also use the [My Situations](#) app to display your open situations in an overview table and close situation instances by setting the appropriate status manually.

For more information, see [Situation Handling](#).

Supported Device Types

- Desktop
- Tablet

App Extensibility: Manage In-House Services

As a key user, you can extend the [Manage In-House Services](#) app according to your business requirements.

Custom Fields

You can use the [Custom Fields](#) tab in the [Custom Fields and Logic](#) app to create and maintain custom fields that you can use to enhance applications. You can add fields to the following UI elements using key user adaptation:

Business Context	UI Technology	UI Element
In-house service header	SAP Fiori	In-House Service Details screen

For more information, see [Creating Custom Fields](#).

Business users can display fields that you have enabled in the [Custom Fields and Logic](#) app. The fields that you enable for this app are included in the personalization options for business users for the following UI elements:

UI Element	Business Context
In-House Services table	In-house service header
Service Objects section (on the Manage In-House Service Details screen)	In-house service object

Custom Logic

You can use the [Custom Logic](#) tab in the [Custom Fields and Logic](#) app to create and maintain implementations of custom logic that can be used to enhance applications and change application behavior. You can implement custom logic for the following Business Add-Ins (BADIs):

Business Add-Ins (BADIs)	Business Context	Business Use
CRMS4_REPA_CHECK_BEFORE_SAVE (BAdI: Check In House-Service and Service Object before Saving)	In-house service header In-house service object	You use this BAdI to call the method CHECK_BEFORE_SAVE. This BAdI method checks any value changes in the customer-specific fields (extension fields) for in-house services and service objects before they are saved automatically or actively by the user. The result of this check needs to be documented in messages that have been assigned to a message table.
CRMS4_REPA_CUST_I_MODIFY (BAdI: Modify Service Object Data for In-House Services)	In-house service object	You can use this BAdI to call the method MODIFY. This BAdI method is used to change the values of the customer-specific fields in the service object.

For more information, see:

- [Creating Implementations](#)
- For more information about the individual BADIs, check the BAdI-specific documentation in Customizing for [Service](#) under [Transactions](#) > [Settings for In-House Service](#) > [Business Add-Ins](#) .

Related Information

[Key User Extensibility](#)
[Creating Extension Items and Transporting Them](#)

Searching for an In-House Service

Use the search function on the SAP Fiori launchpad to search for in-house services.

Use

To find in-house services, you can use the search function in the header bar of the SAP Fiori launchpad.

Search

To search exclusively for in-house services, choose the **In-House Service** service object type. Then, enter one or more search terms and confirm.

i Note

To narrow down your search results, you can combine several search terms.

The search for in-house services observes the following fields:

- In-House Service
- Sold-To Party
- Contact Person
- Employee Responsible
- Equipment
- Product
- Serial Number

Search Result

The system displays a list of in-house services that fits the entered search criteria, and additional information about the in-house service here. If required, you can also filter your results using the following criteria:

- Created On
- Employee Responsible
- Sold-To Party
- In-House Service Status

You can then navigate to the in-house service.

Related Information

[Search](#)

Situation Template: In-House Service Not Confirmed

Situation Template ID: SRVC_DELAY_IN_IHR_CONFIRMATION

Consuming App: [Manage In-House Services](#)

Template used for Situation Handling that informs customer service representatives when an in-house service hasn't been confirmed within a defined number of days after its creation.

For background information about the Situation Handling framework, see the Product Assistance for [Situation Handling](#).

Business Value

As a configuration expert for business processes (SAP business role SAP_BR_BPC_EXPERT), you can use this template for the following situation:

You want your customer service representatives to be informed automatically when an in-house service that was created recently hasn't been confirmed within a defined number of days after its creation. Using automatic notifications, a customer service representative can navigate directly to the critical in-house service and confirm it to trigger the service process promptly.

Default Settings

The following sections describe how this standard situation template is preconfigured and which predefined parameters you can configure in the [Manage Situation Types](#) app when you copy the standard template to create your own situation type.

For more information, see [Manage Situation Types](#).

Situation Trigger Object and Anchor Object

These technical settings define for which object a situation is displayed (anchor object) and the object based on which a situation is triggered (trigger object):

- **Trigger Type:** Batch
- **Trigger Object:** In-House Service
- **Anchor Object:** In-House Service

Description: In each batch job run, the system checks for in-house services that have the status **Open** and for which the time that has passed since their creation date exceeds the predefined number of days. If the system detects such in-house services, it sends out automatic notifications to the defined recipients.

For more information on batch jobs, see [Batch Job Scheduling](#).

Conditions

The condition in this standard template defines that a situation instance is created with the status **Open** and a notification is sent to the defined recipients if the system detects an in-house service that meets the criteria specified above.

For more information, see [Conditions](#).

This template provides the following important filters that you can use to define your own conditions when you create your situation types:

Filter	Description
In-House Service	The ID of an in-house service that represents a record containing one or more in-house service object (service objects) that have been sent in by a customer for maintenance or repair
In-House Service Status	The status of an in-house service. Possible values are:

Filter	Description
	• A: Open • B: In Process • C: Completed
Number of Days After In-House Service Creation	The number of days that have passed since an in-house service was created

Situation Display

When situations occur, users are informed about them by various texts, such as in-app situation messages or notifications on SAP Fiori launchpad. In this section, you can see all preconfigured texts.

For more information, see [Situation Display](#).

i Note

When you copy the template and adapt it to your needs, you can define the notification settings and change all text elements in this section. After making the changes in the original language, please remember to translate these changes into the other languages that you want to use. The template comes with translations that can be changed directly when you copy the template.

For more information, see [Translate Situation Types](#).

Recipients

For this template, the recipients of the notification can be determined by two predefined rules:

- **Notify creator of the business transaction**
- **Notify employee responsible**

The rules determine that if an in-house service hasn't been confirmed within a defined number of days after its creation, both the creator of the in-house service and the employee responsible for it are informed about this critical business situation.

For more information, see [Recipients](#).

Monitoring Situation Instances

In this template, monitoring of situation instances is disabled by default. If you want to use the **Monitor Situations** app to track the handling of situations in your business, you need to select the **Monitor Instances** checkbox in your situation type.

For more information, see [Monitor Situations](#).

Related Information

[Situation Handling](#)

[Key Concepts in Situation Handling](#)

Situation Template: Processing of In-House Service Overdue

Situation Template ID: SRVC_DELAY_IN_IHR_PROCESSING

Consuming App: [Manage In-House Services](#)

Template used for Situation Handling that informs customer service representatives when the processing of an in-house service took longer than expected.

For background information about the Situation Handling framework, see the Product Assistance for [Situation Handling](#).

Business Value

As a configuration expert for business processes (SAP business role SAP_BR_BPC_EXPERT), you can use this template for the following situation:

You want your customer service representatives to be informed automatically when the processing time that you defined for an in-house service has been exceeded. Using automatic notifications, a customer service representative can navigate directly to a critical in-house service, identify impediments to its processing and bring it to completion quickly.

Default Settings

The following sections describe how this standard situation template is preconfigured and which predefined parameters you can configure in the [Manage Situation Types](#) app when you copy the standard template to create your own situation type.

For more information, see [Manage Situation Types](#).

Situation Trigger Object and Anchor Object

These technical settings define for which object a situation is displayed (anchor object) and the object based on which a situation is triggered (trigger object):

- **Trigger Type:** Batch
- **Trigger Object:** In-House Service
- **Anchor Object:** In-House Service

Description: In each batch job run, the system checks for in-house services that have the status **Open** and **In Process** and for which the time that has passed since their creation exceeds the predefined number of days. The predefined number of days is currently set to five, but you are free to change it according to your needs. If the system detects such in-house services, it sends out automatic notifications to the defined recipients.

For more information on batch jobs, see [Batch Job Scheduling](#).

Conditions

The condition in this standard template defines that a situation instance is created with the status **Open** and a notification is sent to the defined recipients if the system detects an in-house service that meets the criteria specified above.

For more information, see [Conditions](#).

This template provides the following important filters that you can use to define your own conditions when you create your situation types:

Filter	Description
In-House Service	The ID of an in-house service that represents a record containing one or more in-house service objects (service object) that have been sent in by a customer for maintenance or repair
In-House Service Status	The status of an in-house service. Possible values are: - A: Open - B: In Process - C: Completed
Number of Days After In-House Service Creation	The number of days that have passed since an in-house service was created

Situation Display

When situations occur, users are informed about them by various texts, such as in-app situation messages or notifications on SAP Fiori launchpad. In this section, you can see all preconfigured texts.

For more information, see [Situation Display](#).

i Note

When you copy the template and adapt it to your needs, you can define the notification settings and change all text elements in this section. After making the changes in the original language, please remember to translate these changes into the other languages that you want to use. The template comes with translations that can be changed directly when you copy the template.

For more information, see [Translate Situation Types](#).

Recipients

For this template, the recipients of the notification can be determined by two predefined rules:

- **Notify creator of the business transaction**
- **Notify employee responsible**

The rules determine that if an in-house service hasn't been set to status **Completed** within a defined number of days after its creation, both the creator of the in-house service and the employee responsible for it are informed about this critical business situation.

For more information, see [Recipients](#).

Monitoring Situation Instances

In this template, monitoring of situation instances is disabled by default. If you want to use the **Monitor Situations** app to track the handling of situations in your business, you need to select the **Monitor Instances** checkbox in your situation type.

For more information, see [Monitor Situations](#).

Related Information

Manage In-House Service Objects

App ID: F7695

You can use this app to access the in-house service object (service object) and to create the service transactions and documents to perform the maintenance or repair. You can filter, sort, and group a list of all service objects in the system.

Key Features

You can use this app to do the following:

- Search for service objects using filter criteria, such as the employee responsible.
- Filter for service objects that are in a specific processing phase within the in-house service process.

To do so, filter for service objects that are related to service orders and service quotations in a specific status.

- Create service transactions, such as service quotations and service orders for multiple service objects in the same in-house service simultaneously (**mass decision**) or for individual service objects.
- Launch logistics-related process steps for multiple service objects in the same in-house service simultaneously (**mass decision**) or for individual service objects.

You can, for example, create returns orders or transfer the service objects to other service centers at which the service is to be performed.

- Add serial numbers and equipment to service objects with a quantity greater than one.

You can do this if the service object is in status **Open**.

- Display the details of individual service objects.
- Get an overview of all service transactions and documents that are used to process the service of the service object.
- Complete the service object when all activities for the service object have been finalized.

You can also complete multiple service objects in the same in-house service simultaneously (**mass decision**).

Supported Device Types

- Desktop
- Tablet (without navigation to service orders, service confirmations, and service quotations)

App Extensibility: Manage In-House Service Objects

As a business expert, you can extend the [Manage In-House Service Objects](#) app according to your business needs.

Custom Fields

You can create and edit custom fields to enhance applications using the [Custom Fields](#) app. You can add fields to the following UI elements using key user adaptation:

Business Context	UI Technology	UI Element Type
In-house service object	SAP Fiori	General Information section under Manage Service Object Details i Note You need to display this section by choosing Add: Section.

Additional Standard Fields

You can add standard fields that are not shown on the UI by default using key user adaptation. To check which standard fields you can add to the app, start adding a field to the UI element that you want to enhance. You then see a list of all fields that are available (both standard fields and custom fields).

For more information, see:

- [Custom Fields](#)
- [Adding Extension Fields to Apps with Different UI-Technologies](#)

Custom Logic

With the [Custom Logic](#) app, you can create and maintain implementations of custom logic that can be used to enhance applications and change application behavior. You can implement custom logic for the following business add-ins (BAIs):

BAdI Description	Business Context	Business Use
CRMS4_REPA_CHECK_BEFORE_SAVE (BAdI: Check In House-Service and Service Object before Saving)	In-house service object	You use this BAdI to call the method CHECK_BEFORE_SAVE. This BAdI method checks any value changes in the customer-specific fields (extension fields) for in-house services and in-house service objects before they are saved automatically or actively by the user. The result of this check needs to be documented in messages that have been assigned to a message table.
CRMS4_REPA_CUST_I_MODIFY (BAdI: Modify Service Object Data for In-House Service)	In-house service object	You can use this BAdI to call the method MODIFY. This BAdI method is used to change the values of the customer-specific fields in the in-house service object.

For more information, see:

- [Custom Logic](#)
- BAdI-specific documentation under [Custom Logic](#) > [your implementation](#) > [Extension Point](#) > [View Documentation](#)

Alternatively, see the documentation in Customizing for [Service](#) under [Transactions](#) > [Settings for In-House Service](#) > [Business Add-Ins](#).

Related Information

Situation Template: Follow-Up Step of Precheck Pending

Situation Template ID: SRVC_IHRITM_PRECHECK_DELAYED

This is a situation template delivered by SAP for [Situation Handling](#).

Business Value

You can use this template to notify your service managers when the precheck for a newly created in-house service object (service object) hasn't been completed within a specified period. In this way, they can take the necessary action without delay, for example, create a service order or a service quotation, initiate an outbound delivery, or mark the service object as completed.

Default Settings

The template comes with predefined settings. For the settings that aren't visible or self-explanatory, you can find information in the following sections. For generic information about how to configure situations based on this template, refer to the documentation of the [Manage Situation Types](#) app with which you can display and use the template.

Situation Trigger Object and Anchor Object

These technical settings define for which object a situation is displayed (anchor object) and the object based on which a situation is triggered (trigger object):

Trigger Object	Service Object
CDS View for Trigger Object	I_REPROBJPRECHECKDELAYEDSITN
Anchor Object	Service Object
CDS View for Anchor Object	I_REPROBJPRECHECKDELAYEDSITN
Trigger Type	Batch

The situation is triggered when the system finds service objects for which the precheck hasn't been completed after a number of days you define. That means no service order, service quotation, or outbound delivery has been created and the service object hasn't been marked as completed yet.

i Note

These settings aren't visible in the template in the [Manage Situation Types](#) app and cannot be changed when you create a situation type based on this template. All of the following settings are visible in the template and you can adapt them when you copy the template to create a situation type.

Conditions

The condition in this template defines that a situation is created with the status **Open** and a notification is sent to the defined recipients if the system detects a service object that meets the criteria specified above.

For more information, see [Conditions](#).

Recipients

In this section, you see the predefined settings for determining who is responsible for a situation instance. Based on these, you can define who receives a notification when a situation instance is triggered (if enabled) or who sees the instance in the [My Situations](#) app.

The recipients of the notifications can be determined by the following predefined rules:

- **CREATOR (Notify creator of the business transaction)**
- **RESPEMPLOYEE (Notify employee responsible)**

These rules determine that if a follow-up step of the precheck is still pending for a service object, the user who created the service object or who is specified as the employee responsible for this service object is informed about this critical business situation.

For more information, see [Recipients](#).

Situation Monitoring

Monitor Instances: Monitoring (the tracking of changes to situation instances) isn't enabled by default for situation templates. When you copy the template to create a situation type, you need to enable this option if you want to capture data for this situation type and enable the creation of [business situation events](#). You can view the data in the [Monitor Situations](#) app.

Related Information

[Situation Handling](#)

[Perform Precheck](#)

Situation Template: Inconsistency in Logistics Information for Service Object

Situation Template ID: SRVC_IHRITEM_EQUIP_LOGS_ERROR

This is a situation template delivered by SAP for [Situation Handling](#).

Consuming App: [Manage In-House Service Objects](#)

The consuming app is a business app for which this template is designed and which displays a situation message.

Business Value

You can use this template to inform service managers that the logistics information originally entered for a service object has changed for the related serial number (serialized product or equipment). This can be the following:

- One or more serial numbers are no longer in the plant and at the storage location that have been entered for the related service object.
- The stock type and the special stock indicator that have been specified for the service object have changed for the serial number.

This situation can occur, if, for example, the logistics process for serialized products or equipment is launched in the In-House Service apps and logistical follow-steps are performed in the logistics processes, which is outside of In-House Service. In this case, the logistics information for the related service object is not updated after the change, because service objects are not handled as stock-keeping objects.

How to Solve this Situation

If such a situation occurs, contact the logistics expert to align the logistics information for the serialized products or equipment with the logistics information originally specified for the related service object.

If you start the logistics process for a serialized product or piece of equipment in In-House Service, carry out all subsequent logistics activities exclusively in In-House Service. This ensures that the logistics information is consistent for both the serialized products or pieces of equipment, and the related service object. If you do not keep the logistics information consistent in the service object and the related serialized products or pieces of equipment, you might not be able to perform the logistics processes in In-House Service anymore (see [Logistics Process for In-House Service Objects](#)).

Default Settings

The template comes with predefined settings. For the settings that aren't visible or self-explanatory, you can find information in the following sections. For generic information about how to configure situations based on this template, refer to the documentation of the [Manage Situation Types](#) app with which you can display and use the template.

Situation Trigger Object and Anchor Object

These technical settings define for which object a situation is displayed (anchor object) and the object based on which a situation is triggered (trigger object):

Trigger Object / Anchor Object	In-house service object (service object)
CDS View for Trigger Object / Anchor Object	I_InhRepairItmSrlzdProductSitn
Unique ID of Trigger Object	InHouseRepairItemCharUUID
Unique ID of Anchor Object	InHouseRepairItemUUID
Trigger Type	Batch

The situation is triggered when the logistics information specified for a service object, is subsequently changed for the related serialized products or pieces of equipment.

Logistics information in service objects include the following:

- Plant
- Storage location
- Special stock indicator
- Stock type

i Note

These settings aren't visible in the template in the [Manage Situation Types](#) app and cannot be changed when you create a situation type based on this template. All of the following settings are visible in the template and you can adapt them when you copy the template to create a situation type.

Conditions

The condition in this standard template defines that a situation instance is created with the status **Open** and a notification is sent to the defined recipients if the system detects a service object that meets the criteria specified above.

For more information, see [Conditions](#).

This template provides the following important filters that you can use to define your own conditions when you create your situation types:

Filter	Description
Availability at Service Center	Specifies the availability of the service object that is related to the serialized products or pieces of equipment at the service center Use the value Received by Service Center (I1213) to exclude the service objects that are not to be checked by this situation and for which the logistics process has not been launched in In-House Service.
Logistics Information Has Changed for Serialized Products	Boolean value that indicates whether the system checks the logistics information for individual service objects against the logistics information for the related serialized products or pieces of equipment. Enter x to trigger the situation if logistics information for serialized products or equipment differs from the logistics information entered for the related service object. If you do not enter a value, the system does not perform this check. In this case, a situation is not to be triggered by the system.
In-House Service	Specifies the in-house service to which the affected service object is assigned.
Service Object	Specifies the service object to which the serialized products or pieces of equipment with the changed logistics information are assigned.

Recipients

In this section, you see the predefined settings for determining who is responsible for a situation instance. Based on these, you can define who receives a notification when a situation instance is triggered (if enabled) or who sees the instance in the [My Situations](#) app.

The recipients of the notifications can be determined by the following predefined rules:

- **Notify creator of the business transaction**
- **Notify employee responsible**

The rules determine that if the logistics information for serialized products or pieces of equipment specified in the In-House Service process has changed, both the user who created the service object or who is specified as the employee responsible for this service object are informed about this critical business situation.

For more information, see [Recipients](#).

Situation Monitoring

Monitor Instances: Monitoring (the tracking of changes to situation instances) isn't enabled by default for situation templates. When you copy the template to create a situation type, you need to enable this option if you want to capture data for this situation type and enable the creation of [business situation events](#). You can view the data in the [Monitor Situations](#) app.

i **Note**

Customers are responsible for the data, including the adherence to legal restrictions in their judicial area. Neither SAP nor its associated companies assume any responsibility for the adherence to authorizations or data protection rules for data captured and processed by its customers. This means that customers alone must ensure adherence to these rules and regulations.

Situation Template: Stock Inconsistency for Service Object

Situation Template ID: SRVC_IHRITM_PROD_LOGS_ERROR

This is a situation template delivered by SAP for [Situation Handling](#).

Consuming App: [Manage In-House Service Objects](#)

The consuming app is a business app for which this template is designed and which displays a situation message.

Business Value

You can use this template to inform service managers that the available quantity of a non-serialized product at the specified plant and storage location is less than the quantity originally entered for the related service object. In addition, the stock type and the special stock indicator originally entered for the service object might have changed for the non-serialized product.

This situation can occur, if, for example, the logistics process for the non-serialized product is launched in the In-House Service apps and logistical follow-up steps are performed in the logistics processes, which is outside of In-House Service. In this case, the quantity in the defined plant and storage location, the stock type and/or the special stock indicator for the service object are not updated after the change, because service objects are not handled as stock-keeping objects.

How to Solve this Situation

If such a situation occurs, contact the logistics expert to change the quantity entered for the non-serialized product at the plant and/or storage location to the quantity originally specified for the related service object in In-House Service.

If you start the logistics process for non-serialized products in In-House Service, carry out all subsequent logistics activities exclusively in In-House Service. This ensures that the quantity in the plant and at the storage location, the stock type, and the special stock indicator is consistent for both the non-serialized product and the related service object. If you do not keep the quantities consistent in the service object and the related non-serialized product, you might not be able to perform the logistics processes in In-House Service anymore (see [Logistics Process for In-House Service Objects](#)).

Default Settings

The template comes with predefined settings. For the settings that aren't visible or self-explanatory, you can find information in the following sections. For generic information about how to configure situations based on this template, refer to the documentation of the [Manage Situation Types](#) app with which you can display and use the template.

Situation Trigger Object and Anchor Object

These technical settings define for which object a situation is displayed (anchor object) and the object based on which a situation is triggered (trigger object):

Trigger / Anchor Object	In-house service object (service object)
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CDS View for Trigger Object / Anchor Object	I_InhRepairItemLogsErrorSitn
Unique ID of Trigger Object	InHouseRepairItemCharUUID
Unique ID of Anchor Object	InHouseRepairItemUUID
Trigger Type	Batch

The situation is triggered when the available quantity of the non-serialized product in the specified plant and storage location is less than the quantity entered for the related service objects at this plant and at this storage location.

i Note

These settings aren't visible in the template in the [Manage Situation Types](#) app and cannot be changed when you create a situation type based on this template. All of the following settings are visible in the template and you can adapt them when you copy the template to create a situation type.

Conditions

The condition in this standard template defines that a situation instance is created with the status **Open** and a notification is sent to the defined recipients if the system detects a service object that meets the criteria specified above.

For more information, see [Conditions](#).

This template provides the following important filters that you can use to define your own conditions when you create your situation types:

Filter	Description
Availability at Service Center	Specifies the availability of the service object that is related to the non-serialized product at the service center Use the value Received by Service Center (I1213) to exclude the service objects that are not to be checked by this situation and for which the logistics process has not been launched in In-House Service.
Check Individual Service Objects	Boolean value that indicates whether the system checks the quantities of individual service objects against the quantities of the related non-serialized product at the specified plant or storage location. If X is entered, the situation compares the quantity of the non-serialized products at the specified plant and storage location with the individual service objects. If no value is entered, the situation compares the quantity of the non-serialized product at the specified plant and storage location with the total quantity of all related service objects at the specified plant and storage location.
Non-Serialized Product Has Lower Quantity at Storage Location	Boolean value that indicates whether the available quantity of the non-serialized product at the specified plant and at the specified storage location is less than the quantity originally entered for the related service object. If X is entered, the situation checks if the available quantity of the non-serialized product at the specified plant and at the storage

Filter	Description
	location is less than the quantity originally entered for the related service object. i Note You must not enter a value in this field if you have entered a value in the Check Individual Service Objects field. However, we recommend setting the value x in this field.
In-House Service	Specifies the in-house service to which the affected service object is assigned.
Service Object	Specifies the service object to which the non-serialized products with the changed quantity at the plant or storage location are assigned.

Recipients

In this section, you see the predefined settings for determining who is responsible for a situation instance. Based on these, you can define who receives a notification when a situation instance is triggered (if enabled) or who sees the instance in the [My Situations](#) app.

The recipients of the notifications can be determined by the following predefined rules:

- [Notify creator of the business transaction](#)
- [Notify employee responsible](#)

The rules determine that if the logistics information for non-serialized products specified in the In-House Service process have changed, both the user who created the service object or who is specified as the employee responsible for this service object are informed about this critical business situation.

For more information, see [Recipients](#).

Situation Monitoring

Monitor Instances: Monitoring (the tracking of changes to situation instances) isn't enabled by default for situation templates. When you copy the template to create a situation type, you need to enable this option if you want to capture data for this situation type and enable the creation of [business situation events](#). You can view the data in the [Monitor Situations](#) app.

i Note

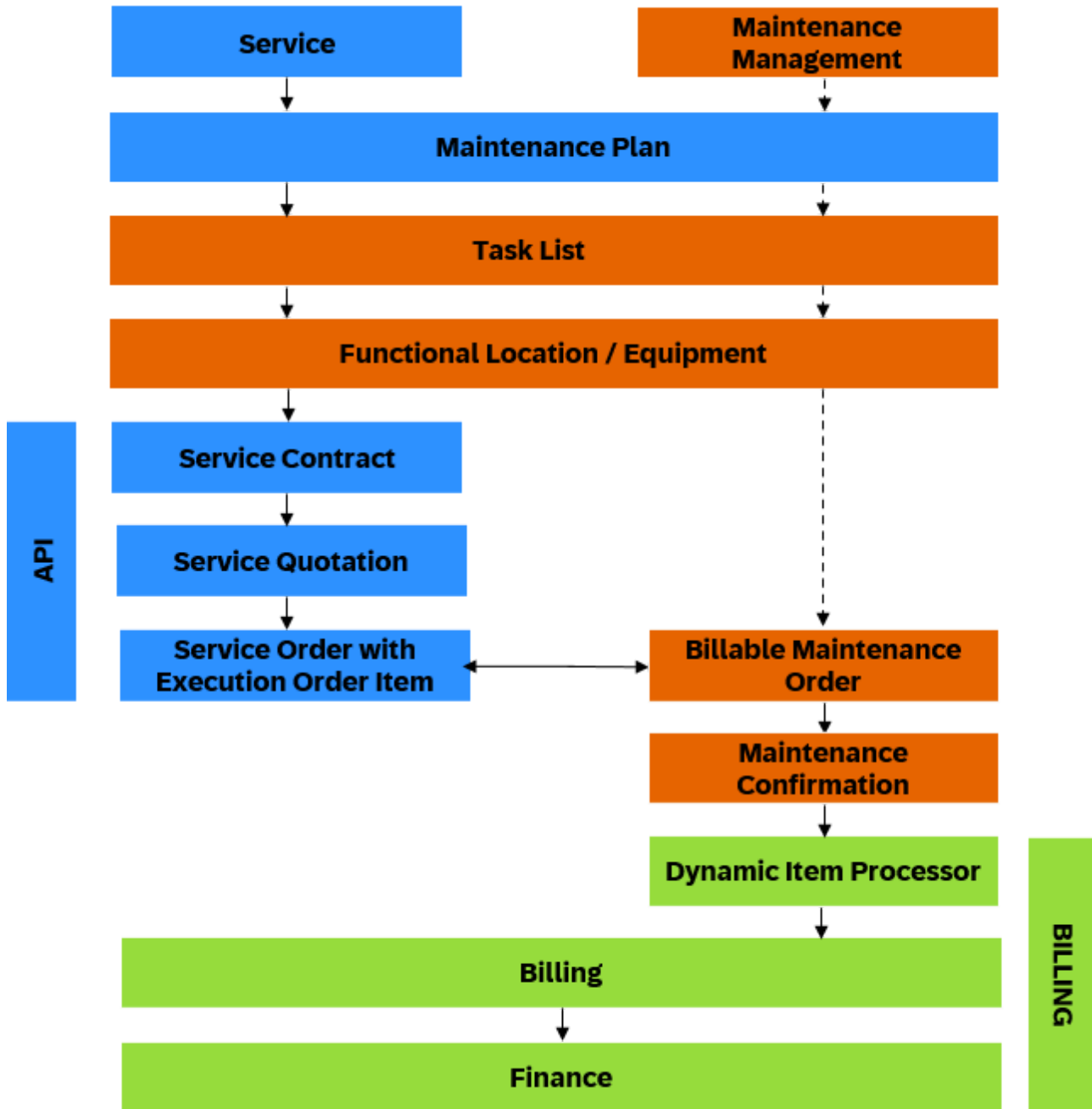
Customers are responsible for the data, including the adherence to legal restrictions in their judicial area. Neither SAP nor its associated companies assume any responsibility for the adherence to authorizations or data protection rules for data captured and processed by its customers. This means that customers alone must ensure adherence to these rules and regulations.

Service with Advanced Execution

Service with Advanced Execution is a process that is suitable for services which require extensive planning and execution. They generally have longer service times with a requirement to bill the time and service parts used for the service. The core characteristic of this process is the integration of the commercial aspects of Service with the planning and execution capabilities of Maintenance Management.

In the Service with Advanced Execution process both Service and Maintenance Management have specific roles to play. Service is concerned with understanding the scope of the service requirement, negotiating the price for the service, ensuring customer satisfaction, and determining profit margins. Maintenance Management is responsible for service execution planning, understanding the technicalities of the equipment to be serviced, scheduling, and executing the service as planned. This interaction between Service and Maintenance Management brings the best of both in one process.

Items created in the Service with Advanced Execution process are called **execution order items**. In corrective service, these items can be entered as part of a service quotation or a service order, with or without reference to a service contract. Execution order items can also be the result of a maintenance plan, which will facilitate preventive maintenance. Each execution order item creates a corresponding maintenance order. The maintenance order that is created enables the planning and execution of the service.



The above diagram provides an overview of the Service with Advanced Execution process. The process can be understood as interactions between elements of Service, Maintenance Management, and Billing.

Service

Service contracts are used to determine the terms of the service. A service quotation is used to provide a cost estimate for a requested service, and a service order is used to manage the service. A service quotation or service order can have multiple execution order items.

The maintenance order that is created when the execution order item is released is used to plan and execute the service. Maintenance plans in recurring service are used to schedule periodic service orders, thereby facilitating preventive maintenance.

Maintenance Management

Maintenance orders are used for the detailed planning of tasks and for monitoring their execution. The planning process involves task lists, scheduling, availability checks for material, and assignment of work centers. After the planning is completed, the actual service is performed and costs are recorded.

Billing

The billing process is carried out through the Dynamic Item Processor (DIP). A debit memo request is used to generate an invoice. The entire process works using item-based accounting in which service transaction items are recognized as account assignment objects to which costs and revenues are posted.

Use Cases

The Service with Advanced Execution process is typically suited to support the following scenarios:

Preventive Service

A service manager needs to be able to set up regular visits to the customer site based on service agreements. The service manager should be able to predefine a set of tasks that needs to be performed at different points in time to ensure the equipment is up and running.

In this way, a maintenance planner will be well informed in advance about an upcoming service so that tasks and resources can be planned accordingly.

Corrective Service

A service manager needs to be able to understand and capture customer service requirements, make a price proposal either as a cost estimate or as a final cost document. If the services requested are to be tailored to the customer's requirements, there needs to be a collaboration with the maintenance planner to come up with the cost estimate.

When the customer accepts the quotation, the service manager creates the service order, and the maintenance planner can go ahead with the planning and execution.

Billable Maintenance

Sometimes when a technician reports an issue during a regular visit or when an asset reports a malfunction, the maintenance planner is informed about the equipment not functioning correctly. The planner may not initially be aware whether the equipment is maintained by a customer or by the company. In such cases, the maintenance planner starts planning the work for maintaining the equipment. If the planner realizes that this equipment is owned by the customer, or the issue is due to customer misuse, then the planner needs to be able to bill the cost to the customer. In such a case, the maintenance planner needs to indicate that the work is chargeable so that the service manager can get involved in the process. The service manager can negotiate the price with the customer.

Prerequisites

The Service with Advanced Execution scenario can be setup and tested with the following scope items:

- 6AU (Service with Advanced Execution)

- 6F1 (Recurring Services with Advanced Execution)

Integration between Service and Maintenance Management

In the Customizing activity [Map Order Types](#) you need to maintain a mapping between transaction type and item category in Service and the maintenance order types in Maintenance Management. This mapping must be maintained to specify which order type would be used to create the maintenance order corresponding to an execution order item depending on the transaction type and item category.

► [Service](#) ► [Transactions](#) ► [Settings for Service Transactions](#) ► [Integration](#) ► [Plant Maintenance Integration](#) ► [Map Order Types](#) ►

To map the priorities of the service order and maintenance order, you must carry out the Customizing activity [Map Priorities](#) under ► [Service](#) ► [Transactions](#) ► [Settings for Service Transactions](#) ► [Integration](#) ► [Plant Maintenance Integration](#) ► [Map Priorities](#) ►.

To transfer information between the service order and the maintenance order, you must carry out the Customizing activity [Define Text Type](#) under ► [Service](#) ► [Transactions](#) ► [Settings for Service Transactions](#) ► [Integration](#) ► [Plant Maintenance Integration](#) ► [Define Text Type](#) ►.

i Note

In Service with Advanced Execution, and also for Service in general, it is recommended to use service products of material type [SERV](#).

Item-Based Accounting

The Customizing activity [Enable Item-Based Accounting for Service](#) needs to be enabled to allow you to use Service with Advanced Execution. This activity can be found in the Customizing under ► [Service](#) ► [Transactions](#) ► [Settings for Service Transactions](#) ► [Integration](#) ► [Enable Item-Based Accounting for Service](#) ►.

Related Information

[Item-Based Accounting](#)

Basic Concepts

[Terminology](#)

[Execution Order Items](#)

[Billable Maintenance](#)

[Service Contracts](#)

Terminology

List of terms used in Service with Advanced Execution

- Preventive Service: Process concerned with the service of equipment on a regular basis with predefined time intervals or counter milestones. Regular service is used to keep equipment in optimum working condition and to avoid breakdowns and production downtime.
- Corrective Service: Process concerned with the service of equipment when a breakdown or failure or malfunction occurs.

- **Service Quotation:** Used to provide an estimate of the time required and the price to be paid for a particular service. The customer can accept or reject the service quotation.
- **Service Order:** Used to manage a service agreement between a service provider and a service recipient. It can be created as a standalone order, a follow-up from a service quotation, or as a call object from a maintenance plan.
- **Maintenance Order:** Used to plan and monitor execution of maintenance tasks that need to be performed.
- **Billable Maintenance Order:** Type of a maintenance order that allows billing and is used in Service with Advanced Execution.
- **Execution Order Item:** An item in a service transaction used in Service with Advanced Execution. The execution order item is generally associated with a billable maintenance order which supports in the planning and execution of the service.
- **Dynamic Item Processor (DIP):** A tool that the system uses to summarize data, such as planned costs or actual costs, into dynamic items in detailed quotation planning or resource-related billing.
- **Task list:** A sequence of individual maintenance activities that must be repeatedly performed by the service technicians.
- **Item-based accounting:** A type of controlling where service transaction items are recognized as account assignment objects to which costs and revenues are posted.
- **Distributed Execution:** Process by which the planning and execution of a service requirement is split into multiple billable maintenance orders.

Parent topic: [Basic Concepts](#)

Related Information

[Execution Order Items](#)

[Billable Maintenance](#)

[Service Contracts](#)

Execution Order Items

An execution order item is a service item that has capabilities like pricing, status management, assignment of organizational data, and reference object. It is used to define the scope and terms of a service requirement as part of the Service with Advanced Execution process.

Use

You can assign one or more execution order items to a service order template, service quotation, or service order. In general, an execution order item will be associated with a corresponding billable maintenance order. The maintenance order facilitates the planning and execution of the service.

Features

Billing Relevance

You can set the billing relevance for an execution order item depending on the type of billing scenario.

- **Not Relevant for Billing**

This billing scenario is specific for cases where the service will not be billed. For example, services under warranty or performed as goodwill.

- **Summarized Billing**

This billing scenario is specific to service with advanced execution and uses the dynamic item processor to bill services. In this scenario, a service is billed as a single main item. The billed value is determined from the aggregated cost of individual subitems together with a markup.

- **Itemized Billing**

This billing scenario is specific to service with advanced execution and uses the dynamic item processor to bill services. In this scenario, a service is billed as a list of subitems. The net price is the aggregated price of these subitems determined by pricing conditions.

- **Billing on Completion**

This billing scenario is used when the price for the service is determined from existing service contracts or pricing conditions irrespective of time and material costs.

- **Billing via Billing Plan After Release**

This billing scenario is used when billing dates are to be scheduled based on agreed milestones. This billing scenario can only be used for ad hoc billing.

Dynamic Items

Dynamic items are the data summarized according to defined criteria. Data is summarized - from defined sources, such as planned cost totals records or actual cost line items - using the dynamic item processor (DIP) and is regulated by the settings you enter in the dynamic item processor profile (DIP profile).

DIP profile is mandatory when the billing relevance is Summarized or Itemized. During billing, the DIP determines the items or lines in the debit memo request, based on the DIP profile entered in the execution order item in the service order.

When using service quotation for detailed planning and the planning is completed in the maintenance order, DIP-based subitems are entered in the service quotation under the higher-level execution order item. These subitems can be generated only on the basis of the DIP profile entered.

i Note

The DIP Profile for execution order item must have Usage **11 Quotation creation and sales pricing**. This usage also allows the creation of [Resource-Related Cost item](#).

The value of the DIP Profile can be defaulted from the predecessor document such as service contract or service quotation or from the item category customizing. It is also possible to manually enter or overwrite the default values.

Planning Plant and Service Organization

Planning plant and work center are essential in maintenance management transactions to plan and execute maintenance activities. Service organization and service team play a similar role within Service and can be treated as counterparts of planning plant and work center.

Planning plant and work center are available as additional fields in the execution order item of the service quotation and service order. These fields are mandatory in the maintenance order and are copied over from the execution order item. The planning plant and work center can be determined from the service organization unit and service employee group respectively. It is also possible to manually enter or overwrite the default values. The planning plant and work center are copied from the service quotation to the service order.

i Note

The planning plant entered in the execution order item must belong to the same company code as the sales organization of the service quotation or service order.

Maintenance Order Assignment Block

The billable maintenance order is linked with the execution order item and the maintenance order number is displayed in the Transaction History of the item.

When the billable maintenance order is created, the execution order item page displays an assignment block called **Maintenance Order**. This assignment block displays key attributes from the maintenance order as a quick reference.

Advanced Variant Configuration

You can use this function to configure execution order items in service orders. You can combine advanced variant configuration with configurable task lists to configure and manage customizable products.

To configure an execution order item, it should be marked as configurable in the material master data. You configure the product by assigning values to its characteristics. Dependencies for the characteristics, characteristic values, and the configuration profile are processed as you enter characteristic values. The characteristics can be mapped to the operations of a task list in a billable maintenance order. For more information, see [Configurable Products and Product Variants](#) and [Configurable General Maintenance Task Lists](#).

Credit Check

You can use the credit check feature for execution order items in a fixed price billing scenario. The credit check is automatically triggered when the service order status is changed to **Released** and saved. If the item passes the check, the service order has the status **Released**. If the item fails the credit check, it remains in the **In Process** status until it is released in the **Manage Documented Credit Decisions** app. For more information, see [Perform Credit Checks in Service Orders](#).

The credit check only observes the total credit exposure for a business partner. Credit exposure is the combined value of all items that are related to a business partner. For more information, see [Managing Credit Exposures](#).

For execution order items, credit check is only supported for the billing relevance **Billing on Completion**.

Intercompany Service Orders

An intercompany service order is a business transaction used to distribute a service between two companies with different company codes that belong to the same organization. Execution order items can be created as part of intercompany service orders. For more information, see [Intercompany Service Orders](#).

Integration with SAP GTS

Service with advanced execution can be integrated with SAP Global Trade Services (SAP GTS) to help you comply with global trade rules and reduce risk of non-compliance. With this integration, the creation and update of a billable maintenance order is prevented if the execution order item fails the trade compliance check. For more information, see [Integration with SAP Global Trade Services](#).

Other Aspects of Service with Advanced Execution

Service Order Templates

You can use service order templates to define reusable sets of service-related data that minimize the amount of time required to create a service transaction. A service order template describes only the scope of planned services and not the actual execution (customer and time frame). Execution order items can be added in service order templates. For more information, see [Service Order Templates](#).

Person Responsible

Person Responsible is available as an additional partner function in the execution order item of the service quotation and service order. It is an optional partner in the standard delivery. This can be determined based on the main reference object entered in the execution order item. The person responsible entered in execution order item is used to default the person responsible in the corresponding billable maintenance order. This is dependent on the Customizing in [►Service ► Basic Functions ► Partner Processing ► Map Partner Functions ►](#).

Planned Cost and Revenue

The costs for the service execution are determined from the billable maintenance order. However, the revenues of the service are determined on the service order.

The baseline planned cost is determined when the billable maintenance order is released. The baseline planned revenue in all cases, is always determined when the service order is released.

In addition, the continuous planned costs and revenues are also displayed in the fields called [Ongoing Planned Cost](#) and [Ongoing Planned Revenue](#). These fields are updated with every change in planning data of the billable maintenance order and every change of pricing data in service order.

When the billing is fixed price, the planned revenue is taken from the net value of the item. When the billing is resource-related, the planned revenue is determined from the planned cost of the billable maintenance order by using the [Get Calculated Cost](#) button.

Billing Status and Documents

Once the billing is started for the service order, then the fields [Transfer to Billing](#) and [Billing status](#) continuously reflect the latest status of the billing process. The billing documents such as Billing Document Request, Debit Memo Request and Invoice are also displayed in the [Transaction History](#) of the execution order item.

Parent topic: [Basic Concepts](#)

Related Information

[Terminology](#)

[Billable Maintenance](#)

[Service Contracts](#)

[Item-Based Accounting for Service with Advanced Execution](#)

Resource-Related Cost item

Resource-related cost items (RRCO items) help to determine the cost estimate for the service.

Use

Based on service requirements, operations and components are planned in the billable maintenance order. This planning on the billable maintenance order dynamically determines line items in the service quotation and service order as per the DIP profile on execution order item. These line items are automatically added as sub-items with item-category usage RRCO to the execution order item. Such sub-items are referred to as Resource-Related Cost (RRCO) items.

In Service Quotation

When you select [Send to Planning](#) in a service quotation for an execution order item, a billable maintenance order is created. This allows the maintenance planner to start planning for the service. When the planning is completed, the labor and material that is required for the service, is displayed in the [Items](#) assignment block of the service quotation as Resource-Related Cost (RRCO) items. The charged cost displayed in the [Price Details](#) of the RRCO item is taken from the billable maintenance order. Each RRCO

item is a subitem of the execution order item and all charged costs of statistical RRCO subitems are aggregated to the main execution order item.

The **Refresh Planning** button can be used to delete the existing RRCO items and create new ones depending on the latest changes made to the planning of the billable maintenance order and the current DIP profile maintained in execution order item.

In Service Order

When a service order is created as a follow-up from a service quotation, you have the option to copy RRCO items into the service order. However, the execution order item containing the RRCO items has to be copied as well. You also have the option to select only the execution order item and leave out the RRCO items. Furthermore, you can also select only some of the RRCO items (provided that you select the main execution order item). The unselected RRCO items will be set to the status **Completed** and will not appear in the pop-up window if the quotation is still in-process and the create follow-up function is triggered again. When you use the **Refresh Planning** button the existing RRCO items will be deleted and the updated items from the billable maintenance order will be generated.

When you add a new execution order item to a service order that has not been created as a follow-up, you can use the **Refresh Planning** button to generate the RRCO items directly in the service order.

Configuration

You need to select the item category usage **RRCO (Resource-Related Cost)** for the determination of resource-related cost items. You do this in the Customizing: **►Service ► Transactions ► Basic Settings ► Define Item Category Determination ►**

Properties

Resource-related cost items have the following properties:

- Their life cycle status mirrors that of the main execution order item and can't be changed manually.
- They can be copied from service quotation to follow-up service order.
- They will inherit the controlling object of the main execution order item.
- They can be included in pricing and planned revenue calculations.
- Their charged cost is aggregated to the main execution order item.
- They are not relevant for contract determination and can't be assigned with any service contract. They inherit the effects of the contract assignment on the main item.
- They are deleted on replanning.
- They can't be used as price items.
- They will not appear in the output control PDF file.
- They have a limited number of assignment blocks in their overview page.

Example

1. A customer has requested you to provide a quotation for the maintenance of a centrifugal pump.
2. You start by creating a service quotation with the relevant details and adding the maintenance of the pump as **Item 10** with **Execution Order Item** as the item category.
3. After setting **Item 10** to **In Process** and selecting the **Send to Planning** checkbox, you save the quotation.

4. A billable maintenance order has now been created and the maintenance planner can begin planning the service in the [Change Order \(IW32\)](#) application.
5. The maintenance planner determines that this maintenance involves the following three operations and adds it to the [Operations](#) tab in the billable maintenance order:
 - dismantling the pump – 2 hrs
 - inspecting the pump – 4 hrs
 - reassembling the pump – 2 hrs
6. Considering the age and operational life cycle of the pump, the planner determines that the bearings and the mechanical seals need to be replaced. This is added under the [Components](#) tab.
7. Once all the planning information is entered, the planner selects [Planning Completed](#). This sets the maintenance order status to [ORAI \(Order Approval in Process\)](#).
8. When you refresh the service quotation you will see that three new items have been added. [Item 20](#) represents the total time for the service, 8 hrs, [Item 30](#) represents the replacement bearings and [Item 40](#) represents the replacement seals. These three items are RRCO items of [Item 10](#).
9. You can use the resource-related cost item information together with the charged cost to provide the best price estimate for the customer.
10. When you set [Item 10](#) to [Released](#), the status of the resource-related cost items will also change to [Released](#).
11. If the customer accepts the quotation and you create the follow-up service order, you will have the option to copy the execution order item along with the resource-related cost items into the service order.

Billable Maintenance

Maintenance Management involves the inspection, maintenance, and repair of a technical object. Maintenance Management does not usually include the capability to bill a service provider. However, in certain maintenance scenarios a maintenance order needs to allow the possibility of billing. But this billing requirement comes after the maintenance order has been created. So to allow for the possibility of billing, a corresponding service order is created. Using the service order you can generate a billing document on completion of the service.

You trigger the service order creation process by selecting the [Billable](#) checkbox in the top right corner of the [Change Maintenance Order \(IW32\)](#) screen.

When the [Billable](#) checkbox is selected you can enter the sales data and service details. The fields in these sections allow you to enter details such as sold-to party, sales organization, distribution channel, service product, and quantity. Once the planning has been completed the maintenance order is released and the work can be done by the technician. After confirmation that the service is done, the order can be set to completed. The billing process can then be initiated.

i Note

The sales data and the service sections appear only when the [Billable](#) checkbox is selected. You cannot deselect the Billable checkbox once selected.

Parent topic: [Basic Concepts](#)

Related Information

[Terminology](#)

Service Contracts

Service contracts are outline agreements with business partners that define the services offered for a particular period. Service contract management has been integrated with Service with Advanced Execution to support the execution order items in service quotations and service orders.

Item-Based Accounting

Item-based accounting is supported in service contracts. Service contract items are recognized as account assignment objects to which costs and revenues can be posted. Item-based accounting must be enabled as a prerequisite for the Service with Advanced Execution process.

Contract Determination

For all execution order items, you can determine service contracts based on the product in a service contract item. The service contracts whose items contain the product that you have specified in an execution order item can be determined for your selection.

DIP Profile

In the Service with Advanced Execution process, service contracts contain an additional field of information called the dynamic item processor (DIP) profile. The dynamic item processor is a tool that the system uses to summarize data into dynamic items in resource-related billing.

How to display the DIP profile

1. Launch the WebClient User Interface and select the business role **Service Professional** (S4C_SRV_PRO).
2. Choose the **User** icon and select the **Personalize** option.
3. Choose **Personalize** Settings.
4. Choose **Enable configuration mode**.
5. Enter a service contract.
6. Choose **Billing** and select the **Show Configurable Areas** icon.
7. Choose the DIP profile from **Available Fields** and move it to the displayed area.
8. Choose **Save and Close**.

Parent topic: [Basic Concepts](#)

Related Information

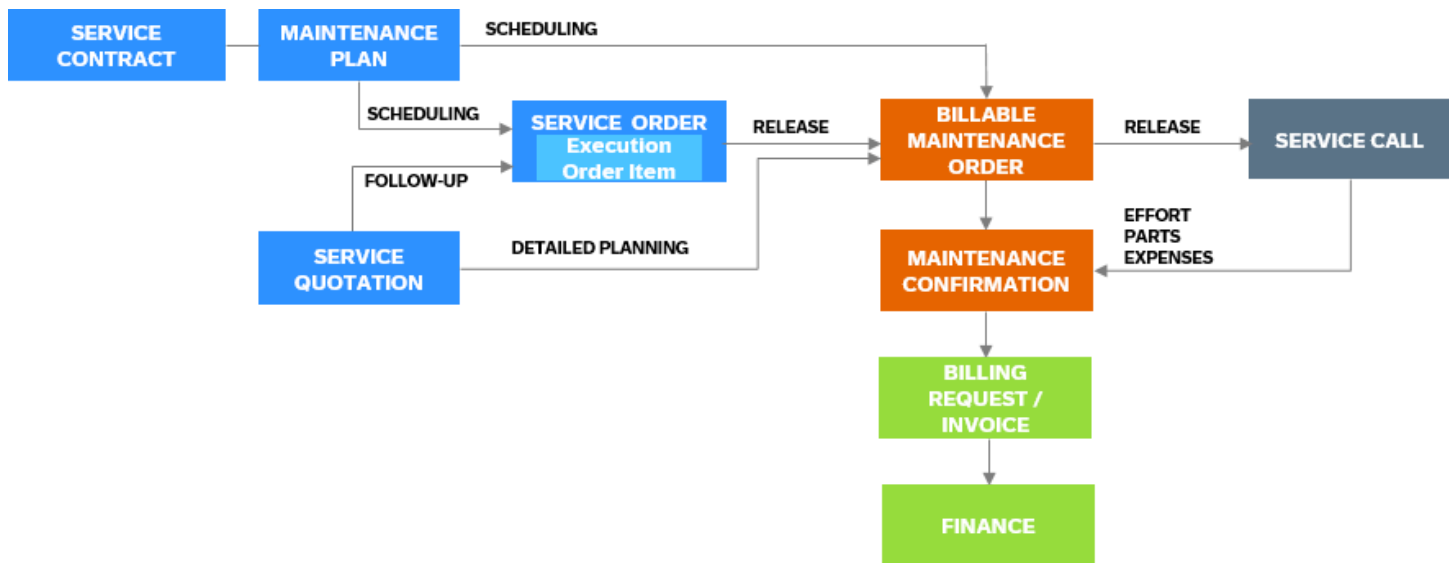
[Terminology](#)

[Execution Order Items](#)

[Billable Maintenance](#)

Processes in Service with Advanced Execution

Get an overview of the various processes in Service with Advanced Execution.



The above figure is an illustration of the Service with Advanced Execution process across Service and Maintenance Management.

The core aspect of Service with Advanced Execution is the connection between the execution order item and the maintenance order. This can be seen in the following scenarios.

- [Service with Advanced Execution in Service Quotation](#)
- [Service with Advanced Execution in Service Order](#)
- [Service with Advanced Execution in Recurring Service](#)
- [Service with Advanced Execution in Maintenance Management](#)

Service with Advanced Execution in Service Quotation

You can use Service with Advanced Execution to create a service quotation to provide customers with an estimate of the time and cost of a service.

Use

When customers want to get an estimate of the time and cost involved in the service of a technical object, you can create a service quotation to provide this information. The service quotation helps customers and service providers to plan the anticipated services at an early stage. Customers can plan their budget and their production capabilities, while service providers can plan their revenue and service resources such as labor and service parts.

You use the [Manage Service Quotations](#) app to create a quotation.

Extent of Planning

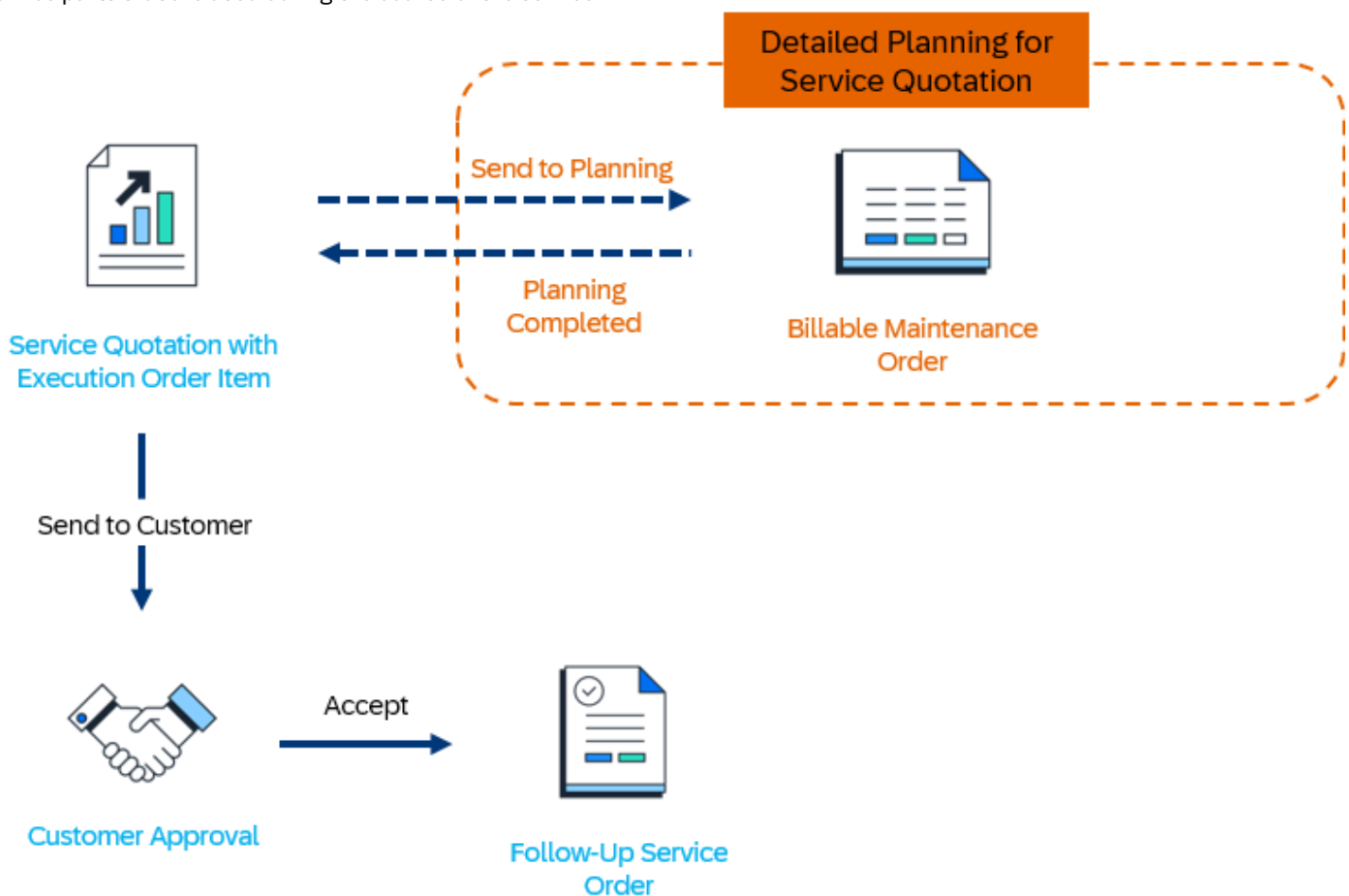
The first step to creating a service quotation is to determine the extent of planning involved in the service. For a service that requires less detailed planning, a quotation can be created and sent to the customer, who will then approve or reject the proposal. But for a service that requires extensive planning it is necessary to account for this planning in the quotation. To that effect a maintenance order is created for each item in the quotation. The maintenance order provides the information on the planning and

execution of the actual service. Once this is factored into the quotation, the quotation can be sent to the customer. For more information, see [Detailed Planning in Service Quotations](#).

When you create a quotation you can indicate the type of billing by using the **Billing Relevance** attribute. However, when creating a service quotation the billing relevance does not carry much significance, irrespective of whether detailed planning is required or not. It only offers a possibility to propose at an early stage what billing relevance can be used eventually. Depending on the billing relevance, the app that is used for billing can also be different.

Billing Relevance	Type of Billing	Billing App
Billing on Completion	Fixed price billing	Release for Billing
Summarized Billing or Itemized Billing	Resource-related billing	Create Resource-Related Billing Request: Service Order

In fixed price billing, the cost of the service is agreed upon with the help of an existing price master. Irrespective of the labor or service parts used, the cost will not change. In resource-related billing the cost of the service is determined based on the labor and service parts that are used during the course of the service.



Creating a Service Quotation

You can create a service quotation using the [Manage Service Quotations](#) app.

In the service quotation, make the following entries:

1. Enter general data such description and sold-to party.
2. Enter the sales organizational data.

i Note

This is custom documentation. For more information, please visit [SAP Help Portal](#).

Depending on customizing, automatic selection of organizational data is possible.

3. Under **Parties Involved** enter the service employee group and employee responsible.
4. Add items and select **Exec. Ordr Itm (Qtn)** (Execution Order Item (Quotation)) as the item category.
5. For each execution order item, you need to select a billing relevance. You need to enter a DIP profile if you select summarized billing or itemized billing as the billing relevance.

i Note

- You can create a maintenance order from a service quotation to facilitate the early planning of a service. For more information, see [Detailed Planning in Service Quotations](#).

Send to Customer

Once you have finalized the service quotation, you can choose **Send to Customer**. The service quotation is sent to the customer using output management. The status of the service quotation changes to **Sent to Customer**, the status of the items is set to **Released**.

Accept the Service Quotation

Once the customer agrees to the service quotation, choose **Accept** to record the customer's decision. The status of the quotation changes to **Completed**, the items accepted are set to **Completed**. A service order is automatically created with an **In Process** status for each execution order item.

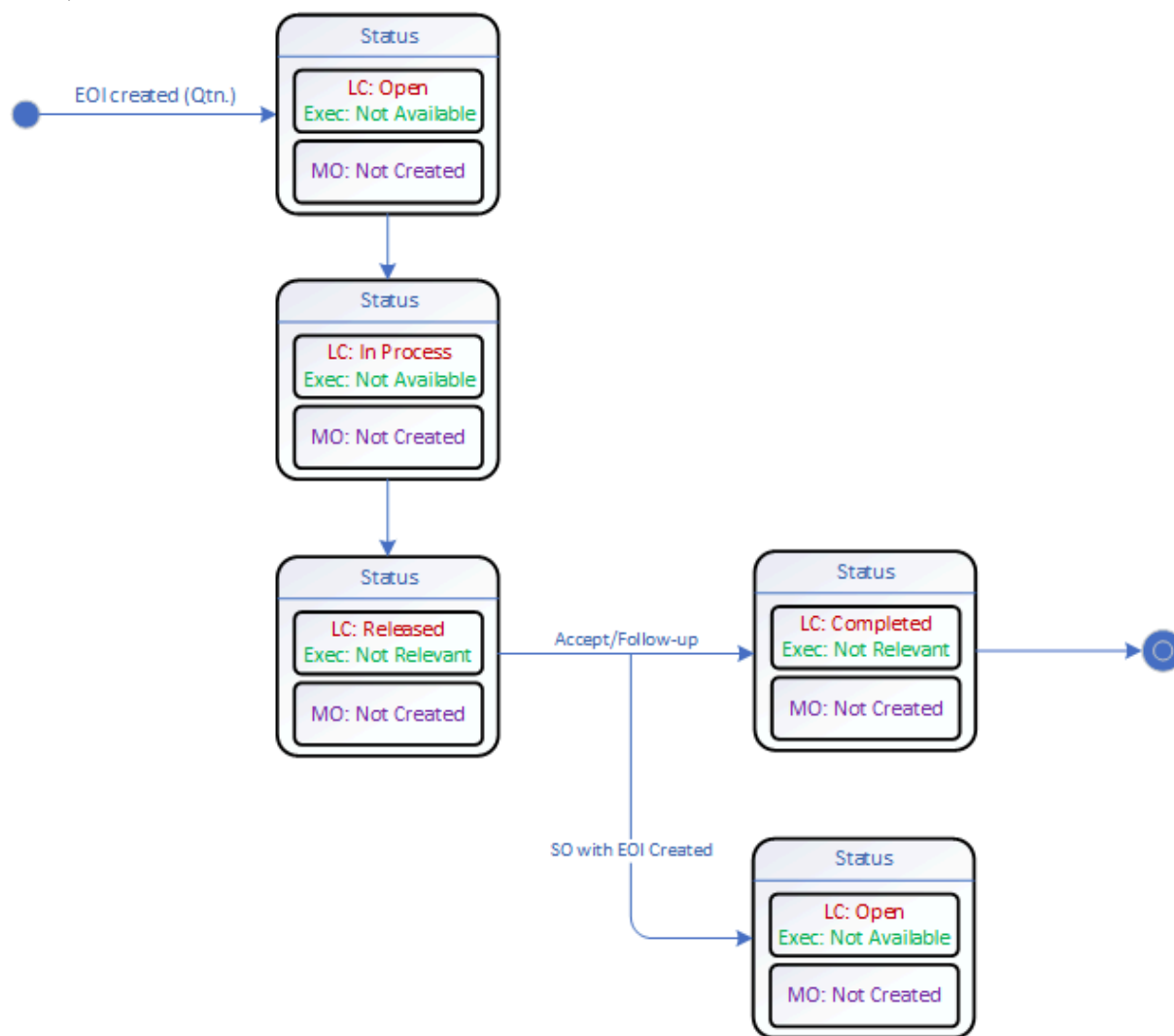
Reject the Service Quotation

If the service quotation is rejected, its status changes to **Completed** and the items are set to **Completed** and each item execution status is set to **Canceled**.

Replanning

After the service quotation is shared with the customer, there could be a need to revisit the service quotation. When such a need arises, you can change the status of the quotation from **Released** to **In Process**. This allows you to make changes without having to create a new service quotation.

Overview of Service Quotation Status



Note:
1. Accept or Create Follow-up creates SO with one EOI.

SO	Service Order
Qtn	Quotation
MO	Maintenance Order
LC	Lifecycle
Exec	Execution

Status	Description
Open	This is the default status that all service quotations get after they are created. It indicates that you are free to edit to the quotation.
In Process	In this status the quotation and its items can be edited.
Released	When the quotation items have been finalized it is set to Released .
Completed	When the service quotation has been accepted by the customer, the status is set to Completed . This status can also be manually set if the quotation is rejected by the customer.

An item execution status indicates in which stage of execution planning or execution the item actually is in. This status is important when a maintenance order has been created for an item.

Related Information

Detailed Planning in Service Quotations

Sometimes it is necessary to begin detailed planning for a service in advance. Detailed planning can provide better cost and time estimates for the actual service. The price can be set based on the cost calculations and the service quotation can be sent to the customer for approval. This is especially useful in service jobs that require a lot of time and service parts to complete. You can initiate this planning with the creation of a billable maintenance order corresponding to an execution order item.

The maintenance order enables the detailed planning for the service requirements in terms of individual tasks, operations and components and thus provides cost estimations. The process also supports an approval workflow for the planning.

Each execution order item is capable of creating its own billable maintenance order. By choosing **Plan All Items** you can create billable maintenance orders for all the items in the quotation. You can also select items for which you want to create billable maintenance orders by selecting the checkbox under **Send To Planning**.

The execution order item must be in **In Process** status to create the maintenance order. You can also trigger **Send To Planning** automatically by selecting the **Send to Planning Immediately** checkbox in the Customizing activity **Map Order Types**.

Send to Planning

The **Send to Planning** action triggers the creation of the corresponding billable maintenance order. Some data is automatically copied into the maintenance order from the execution order item in the service quotation. For example, planning plant, work center, header reference object, and object list. Partners and texts are copied based on the Customizing.

Successful creation of the billable maintenance order sets the maintenance order status to **ORAR (Order Approval Required)**. However, in case of failure to create maintenance order, errors will be raised in the service quotation and the **Send to Planning** checkbox is reset automatically. In this case, the errors can be corrected by adjusting the master data or configuration data and you can retry the Send to Planning function again.

Planning Completed

When the planning is completed, you can select the **Planning Completed** button and set the status of the maintenance order to **ORAI (Order Approval in Process)**.

With the planning completed, you can use the cost estimates to negotiate and finalize a price for the service in the service quotation. You can then either send the quotation to the customer for acceptance or could trigger an additional cycle of planning.

Re-planning

If you want to request an adjustment in the planning data, you can use the **Send to Planning** function again. The same billable maintenance order which was created before is editable again. The status of the maintenance order is now set to **ORRJ (Order Rejected)**. It is still not possible to release the maintenance order and the service quotation cannot be sent to customer since it is in planning.

The planning is adjusted based on the requirements of the service, and the **Planning Completed** button is selected again. This sets again the status of the maintenance order to **ORAI (Order Approval in Process)**.

This cycle of planning adjustments and corrections in maintenance order can be performed multiple times, until the planning data is addressing the requirements of the service and the requests of the customer.

Send to Customer

When the service quotation is sent to the customer, the execution order item status is set to **Released**. It is possible to configure the output management such that the DIP-based subitems can also be included in the service quotation document which is sent to the customer.

At this stage, the customer could request for an adjustment in the planning data or the terms of the quotation. You have the option to revoke the release of the execution order item and set the status back to **In Process**. With this, it is now possible to do one more cycle of Send to Planning, Planning Completed and Send to Customer.

If the customer rejects the service quotation, it is set to status **Canceled** and the corresponding maintenance order is set to **Do not execute**.

If the customer accepts the service quotation, it is set to status **Completed** and the maintenance order is set to status **ORAP (Order Approved)**. Simultaneously, the follow-up service order is created and the reference of the existing maintenance order is carried over into the service order execution order item so that the planning data can be retained as part of the same maintenance order.

With the status ORAP (Order Approved) on the maintenance order, the order is editable again and it is open for release. It is important to note that the Accept action on the service quotation and the subsequent setting of the ORAP status, sets the baseline planned cost.

Refresh Planning

During the planning phase of the service, the time and materials required for the service are planned in the billable maintenance order. Through the dynamic item processor (DIP) this planning information is retrieved by the service quotation as subitems of the execution order item when the maintenance order is set to **Planning Completed**. Sometimes, an error may prevent the subitems from being displayed in the service quotation or the DIP profile may need to be changed. By using refresh planning you can retrieve the planning information once more.

To use refresh planning you need to ensure that:

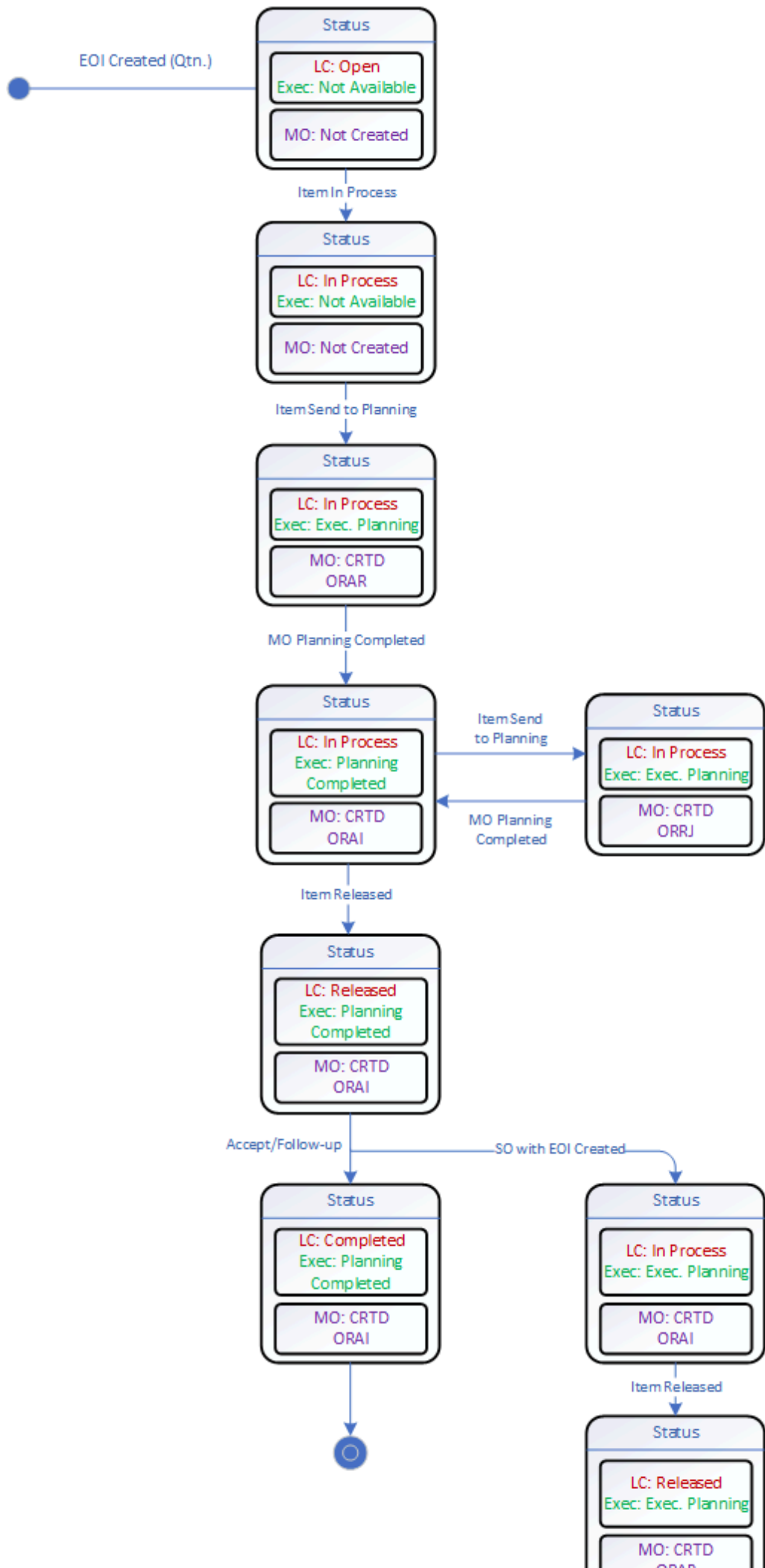
- Life cycle status of the execution order item is **In Process**
- Actual planning is completed and the execution status is **Planning Completed**

i Note

The **Refresh Planning** button is disabled when the service quotation is accepted.

Overview of Status

The following diagram and table provide an overview of the maintenance order status and the corresponding item execution status.



1. EOI cannot be released when execution status is Execution Planning.
2. Accept or Create Follow-up creates SO with one item.

SO	Service Order
Qtn	Quotation
MO	Maintenance Order
LC	Lifecycle
Exec	Execution
CRTD	Created
ORAR	Order Approval Required
ORAI	Order Approval in Process
ORAP	Order Approved

Status on Maintenance Order	Item Execution Status	What Happens in this Status
CRTD (Created)	Execution Planning	All maintenance orders are created in this default status.
ORAR (Order Approval Required)	Execution Planning	The billable maintenance order corresponding to the item has been created and the maintenance planner can start with the planning. In this status, the maintenance order can't be released, and the service quotation can't be sent to the customer.
ORAI (Order Approval in Process)	Planning Completed	The maintenance planner completes the planning and reverts to the service representative to accept or reject. In this status, the maintenance order can't be edited.
ORRJ (Order Rejected)	Execution Planning	The customer service representative rejects the planning and sends the item back for re-planning. In this status, the maintenance order can't be released, and the service quotation can't be sent to the customer.
ORAP (Order Approved)	Planning Completed	The customer accepts the service quotation. A service order is created for the item. The maintenance order can be edited or released.

Service with Advanced Execution in Service Order

You can use Service with Advanced Execution to create service orders that specify the details of a service between a service provider and recipient.

Use

You can use service orders to record information about a service that has been agreed between the service provider and service recipient. Service orders provide the necessary information regarding the service process and the cost that is incurred. Execution order items in the service order create a corresponding billable maintenance order that facilitates the planning and execution of the service.

In Service with Advanced Execution, you can create a service order:

- As a standalone service order

The service order is created in the **Open** status and can be set to **In Process**. When the execution order item is released the billable maintenance order is created.

- As a follow-up transaction, from an accepted service quotation

The service order is created in the **Open** status and can be set to **In Process**. When the execution order item is released the billable maintenance order is created.

- As a follow-up transaction, from an accepted service quotation with detailed planning

The service order is created in the **In Process** status. The billable maintenance order that was created during detailed planning of service quotation is carried over to the service order.

- As a call object in a service plan

The service order is created in the **In Process** status simultaneously with the billable maintenance order during scheduling of maintenance plan.

- As a billable maintenance order as part of billable maintenance

When a maintenance order is retrospectively marked as billable, a service order is created with an execution order item. The service order is created in the **In Process** status.

Creating a Service Order

You can create a service order using the **Manage Service Orders** app.

In the service order, make the following entries:

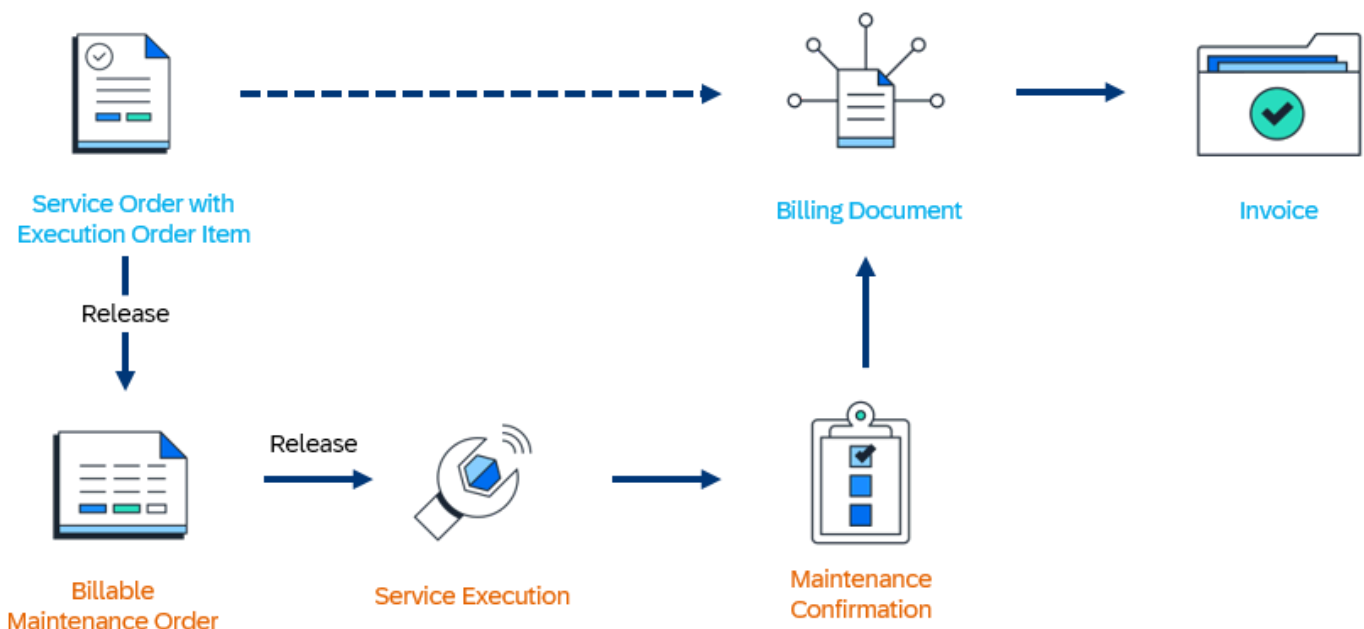
1. Enter general data such description and sold-to party.
2. Enter the sales organizational data.

i Note

Depending on customizing, automatic selection of organizational data is possible.

3. Under **Parties Involved** enter service employee group and employee responsible.
4. Add items and select **Execution Order Item** as the item category. Each execution order item creates a maintenance order. The maintenance order is used for the execution planning and execution of the service.
5. For each execution order item, you need to select a billing relevance. You need to enter a DIP profile if you select summarized billing or itemized billing as the billing relevance.

Whenever the billable maintenance order is created corresponding to an execution order item certain data is copied over from the service order to the maintenance order. This includes information such as shorts texts, planning plant, main work center, profit center, reference objects, and basic start and finish dates. Specific partners can also be entered automatically in the maintenance order based on Customizing settings.



Service Execution Planning

The actual service tasks that are to be performed usually require detailed advance planning, in order to minimize the time that the object is not productive. The more accurately you plan a task, the more efficiently it can be performed by the employees on site. This is extremely important in the case of objects that cause bottlenecks and may lead to a shutdown when they are out of order. The billable maintenance order facilitates this planning and execution.

Planning

Part of the planning involves using operations to describe the individual maintenance tasks to be performed. The operation includes the time, the work center, and other controlling information for an individual maintenance task. Another important aspect is the planning of materials. This can be a spare part or repairable spare or an activity. Materials can be reserved in a warehouse and later withdrawn by the technician or delivered to the customer. For more information, see [Planning of an Order](#).

- Scheduling

Scheduling is used to determine actual execution dates, the capacity needed to execute the order, and the date at which materials should be made available. For more information, see [Scheduling](#).

- Task Lists

Task lists describe a sequence of individual maintenance activities which must be performed repeatedly within a company. They are used as a reference and an input tool when processing orders and considerably reduce the time required during work scheduling.

The task list of the billable maintenance order is determined from configuration **OISD**. From the planning plant entered in the execution order item along with the service product, the system determines a unique task list. This task list is entered automatically in the planning data of the maintenance order. With this, the operations and components are also determined.

For more information, see [Working with Task Lists](#).

Order Release

When all the planning has been completed the billable maintenance order is released. Without releasing the order, the employees on site cannot start with the defined activities. For more information, see [Order Release](#).

Service Execution and Completion

Execution

Once the billable maintenance order is released you can move into the execution stage. Here the actual service or maintenance takes place. This stage involves the withdrawal of material, arrival of technician, actual maintenance process including replacement of parts or equipment and ends with the completion of the order. For more information, see [Execution of an Order](#).

Completion

Completion of the billable maintenance order indicates that the technician has finished the job and various confirmations have been posted. Some examples of confirmations are time completion, material completion, confirmation of measurement and counter readings, and so on.

You usually complete an order technically once the maintenance work planned in the order has been performed. The status is set to **Technically Completed**. For more information, see [Completion of an Order](#) and [Technical Completion of an Order](#).

Billing

When the technician has completed the service the billable maintenance order is set to **Technically Completed** (TECO). The technician enters the materials used and the time taken for the completion of the service. The order can then be sent for billing. Depending on the scenario the order can be billed as ad hoc, fixed price, or resource-related. For more information, see [Billing](#).

Early Creation of Billable Maintenance Order

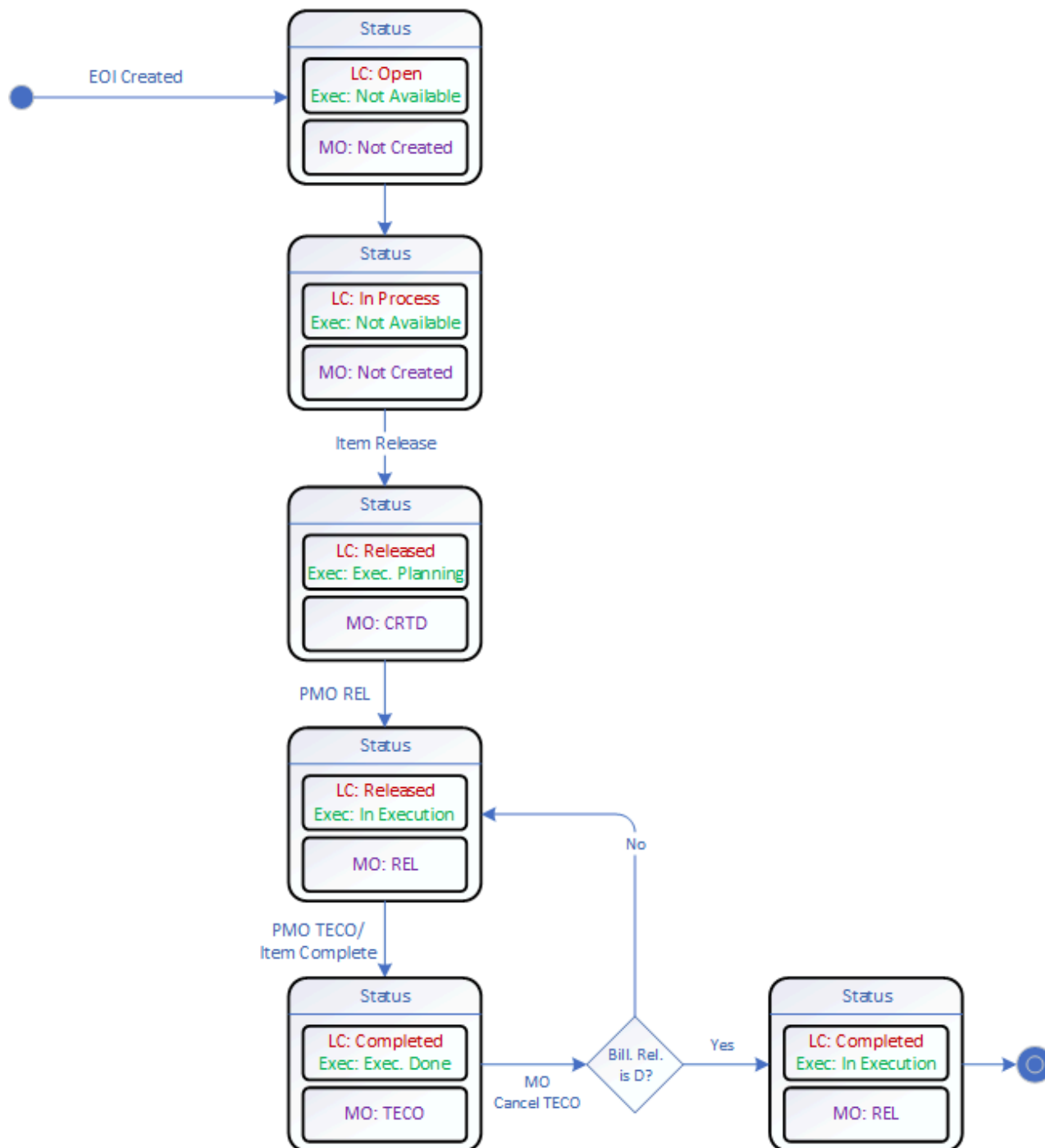
Usually, a billable maintenance order is created when the execution order item is set to **Released**. However, it is possible to create a billable maintenance order already when the execution order item is in the status **In Process**. This is particularly useful when there is no preceding document, in this case no service quotation and therefore no possibility of prior service planning. By allowing the early creation of a billable maintenance order, the maintenance planner can plan for the service execution before the actual start of service. For the early creation of a billable maintenance order the following conditions must be met:

- The execution order item must be in the status **In Process**.
- There is no corresponding billable maintenance order for the execution order item.
- There are no errors in the creation of the execution order item.

If the above conditions are met, then you can use the **Plan All Items** function or select the checkbox **Send to Planning**. You can also trigger **Send To Planning** automatically by selecting the **Send to Planning Immediately** checkbox in the Customizing activity **Map Order Types**.

Overview of Status

The following diagram and table provide an overview of the statuses that service orders can have.



Note:
1. Bill. Rel. D means Billing on Completion (Fixed Price scenario).

EOI	Execution Order Item
MO	Maintenance Order
LC	Lifecycle
Exec	Execution
DNE	Do Not Execute
CRTD	Created
REL	Released
TECO	Technically Complete
CLSD	Closed
NCMP	Not Complete
Bill. Rel.	Billing Relevance

Status	What It Means
Open	This is the default status that all service orders take after they are created. It indicates that you are free to make changes to the order.
In Process	In this status the service order and its items can be edited.
Released	When the service order has been finalized it is set to Released .
Completed	This status is automatically set when the service has been completed and the maintenance order is set to Technically Completed . The Completed status can also be set manually on the service order which will in turn set the status on the billable maintenance order to Technically Completed .

The following table provides an overview of the billable maintenance order status and the corresponding item execution status.

Status on Maintenance Order	Item Execution Status	What Happens in this Status
Created	Execution Planning	All billable maintenance orders are created in this default status.
Released	In Execution	The billable maintenance order has been released.
Technically Completed	Execution Done	This status is set when the service has been performed. The item life cycle status is Completed .
Cancel Technically Completed System status reverts back to Released	In Execution	If Technically Completed is canceled the status of the billable maintenance order reverts to Released along with the item life cycle status. The execution status reverts to In Execution from Execution Done .
Do Not Execute	Canceled	This status indicates that the service has been stopped or canceled.

If the status of a service order is changed from **Completed** to **In Process**, the maintenance order is set to **Released**.

Related Information

[Service Orders](#)

Service with Advanced Execution in Recurring Service

You can use Service with Advanced Execution to create maintenance plans for execution order items.

Use

You can use maintenance plans to generate regular services that have to occur repeatedly at certain intervals. These can be part of regular maintenance or based on usage or performance as agreed by means of a service contract. Regular maintenance helps to keep the customer's assets in optimum working conditions and minimizes the breakdown and repair costs. You can create time-based or performance-based single cycle plans, strategy plans or multiple counter maintenance plans.

Types of Maintenance Plans

- Single cycle plan

This is the simplest form of maintenance plan with or without a task list. It is defined for a one time-based or a performance-based maintenance with the same set of tasks every time. Here you specify the interval at which the maintenance plan should be executed, for example, the annual maintenance of an air conditioner.

- Strategy plan

This is used for complex maintenance cycles where the asset needs to be maintained with a different set of operations at different intervals. A maintenance strategy contains packages and general scheduling information and can be used for many maintenance plans. For example, it makes sense to use a strategy plan if different maintenance tasks for a car are due in different cycles: oil check every 1000 km, oil change every 7000 km.

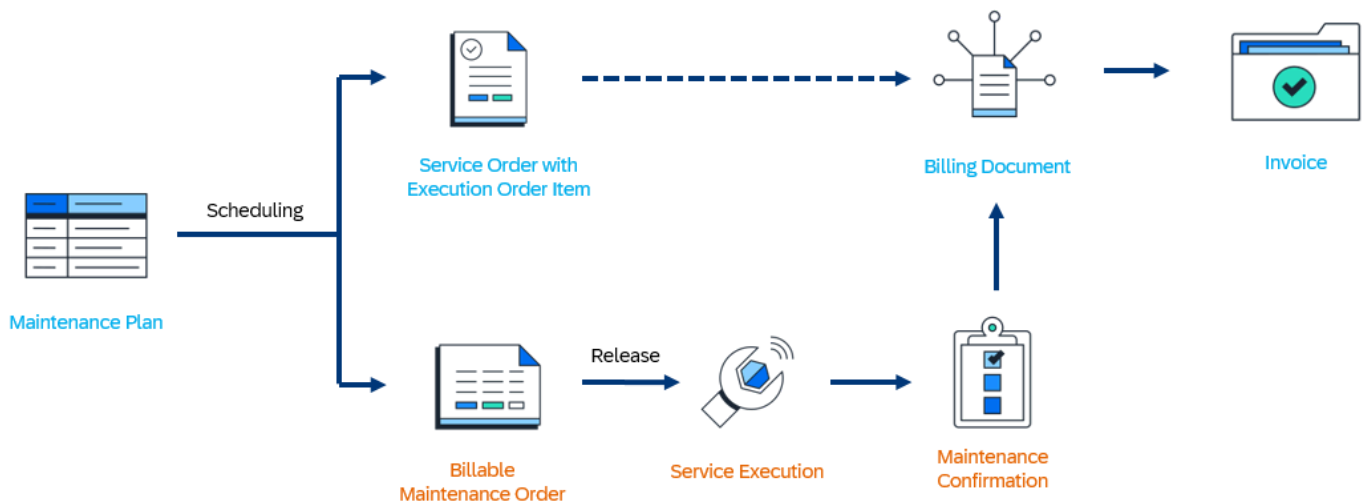
- Multiple counter plan

This is used in time-based maintenance or counter-based maintenance or a combination of both. Here you create maintenance cycles and allocate counters of equipment or functional locations with different dimensions to them. For example, maintenance of aircraft based on either flight hours, number of take-offs and landings, calendar time, or kilometers traveled.

In the Service with Advanced Execution process a service order with a billable maintenance order is created on the scheduling of a maintenance plan.

Call Object

Once you create a maintenance plan, you can schedule the plan to generate maintenance call objects when the system is due for maintenance. The call object is determined based on the maintenance plan category. In Service with Advanced Execution, service order with a corresponding billable maintenance order is used as the call object.



Creating a Maintenance Plan

You can create a maintenance plan using the **Add Single Plan**, **Add Strategy-Controlled Plan**, or **Add Multiple Counter Plan** apps depending on your requirement. Before creating a maintenance plan, ensure that you have prepared the related service contract.

In the app, make the following entries:

1. Select the maintenance plan category.
2. On the **Item** tab, enter a new item by specifying item description and technical object. In the planning data section enter service contact, service contract item, and order type.

3. Provide a name for the maintenance plan and enter the cycle. Enter a task list.

i Note

If you have created a plan without task list, you can use the [Change Maintenance Plan](#) app to assign a task list later.

4. Enter the [Scheduling Period](#) under [Maintenance plan scheduling parameters](#) to specify the number of days, months, or years over which scheduling will take place.

Scheduling a Maintenance Plan

Once you have created a maintenance plan, you need to schedule it. In the [Schedule Maintenance Plan](#) app make the following entries:

1. Enter the plan number you have created.
2. When you choose [Start](#), the scheduling list gets populated with all scheduled calls and the dates for which they have been scheduled.
3. Once you select [Save](#) the service order and its corresponding maintenance order is generated.

Mass Scheduling of Maintenance Plans

To schedule multiple maintenance plans you can use the [Mass Schedule Maintenance Plans - Service](#) app. For more information, see [Mass Schedule Maintenance Plans](#).

Call Object Generation

When the maintenance plan is scheduled, the service order and the connected billable maintenance order are generated as call objects. In the billable maintenance order you can view the task list that was entered when creating the plan. From the maintenance order you can navigate to the service order.

i Note

When the service order is created as the call object, the execution order item has the status [In Process](#).

Maintenance plans for execution order items can be created from:

- Service contract: A service contract is for a single execution order item.
- Service order template: A service order template can have more than one item. It can also be a mix of service items and execution order items.

Related Information

[Recurring Service](#)

BAdI: Reuse Service Transaction in Recurring Service with Advanced Execution

Use

This Business Add-In (BAI) is used in the Recurring Service (CRM-S4-SRV-MNP) component. You can use this BAI in Recurring Service with Advanced Execution (Maintenance plan) to reuse a service transaction. This BAI allows the addition of execution order items into an existing service transaction. You can find this BAI in Customizing under [Service > Transactions > Settings for Recurring Service > Business Add-Ins > BAI: Reuse Service Transaction in Recurring Srv with Advanced Execution >](#) .

The BAI uses the method GET_SERVICE_TRANSACTION_REUSE to determine the service transaction to which service items will be added.

Import Parameters

- IS_MPOS: Maintenance plan item details such as service contract, service transaction type, reference object and so on.
- IS_MHIS: Maintenance plan call history data such as call number, call horizon and so on.

Changing Parameters

- CV_SERVICE_TRANSACTION_GUID: Used to enter service transaction GUID based on the maintenance plan item details.
- CS_OPTIONAL_CHECK - no_service_org_check: Used as an optional check to ignore the validation of the service organization.
- CV_GEN_NEW_ORDER_IF_ERROR: If the validation for the service transaction from the BAI fails and the flag is not set, partial scheduling takes place. If the validation for the service transaction from the BAI fails and the flag is set to 'X' then the standard behavior takes over and new service order is created for the maintenance plan item.

Effect in Mass Scheduling

Mass scheduling is used to schedule many maintenance plans through background batch jobs which can be set to run in certain intervals. For each scheduling run, the system goes through maintenance plans that match the selection condition of scheduling and generates the maintenance call object for each due maintenance item. This is usually done by parallel processing where each parallel process runs multiple maintenance plans simultaneously.

During parallel processing there is a possibility of the service order getting locked. This occurs when multiple maintenance plans access the same service order. To prevent this, the BAI groups together maintenance plans that might share the same service order. Grouping is based on a combination of the sold-to party and the sales area of the service contract item that is linked to the maintenance item.

Service with Advanced Execution in Maintenance Management

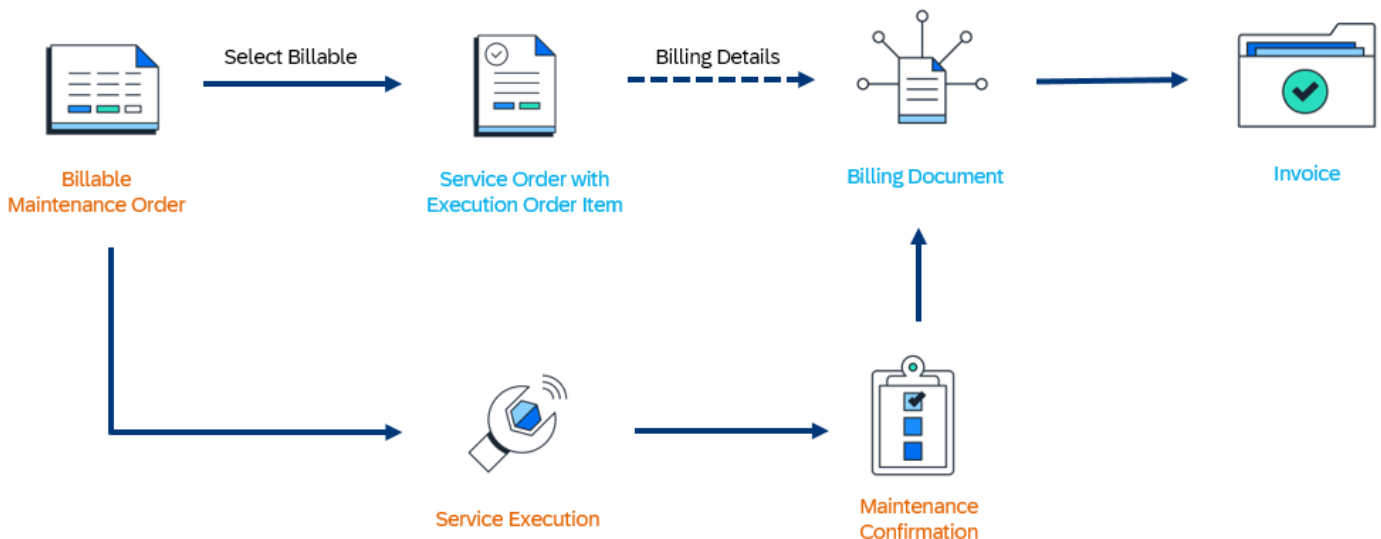
You can use Service with Advanced Execution to create billable maintenance orders to assist in the planning and execution of a service.

Use

Maintenance orders are created by maintenance planners and are used to plan and execute maintenance tasks. These tasks are generally not billed since the maintenance is performed on their own equipment. However, in some cases the customer needs to be billed. It may be due to maintenance or malfunction of the equipment attributed to a customer. At the time of maintenance order creation the planner is not aware that this maintenance needs to be billed. When the planner realizes that maintenance needs to be billed, he selects the **Billable** checkbox, which in turn triggers creation of a service order. Upon completion of the service a billing document can be created.

i Note

Enabling the **Billable** checkbox is an irreversible change.



Prerequisites

- You have made Customizing settings related to Service with Advanced Execution.
- You have maintained the required mapping between transaction type, item category, and order type in the Customizing activity **Map Order Types**.
- You have not released the maintenance order that was created.

Creating a Billable Maintenance Order

Once you have determined that the maintenance order needs to be billed, you can use the transaction **Change Maintenance Order (IW32)** to create a billable maintenance order.

1. Enter the order that needs to be changed to a billable maintenance order.
2. Select the **Billable** checkbox from the order header. The sales data and service details sections will appear when the **Billable** checkbox is selected.
3. Enter the relevant sales details:
 - Sold-to party (Mandatory)
 - Sales organization
 - Distribution channel
 - Division
 - Sales office
 - Sales group

i Note

There should be at least one sold-to party in the **Partner** tab if the sold-to party partner function is available in the partner determination procedure for the maintenance order. The sold-to party must be the same in the header and partner sections.

4. Enter the relevant service details:

- Service product (Mandatory)
- Quantity and unit (Mandatory)

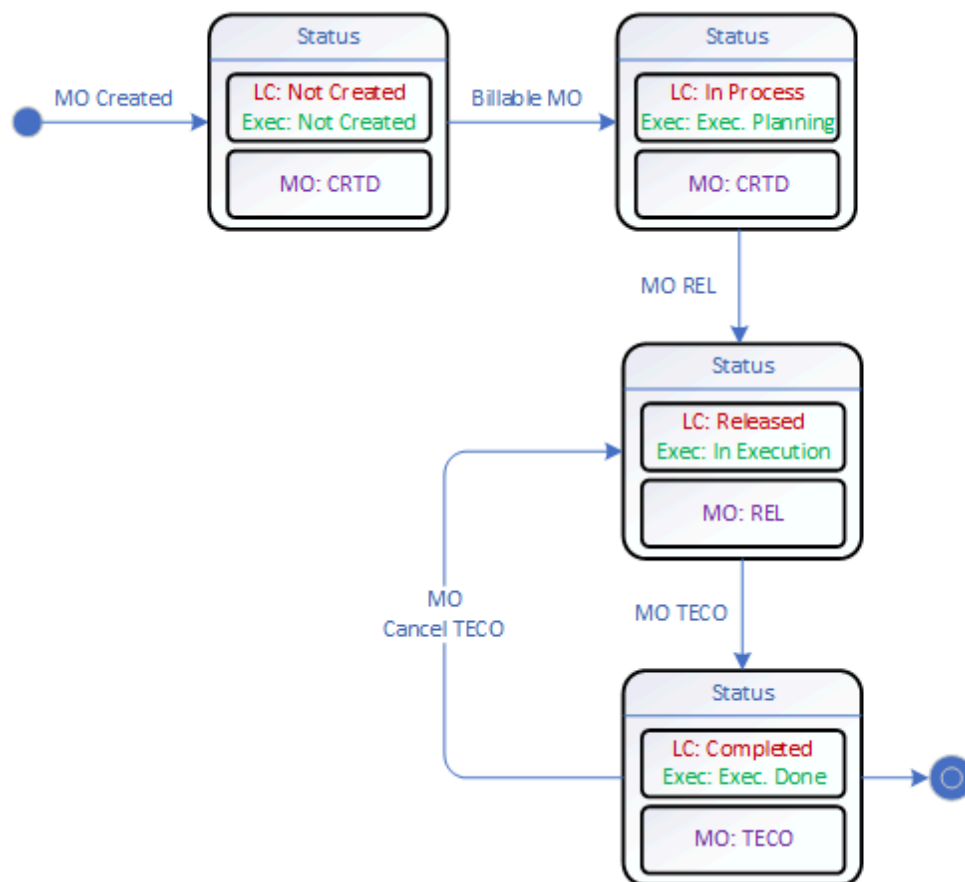
5. Click **Save** to create a billable maintenance order. The system automatically creates a service order with an execution order item

For more information, see [Specifying a Maintenance Order as a Billable Maintenance Order](#).

Synchronization with Service

After the billable maintenance order is created, a corresponding execution order item is created in a service order. There is a 1:1 relationship between maintenance orders and execution order items.

The following diagram and table provide an overview of the maintenance order status and the corresponding item execution status.



MO	Maintenance Order
LC	Lifecycle
Exec	Execution
DNE	Do Not Execute
CRTD	Created
REL	Released
TECO	Technically Complete
CLSD	Closed
NCMP	Not Complete

Status on Maintenance Order	Item Execution Status	What Happens in this Status
Created	Execution Planning	All maintenance orders are created in this default status.
Released	In Execution	The maintenance order has been released and is ready for execution.
Technically Completed	Execution Done	This status is set when the service has been performed. The item life cycle status is Completed .
Cancel Technically Completed The status reverts back to Released	In Execution	If the status Technically Completed is canceled the status of the maintenance order reverts to Released along with the item life cycle status. The execution status reverts to In Execution from Execution Done .
Do Not Execute	Canceled	This status indicates that the service has been stopped or canceled.

Related Information

[Processing a Billable Maintenance Order Created in Maintenance Management](#)

[Planning and Executing a Billable Maintenance Order](#)

Advance Shipment Using Billable Maintenance Order

You can use the billable maintenance order with this logistics process to ship service parts to a technician or customer.

Use

This logistics process enables advance shipment of service parts to the technician or customer as part of service processing.

The shipment of service parts is processed using the billable maintenance order integrated with a sales order. The billable maintenance order is also integrated with the execution order item of the service order.

The maintenance order enables a service-specific results view from a controlling perspective and improves coordination between delivery dates and deployment dates for the service technician. The sales order allows customer consignment stock to be created. As a result, parts that are actually required can be invoiced after their use has been posted to the maintenance order.

The service parts to be shipped are issued from the warehouse stock. The technician can return the service parts that were not required to the warehouse.

i Note

Billable maintenance order refers to the order that supports planning and execution of execution order item represented as a product in the service order.

Process Flow

Creating and Releasing a Billable Maintenance Order for Advance Shipment

When you create a service order for an execution order item a corresponding maintenance order is created. You can plan the service parts as components in the maintenance order to be delivered to the customer and then you can deploy a service technician.

From the maintenance order, you can create a sales order (consignment fill-up type order). You have the option of copying the service parts already planned in the maintenance order to the sales order. You can transfer service parts from the bill of material of the reference object in the maintenance order in case no components are planned in the maintenance order level.

Performing a Delivery for a Sales Order Item

After releasing the maintenance order, you create the delivery note for the sales order and post the goods issue for the delivery.

The sales order type can be configured in such a way that the delivery is created automatically for the sales order item (immediate order) when the maintenance order is saved.

Performing Completion Confirmations and Returning Service parts

After the service has been performed, the technician performs completion confirmation for the maintenance order by entering the service parts used and the working time in the maintenance order. In the completion confirmation, the technician can also change the accounting indicator.

The service technician sends back the service parts that were not used. Since the return delivery does not affect billing, it can take place before or after billing. For the return delivery of these materials, they must be included as items in the sales order.

Posting Goods Withdrawal from Consignment Stock

At the same time as completion confirmation, the goods withdrawal from consignment stock for the material that were used must be posted manually (movement type 261, special stock W). The consignment stock is then reduced according to the parts used.

Performing Resource-Related Billing

Using resource-related billing the confirmed service parts and working times are invoiced to the customer.

Creating a Sales Order for Advance Shipment

Prerequisites

When you create the sales order from the billable maintenance order, you can enter this in the user master record under the set/get parameter VORAB_AUART.

To be able to send the service parts to a service technician (instead of to a customer) when creating a consignment stock, a customer master record must be created for the service technician.

Procedure

1. In order to plan the service parts that you want to send in advance to the customer, choose **Components**.
2. The component overview is displayed.
3. Enter the service parts, the quantities, and the item category (for example, stock material). Material master records with sales data must be created and have prices maintained for the service parts.
4. If you have entered a reference object in the order header, you can select the service parts to be sent by clicking on the icon **Structure list** and copying them into the component overview.

5. Select the service parts that you want to ship in advance and click on the icon [Advance shipment](#).
6. The dialog box for entering the advance shipment parameters is displayed.
7. You can overwrite the sales order type. Enter a shipping condition and an accounting indicator where necessary and choose [Continue](#).
8. In the next dialog box that is displayed, you can:
 - Perform an availability check
 - Perform a completeness check
 - Make changes to the items proposed from the billable maintenance order (these changes are not copied to the maintenance order)
9. Choose [Back](#) to return to the billable maintenance order and save it.

Result

The system saves the sales order together with the billable maintenance order. You can use the document flow function to see the assignment of maintenance order to the sales order.

Both the maintenance order and the individual service parts are assigned the system status [Advance shipment occurred](#). This means that you can use this status to search for billable maintenance orders and components.

The system also sets automatically the indicator "Non-MRP-relevant" for the service parts. This is because the materials are copied into the sales order and are planned there.

Advance Shipment Status

You can use the [Advance shipment status](#) function in the component overview or detail screen in the billable maintenance order to obtain the current status overview of the sales order. You can see all system and user statuses of the sales order at a glance. This enables you to display the required status information from the billable maintenance order without having to call up the sales order or the delivery note. Furthermore, the current delivery status of the individual components is also recorded here. This means you can use this function to track the delivery dates of the service parts with the deployment date of the service technician.

Distributed Execution

You can use Distributed Execution to create multiple billable maintenance orders for an execution order item.

Use

Distributed execution allows you to break down the execution of a service requirement into multiple smaller parts. By creating multiple billable maintenance orders for the same execution order item, you can plan and execute each part of the service separately.

Consider a piece of equipment that requires a service due to maintenance or breakdown. If the equipment can be broken down into separate components that can be serviced independently by different teams and at different times, you can create a billable maintenance order for each component. You can also create multiple billable maintenance orders for a scenario where you want to separate each aspect of the service using a separate order. For example, a maintenance order for inspection, another one for service execution, a separate order for clean-up and testing and so on.

The breakdown of the planning and execution into multiple maintenance orders are, in general, carried out by the maintenance planner.

Prerequisite

You have selected the checkbox **Allow Multiple Maintenance Orders** for item **BUS2000116** (execution order item). You do this in Customizing under ► **Service** ► **Transactions** ► **Basic Settings** ► **Define Item Categories** ►.

i Note

Distributed execution is only possible with a service order. Other Service transactions are currently not supported.

Process Flow

When you have enabled the creation of multiple maintenance orders, the next step is to create a service order with a corresponding billable maintenance order. Alternatively, you can also start by creating a maintenance order and selecting the **Billable** checkbox to create a corresponding execution order item in a service order.

Once you have created the initial link between Service and Maintenance Management, you can use distributed execution in one of the following three ways:

1. Copy

You can use this feature in the **Create Maintenance Order** (IW31) app to create a billable maintenance order by using an existing billable maintenance order with similar data as reference. You can choose the data you wish to copy over by selecting the appropriate checkbox. This feature is useful for maintenance tasks that are repetitive.

2. Create follow-on

You can use this feature in the **Create Maintenance Order** (IW31) app to create a billable maintenance order as a follow-on of an existing billable maintenance order. You need to select the checkbox **Create Follow-On Order** and enter the reference order number. This can be used in scenarios where the defects reappear or unexpected issues develop after execution of the initial maintenance order.

3. Create suborder

You can use this feature in the **Create PM Sub-Order** (IW36) app when a particular maintenance requirement needs to be broken down into smaller parts. A hierarchical relationship exists between the initial billable maintenance order (superior order) and the suborders. A Superior maintenance order is mandatory to create suborders and the superior order can be set to technically completed only after all its suborders are set to technically completed. Also when resource-related billing process is started for the superior order, the system automatically aggregates costs from its suborders.

You need to select the **Service** checkbox in the **Create Maintenance Order** (IW31) app or the **Create PM Sub-Order** (IW36) app to create multiple billable maintenance orders. This will ensure that the new maintenance order is a billable maintenance order. Additionally, the service order and execution order item reference are carried into the new billable maintenance order.

The new billable maintenance orders and suborders can be viewed in the **Maintenance Orders** assignment block in the execution order item overview page of the service order.

Restricting Synchronization

If the checkbox **Allow Multiple Maintenance Orders** is not selected, only one billable maintenance order can be created corresponding to the execution order item. In addition, changes made in the service order are copied over to the billable maintenance order. Similarly, changes made in the billable maintenance order are copied over to the service order. This is applicable for fields like planning plant, work center, reference objects, and so on.

However, if the checkbox has been selected to enable multiple maintenance orders, then the synchronization of fields as described above does not take place. The only exception to this is the accounting indicator and profit center fields.

Selecting the checkbox allows more flexibility during the planning and execution of the service. Sometimes with multiple billable maintenance orders the planner will want to change attributes and differentiate the maintenance orders as required. For example, if there are three maintenance orders, and these need to be executed at three different work centers, in this case the maintenance planner can change the work center in the various maintenance orders.

Related Information

[Distributed Execution](#)

Billing

Get an overview of the billing process in Service with Advanced Execution.

Billing represents the final processing stage for a service transaction in Service with Advanced Execution.

Types of Billing

- Not Relevant for Billing

This type of billing is used when the customer is not being billed for the service. Goodwill, under warranty, free-of-charge service may be some of the business decisions for the use of this billing type.

- Ad Hoc Billing

This type of billing is used to define billing values and dates freely for fixed price execution order items. Billing document requests (BDRs) are generated for fixed price items with long duration that need to be billed on specific dates and not just when the execution order item is completed.

- Fixed Price Billing

In fixed price billing, the service is billed at a predetermined price irrespective of the actual cost incurred during the service. It is based on the price master data of the service product. The price master data (along with discounts) is maintained in Sales.

- Resource-Related Billing

Here customers are billed for the resources used (for example, labor, material, travel) to carry out services. Resource-related billing can be either itemized billing or summarized billing.

Related Information

[Billing](#)

Ad Hoc Billing

You can use ad hoc billing for execution order items in Service with Advanced Execution.

Use

Ad hoc billing is a structured method of billing that doesn't conform to periodic or milestone billing plan rules, but instead enables you to define billing values and dates freely.

Ad hoc Billing Plan

You can use an ad hoc billing plan to define the dates on which billing is to occur and the value that is to be billed. For each billing date, the system generates one billing request line. In addition, you can optionally specify the settlement period that an individual billing request line covers.

Billing of Remaining Net Value

This feature enables the system to automatically generate a billing request line with all the remaining net value of the execution order item. The billing date of this billing request line is set to the current date by default.

Prerequisites

1. You have made the necessary settings for ad hoc billing plans in Customizing under **Service > Transactions > Basic Settings > Billing Plan > Define Billing Plan Types**. In the **BilPlanUse** field, choose **Ad hoc Billing**.
2. The billing plan **Ad hoc Billing** is assigned to the relevant item category. You do this in Customizing under **Service > Transactions > Basic Settings > Billing Plan > Assign Billing Plan Type to Item Category**. The predefined Customizing settings include the transaction type SRVO and the item category MS11.
3. You have maintained the billing relevance for the execution order item as **Billing via Billing Plan After Release**.

Process Flow

Service Order

1. Using the **Create Service Order** app create an execution order item with the billing relevance **Billing via Billing Plan After Release**.
2. On setting the above billing relevance, the **Billing Plan** assignment block automatically appears.
3. Within the **Billing Plan** assignment block you can choose the billing date and the billing value. You can then view the remaining value to be billed.
4. Save the service order.

Related Information

[Ad Hoc Billing for Service Orders](#)

Fixed Price Billing

You can use fixed price billing for execution order items in Service with Advanced Execution.

Use

In fixed price billing, the completed service is billed based on the price master data of the service product. The price master data (along with discounts) is maintained in Sales. This price is proposed for the service order.

The billing relevance used for fixed price billing is always **Billing on Completion**. When the execution order item status is **Completed**, a billing document request (BDR) can be created through the **Release for Billing** app.

With the **Release for Billing** app, you can release maintenance service order items for billing. After you have released the items, the you can create the billing document requests (BDR).

i Note

If an item belongs to a service bundle, you must release all items of the bundle simultaneously.

Process Flow

1. Open the [Release for Billing](#) app.
2. Enter the service order ID and search.
3. From the results, select the service order ID and then choose **Release for Billing**.

i Note

Once you choose [Release for Billing](#), the service order disappears from the search results.

4. Open the [Create Billing Documents](#) app and search using the service order ID.
5. Select the order from the search result and choose **Create Billing Documents**.
6. The invoice is generated.

Related Information

[Billing of Service Transactions](#)

Resource-Related Billing

You can use resource-related billing for execution order items in Service with Advanced Execution.

Use

The price for customer-specific services is not always defined in a service contract as a fixed price nor can it always be determined using standard pricing.

Typical examples of this are:

- External plant maintenance
- Specific services, such as consulting

In these cases customers are billed for the resources used (for example, personnel, material, travel) to carry out services. In resource-related billing, the system creates dynamic items from the expenditure information on the maintenance order.

Resource-related billing can be either [itemized billing](#) or [summarized billing](#). They are identified with the billing relevance field attribute at the item level.

- Itemized billing

A service is billed as a list of subitems. The net price is the aggregated price of these subitems determined by either pricing conditions or by their costs. Itemized billing can have multiple debit memo requests depending on number of visits, change in tasks, requirement of additional parts etc. Itemized billing also supports event-based revenue recognition.

- Summarized billing

A service is billed as a single main item. The net price is determined from existing pricing conditions of the main item or as the aggregated cost of individual subitems. There is only one debit memo request created at the end of the service. Summarized billing does not currently support event-based revenue recognition.

Resource-related billing is carried out in the [Create Resource-Related Billing Request - Service Order](#) app.

i Note

You can use the transaction code [DP92](#) to launch the app from SAP GUI.

DIP Profile

The dynamic item processor (DIP) profile is mandatory in the case of resource-related billing. During billing, the dynamic item processor determines the items or lines in the debit memo request, based on the DIP Profile entered in the execution order item in the service order.

The value of DIP Profile can be defaulted from the predecessor document like service contract or service quotation or from the item category customizing. It is also possible to manually enter or overwrite the default values. For more information, see [Dynamic Items](#).

i Note

The DIP Profile entered in the service order must have Usage **1 Billing**.

Prerequisites

In Customizing

You have maintained a dynamic item processor (DIP) profile.

In the Service Order

- You have created one or more execution order items in the service order that are supported by a corresponding maintenance order.
- The billing relevance is selected either as Itemized billing or Summarized billing.
- You have maintained the following data in the service order
 - Customer
 - Sales organization
 - Distribution channel
 - Division
- You have entered a DIP profile for the execution order item.
- You have released the maintenance order and booked actual costs.

Process Flow

1. Open the [Create Resource-Related Billing Request - Service Order](#) app.
2. Enter the service order ID and the execution order item. Alternatively, you can use the help to search for a service order or item. Then select **Create**.

- 3. The system determines the items in the expenditure view using the DIP profile and actual cost from the maintenance order. The prices are determined using Sales and Distribution Pricing. The required pricing procedure is determined using the document category specified in the DIP profile.
- 4. You can set the quantity or the amount to be billed. You can also reject or postpone by quantity or amount.
- 5. Select the **Billing Request** button to generate a debit memo request (DMR).

Related Information

[Event-Based Revenue Recognition for Service with Advanced Execution](#)
[Resource-Related Billing](#)

Apps Used in Service with Advanced Execution

You can use the following apps in Service with Advanced Execution.

Apps Used in Service Quotations and Service Orders

App Name
Manage Service Quotations
Manage Service Orders

Apps Used in Recurring Service

App Name
Create Maintenance Plan (IP01)
Change Maintenance Plan (IP02)
Display Maintenance Plan (IP03)
Create Maintenance Item (IP04)
Change Maintenance Item (IP05)
Display Maintenance Item (IP06)
Schedule Maintenance Plan (IP10)
Add Single Plan (IP41)
Add Strategy-Controlled Plan (IP42)
Add Multiple Counter Plan (IP43)
Mass Schedule Maintenance Plans - Service

Apps Used in Maintenance Management

App Name
Change Maintenance Order (IW32)

App Name
Display Maintenance Order (IW33)
Change PM Orders (IW38)
Enter PM Order Confirmation (IW41)

Apps Used for Billing

App Name
Release For Billing
Create Resource-Related Billing Request: Service Order (DP92)
Mass Creation of Resource-Related Billing Requests: Service Order (DP92A)

Create Resource-Related Billing Request

App ID: DP92

With this app, you can create debit memo requests for the time and service parts spent on the service.

Key Features

This app provides the following key features:

- Generate a debit memo request for an execution order item. Other service order items are not currently supported. Can also generate credit memo request.
- Items are determined using DIP profile and actual costs are determined from billable maintenance order. Prices are determined using Sales and Distribution Pricing.
- Supports only billing relevance type summarized billing and itemized billing.
- Supports the billing of multiple billable maintenance orders corresponding to a single execution order item.

Supported Device Types

- Desktop
- Tablet

Mass Creation of Resource-Related Billing Requests

App ID: DP92A

With this app, you can process multiple service orders and execution order items to create one or more billing requests.

Key Features

You can use this app to do the following:

- Create credit memo or debit memo requests in bulk that are due over a period for a set of service orders using a combination of sales organization, distribution channel, division or sold-to party.
- Run individual billing request lines of a service order and decide whether to bill now, bill later, or reject the billing request line.
- Save billing lines in an extract that are periodically processed to create debit memo or credit memo requests.
- Process billing requests in the background while working on other tasks in the foreground.
- Process multiple billing requests at the same time to reduce the overall processing time.

Supported Device Types

- Desktop
- Tablet
- Smartphone

Case Management

Case Management enables you to consolidate, manage, and process information about a complex problem or issue in a central collection point, the case. Within a case, you can group diverse information, such as business partners, transactions, products, and documents. This information can reside in different physical systems.

Use

You can also use Case Management to process problems and issues that involve multiple processing steps or multiple processors. Case Management therefore supports the processing and communication flow between organizational units and helps you to increase processing efficiency.

Case Management contains various [functions and features](#) that are applicable to different scenarios.

SAP provides the framework for processing issues in different industries and scenarios for Case Management:

- Incidents relating to complex or expensive products for which you provide technical services
- Issues in the utilities industry, such as power outages
- Insurance claims cases
- Services in government agencies

SAP provides predefined case types to support the following scenarios:

- General incident tracking
- Disputes regarding customer invoices and payments in the telecommunications industry

Prerequisites

You have made all the necessary settings in Customizing under  **Service**  **Case Management** .

SAP provides standard Customizing settings that enable you to run Case Management with few or no modifications. Depending on your business processes and system landscape, however, you may need to make further settings.

The section **Enhanced Customizing** contains more specific settings if you have special requirements. For example, define a case closing profile, or set up case archiving.

You only need to maintain the **Special Settings** section under **Enhanced Customizing** in exceptional cases. To do this, you require assistance from an experienced consultant.

If you want to use reference categories, you must have already set up [Multilevel Categorization](#). You do this in the Customizing for Service under **►Cross-Application Components ► Multilevel Categorization ►** and you need to perform additional tasks in the WebClient UI for the IC Manager role under **Process Modeling**.

For details on how to set up Case Management, see the documentation of the individual Customizing activities under **►Service ► Case Management ►**.

Features

- Linking existing business objects to a case: You can link business object types to a case, for example:
 - Business partner
 - Products, objects, installed base components
 - Sales and service business transactions
 - Problems and solutions from the knowledge base
- Assigning and displaying the involved parties: When you assign or link involved parties with a case, you can assign partner functions and specify the main business partner. In Customizing you can define partner determination procedures using a Business Add-In (BAI) for each case type and assign partner functions only using the existing BAI. Furthermore, you can define a minimum and maximum number of partners that must be entered for the various partner functions for each case type.
- Linking electronic documents, such as forms and policies, to a case: Content Management functionality enables you to structure documents, and link archived documents. For more information, see [Content Management](#) and [Document Management](#).
- Linking activities to a case: From Case Management you can directly access activity management. You can create multiple activities to plan and track the required processing steps for a case. You can assign an activity to a person responsible who is not necessarily the case processor. Processors can generate a list of activities for which they are assigned as person responsible. You can also link existing activities to a case. For more information, see [Activity Management](#).
- Creating and linking actions: You can create and link actions from within a case. For example, you can define actions for sending e-mail messages, creating and linking service orders. Actions allow you to process cases more efficiently.
- Entering case notes: You can enter notes if you want to record case-specific information. These notes are always assigned directly to the case and facilitate communication between agents. Case notes are stamped with the name of the user who created the note and the date and time of creation. You can categorize notes by assigning them a note type. You can also make a note of the reason for linking objects to a case next to the objects themselves.
- Change history: Changes to a case are automatically recorded in a log, with a user, date and time stamp. This change history helps you to track changes to case attributes, linking of business objects to a case, and unlinking of business objects in a case.
- Status management: You can assign customer-defined statuses to cases. You define the system status in Customizing where you specify which status changes are permitted.

- Using case categories: You can categorize cases and analyze the data. You define categorization in using [multilevel categorization](#).
- Using reference categories: You can assign a reference category for multilevel categorization at case header level, which allows you to control case processing. You can use reference categories to include various business objects and functions in your case processing. You can use activity templates that are linked to categories to create additional activities and to link them to the case. You can use document templates that are assigned to categories to create documents within a case. By assigning extended attributes, you can define a time version of the reference category.
- Case hierarchies: You can manage associated cases in a hierarchy. To do this, you need to define subordinate cases.
- Creating new cases: You can create a new case from an existing one and use it as a template, or you can choose to create a new case type from a list of case types. When you copy a case, a case of the same case type is created.

Case

Definition

A structured collection of information related to an issue, incident, or problem that requires processing. The processing flow of a case is defined by activities and processors that are assigned to the case.

Use

Cases allow you to group related objects in a central collection point that serves as a single point of access for processing an issue. You can create a case when you establish that an issue extends beyond a single business partner or transaction, or involves many processing steps.

Structure

A case can contain the following information:

- Case attributes

These include case type, case category, priority, status, processor, employee responsible, cause, escalation reason, creation date, and change date.
- Case notes
- Parties involved
- Case category
- Business objects:
 - Transactions
 - Products, objects, installed base components
 - Sales and Service business transactions
 - Attachments
 - **SAP Contract Accounting** documents

You can enable users to link additional business objects by extending the record model for the case. Record models are part of **SAP Records Management**, on which generic SAP Case Management and Case Management in Service are based.

- Electronic documents
- Activities
- Log of changes made to the case
- You can show the relationships between cases by defining a hierarchy of cases. In addition to the existing hierarchical case structures (subcases, higher-level cases), you can [link cases](#) which are related, but do not form a hierarchy.

Example

You receive recurring complaints about a product defect and decide to create a case to monitor these. You then link the affected case business partners and product involved to the case, and also the complaints documents already created to record complaints about the product. You also create an initial activity **Perform technical analysis**, write a note describing the reason for opening a case, and assign the case to a processor.

Case Authorizations

You can use various authorizations in Case Management to control user access to case information. You set up authorizations by defining values for authorization objects in authorization profiles and then assigning them to users or user groups via roles.

You maintain authorization objects for user or roles in the SAP menu ([Architecture and Technology](#) > [System Administration](#) > [User Maintenance](#) >).

For more information about authorizations, see [SAP Authorization Concept](#).

The following authorization objects are specific to Case Management. In addition to these, you also require the SAP Records Management authorization objects. These are also included in the predefined roles. For a description of the SAP Records Management authorization objects, see [The Authorization Concept for Working with Records Management](#) (and subsequent sections) and [The Authorization Concept for Customizing](#).

S_SCMG_CAS (Case Management: Case)

This object defines authorizations for entire cases. It contains the following authorization fields for which you can define values to control authorization:

- CASETYPE: Case type
- SCMG_ACT: Case processing activity, for example, create, change, or delete
- SCMG_KEY: Key for authorization check

This field is a user exit that is relevant for authorization checks using the Business Add-In (BAI) SCMG_AUTHORITY_S (method DETERMINDE_KEY). In this field, you can enter a current value for a field of an application. The BAI generates a new key each time the authorization check is run.

This field is only checked if this BAI is implemented.

- SCMG_LVL: Authorization level

This field controls user access and processing activities dependent on the authorization level assigned to a case.

The values that you can enter for this field are specified in the Case Management Customizing activity [Basic Settings](#) > [Determine Permitted Values for Case Attributes](#) > [Create Values for Authorization Levels](#) > . At runtime, the system checks whether the user has authorization for the authorization level currently entered in the case. If the user does not have the correct authorization level, they cannot execute the activity.

- SCMG_ROLE: Role of the user in relation to the case (see above)

This field controls user access and processing activities dependent on the user's current role as defined in the case header data.

You can enter the value P (Processor) or R (Responsible) for this field. Note that the value C is only relevant for [SAP Dispute Management \(FIN-FSCM-DM\)](#).

Case processors have the role [Processor](#) or [Responsible](#) if they are currently named in the case header as in the processor or person responsible respectively. At runtime, the system checks which, if either, of these roles the case processor currently has. If the user does not have authorization for a role or is neither a processor or person responsible, they cannot execute the activity.

- SPS_ID: Element type within a case

S_SCMG_FLN (Case Management: Field-Level Authorization)

This object defines authorizations for the attribute fields in a case header. It contains the following authorization fields for which you can define values to control authorization:

- ACTVT: Activity ([Change](#), [Display](#))
- CASETYPE: Case type
- SCMG_FIELD: Attribute field

This field controls whether a user can display or change a case attribute.

- SCMG_ROLE: Role of the user in relation to the case (see above)
- SPS_ID: Element type within a case

S_SCMG_STA (Case Management: Status)

This object defines authorizations to change the case status. It contains the following authorization fields for which you can define values to control authorization:

- ACTVT: Activity ([Change](#))
- CASETYPE: Case type
- ORDER_NO: Status

The status that you enter here is the user-defined status, as defined in the Case Management Customizing activity [Extended Customizing](#) [Create Status Profile](#). [If you select values for this field using F4 Help, you can also see the details of the relevant status profile.](#)

- SPS_ID: Element type within a case

S_SCMG_TXT (Case Management: Notes)

This object defines authorizations for case notes. It contains the following authorization fields for which you can define values to control authorization:

- ACTVT: Activity ([Add or Create](#), [Change](#), [Display](#))
- CASETYPE: Case type
- SPS_ID: Element type for case notes

- TEXTID: Text ID

The values that you can enter for this field are specified in the Case Management Customizing activity [Extended Customizing](#) [Create Text Profile](#) [Create Text IDs](#). At runtime, the system checks whether the user has authorization for the relevant type of text to determine whether the user can perform the desired activity for the text.

Document Management

Use

Case Management supports a variety of functions to make documents in electronic formats available in cases. When you create a case, you can upload attachments from your hard drive or you can link to documents on your hard drive. All documents that are linked to cases are physically saved in Content Management. You can always access and manage the documents by using Content Management or SAP Product Lifecycle Management.

Features

Records

You can use the case record structure to link record documents to the model nodes provided in the record. The assigned record model predefines the respective object (record documents or other business objects such as business partners) for each model node. You can edit the list of records linked to the case and the type of record.

Each record document occupies its own instance node in the record hierarchy. Changes to a document are saved as a new version of the document in Content Management.

Documents

In Case Management, you can process case documents using an unstructured collection of case documents, or you can also create folders or structures of folders to flexibly structure existing documents. You can even work with document templates from Content Management. You can upload these from the Content Management system or set a link to them.

Activities

You can assign activities with case notes to a case. A notes document is attached the case but does not appear in the record structure. It is a separate element.

Attachments

In Case Management, you can upload documents from SAP PLM, or from the Content Management system. If you want, you can attach the document as a link or append the actual document to the case. If you upload a document from Content Management and you edit the document, it is saved as a new version. You can also upload a document from a local hard drive and attach it to the case. If there is an existing attachment linked to the case with the same name, you can overwrite it with the most current one from your hard drive. However, in this case no version management occurs.

You can upload a URL as an attachment. It is then a link to any valid external or internal Web site.

Extensibility for Case Management

There are two options for extensibility in Case Management:

- Field extensibility

- Table extensibility

Features

Field Extensibility

Key users use the [Custom Fields and Logic](#) app to extend fields.

Business Context

There are two business contexts for fields:

- Case attribute extension
- Case note attribute extension

All of the standard data types available in the [Custom Fields and Logic](#) app can be used for field extension in Case Management.

Table Extensibility

You can use the Application Extensibility Tool (AET) to extend tables. For more information on the AET, see [Application Enhancement Tool](#).

Related Information

[Key User Extensibility](#)

[Custom Fields App and Custom Logic App](#)

Enhanced Case Management in Service - Features

There is an expanded set of functions available in Case Management. The functions cover:

- [Automatically created case notes](#). A case note can now be created automatically to inform a case processor about events that occurred and affect the case. You can define additional attributes for your case notes to store additional information.
- [Reassignment of business partners in linked objects](#). If you transfer an ongoing case to another case processor or department, all outstanding case business partners and their associated activities are automatically transferred with the case. You can use the delivered Business Add-In (BAI) to influence this behavior.
- [Configuring recurring tasks](#). Enhanced Case Management provides functions that enable you automate recurring tasks and activities that are part of a case. You can also define the parameters required to repeat these tasks and activities. For example, there may be a report that must be run weekly over the life of an open case to monitor its progress.
- [Sending e-mail directly from a case](#). You can now send an e-mail directly from within a case that includes one or multiple attachments.
- [Linking related cases](#). Cases are normally structured in a hierarchy of higher-level cases and sub-cases. You can now link related cases that are not part of the same hierarchy to obtain a different overview.
- [Defining case closing profiles](#). You can create profiles that define how cases with open dependent transactions are to be closed. These profiles allow you to handle different transaction types and to define system behavior dependent on the transaction type.
- [Case archiving](#). You can define checks before archiving a case, archive (write) cases to the database, and delete cases from the database once they have been archived. You can also call up archived cases if you want to review an archived cases for

Configuring Recurring Tasks

Use

You can create recurring tasks in Activity Management. Recurring tasks are actions that a user executes in a defined period of time on a regular basis. You can use a recurring task, for example, if a user needs to make a weekly report for an activity that is meant to run over a long period of time.

A default recurring task is part of the standard delivery, however, you can create your own recurring tasks. To see how the default is configured and use it as a reference for your own configuration, see the explanation of the process in Customizing under ► [Service](#) ► [Transactions](#) ► [Settings for Activities](#) ► [Configure Recurring Tasks](#) ►.

Prerequisites

A recurring task must have the following date fields:

- Start Task Date
- Recurrence, determining the frequency at which a task is to be created
- Recurrence Rule, calculating when the next task is to be created
- Next Recurrence
- Recurrence End
- Completion Date

Activities

There are various activities you need to perform to create a repetitive task:

1. Create a date profile.
2. Create an action profile to determine the action the system performs when the task is executed.
3. Define the conditions under which the action is executed.
4. Set a callback using the function module CRM_CMG_SET_COMPLETION_DATE.
5. Define a transaction type.

Displaying and Editing Time-Dependent Data

Use

You can display and edit time-dependent information related to the primary person of a case without having to navigate away from the main information to the business partner.

Time-dependent information refers to any historic case information that gives an overview of changes over time. The primary person is the person who is directly related to this case, such as the claim applicant.

This information is important to determine eligibility and the amount of a certain benefit. By showing all case-related information, which was valid at a point in the past, the case processor can understand what decisions were made based on different assumptions.

Features

A case processor can display and edit the nationality, income, name, and date of birth of the primary person.

A case processor can also identify if:

- overlapping records exist, for example, whether a specific point in time two or more records for the same data set existed
- gaps have occurred, for example, if data changes to the applicant's status have occurred that require a recalculation of benefits

Changes to data in the past can be restricted as such changes could invoke complex re-calculation of benefits.

Example

In some instances, information changes regarding the primary person associated with the case. For example, a person receives a state benefit, but there has been a change to the marital status, income earned, place of residence, or any other factor that affects the eligibility of a claimant. If changes occur that must be documented, you can make them directly in the application.

Sending E-Mail from a Case

Use

You can send e-mail when working in a case and attach relevant documents from the case. When the e-mail is sent, an automatic link to an activity can be set for the case. For example, if an e-mail is sent to a customer stating that action will be taken regarding a problem in order to correct it, a case processor could assign an activity such as a service ticket or a follow-up call.

This function appears in the [Activities](#) assignment block and the [Record](#) assignment block.

Features

In the [Case Details](#), when you click [Create E-Mail](#), a new page opens allowing you to send an e-mail. You can:

- Create and send a new e-mail automatically
- Specify references regarding the e-mail such as the account, contact, employee responsible for handling this e-mail and an additional reference, for example, if it is a lead, an activity or an opportunity
- Specify the priority (high, low, or normal) and the sensitivity (normal or private)
- Upload attachments related to the case from your PC or network
- Use plain text or HTML formats and format the appearance of the e-mail (HTML only)
- Create and use an e-mail signature for plain text or HTML formats

Reassigning Business Partners in Linked Objects

When a business partner at the case level is changed, business partners in linked objects of a case can be automatically reassigned. For example, if a new case processor or a different department takes over the responsibility for an existing case, the former case processor or department are also replaced by the new processor in all dependent, open activities.

i Note

You must use this Business Add-In (BAdI) implementation to be able to reassign business partners.

This behavior is applicable to business partners for all partner functions and all linked objects, which have their own business partners.

In Customizing, you can use a BAdI to influence how the business partners are reassigned. Choose **►Service ►Case Management ►Process Partners ►BAdI: Reassign Business Partners in Linked Objects ►**.

The BAdI contains the following methods:

- **GET_PARTNER_FCT_TO_CHANGE:** Receives the list of partner functions that can exist in the case; includes only the partner functions with the categories Sold-to, Person Responsible, and Organization Responsible
- **GET_OBJECT_TYPE_TO_CHANGE:** Returns the list of object types to process for changes to business partners; includes only the object types Activity Task, Business Activity, Service Contract, and Service Transaction
- **PROCESS_BPCHG_IN_CASE:** You can use this method to change the processor attribute of a case when the person responsible in the partner function is changed.
- **CHECK_BEFORE_BPCHG_FOR_TX:** You can use this method to decide whether or not specific, open, dependent objects of the case are updated.

i Note

The default implementation contains coding for only the first two methods.

Case Notes

Use

You can store information in the form of case notes within case processing. You can manually create case notes. The system also supports event-controlled, automatic generation of predefined case notes.

A case note consists of a short text and a long text. If you don't create a long text, the system uses the short text as long text information when you save a note. You can define text IDs in Customizing to categorize cases note according to their content.

i Note

Users who create or change case notes aren't recorded with a time stamp. Case notes are saved language-independently. The language specification for case notes is purely informative.

Prerequisites

- You've made the Customizing settings for case notes under **►Service ►Case Management ►Basic Settings ►Create Text Profile ►**.
- You've defined the relevant text profile for each case type in Customizing for Service under **►Case Management ►Basic Settings ►Define Case Types ►**.

Features

Automatic Creation of Case Notes

The RFC-enabled function module CRM_CMG_ADDNOTES supports the automatic creation of case notes. These automatic case notes are either triggered in relation to events within case processing or by applications that are connected to Case Management. For more information, see [Configuring Automatically Created Case Notes](#).

The module supports the following possible sources for automatic case notes:

- Case notes from T100 messages

In this connection, you can make full use of the features of messages, for example, the use of up to four message variables or language-dependency (see Maintaining Messages).

- Case notes from applications or systems

The import parameters IV_TDTITLE and IV_LONGTEXT enable you to transfer short and long texts from other Service applications. You can also use Remote Function Calls to import short and long texts from other systems.

The Business Add-In (BAI) CRM_CMG_NOTES_ADD supplements the function module. It enables you to control the automatic creation of case notes in relation to the case type.

Configuring Automatically Created Case Notes

Use

You can customize your system so that case notes are automatically created. You can either use T100 messages, or you can integrate a text editor if you wish to support manual input. You are responsible for implementing an appropriate text editor for this purpose.

T100 messages are already fully integrated in SAP systems and support functions such as multilingual capability and using variables.

Process

The following is the sequence of the activities you need to perform or consider when configuring automatically created case notes using T100 messages.

1. Create T100 messages

Create the required case notes as T100 messages in the customer name space.

i Note

To support multilingual case notes, create the additional language versions for each message (see [transaction SE63](#)).

2. Create text IDs

In Customizing for Service, choose  **Case Management**  **Extended Customizing**  **Create Text Profile**  **Create Text IDs**.

Create the text IDs you require for the text object SCMG_CASE. The text IDs are displayed as text types on the user interface for case notes and classify a case note according to its content.

Example

The standard system contains the predefined profile DEMO SAP with the text IDs [Description](#), [Internal Note](#), [Concluding Remark](#), and [Reply](#).

i Note

To support multiple languages, create the required language versions of the text IDs using the SAP translation tool (see [transaction SE63](#)).

3. Create text profiles and assign the text IDs they contain

In Customizing for Service, choose ► [Case Management](#) ► [Basic Settings](#) ► [Create Text Profile](#) ► [Create Text Profile](#). ►

4. Assign text profile to case type

In Customizing for Service, choose ► [Case Management](#) ► [Basic Settings](#) ► [Define Case Types](#) ►, and determine the relevant text profile for each case type. You can use the same text profile for each case type (n:1).

5. Call the function module CRM_CMG_ADDNOTES

Implement the call of the function module CRM_CMG_ADDNOTES at the desired point in time of an application. You can do this using a corresponding BAdI implementation or BRF event. To find out which exit you can use for this purpose, refer to the documentation for the respective application.

More Information

[Case Notes](#)

Customizing Case Note Attributes

Use

You can customize case note attributes to modify the standard structure and fields, or you can add two customer-defined fields. This is done using the standard structure delivered with Case Management CRMT_CMG_NOTES_GEN.

i Note

You can currently only add two fields to the user interface. If you create more than two attributes, the additional ones will not be displayed.

Activities

1. Run transaction SE11.
2. Select the [Data type](#) radio button, enter **CRMT_CMG_NOTES_GEN** and click [Change](#).
3. Under the [Components](#) tab create the attribute extensions and save your work.

More Information

[Case Notes](#)

[Configuration of Automatically Created Case Notes](#)

Defining a Case Closing Profile

The case closing profile lets you define how open, dependent transactions are to be handled that are in a case that is to be closed. In addition, you can define case closing profiles for specific business transactions that may differ from standard business transactions. After a case is closed, you can extract the case from the database and [archive the information](#).

The case closing profile contains a default reaction for all open transactions and reactions for open transactions of a specific transaction type. The default case closing profile is S_CLACT (**Do not close case if open transactions exist**). The case closing profile influences processing before the save when a case is closed (performing checks), and after the save when a case is closed (automatic closing). Depending on the profile and the case type, different system actions occur when the case is closed.

The action CRM_CMG_CASE_CLOSE is included as an example in the demo case action profile CRM_CMG_DEMO.

In Customizing choose ► **Service** ► **Case Management** ► **Basic Settings** ► **Define Case Closing Profile** ►.

After you have created the profile, you can assign it to a case type under ► **Service** ► **Case Management** ► **Basic Settings** ► **Define Case Types** ►.

Additional Activity

There is an Business Add-In (BAI) available that you can use to define additional tasks for the case closing profile. Tasks can be performed on the case or its dependent actions either at the check level or when processing the automatic close function. In Customizing choose ► **Service** ► **Case Management** ► **Basic Settings** ► **BAI: Define Additional Tasks for Case Closing Profile** ►.

Archiving Cases

Use

After cases have been closed or completed, you can archive them. There are several Business Add-Ins (BAIs) you can use to set up case archiving.

In Customizing choose ► **Service** ► **Case Management** ► **Extended Customizing** ► **Case Archiving** ►.

Using the available activities, you can:

- Define custom checks that run before a case is archived so that only cases meeting specific criteria are archived
- Archive (write) cases to the database directly
- Delete cases from the database once they've been archived

Central to the data archiving process is the archiving object. In Case Management, this object is called **SCMG**. For more information about the archiving process, see ► **Service** ► **Case Management** ► **Extended Customizing** ► **Case Archiving** ► **Case Data Archiving Process** ►.

Prerequisites

An SCMG object must also fulfill the following conditions to be archivable:

- The business transactions related to the case are in status **Closed** or **Completed**
- All the lower-level objects of the case are already archived

Linking Related Cases

You can link related cases that don't form a hierarchy. Normally in Case Management, cases are structured in a hierarchy where they're independent from each other. However, cases can contain similar data or features or have a common element. As a result you can link these cases to gain a better overview of related cases.

You can display all related cases and from the overview page, you can edit the list to add, modify, or delete a related case, as well as give a reason for linking it.

You can't set a link to cases that have been archived. However, you can edit nonarchived, related cases. To edit information, click the hyperlink associated to the case, which brings you to the active case information where you can edit it.

Multilevel Categorization in Case Management in Service

Use

Case Management in Service supports multilevel categorization, allowing you to assign cases to a category of a multilevel categorization schema. Such a category is called a "reference category".

For general information about multilevel categorization, see [Multilevel Categorization](#).

Integration

Multilevel categorization in Case Management in Service uses the Multilevel Categorization application. If you want to use document templates assigned to a reference category, you must first define document template profiles in Content Management. You can assign a profile to one or more categories in Service on the WebClient UI. Select the role **Service Professional** then goto ► [Home](#) ► [Service Operations](#) ► [Categorization Schemas](#) ► under [Search](#) or [Create](#).

If you want to use activity templates assigned to a reference category, you must define activity templates in Activity Management under [Sales](#). You can then assign the templates to one or more categories.

Activities

Setting up multilevel categorization for Case Management in Service consists of the following activities:

- [Creating Multilevel Categorization Schemas for Case Management](#)
- [Deploying Categorization Schemas for Case Management in Service](#)
- [Assigning Linked Objects](#)

The information provided here is specific to creating multilevel categorization schemas for case management.

More Information

[Using Multilevel Categorization Schemas in Case Management in Service](#)

Creating Multilevel Categorization Schemas for Case Management

Use

In Case Management in [Service](#), a multilevel categorization schema lets you do the following:

- Create a case using a case type that is assigned to a deployed categorization schema
- Assign a reference category to a case. Reference dates present different views on your categorization schema, depending on the validity periods of the single categories.
- Use activity templates assigned to this category to create new activities within the case
- Use the document templates of the document template profile assigned to this category to create new documents within the case

You create multilevel categorization schemas on the WebClient UI in [Service](#). Select the role [Service Professional](#) then goto ► [Home](#) ► [Service Operations](#) ► [Create](#) ► [Categorization Schemas](#) ►.

i Note

A categorization schema can be assigned to several case types, but a specific case type can be assigned to only one categorization schema.

You can perform your Customizing for [Service](#) by choosing ► [Cross-Application Components](#) ► [Multilevel Categorization](#) ►.

Deploying Categorization Schemas for Case Management in Service

Use

Only categorization schemas with status [Released](#) , [Deployed](#) or [Active](#) can be used in Case Management in [Service](#).

Integration

Multilevel categorization allows for changes to deployed categorization schemas to be reflected in different schema versions. Case Management in [Service](#) requires a new schema version whenever an existing categorization schema version undergoes changes to any of the following:

- Hierarchy structure, for example, adding new categories
- Validity intervals of single categories, that is, deactivation of categories by setting a new [valid to](#) date for a single category
- Linked objects of a category, depending on the setup of the authorization profile for authorization object **CRM_CATEGO**

New categorization schema versions cannot be released with a past [Valid From](#) date/time. When a new categorization schema version is released, the [Valid To](#) date/time of the currently valid schema version is adjusted to the [Valid From](#) date/time of the new schema version.

When a case is first assigned to a reference category, it is bound to the currently valid categorization schema version. For more information see the section [Setting a Schema Reference](#) in [Categorizing Cases in Case Management](#).

Defining Extended Attributes

Use

When you create categorization schemas in the WebClient UI in [Service](#) you can define extended attributes, such as validity periods, for categories.

Activities

- Defining validity periods

You can define validity periods on a single category level, making it possible to specify validity intervals that are independent of the technical validity period of the categorization schema. This functionality allows flexible activation of categories independent of the release time of the schema.

- Inheritance concept of validity periods

The validity period of a category is passed on to its subcategories. Additionally, it is possible to locally define category values. Therefore, the effective validity period of a single category is always a combination of its inherited values and the locally defined values. In the case of conflicting values, the more restrictive value is the valid one.

- Deactivation of validity periods

A category can be removed from a current schema version in the following two ways:

- Create a new schema version and delete the category from the new schema version.
- Create a new schema version and deactivate the category by changing its **valid to** date.

A deactivated category can no longer be used in newly created cases that have a reference date higher than the deactivation date. For more information, see the section [Maintaining a Reference Date](#) in [Categorizing Cases in Case Management in Service](#).

A deactivated category remains available for searching.

- Defining additional extended attributes

You can define category attributes for customer-specific requests.

Assigning Linked Objects

Case Management currently supports assigning activity template linked objects to categories. You can assign one or more activity templates (business activities or tasks) to a category to enable creating new activities.

Subcategories inherit the activity templates of their parent categories. You can also assign additional activity templates on a subcategory level.

For more information on Customizing settings for Multilevel Categorization in [Service](#), choose [Cross-Application Components](#) [Multilevel Categorization](#).

Using Multilevel Categorization Schemas in Case Management in Service

Use

You can use multilevel categorization to set up reference categories for use in Case Management.

The main business reason for assigning a reference category to a case is process control.

Prerequisites

- You have set up a categorization schema in the WebClient UI in [Service](#).
- You have assigned the categorization schema to a corresponding case type.
- You have released and deployed the categorization schema.

Activities

The following are either integral or possible activities associated with case processing when using multilevel categorization schemas:

- Categorizing the case to reference a specific schema version. For more information see [Categorizing Cases in Case Management in Service](#).
- [Searching for Cases by Reference Category](#).

Example

- A case worker can use the activity templates that are assigned to a reference category when creating new activities within the case.
- Assigning reference categories to cases enables companies to produce hierarchical reports based on the multilevel reference categories. The reports provide an accurate picture of how many cases have been created in the different business areas, as represented by the reference categories.

Categorizing Cases in Case Management in Service

Use

The first time a case is categorized, it receives a fixed date/time reference, which cannot be changed, to a specific schema version. This prevents inconsistent case data resulting from schema changes, such as new schema versions.

Before assigning a reference category to a case, you must maintain a reference date.

Both the [General Data](#) page block and the result lists contain a [Reference Category](#) field for assigning a reference category to a case. The field always displays the category label.

The fields that are displayed in the [General Data](#) page block are the result of Customizing. The result lists that are produced by search operations let you do blueprint personalization to define which available fields to display. Through blueprint personalization, you can also display the [Ref. Category ID](#) field.

Activities

- Setting a schema reference

The following table illustrates how the schema reference mechanism allows you to categorize cases that were created before a multilevel categorization schema was set up for Case Management in [Service](#).

Categorizing cases using the schema reference function

Prior to MLC schema	After MLC schema
1. You create a new case. The reference date/time of the schema is (internally) set to the point in time when the case is created. 2. You save the case without assigning a reference category to it. No schema reference date/time is set for this case.	1. You reactivate an existing case. The reference date/time of the schema is (internally) set to the point in time when the case is reopened. 2. You assign a reference category to the case. 3. You save the case. The schema reference date/time is persisted in the database; a case worker cannot change it in the future. Dependent linked objects can now be used during case processing.

- Maintaining a reference date

You can use the [Reference Date](#) field to predate a case to the past.

❖ Example

Public Service Scenario

A citizen calls the local municipality office to apply for sickness benefits. The reference date of this case is not the date of the call but the date when the citizen first became ill, two weeks earlier. The case worker at the municipality office can create a case whose reference date is in the past.

When assigning a reference category to this case, the case worker can only assign a category that has an effective validity period containing the reference date. These categories might be different from the categories that are valid when the citizen calls the municipality office.

The [Reference Date](#) field is initially set to the date when a case is created; however, the reference date can be changed during case processing.

- Assigning a reference category

You assign a reference category to a case in one of the following two ways:

- Manually entering either a category ID or a category label in the [Reference Category](#) field (combined input field). A wildcard input is currently not supported.

The category ID is checked first because it is always unique in a hierarchically structured categorization schema.

- Using F4-help (only available for the [Reference Category](#) field)

When you call the F4-Help for the [Reference Category](#) field, the case type and the reference date are transferred to the F4-Help to define the limits of the search.

The F4-help provides two different kinds of search:

- Structure Search

Clicking [Go](#) without entering any further search criteria for the [Ref. Category ID](#) and [Reference Category](#) fields, displays a hierarchical tree that reflects the valid schema version for the current case. You can expand the different hierarchy levels and search for the category.

Selecting the corresponding category line in the hierarchy transfers the category information back to the Case Management application.

■ Category Search

If you already know the label or parts of the ID of the category you want to assign to your case, you can use either the [Reference Category](#) or [Ref. Category ID](#) field to do a wildcard search on the categories contained in the schema version that is valid for the current case.

Clicking [Go](#) displays a flat list of the categories that fit your search criteria.

Selecting the category that should be assigned to the case transfers the category information back to the Case Management application.

Changing the reference date within the F4-help produces a result list that contains only categories **that** are valid for that date. After you choose a category, the reference date is also transferred back to the Case Management application.

Whenever a reference category-relevant field—the reference category itself or the reference date—is changed, a categorization check is executed to see whether or not the entered/deleted values invalidate the case. For example:

- A category ID is entered manually but the category is not valid for the entered reference date
- The reference date is changed but the assigned reference category is no longer valid for this reference date
- A category value has been entered without first specifying a reference date

Searching for Cases by Reference Category

Use

You can search for cases by specifying the [Ref. Category ID](#) field as the selection criteria within the [Advanced Search](#) tool of Case Management in [Service](#).

In addition, selecting the checkbox [Incl. Subcategories](#) lets you search for cases assigned to any sub-categories of the category defined in the [Ref. Category ID](#) field.

Caution

If the number of subcategories is significant, the search may not be executed within a reasonable time frame.

The [Ref. Category ID](#) field supports an F4-help, which searches for the reference category that should be used for the case search. This F4-help is similar to the F4-help described in the section [Assigning a Reference Category](#) in [Categorizing Cases in Case Management in Service](#).

Only currently valid schema versions are used to search for the category; old schema versions are not displayed here.

Using Additional Object Types with Case Management in Service

Use

The case record model specifies which object types can be linked to a case and how these objects are structured. You must assign case record models to a case type. In Customizing, choose [Service](#) > [Case Management](#) > [Extended Customizing](#) > [Special Settings](#) > [Define Processes](#) > [Define Process-Dependent Parameters](#).

You can specify per case type which object types are used and how these object types are structured. You use case records to define the usage and structure of object types. Before you can use a case record, however, you need a **case record model**.

Depending on the business process and the scenarios, the case record model has a different structure and different object types.

The service provider and the element type are central to SAP Records Management and record models in Service.

- **Service provider:** For each business object type that you want to integrate into this framework, you must define and implement a service provider. If a service provider already exists for an object/element – for instance, there's a service provider for business object types – you can reuse the service provider. You therefore don't have to implement a separate service provider for each business object type.
- **Element type:** The element type is a form of service provider and is the starting point for users to integrate elements/objects into the case record. A service provider can only be used when at least one element type for this service provider exists. Each element type belongs to only one service provider. A service provider can have any number of element types. You create and edit element types in Customizing.

Example

In standard Customizing, the service provider SCMG_SP_BOR is delivered. It supports the integration of business object types into the framework of SAP Records Management. For this service provider, the element type CRM_BUSINESS_PARTNER is shipped with the standard delivery. When you use this element type, it specifies that only business partners can be integrated into the framework. You can use an element type to specify that objects from external systems can be integrated.

The elements of a case record model can be hierarchical, with the following elements available:

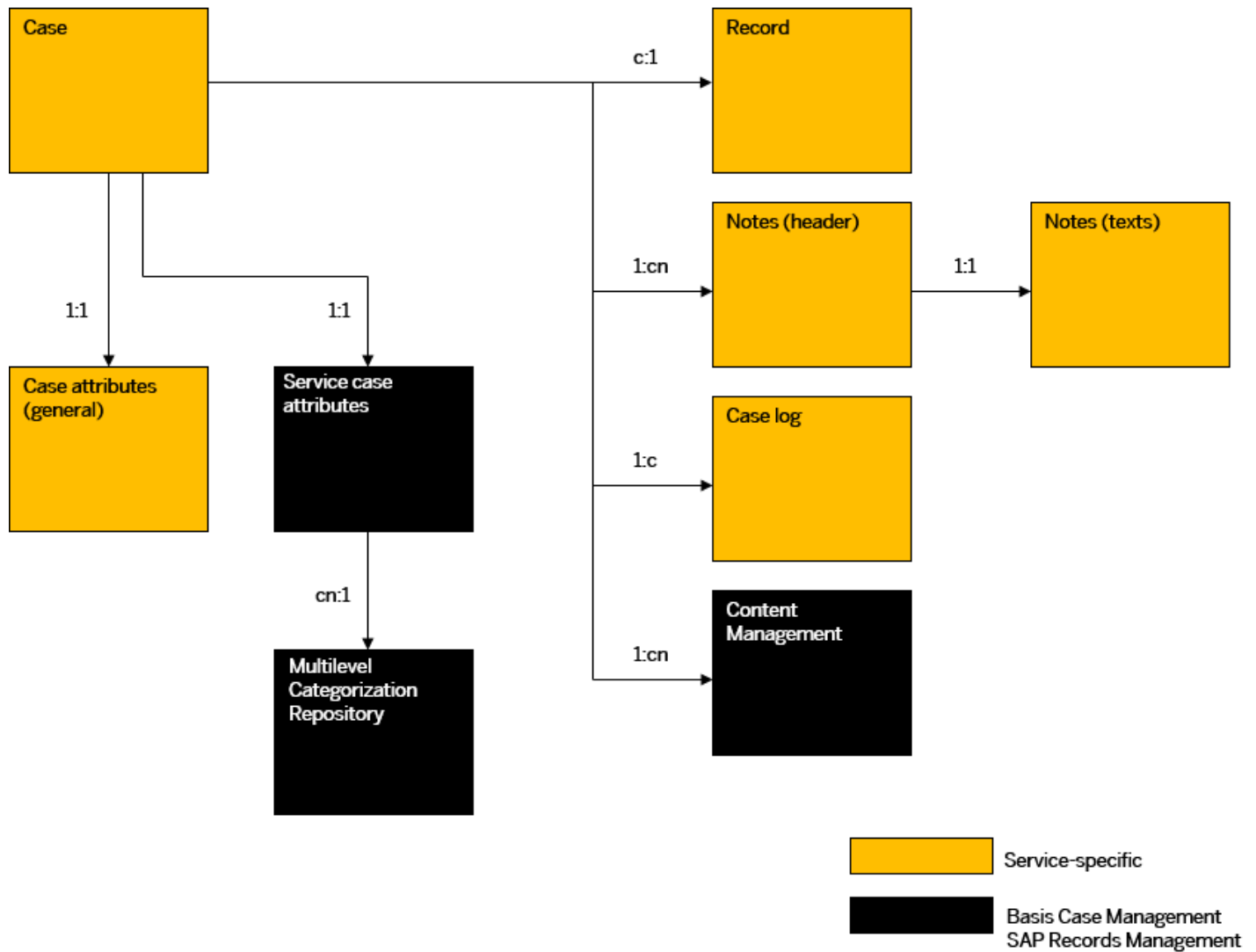
- **Structure node:** Structure nodes act as headers for other nodes and enable you to structure case records. You can't assign elements to structure nodes.
- **Model node:** You can assign elements in case records to model nodes. Model nodes act as placeholders for the objects that can be assigned later. For each model node, you specify the possible object types (for example, business partner, product) and how many of these objects can be assigned.
- **Instance node:** An instance node already has an element/object assigned to it in the record model. There can only be one instance node in a case record. Using instance nodes, you can define information that must be provided in every case record.

More Information

[Creating a Custom Case Record Model](#)

Architecture of Case Management in Service

Case Management in Service is an extension of Basis Case Management, which in turn uses the SAP Basis Records Management. The following diagram illustrates the specific components of a case in Service.



Case Management in Service – Architecture

The [Case](#) in Service consists of attributes such as ID, short text, status, priority, and other attributes, as well as of several subcomponents. These subcomponents are the case record, case notes, case log, and the integration into the document management system.

Case Attributes

A case consists of attributes from the Basis case and can be enhanced with Service attributes. Basis Case Management allows for other applications, for example, Case Management in Service, to add application-specific attributes. To be able to use this feature, you must set an additional table in the attribute profile in Case Management Customizing. In Service, the table [CRMD_CMG_CASE_AT](#) is used for this task. As an additional attribute in Service, only the attribute [Short text in upper case](#) is used. This additional attribute causes upper case short texts to be stored twice – once in a Basis case table and once in the Service table. This feature enables non-case-sensitive searches.

Creating a Custom Case Record Model

Context

The purpose of this procedure is to create a custom case record model on the basis of existing business object types. You create a custom case record model by copying the default element types. Element types are defined in the registry of SAP Records Management.

Procedure

1. Open the registry using transaction **SRMREGEDITC** and choose **S_AREA_CMG**.
2. Create a new element type for the case record itself:
 - a. Expand the application area **S_AREA_CMG**.
 - b. Expand **SCMG_SP_CASE_RECORD** and right-click on the element type **CRM_SPS_CASE_RECORD** and choose **Copy** from context menu.
 - c. In the **Copy To** field, enter an element type ID and a short description. Save your data.

i Note

Using this procedure, you can copy other element types. You can find the element types for the business object types under the service provider **SCMG_SP_BOR**. For each business object type that you want to use, you have to create a separate element type – preferably through copying. After you perform these steps, you can create the case record model using the Records Modeler.

3. Run transaction **SCASE** and choose the Records Management System (RMS) ID. For Case Management in Service, use **CRM_CMG_DATA**.
4. Open the folder **Record and Case Record Models** and right-click the entry **Modeler for CRM Case Records**, and then choose **Create**. A case record model with the default entry **Worklist** as the structure node and **Worklist Element** as the model node now appears in the work area.
5. Change these default entries and create new nodes to set up the model for your own requirements by choosing the **Worklist Element**. The element details are displayed on the right of the screen. For structure nodes, you only define the description. For a model node, you must maintain the following properties:
 - **Description**, for example, **Service Order**
 - **Element Type**: By setting the element type, you determine which object types you can insert into the case record at this point, and can then also be controlled. After you save, the element types created under Step 2 can be used.
 - Number of objects that can be integrated: **At Least** and **Maximum** fields. If an undetermined number of objects can be integrated into a case, enter an asterisk (*) in the field **Maximum**.
 - **Relationship**: This value specifies the relationship between the element/object and the case record. To select the relation, use the F4 help.
 - Attribute **ANCHOR**: This attribute is used to insert objects/elements into a case by using the Basis Case Management API. Since Case Management in Service uses this API, you must maintain this attribute. Use the same value as the one for the corresponding relation. For example, the ID of the relation service order is **CS**, so the corresponding attribute **ANCHOR** must have the value **CS** as well.
6. After the case record model has been created, save your work, and then activate the hierarchy nodes.
7. Release the case record model by clicking the **Model** button, then from the list choose **Change Status** and select **Status Released**. Enter a short text for the case record model as well as a unique ID. You can generate a unique name by clicking the icon at the end of the field.
8. Now, you insert the model within the element type that you created. You do this in the Records Management registry. Run transaction **SRMREGEDITC** and choose **S_AREA_CMG**.
9. Right-click on the element type that you created previously, and in the context menu choose **Change**. Under **Connection Parameter Values**, choose the parameter **MODEL_ID** and click **Change**. Enter **CRM_CMG_DATA** and then press **Enter**. Search for the case record model you created using the F4 help, choose it from the list and confirm the entry. Save the settings.
10. To be able to use the case record model, you must assign both the record model and the element type you created to the case type that you want to use. You do this in Customizing under **Service > Case Management > Basic Settings > Define Case Types**.
11. After creating a new element type, you assign it to the process **C_01**. You do this in Customizing under **Service > Case Management > Extended Customizing > Special Settings > Define Processes > Define Process-Dependent Parameters > Assign Element Types and Business Object Types to a Process**.

Results

You've created a custom case record model and element type that you can use in Case Management.

Integrating Business Object Types With the Case Record Model

Use

You can integrate additional business object types with Case Management in Service. To do this, you must create a new element type.

Prerequisites

You can create element types in Records Management. To create a new element type, you must perform the steps relating to element types described in [Creating a Custom Case Record Model](#).

i Note

Objects and/or elements are integrated with a case using relations or anchors. For each object type, a separate relation exists, for example, the relation and anchors C1 for the business object type **Business Partner**. The same also applies to new object types. As creating new relations is a modification, we recommend that you use the relations X1, X2, X3, X4, X5, X6, X7, X8, and X9.

Process

1. Run transaction **SRMREGEDIT**.
2. Choose **S_AREA_CMG**.
3. Right-click on the service provider **SCMG_SP_BOR** and choose **Create Element Type**.
4. Enter an element type ID and a short text and then press .
5. Select whether the object is local or transportable and save your work. If it's transportable, enter a transport package.
6. Right-click on the element type you created and choose **Change**.
7. Under the **Connection Parameter Values** tab, set the values for **BOR_OBJECT_TYPE** to the desired business object, and leave the setting for **LOGICAL SYSTEM** as **NONE**.
8. Go to the **Classification** tab and maintain the parameter IDs using the values for **TYPE** and **RMS_ID**.
9. Save your changes.

Result

You've created and integrated the element type. To use the new business object type in cases, you must create a new model node for the element type in the case record model you're using.

Additional Customizing Configuration for Case Management in Service

The following is a list of Customizing configuration activities using BAdIs that you can use to enhance Case Management in Service.

- Integrating customer-specific validations using BAdIs SCMG_VALIDATE_C and SCMG_VLDT_BFR_STR_C
- Specifying default field values when creating a case. There are two scenarios: Creating a case from a case management application using BAdI SCMG_INITIALIZE_C, or creating a case from a business transaction using BAdI CRM_CMG_ON_CREATION
- Storing additional link attributes using the enhancement spot /CRMS/CMG_REC_ATTR

For more information in Customizing, choose ► [Service](#) ► [Case Management](#) ► [Extended Customizing](#) ► [Special Settings](#) ► [Define Processes](#) ► [Business Add-Ins \(BAdIs\)](#) ►.

Integrating Multilevel Categorization

Use

Multilevel categorization enables you to individually design the categorization and integration of specific functions in your applications. Case Management in Service supports the multilevel categorization function, making it possible to assign a case to a category of a multilevel categorization schema.

More Information

[Multilevel Categorization in Case Management in Service](#)

[Multilevel Categorization](#)

Additional BAdIs for Case Management in Service

Use

There are various additional enhancements and Business Add-Ins (BAdI) that you can use for Case Management in Service.

Features

- Inserting new object types in a case record
- Initializing case attributes
- Changing a case before the check
- Additional checks to validate cases and case attributes
- Triggering an event after saving a case

In the Customizing choose ► [Service](#) ► [Case Management](#) ► [Extended Customizing](#) ► [Special Settings](#) ► [Define Processes](#) ► [Business Add-Ins \(BAdIs\)](#) ►.

Postprocessing Actions for Cases

Using actions (in the Post Processing Framework) you can schedule and start any postprocessing steps depending on the case type, as well as the various scheduling and start conditions. Examples of postprocessing steps are sending e-mail or starting workflows. You can also stipulate that a specific ABAP coding is executed.

Actions

The actions you can perform are:

- Creating an action profile with action definitions; in the action definitions, you define the action characteristics, such as the processing time or whether the action is modifiable in the document (case)
- Defining conditions for the action definitions (scheduling and start conditions)
- Assigning the action profile to the desired case type
- Assigning the partner determination for actions using a Business Add-In (BAI)

For more information, in Customizing choose [Service](#) ► [Case Management](#) ► [Extended Customizing](#) ► [Process Actions](#) ►.